Design, Fabrication, and Control of Soft Robots with Fluidic Elastomer Actuators

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Submitted to the Department of Electrical Engineering and Computer Science
in partial fulfillment of the requirements for the degree of

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Abstract

The goal of this thesis is to explore how autonomous robotic systems can be created with soft elastomer bodies powered by fluids. In this thesis we innovate in the design, fabrication, control, and experimental validation of both single and multi-segment soft fluidic elastomer robots. First, this thesis describes an autonomous fluidic elastomer robot that is both self-contained and capable of rapid, continuum body motion. Specifically, the design, modeling, fabrication, and control of a soft fish is detailed, focusing on enabling the robot to perform rapid escape responses. The robot employs a compliant body with embedded actuators emulating the slender anatomical form of a fish. In addition, the robot has a novel fluidic actuation system that drives body motion and has all the subsystems of a traditional robot on-board: power, actuation, processing, and control. At the core of the fish's soft body is an array of Fluidic Elastomer Actuators (FEAs). The fish is designed to emulate escape responses in addition to forward swimming because such maneuvers require rapid body accelerations and continuum body motion. These maneuvers showcase the performance capabilities of this self-contained robot. The kinematics and controllability of the robot during simulated escape response maneuvers are analyzed and compared to studies on biological fish. During escape responses, the soft-bodied robot is shown to have similar input-output relationships to those observed in biological fish. The major implication of this portion of the thesis is that a soft fluidic elastomer robot is shown to be both self-contained and capable of rapid body motion.

Next, this thesis provides an approach to planar manipulation using soft fluidic elastomer robots. That is, novel approaches to design, fabrication, kinematic modeling, power, control, and planning as well as extensive experimental evaluations with multiple manipulator prototypes are presented. More specifically, three viable manipulator morphologies composed entirely from soft silicone rubber are explored, and these morphologies are differentiated by their actuator structures, namely: ribbed, cylindrical, and pleated. Additionally, three distinct casting-based fabrication processes are explored: lamination-based casting, retractable-pin-based casting, and lost-wax-based casting. Furthermore, two ways of fabricating a multiple DOF manipulator

are explored: casting the complete manipulator as a whole, and casting single DOF segments with subsequent concatenation. An approach to closed-loop configuration control is presented using a piecewise constant curvature kinematic model, real-time localization data, and novel fluidic drive cylinders which power actuation. Multi-segment forward and inverse kinematic algorithms are developed and combined with the configuration controller to provide reliable task-space position control. Building on these developments, a suite of task-space planners are presented to demonstrate new autonomous capabilities from these soft robots such as: (i) tracking a path in free-space, (ii) maneuvering in confined environments, and (iii) grasping and placing objects. Extensive evaluations of these capabilities with physical prototypes demonstrate that manipulation with soft fluidic elastomer robots is viable.

Lastly, this thesis presents a robotic manipulation system capable of autonomously positioning a multi-segment soft fluidic elastomer robot in three dimensions while subject to the self-loading effects of gravity. Specifically, an extremely soft robotic manipulator morphology that is composed entirely from low durometer elastomer, powered by pressurized air, and designed to be both modular and durable is presented. To understand the deformation of a single arm segment, a static physics-based model is developed and experimentally validated. Then, to kinematically model the multi-segment manipulator, a piece-wise constant curvature assumption consistent with more traditional continuum manipulators is used. Additionally, a complete fabrication process for this new manipulator is defined and used to make multiple functional prototypes. In order to power the robot's spatial actuation, a high capacity fluidic drive cylinder array is implemented, providing continuously variable, closed-circuit gas delivery. Next, using real-time localization data, a processing and control algorithm is developed that generates realizable kinematic curvature trajectories and controls the manipulator's configuration along these trajectories. A dynamic model for this multi-body fluidic elastomer manipulator is also developed along with a strategy for independently identifying all unknown components of the system: the soft manipulator, its distributed fluidic elastomer actuators, as well as its drive cylinders. Next, using this model and trajectory optimization techniques locally-optimal, open-loop control policies are found. Lastly, new capabilities offered by this soft fluidic elastomer manipulation system are validated with extensive physical experiments. These are: (i) entering and advancing through confined three-dimensional environments, (ii) conforming to goal shape-configurations within a sagittal plane under closed-loop control, and (iii) performing dynamic maneuvers we call grabs.

Thesis Supervisor: Daniela Rus
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Chapter 1

Introduction

1.1 Vision

As roboticists, we often use nature as inspiration for the way robots should look, act, or think. For example, we have robots that attempt to reason like humans, run like cheetahs [Park et al., 2014a], grasp with the dexterity of human hands [Deimel and Brock, 2014], and fly with the agility of birds [Moore et al., 2014]. Accordingly, we have a tendency to benchmark the performance of these robotic systems against their biological counterparts. The manufacturing industry has demonstrated the ability of robots to outperform humans when tasks are well-defined, uncertainty is negligible, and the environment is sufficiently controlled. However, outside of these conditions the capabilities of robots are often underwhelming with respect to nature. From a technical perspective, there are many reasons for this apparent performance discrepancy (e.g. limitations in design, fabrication, sensing, control, and motion planning). One salient difference between the majority of current robots and natural systems is the degree of body elasticity, and soft roboticists believe this material mismatch may be a significant technical barrier inhibiting robots from reaching their full potential [Trimmer, 2014, Majidi, 2014]. Natural systems frequently leverage body elasticity to resiliently accommodate environmental variation (Fig. 1-1a), passively conform to spatial uncertainty (Fig. 1-1b), and continuously deform during dexterous tasks (Fig. 1-1c). The goal of this thesis is to explore how autonomous robotic systems

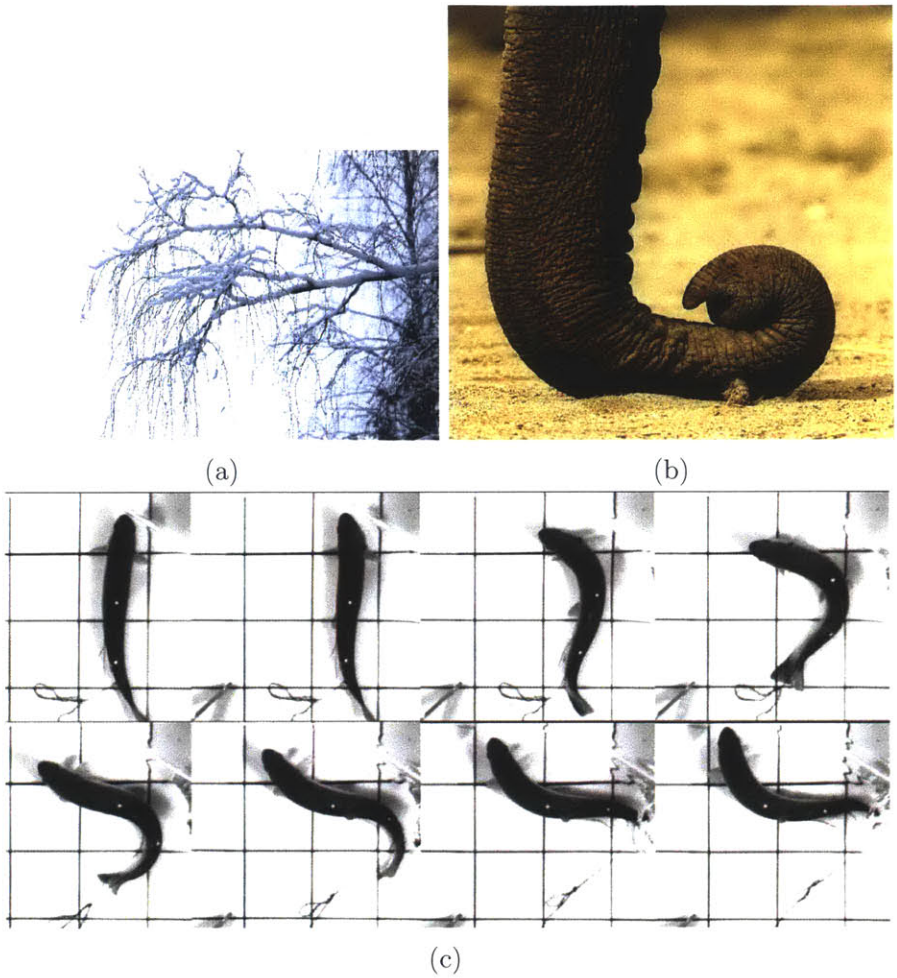


Figure 1-1: Nature utilizes body elasticity to: (**a**) resiliently accommodate environmental variation as illustrated by a tree branch bending to accommodate heavy snow, (**b**) passively conform to spatial uncertainty as shown by an elephant's trunk conforming to flat ground, and (**c**) continuously deform during dexterous tasks exemplified by a fish contorting its body during an escape-response. The image in (**a**) is attributed to Ville Turkkinen of Tampere, Finland and licensed under Creative Commons Deed CC0. The image in (**b**) is "An elephant trunk" attributed to Jenny Downing of Geneva, Switzerland and licensed under Creative Commons Attribution 2.0 Generic. The image at (**c**) is reproduced with permission from Figure 1 A of Goldbogen et al. [2005]

can be designed to also incorporate and leverage *softness*. To do this, we develop extremely soft robot morphologies that are radically different from today's mainstream rigid-body robotic platforms in an effort to break the mold on how we think about designing, fabricating, and controlling such systems. That is, we build robotic sys-

tems with bodies made entirely from soft silicone elastomer and power these bodies with pressurized compressible fluids; the robots in this thesis have approximately five orders of magnitude, or 100,000 times, greater inherent elasticity than traditional rigid-body robots (please refer to Fig. 1-2). These robots serve as archetypical soft autonomous systems. By creating radically different platforms we can begin to solve hard problems arising from the introduction of deformable materials into autonomous systems and thus inform a future where robots are destined to be softer.

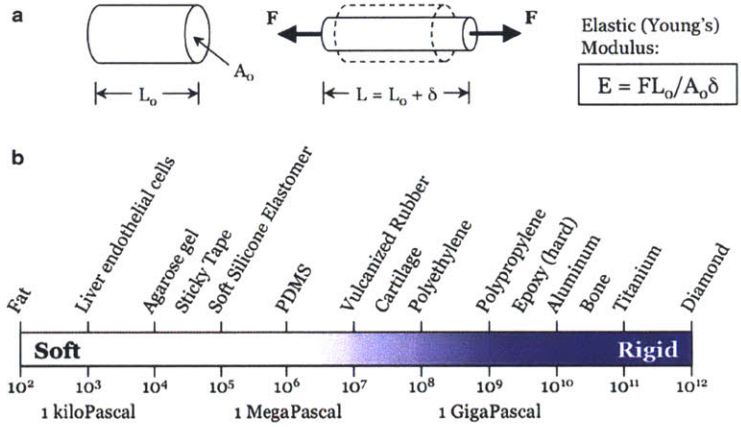


Figure 1-2: (**a**) "The elastic (Young's) modulus scales with the ratio of the force F to the extension d of a prismatic bar with length L_0 and cross-sectional area A_0. (**b**) Young's modulus for various materials (adapted from Autumn et al. [2006])." Reprinted with permission from SOFT ROBOTICS, Volume 1, Issue 1, 2014, pp. 5-11, published by Mary Ann Liebert, Inc., New Rochelle, NY. [Majidi, 2014].

1.2 New Capabilities

1.2.1 Safer Interactions

Imagine a future where robots work alongside humans to cooperatively perform tasks [Edsinger and Kemp, 2007]; safety becomes an immediate concern [Markoff and Miller, 2014]. Although industrial-style manipulators have been transformative for structured repetitive tasks, these robots are often considered too rigid for human-centered environments where the tasks are unpredictable and the robots have to ensure that their interaction with the environment and with humans is safe. At the moment, robots

are isolated from humans and confined to operate behind guarding in industrial environments. Nevertheless, roboticists are constantly balancing the competing goals of safety and performance [Wyrobek et al., 2008]. Much research is aimed at equipping such hard robots with soft capabilities [De Santis et al., 2008]. For example, the inclusion of compliant transmissions function to decouple actuator and link inertia when necessary to minimize collision forces [Bicchi and Tonietti, 2004]. Common approaches to variable-impedance actuation, reviewed by Vanderborght et al. [2013], include series elastic actuators [Pratt and Williamson, 1995] and variable stiffness actuators [Tonietti et al., 2005]. However, despite these safer design morphologies, robots are still fundamentally composed of rigid components and rely on control software to guarantee safety if collisions with humans or environments occur. Soft robots offer an alternative approach. By incorporating highly deformable materials, soft robots offer the potential for mechanical compatibility between robots and humans and this offers better safety margins, as articulated by Lipson [2014]. The time is ripe for inherently soft machines.

1.2.2 Mitigating Uncertainty

Roboticists have optimal, time-tested solutions when tasks are well-defined and a machine's motions and interactions with its environment are predictable. However, outside of structured environments, robots must constantly deal with uncertainty. For example, if a humanoid robot were to misperceive a flight of steps within a residential home, it will likely fall and require a costly repair. Commonly we rely on tools such as a suite of sensors [Kammel et al., 2008], state-estimation [Smith et al., 1990], Bayesian models [Cassandra et al., 1996], robust controllers [Tedrake, 2009], and robust probabilistic reasoning [Thrun et al., 2006] to mitigate uncertainty. These are all very good but computationally complex solutions. An alternative approach is to develop robust and durable machines that can mitigate some of this uncertainty at the hardware level. Autonomous systems can offload computational complexity to mechanical components by incorporating soft, elastic materials into their structure. It is possible to then combine these machines with relatively simple models and control

algorithms to achieve performance.

1.2.3 Continuous Deformation

What if robots could exhibit the dexterity and rapid continuum motion displayed by natural creatures? For example, fish can perform escape responses, or energetic bursts characterized by rapid accelerations (16 - 151 m s^{-2}) over very short durations (30 - 210 ms). This often involves the fish's body initially bending into a "C" shape exceeding 100 degrees [Domenici and Blake, 1997]. Among vertebrates, these are some of the most rapid maneuvers [Jayne and Lauder, 1993]. Although biomimetic robots with finite degree-of-freedom (DOF) bodies and elctro-mechanical actuators show promising capabilities, they often cannot match the speed nor the dexterity of their natural counterparts. Such approaches only approximate naturally continuous body motion with multiple discrete links separated by fixed joints. Soft robots offer the potential to lift the limitations imposed by rigid-body kinematics, as their bodies can deform continuously under actuation. Furthermore, fluid energy can be stored and subsequently released directly into soft actuators without a costly energy conversion stage. These features make soft robots well-suited to emulate the kinematics and dexterity displayed by some natural systems.

1.2.4 Natural Form

Soft materials and fabrication processes allow soft robots to realize complex, amorphous forms [Lipson, 2014]. It is prohibitively difficult to realize naturally occurring features such as continuously varying spatial surfaces and internal non-convex cavities with rigid materials and standard power transmission components. Soft materials can be casted into arbitrary shapes using similar processes to that which an artist uses to create sculptures. Fluidic channels can be continuously embedded throughout these soft machines to provide form-independent power transmission and actuation. Such soft technologies profoundly expand the robotics community's ability to emulate complex biologically inspired morphologies.

1.3 Challenges

Although soft robots offer a promising range of new capabilities, there are surprisingly few soft machines, and even fewer soft autonomous systems. What are the technical challenges inhibiting the growth of soft robotics? To begin answering this question we can look at recent reviews of the field. As Trivedi et al. [2008] notes, soft robots are designed with either a continuously deformable backbone or no backbone at all, and although this feature provides these robots with theoretically infinite degrees of freedom, it presents a variety of technical challenges. To paraphrase Trimmer [2014], the engineering community lacks experience working with highly deformable materials. Our current tools are well-suited for applications using rigid materials; soft, nonlinear materials break many of the underlying assumptions. To paraphrase Lipson [2014], eliciting benefits from soft robots is difficult for several reasons: (i) we lack computational tools that can simulate the many DOF and nonlinear effects of soft materials; (ii) we have limited intuition when designing soft systems and few automated tools at our disposal; (iii) soft actuation methods are relatively inefficient, specifically pneumatic systems require substantial supporting hardware; (iv) the low structural impedance introduced by soft materials makes feedback control difficult and new control strategies are likely needed; (v) lastly, manufacturing processes are tailored for rigid rather than soft systems, and standardization of components is very challenging.

In short, the softer we make robots the less predictable their motions become. To combat this, we need to address technical challenges on many fronts. Specifically, this thesis addresses challenges arising in the areas of (i) device design, (ii) manufacturing processes, (iii) kinematic and dynamic modeling, as well as (iv) algorithms for control and planning. By studying extreme examples of soft robots, i.e. ones made entirely from soft elastomer and powered by fluids, this thesis begins to identify appropriate morphologies, fabrication processes, motion models, computational tools, and control strategies for a growing class of robots that are designed to incorporate *softness*.

1.3.1 Devices

The concept of using very soft elastic materials to construct autonomous robots is relatively new compared to and radically different from the time-tested form and structure of traditional robotic systems. As soft roboticists, we are in the process of defining long-lasting morphologies for soft machines. Accordingly, a considerable amount of innovation is required to design functional soft machines as well as mechanisms for driving their actuation.

Performance and autonomy are competing goals in soft mobile fluidic elastomer robots. Some fluid-powered soft machines show promising capabilities like walking [Shepherd et al., 2011] and leaping [Shepherd et al., 2013a] but are primarily driven by cumbersome external hardware limiting their practical use. Conversely, there are instances of self-contained fluidic soft robots [Onal et al., 2011] [Onal and Rus, 2013]; however, because of the constraints imposed by bringing all supporting hardware onboard, the performance of these robots is severely limited when compared to rigid-bodied robots. Accordingly, one of the primary technical challenges addressed by this thesis work is:

> *How do we advance soft-bodied fluidic robots to be capable of rapidly achieving continuum body motion while simultaneously being self-contained?*

Next, we address device challenges associated with soft manipulation. Although in this design space we can relax the constraint of on-board supporting fluidic hardware, we encounter the competing goals of task precision and body compliance. In general, the designs of existing soft position controlled manipulators are not very soft. A fundamental limitation in designing robots to be softer and more compliant is that the robots become increasingly unconstrained, making predictable and controlled movement difficult. In more traditional manipulator morphologies there is a balance between compliance and internal kinematic constraints that make controlled movement feasible; however, in soft robots low durometer elastic materials effectively lower the systems' structural impedance.

> *What is the morphology of a manipulator whose body and actuators are*

composed entirely of soft elastomer but that is used for tasks requiring task-space control?

Even if we can devise an appropriate manipulator morphology, what mechanisms do we use to drive its actuation? In order to continuously vary the curvature of a soft fluidic elastomer robot, input fluid energy needs to be continuously varied. Most soft robots use valves to pressurize and sequence their actuation. A common strategy is to rely on the fluidic actuator's relatively long time constant in combination with high frequency valve switching to approximate continuous fluid delivery. This strategy falls within the domain of morphological control (see Section 2.2.2) and is fundamentally limited by the fact that it uses a discrete pulse width modulation approach to control continuous motion. Furthermore, this strategy can be prohibitive as it is difficult to recover fluid energy once it is delivered to the actuator. To enable precise curvature control for soft fluidic elastomer robots this thesis addresses the following challenge:

How do we deliver continuous closed-circuit fluid flow to a soft robot in order to enable continuum configuration control?

1.3.2 Hardware Processes

Such radically different robot morphologies cannot be built using the same processes by which engineers build traditional, rigid-body robots. Innovative approaches to fabrication are required in order to build robots composed of soft rubber and deform under fluid pressure. In the absence of fasteners, hard chassis, mechanical linkages, and other standardized components we generally rely on casting and lamination processes to realize soft robots. Cho et al. [2009] review manufacturing processes for soft robots and provide only one reference to the use of embedded molding by Dollar and Howe [2006]. Perhaps this is testament to the novelty of such processes for robotics. More recent work by Correll et al. [2010] and Onal and Rus [2012] suggests this process is well-suited for creating fluidic elastomer robots. However, before such robots can attain mainstream usage, it is necessary that we build on these contributions and devise repeatable and general methods for constructing soft machines.

Traditionally, roboticists sequentially construct robots. First, the frame of a robot is built and then components (e.g. motors, gears, pulleys, cables, etc) are installed on the frame to provide actuation. Fluid-powered soft robots provide the unique challenge of requiring the robot's body and actuators to be integrated, both in form and function, into one seamless system. This means at the time of forming the body we must simultaneously form actuated regions that have specific material properties and geometric profiles. The fabrication constraints and requirements of the actuators must fit within the fabrication constraints of the body such that both sets of constraints can be simultaneously satisfied. Accordingly, this thesis addresses the technical challenge of:

> *How do we continuously integrate and embed fluidic actuators throughout a soft-bodied robot?*

In order for soft robots to migrate from research environments to real-world operations, we must also devise a way for their bodies to take task-specific, three-dimensional forms. As within other engineering disciplines, form-function relationships are important in robotics. For example, in the case of building biomimetic robots, it is often necessary to emulate the anatomical form of the robot's natural counterpart to achieve proper functionality (i.e. a fish needs a slender form to reduce hydrodynamic forces such as drag). Additionally, in the case of robotic manipulators, it is often necessary to adjust the manipulator's mass, volume, and shape link by link to accomplish certain manipulation tasks (i.e. a base link may be larger than a distal link to minimize the effects of gravity). A major technical challenge addressed in this work is:

> *How do we produce soft elastomer bodies that take on task-specific, three-dimensional forms by means of casting and lamination processes?*

As we desire more functionality from a soft robot, we inevitably need to add more actuated DOF to their bodies. Such functional requirements increase the robot's kinematic capabilities but also add considerable complexity to the fabrication process. For example, we need ways to independently supply fluid to each actuator within each

body segment while not artificially constraining the robot's spatial mobility. The process is analogous to adding multiple integrated circuits (ICs) to a printed circuit board. Here, the board designer must route traces to each IC in order to supply power and connect signals while minimizing the board's overall footprint. It follows that, a challenge addressed by this thesis is:

> *What is a scalable approach to fabricating multi-body soft fluidic elastomer robots?*

1.3.3 Models

In a 2008 review by Trivedi et al. [2008] the challenges associated with modeling soft robots were articulated:

> "Accurate control of soft robots requires model-based prediction of the set of possible configurations. Dynamic models that accurately describe large-scale deflections of soft robots and cover their entire workspace are currently too complicated to be used for control. Current control approaches, based on simpler models, are not guaranteed to be stable or effective for large deflections (Gravagene et al. 2001). Also, including distributed forces such as gravity, and structural stability of multiple section robots into control schemes is a challenging problem."

This problem is further compounded by the fact that the soft robots in this thesis have body segments composed entirely from low durometer elastomer and are actuated by fluids. This means the bodies of these soft robots undergo large and continuous circumferential and longitudinal deformation due to the low elastic modulus of their material composition. Accordingly, a primary technical challenge addressed in this thesis is:

> *What is an appropriate static model for the large-scale elastic deformation of a soft fluidic body segment?*

Next, in order to autonomously and accurately perform tasks such as point-to-point movements, pick-and-place operations, and trajectory following we must develop reasonably accurate multi-segment kinematic models of multi-body soft fluidic elastomer robots. Forward and inverse kinematic models are vital to virtually all manipulation motion primitives. Although these models must be accurate enough to capture the complexity of a highly compliant and highly deformable multi-segment manipulator, they must be simple enough to implement in real-time control routines. Consequently, this thesis addresses the following technical challenges:

> *What are appropriate approaches to modeling the forward and inverse kinematics of multi-segment soft fluidic elastomer manipulators?*

As articulated by Trivedi et al. [2008], to really make the concept of soft robotics a game changer we have to be able to model the dynamics of multi-body soft robots subject to gravity. This problem is exemplified by this thesis because again we are working with robots on the extreme soft end of the "soft" robotics spectrum.

> *What are appropriate approaches to modeling the dynamics of multi-segment soft fluidic elastomer manipulators?*

1.3.4 Algorithms

Soft robots are in need of automation. In order for soft fluidic elastomer robots to autonomously perform tasks we need to first develop appropriate motion control and planning algorithms for these robots. The extreme elasticity, body compliance, and fluidic power of this class of soft robots makes developing such algorithms a challenge.

A fundamental requirement for automating the aforementioned soft robots is both open-loop and closed-loop control of body segment curvature. In the context of soft fluidic elastomer robots, open-loop control techniques are well-suited when desired body motions are required to be fast but not necessarily precise, whereas closed-loop techniques are favorable when desired body motions are required to be precise but not necessarily fast. Thus, this thesis must address the following challenge:

How do we provide open-loop and closed-loop body segment curvature control for soft fluidic elastomer robots?

After the challenge of controlling a single segment's curvature is met, the next technical challenge is to have these robots position themselves within a Cartesian task-space. We need to develop algorithms that build on the capability of configuration control and leverage appropriate multi-body kinematic models to enable position control. That is:

How do we repeatably position a multi-body soft fluidic elastomer robot?

One of the primary advantages of a soft robot is that it can harmlessly conform to its environment. To enable this benefit, we must develop algorithms that build on positional controllers, devices that deliver continuously variable flow, as well as that leverage the soft material properties of this class of robots. A major question addressed by this work is:

How do we autonomously allow soft fluidic elastomer robots to navigate confined environments?

Additionally, soft robots should have capabilities beyond those provided by traditional rigid-body robots. Our intent is to develop soft robot manipulators capable of autonomous, safe, and dynamic interactions with people and their environments. Accordingly, we must develop algorithms for dynamically controlling soft robots acting under gravity in 3D environments:

How do we develop algorithms that leverage a soft fluidic elastomer manipulator's dynamics to increase its performance?

1.4 Our Approach

1.4.1 Summary

This thesis addresses the technical challenges presented by soft robots by cyclically innovating solutions as we build multiple autonomous soft fluidic elastomer platforms.

These platforms gradually increase in complexity, and each platform builds upon the subsequent.

First, we work with single segment soft-bodied robots and develop a fundamental understanding of this new technology. We present an autonomous and self-contained soft-bodied robot that is a significant advancement over the state of the art in this field, namely Shepherd et al. [2011] where the main innovation was fluidic actuation for a robot's body. All supporting hardware and computation was external to the mechanism. We provide a complete approach to creating autonomous soft-bodied robots with onboard computation, actuation, power, and control and describe how we achieve this through modeling, design, fabrication, and algorithms. This work brings all systems found in a traditional rigid-bodied robot onboard the soft robot: an actuation system, power system, driving electronics, and computation and control system. We develop a robotic fish to provide an instantiation of our approach to creating autonomous soft-bodied robots capable of rapidly achieving continuum body motion. In this system, soft muscle-like actuators generate curvature in a continuously deformable, vertebrate-like body. Novel, form-independent actuator technology as well as miniaturization of supporting hardware enable the robot to take on the fundamental anatomical structure of a fish while being self-contained and unconstrained.

Next, we extend these concepts and create multi-segment planar soft fluidic elastomer robots. We outline an approach to designing, fabricating, and controlling pressure-operated soft robotic manipulators. Three alternative actuator morphologies and three fabrication processes are explored. Forward and inverse kinematic models are presented and we show how they integrate into an autonomous control system for these robots. Arms consisting of six independently controllable segments are analyzed on their (i) single section curvature tracking, (ii) point-to-point movement accuracy, (iii) path tracking accuracy, and (iv) ability to maneuver in confined environments. Then, an arm is combined with a gripper and evaluated on its (v) ability to grasp and place objects.

Lastly, we develop a multi-segment spatial soft manipulation system that operates subject to gravity. We provide the design, fabrication, modeling, and control of

this system, and we explore capabilities enabled by this new soft fluidic elastomer manipulator. The arm extends our modular planar manipulator morphology and fabrication process into three spatial dimensions. We build a prototype consisting of four independently casted and serially concatenated modular segments that each move in three spatial dimensions with two degrees of freedom. We use a piece-wise constant curvature assumption to model the arm and validate this assumption on the the physical prototype. We demonstrate the arm's ability to pass through confined environments, achieved closed-loop configurations, and position itself in three dimensions. Additionally, we provide a dynamic model of the spatial manipulation system under a sagittal plane assumption as well as a process for identifying the model's parameters. We develop planning algorithms that leverage this dynamic model to perform new capabilities like dynamic grabbing. Experimentally, we demonstrate task precision improvement using bracing as well as dynamic positioning accuracy of 4 centimeters outside of the arm's statically reachable envelope.

1.4.2 Single Segment Soft Robots

We address the following hypothesis:

> **Hypothesis 1**: A soft-bodied fluidic robot can be both capable of rapid continuum body motion and entirely self-contained ([Marchese et al., 2014d], [Marchese et al., 2013], and [Marchese et al., 2011]).

We advance soft robotics by providing a method for creating and controlling autonomous self-contained soft-bodied systems. Specifically, we introduce a novel self-contained fluidic actuation system and control algorithms used to deliver continuum motion in soft robots. We demonstrate this soft actuation in a case study by building an autonomous soft-bodied robotic fish powered by an on-board energy source; see Figure 1-3 and 1-4. The fish is novel in that it uses a soft continuum body and an innovative fluidic actuation system for the soft body. Additionally, it has onboard autonomy. That is, all power, actuation, and computational systems are located onboard. The continuum body has an embedded flexible spine and embedded

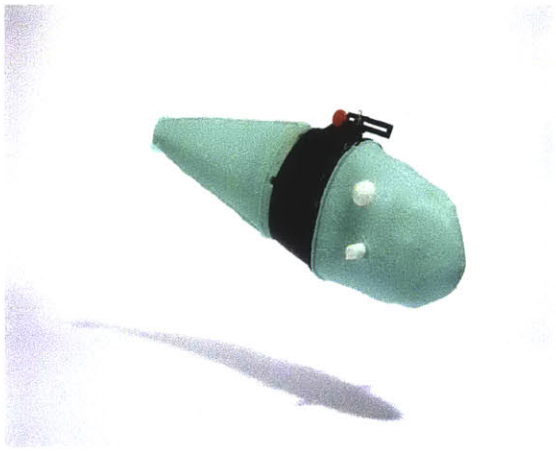


Figure 1-3: An autonomous soft-bodied robot that is both self-contained and capable of rapid, continuum body motion. The robot employs a compliant body with embedded actuators emulating the slender anatomical form of a fish. Photo courtesy of Devon Jarvis for Popular Mechanics.

anatomically proportioned muscle-like actuators. The robot is capable of forward swimming and performing agile maneuvers, scaled versions of an escape response[1].

We illustrate our proposed technical approach by designing and building a soft robot fish capable of emulating the escape response of fish. A fish was chosen as a case study because it naturally exhibits: continuum body curvature, rapid motion during an escape response [Domenici and Blake, 1997, Borazjani et al., 2012], a compliant posterior that bends under hydrodynamic resistance [Wakeling and Johnston, 1999], and an anterior suitable for housing rigid supporting hardware.

We evaluate the forward swimming and escape response maneuver of this soft robot in a suite of experiments. Extensive kinematic data is collected on the escape response and we compare the performance of the robot to various studies on biological fish. We show our robotic system, although on a different time scale, is able to emulate

[1]Escape response maneuvers are characterized by rapid body accelerations over very short durations and that often involve the body initially bending into a "C" shape Domenici and Blake [1997]. Among vertebrates, these are some of the most rapid maneuvers Jayne and Lauder [1993] and subject of frequent study. The extremely agile behavior exhibited by fish during escape response maneuvers is central to predator-prey interactions Webb and Skadsen [1980], and accordingly escape response performance carries marked ecological significance Walker et al. [2005], Gibb et al. [2006], Domenici et al. [2008], Bergstrom [2002]. Furthermore, the behavior serves as a neurophysiological model Eaton et al. [1981, 1991]. Understanding this behavior can give scientists insights on vertebrate evolution Hale et al. [2002] and physiology. Recently, the hydrodynamics of the maneuver have been explored in great detail; see Borazjani et al. [2012].

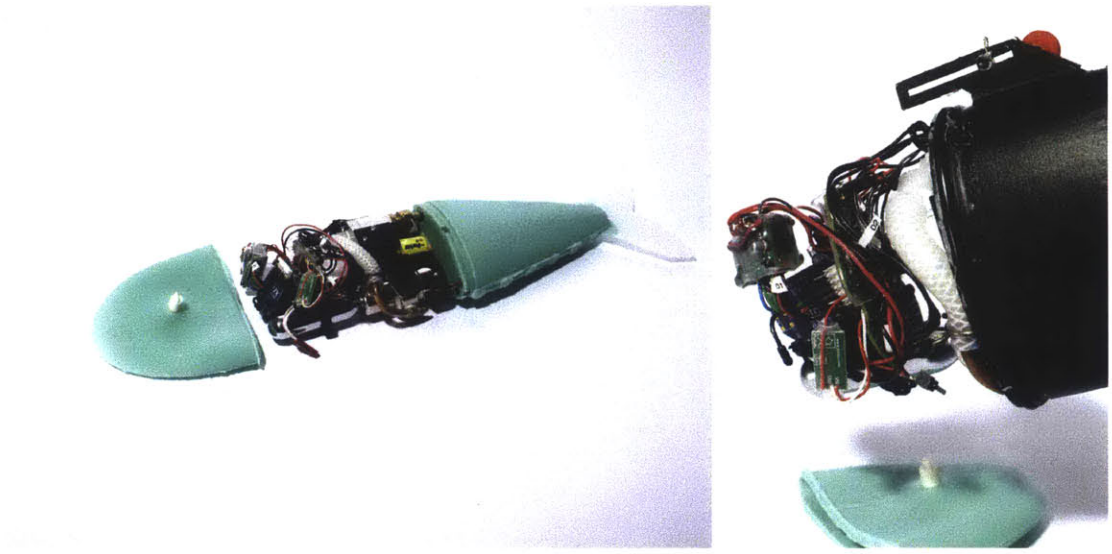


Figure 1-4: Left: Soft-bodied robotic fish with hull and rubber anterior cowl removed exposing the robot's onboard power, actuation, and computational subsystems. Right: A close-up of the robot with its cowl removed showing the wireless communication and control circuitry as well as the central fluid artery. Photos courtesy of Devon Jarvis for Popular Mechanics.

the basic structure of an escape response and that the performed maneuvers have a similar input-output relationship as observed in biological fish.

1.4.3 Multi-segment Planar Soft Robots

Additionally, in this thesis we address another important hypothesis:

> **Hypothesis 2**: Planar manipulation is possible with a soft fluidic elastomer robot. That is, a fluid powered multi-segment planar robot made entirely from soft elastomer can be precisely positioned using a closed-loop kinematic controller ([Marchese et al., 2015a], [Marchese et al., 2014c], [Marchese et al., 2014a], and [Katzschmann et al., 2015]).

This thesis demonstrates that autonomous manipulation with soft fluidic elastomer robots is possible. First, we present the design and characterization of three fluidic elastomer manipulator morphologies. Each of the arm's serially connected body segments are fundamentally constructed from derivatives of fluidic elastomer actuators

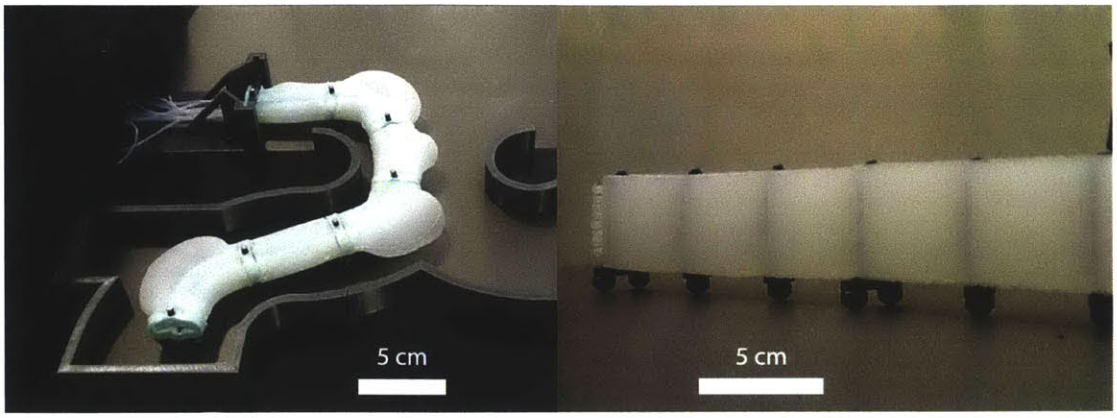


Figure 1-5: Two planar soft fluidic elastomer manipulator morphologies. Left: a manipulator prototype composed of six independently actuatable body segments. Each cylindrical segment has actuated channels embedded in its outer layer enabling the body segment to bend. Right: a six segment manipulator prototype where each rectangular body segment generates curvature using two agonist fluidic elastomer actuators separated by a thin inextensible spine.

(FEAs) [Correll et al., 2010] and these actuators deform by bending about a neutral axis when pressurized [Onal et al., 2011]. Next, we provide three alternative fabrication approaches for reliably fabricating these manipulators. Then, a method for closed-loop positional control of these soft manipulators is developed. This capability requires two critical innovations. First, we solve the previously unaddressed problem of controlling the configuration of an entirely soft and highly compliant pneumatic arm. That is, we develop real-time, closed-loop curvature controllers that drive the bending of the manipulator's soft pneumatic body segments despite their high compliance and lack of kinematic constraints. Specifically, to achieve curvature control we use an array of cascaded PI and PID controllers as well as develop an array of fluidic drive cylinders. Second, we apply a simplifying piece-wise constant curvature (PCC) assumption to model the forward and inverse kinematic relationship between the arm's configuration space (i.e., segment curvatures and lengths) and task space (i.e., the pose of points along its backbone) in a manner consistent with traditional continuum manipulation literature, as reviewed by Webster and Jones [2010]. Under this assumption, we develop forward and inverse kinematics algorithms to transform between configuration and task space.

We combine all these developments into an aggregate system for which we create a suite of planning algorithms, and with this we achieve novel capabilities for this class of robot. First, using a Jacobian-based approach to the inverse kinematics problem, we experimentally evaluate the arm's ability to repeatably move to poses in free-space as well as track linear end-effector trajectories.

Second, we provide an approach for autonomously moving a planar fluidic elastomer arm through a confined, pipe-like environment. We provide a computational approach to whole arm planning that finds a solution to the inverse kinematics problem for this class of arms. The solution considers both the primary task of advancing the arm's end effector pose as well as the secondary task of positioning the whole arm's changing envelope in relation to the environment. Specifically, we find a transformation from the arm's task space to its arc space that is aware of the soft arm's entire shape. We achieve this by posing a series of constrained nonlinear optimization problems and solving for locally optimal arc space parameters. A key feature of our approach is that we do not prevent collisions, but rather minimize their likelihood. In fact, since we have designed an entirely soft and compliant robot, we can tolerate collisions. Often, the arm's ability to passively comply with the environment allows the primary task to be accomplished despite the collision. To experimentally validate the soft robot's ability to successfully advance through a confined environment, we carry out a series of experiments using a six segment soft planar manipulator. The primary goal of these experiments is to validate the whole body planner's ability to incrementally advance the robot through one of four distinct pipe-like sections.

Lastly, we present a fluid powered gripper for these soft manipulators that can conform to variations in object geometry while ensuring encapsulation of a round object. The gripper is inspired by fingers developed by Polygerinos et al. [2013] and is advantageous for grasping because it exhibits high curvature, minimal radial expansion, and remains compliant during actuation. We attach this gripper to a multi-segment soft manipulator to enable grasp-and-place capabilities. We also present a planning algorithm that advances the arm through all necessary states of the grasp-and-place operation. The system first plans concentric approach circles shrinking from the ini-

tial end-effector pose to the object. Next, the system searches for locally optimal manipulator configurations that constrain the end-effector to lie on these approach circles so that the manipulator does not collide with the object. We experimentally validate the system's ability to repeatably and autonomously grasp-and-place randomly placed objects with a 7 DOF planar fluidic elastomer manipulator prototype.

1.4.4 Multi-segment Spatial Soft Robots

Lastly, in this thesis we address the hypothesis:

> **Hypothesis 3**: Spatial manipulation is possible with an arm composed entirely of low durometer elastomer and powered by fluid. That is, an entirely soft fluid-powered multi-segment spatial robot subject to gravity can be autonomously positioned to accomplish tasks ([Marchese and Rus, 2015] and [Marchese et al., 2015b]).

In this thesis we present a complete soft spatial manipulation system. That is, we provide the design, fabrication, and kinematic modeling of a new manipulator morphology: a fluid-powered three-dimensional multi-segment arm composed entirely of soft elastomer. Additionally, we develop a power system as well as processing and control algorithms that enable autonomous closed-loop control of the soft manipulator despite the self-loading effects of gravity. We show how the fluidic elastomer manipulator's continuum kinematics and soft material composition lead to several distinct advantages when compared to traditional rigid body manipulators. First, we show that the manipulator's soft segments deform according to constant curvature. With a constant curvature assumption [Webster and Jones, 2010], we can parameterize this N-link spatial soft manipulator with $2N$ joint variables. Second, the kinematics and extreme compliance of such a soft manipulator enable it to fit within and advance through confined environments. When the boundaries of the environment can be parameterized by curved cylinders and its curvature is non-zero, an idealized soft fluidic elastomer manipulator will be more capable of advancing through a confined environment than a manipulator with rigid links and discrete joints. We demonstrate this

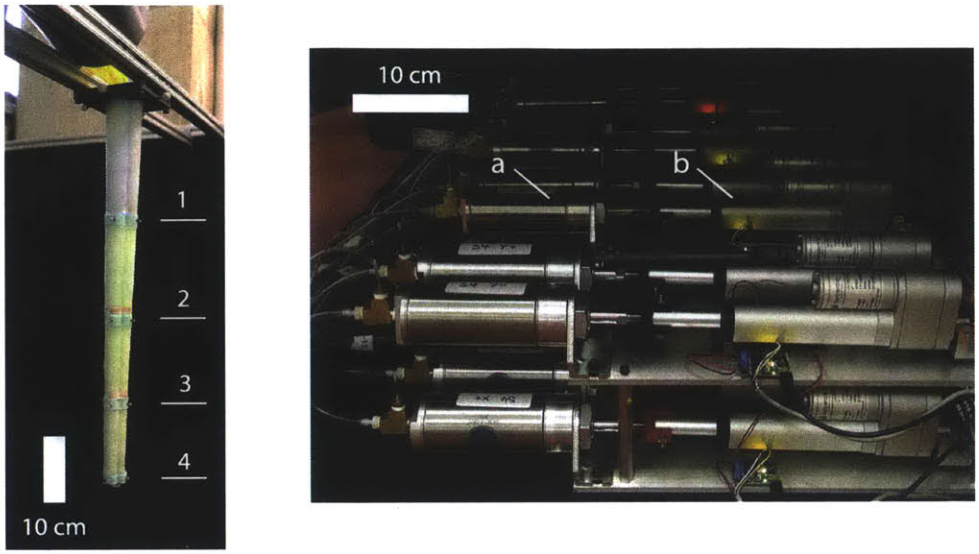


Figure 1-6: A spatial soft fluidic elastomer manipulator composed entirely from low durometer rubber. The manipulator has four independently actuatable body segments, each capable of 2 DOF bending. In this work, an external camera system is used to localize soft connectors between arm segments shown in green. Right: An array of high capacity fluidic drive cylinders are used to drive the manipulator's distributed fluidic elastomer actuators. Each drive mechanism consists of a pneumatic cylinder (a) driven by an electric linear actuator (b). The primary benefits of this drive mechanism are that it is closed-circuit and allows realization of continuously variable flow profiles.

concept experimentally. Third, the continuum kinematics of a soft fluidic elastomer manipulator enable a high degree of dexterity. Specifically, in an environment where a collision-free path is parameterized by a curved path, the continuum kinematics of a fluidic elastomer manipulator can generally fit the curvature of the path better than a rigid link manipulator with discrete joints and rigid links.

In this thesis we also provide an approach for dynamically controlling soft robots. Through simulation and experiments we demonstrate repeatable positioning of the soft fluid-powered multi-segment spatial robot to states outside of the statically reachable workspace in dynamic maneuvers we call *grabs*. Specifically, we begin by developing a dynamic model for such a soft manipulation system as well as a computational strategy for identifying the model. Using this identified model and trajectory optimization routines, locally-optimal dynamic maneuvers are planned through iteration learning control and repeatably executed on a physical prototype. Actuation limits,

the self-loading effects of gravity, and the high compliance of the manipulator, physical phenomena common among soft robots, are represented as constraints within the optimization. For example, consider a soft manipulator that can safely and dynamically interact with humans by quickly grabbing objects directly from a human's hand.

1.5 Thesis Contributions

The results presented in this thesis differ from prior work in design, fabrication, and control and enable new autonomous capabilities for soft robots.

1.5.1 Single-segment Soft Robots

A primary contribution of this thesis is that we show a soft-bodied fluidic robot can be both capable of rapid continuum body motion and entirely self-contained. More specifically this contribution consists of:

1. A soft fluidic actuation system capable of rapidly achieving continuum body motion.

2. A static model for the non-constant bending deformation of a tapered planar fluidic elastomer actuator.

3. A method for fabricating amorphous soft bodies with embedded fluidic channel structures.

4. A self-contained soft-bodied fluidic robot with 3 DOF and 2 bidirectional actuators that embodies our approach to mobile soft robotics and that emulates the planar forward swimming and escape maneuvers of biological fish.

5. Experimental evaluations with this robotic system that demonstrate the energy consumption, motion, and speed of the system and a comparison to biological fish.

1.5.2 Multi-segment Planar Soft Robots

Another contribution of this thesis is that we develop a soft fluidic manipulator capable of grasp-and-place and planned continuous motion in environments with obstacles. This work shows that planar manipulation with soft fluidic elastomer robots is possible and first to provide a comprehensive approach to design, fabrication, closed-loop control, and planning of such manipulators. Specifically this contribution consists of:

1. Three viable multi-segment manipulator morphologies that are (i) composed primarily of soft silicone rubber, (ii) powered by fluids, (iii) suitable for automation;

2. Three fabrication processes for reliably manufacturing these soft fluidic elastomer manipulators;

3. The first method for closed-loop configuration control for a soft fluidic elastomer robot consisting of (i) a kinematic model and an algorithm for estimating the manipulator's configuration in real-time, (ii) a novel device for providing continuous, closed-circuit adjustment of the manipulator's fluid, and (iii) a cascaded curvature controller;

4. Task-space planning algorithms that solve the IK problem and enable these manipulators to autonomously (i) position their end-effector in free-space, (ii) maneuver in confined environments, and (iii) grasp and move objects;

5. Experiments with a soft multi-segment planar arm prototype made of 6 distributed bidirectional rectangular fluidic elastomer actuators with 6 DOF demonstrating repeatable free-space positioning;

6. Experiments with a soft multi-segment planar arm prototype composed of 6 distributed bidirectional cylindrical fluidic elastomer actuators with 6 DOF demonstrating repeatable maneuvering in confined environments;

7. Experiments with a soft multi-segment planar arm equipped with a gripper

prototype having 7 DOF demonstrating repeatable successful grasping demonstrations.

1.5.3 Multi-segment Spatial Soft Robots

Another contribution of this thesis is the first autonomous three-dimensional fluidic elastomer manipulator. That is, we provide:

1. A novel multi-segment manipulator prototype (i) constructed 100% from soft silicone rubber, (ii) powered by four fluidic elastomer actuators per segment, and (iii) designed with a modular morphology suitable for automation;
2. A novel process to repeatably fabricate this manipulator;
3. A novel iterative physics-based model to understand a spatial segment's deformation;
4. A multi-segment kinematic model and processing and control systems that enable the first autonomous capabilities for this manipulator type, e.g. (i) advancing through a confined environment, (ii) following configuration trajectories within a sagittal plane, and (iii) positioning in 3D;
5. Evaluations in both simulation and physical experiments with a four-segment prototype evaluating capabilities i-iii above;
6. Experimental evaluations of one and two segment prototypes that quantify the accuracy of bending angle and center of mass model estimations.

To the best of our knowledge, this thesis also provides the first instance of dynamic motion control for a soft fluidic elastomer robot. More specifically, we provide:

1. A dynamic model for a fluid powered spatial manipulator made entirely from soft elastomer as well as a process for fitting the model to experimental data
2. Dynamic control algorithms that allow such a 3D soft manipulator operating under gravity to be precisely positioned

3. A manipulation primitive built on these dynamic control algorithms, grabbing.

4. Extensive dynamic experiments with a physical prototype demonstrating repeatable grabbing.

1.6 Thesis Outline

This thesis is organized as follows: First, in Chapter 2 we provide a current review soft robotics. Then, in Chapter 3 our work with locomotory soft fluidic elastomer robots composed of a single body segment is detailed. We develop this work in the context of a soft robotic fish, and we focus on addressing the open challenge of creating a self-contained soft robot that is capable of rapid continuum body motion. Then, in Chapter 4 we extend this work to planar multi-segment soft robots. We focus on addressing the open challenges of creating and controlling multi-body fluidic elastomer robots and develop manipulators that can advance through confined environments, grasp randomly placed objects, and track trajectories in free space. In Chapter 5 we extend our work to multi-segment soft robots that can move in three spatial dimensions subject to gravity. Here, we focus on addressing open challenges in design, fabrication, dynamic modeling, and control. We develop manipulators that can advance through confined environments, dynamically grab objects, and position an end-effector in free space. Lastly, in Chapter 6 we conclude with lessons learned and important areas for future research.

Chapter 2

Related Work

Soft robotics is a nascent, interdisciplinary field. Just this year the journal of *Soft Robotics* was created to connect the rapidly growing community. This community is unique in that it merges the seemingly disjoint disciplines of biology, chemistry, robotics, artificial intelligence, material science, and biomedical engineering. Arguably, some of the field's most important work has just been published this year. In the following, we provide a current review of soft robotics, and when necessary, we narrow our focus to soft fluidic elastomer robots. Our review spans both the body and brains of such robots. That is, we review design and fabrication, computation and control, as well as robotic systems and applications.

2.1 Design and Fabrication

2.1.1 Actuation

There are various approaches to actuating the body of a soft robot. One distinguishing feature of many soft robots is that actuators and/or power transmission systems are integrated within and distributed throughout the body. In the following we review four common actuator types, and these are also depicted in Figure 2-1.

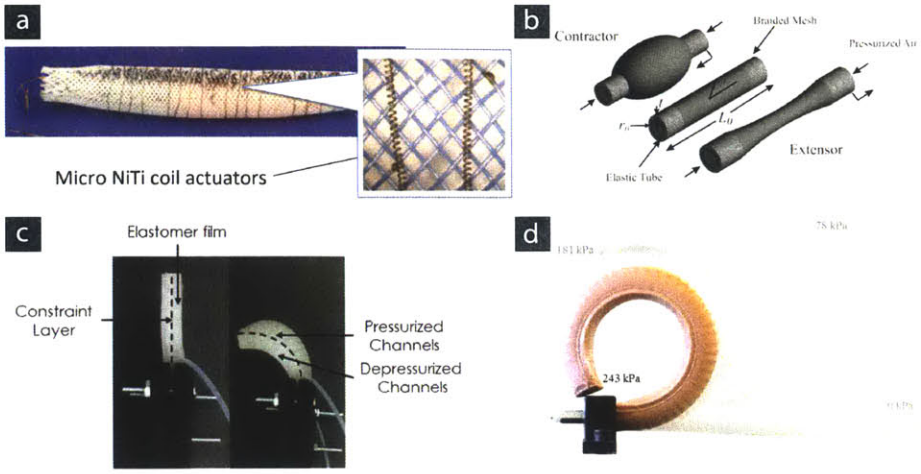


Figure 2-1: Common actuation approaches for soft robots. (**a**) Shape Memory Alloy (SMA) actuators [Seok et al., 2010], (**b**) Pneumatic Artificial Muscle (PAM) actuators [McMahan et al., 2006], (**c**) Fluidic Elastomer Actuators (FEAs) [Onal et al., 2011], and (**d**) Fiber reinforced FEAs [Galloway et al., 2013].

Tendons

Originally, many hard hyper redundant and hard continuum robots [Hannan and Walker, 2003, Cieslak and Morecki, 1999, Buckingham, 2002, Gravagne and Walker, 2002, McMahan et al., 2005, Camarillo et al., 2009] used an array of servomotors or linear actuators to pull cables that move rigid connecting plates located between body segments. Some softer robots have adopted a similar actuation scheme consisting of tendons pulling rigid fixtures embedded within an elastomer body. For example, the elastomer based bio-inspired octopus arm developed in Calisti et al. [2010], Laschi et al. [2012] and Calisti et al. [2011] uses Shape Memory Alloy (SMA) actuation. Further, Seok et al. [2010] use SMA actuators within a worm-like locomotory robot (see Fig. 2-1a). The basic operating principle behind SMA technology is that nickel titanium (NiTi) wire contracts under joule heating. This heating is typically produced by passing electrical current through the wire. The contracting wire can be used as an agonist actuator, similar to the way one's bicep pulls the forearm towards the body during a curl. There are also soft elastomer robots that use more traditional variable tension cables. For example, the soft-bodied fish developed by Valdivia y Alvarado and Youcef-Toumi [2006] as well as the soft arm developed by Wang et al. [2013] use

this actuation approach, but these both consist of only one actuated segment.

Pneumatic Artificial Muscles

Another common actuation scheme for soft robots involves distributed Pneumatic Artificial Muscle (PAM) actuators (see Fig. 2-1b) also known as the McKibben actuator. A PAM is fundamentally composed of an inflatable elastic tube surrounded by a braided mesh. Depending on the weave pattern of the braided mesh the actuator can be designed to contract or extend under input pressure. Typically these actuators are operated with driving pressures between 50 and 100 psi. These actuators have been used and studied extensively in Chou and Hannaford [1996], Tondu and Lopez [2000] and Daerden and Lefeber [2002]. Notable semi-soft robots using PAMs include [McMahan et al., 2006, Pritts and Rahn, 2004] and Kang et al. [2013].

Fluidic Elastomer Actuators

A softer alternative is the Fluidic Elastomer Actuator (FEA), which is used predominantly throughout this thesis. The FEA is a bending actuator composed of low durometer rubber and driven by relatively low-pressure fluid, 3 to 8 psi. Its basic structure consists of two soft elastomer layers separated by a flexible but inextensible constraint. Each of these elastomer layers contains embedded fluidic channels. By pressurizing the fluid entrapped in these channels, stress is induced within the elastic material producing localized strain. This strain in combination with the relative inextensibility of the constraint produces body segment bending (see Fig. 2-1c). FEAs can be powered pneumatically or hydraulically.

Perhaps the earliest application of pneumatically actuated elastomer bending segments to robotics was by Suzumori et al. [1992]. Here fiber-reinforced Flexible Microactuators (FMAs) were developed and shown viable in a manipulator and multi-fingered hand. Recently, these concepts have been extended and developed into the FEA and used to build a variety of robotic systems [Shepherd et al., 2011, Ilievski et al., 2011, Onal et al., 2011, Morin et al., 2012, Martinez et al., 2013, Onal and Rus, 2013, Tolley et al., 2014a]. These robots use elastomers of varying stiffness as

well as cloth, paper, plastics, and even stiffer rubbers for their constraint layers. Furthermore, Mosadegh et al. [2014] and Polygerinos et al. [2013] have investigated more elaborate channel designs in order to reduce elastomer strain.

There are also less flexible, fiber-reinforced FEAs (see Fig. 2-1d) that occupy the soft actuator space between purely elastomer FEAs and PAMs. These actuators operate with driving pressures of between 25 and 35 psi and can accordingly apply higher forces which is an advantage for many applications. There are several notable examples of fiber-reinforced FEAs in the literature Galloway et al. [2013], Bishop-Moser et al. [2012], Deimel and Brock [2013, 2014], Park et al. [2014b] and Suzumori et al. [2007].

2.1.2 Power

Fluidic power sources present many challenges for soft robots. Recently, Wehner et al. [2014] review existing pneumatic energy sources. Besides the use of compressed gas, which was proposed to the community in Marchese et al. [2013] and Marchese et al. [2014d] and detailed in Chapter 3, there are three viable alternatives. These are: (i) microcompressors, (ii) explosive combustion, and (iii) peroxide monopropellants.

Microcompressors

Microcompressors, as used in Tolley et al. [2014b], Katzschmann et al. [2014], and Onal and Rus [2013], are well-suited for low-flow applications where the duration of the robot's operation is of primary concern. Typically, a microcompressor converts electrical energy from lithium polymer batteries into fluid energy, and this approach makes use of the high energy density of commercial battery technology. In general the components of this power source are commercially available and easy to control.

Explosive Combustion

In contrast, explosive combustion, as used in Shepherd et al. [2013a] and Tolley et al. [2014a], is well-suited for high-power, high-speed applications because chemicals like

butane and methane offer unmatched energy densities with respect to batteries and compressed gas. However, considerable supporting hardware is required to prepare, mix, and trigger such reactions. Besides producing noise and heat, the release of energy can be difficult to control.

Peroxide Monopropellant

Alternatively, peroxide monopropellant, as used in Onal et al. [2011], leverages the decomposition of hydrogen peroxide into water and oxygen. By carefully, but passively controlling the introduction of a decomposition catalyst, a portable peroxide pump has been shown to slowly release gas in order to sustain driving pressures suitable for soft robot actuation.

Compressed Gas

Additionally, we show stored compressed gas is well-suited when high driving pressures and high maximum flow rates are required, and accordingly rapid, quiet actuation is required from a robotic system. This is primarily because fluid energy is stored and released onboard, as opposed to generated. Accordingly, no energy is converted between domains. However, because this strategy leverages compressed gas canisters whose energy densities are significantly lower than both lithium polymer batteries and the chemicals within the aforementioned reactions, the total available energy is limited and therefore the system's duration of operation can be limited.

2.1.3 Design Tools

As mentioned in the introduction of this thesis, design tools for soft robots are limited with respect to the availability of design tools for more traditional rigid-body robots. Suzumori et al. [2007] use Finite Element Modeling (FEM) to analyze the bending of fiber reinforced pneumatic tube-like actuators. Specifically, hyperelastic material models are used to capture the nonlinear material properties of rubber, line elements are used to represent radial inextensibility constraints due to fiber reinforcement,

and the simulation is performed using the software MARC. Outside of this example, the community has generally found that iterative nonlinear finite element solvers are limited to small deformations and of limited use when modeling very soft nonlinear materials [Lipson, 2014]. VoxCAD and the Voxelyze physics engine, as used in Cheney et al. [2013] and Lehman and Stanley [2011] and reviewed by Lipson [2014], are simulation tools for very soft nonlinear materials. These tools use the concept of nonlinear relaxation to effectively perform physically correct particle-based material simulation. They have the advantage of allowing the user to individually set the local material properties of each particle. The disadvantage is that many physical parameters of active and passive material types must be experimentally derived.

2.1.4 Fabrication

Cho et al. [2009] review several manufacturing processes for soft biomimetic robots. The vast majority of soft elastomer robots rely on the processes of soft lithography [Xia and Whitesides, 1998] and/or shape deposition manufacturing [Cham et al., 2002]. Specifically, for soft fluidic elastomer robots this fabrication process generally consists of three steps as shown in Figure 2-2: (1) Two elastomer layers are molded through a casting process using pourable silicone rubber. The mold used for the outer layer contains a model of the desired channel structure. When cast, the outer layer contains a negative of this channel structure. The mold used for the constraint layer may contain fiber, paper, or a plastic film to produce the the inextensibility property required for actuation. When the elastomer is poured, this material is effectively embedded within the constraint layer. (2) The two layers are cured, removed from their molds, and their joining faces are dipped in a thin layer of uncured elastomer. (3) Lastly, the two layers are joined and cured together. The primary limitation of this soft lithography fabrication process is that it is fundamentally 2.5D, meaning that the robots are largely constrained to a planar morphology. This process limits a soft robot's ability to achieve amorphous, 3D forms. Additionally, shape deposition manufacturing (SDM) [Cham et al., 2002] is a cyclical, layering fabrication process where material is iteratively deposited, shaped to the desired geometry, and

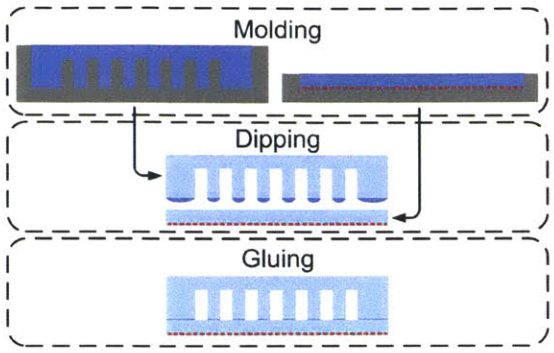


Figure 2-2: Soft lithography fabrication process for soft fluidic elastomer robots. Reproduced with permission from Onal and Rus [2012].

embedded with components as necessary. This process enables multi-material structural assemblies that include actuators and supporting infrastructure. Additionally, Umedachi et al. [2013] provide the first SMA actuated soft robot fabricated using 3D printing. However, although 3D printing allows printing flexible materials in amorphous forms, these materials are relatively brittle with respect to casted rubbers and are not well-suited for FEAs.

2.2 Computation and Control

Computational approaches to motion control and planning assist robots in performing autonomous tasks. Previous research has explored computation-based approaches to motion control and planning in the context of continuum robots with appreciable levels of rigidity, and these are reviewed within this section. In contrast, elastomer-based continuum robots are very well-suited to realize the concepts of *morphological computation* [Pfeifer and Iida, 2005] or *embodied intelligence* [Pfeifer et al., 2007] to perform tasks. Morphological computation is the idea that a robot's morphology (i.e. its form and structure) as well as material composition can be exploited to achieve complex tasks in a computationally "cheap" manner. Embodied intelligence refers to the theory that a robot or creature's physical embodiment (e.g. its morphology and/or material composition) is closely coupled to its intelligence. For example, consider the task of grasping a novel object from a bin. As opposed to precisely

perceiving and reconstructing the object using a suite of sensors and then planning a fully actuated grasp where each joint of the hand is considered, embodied intelligence would suggest combining the approximate location of the object with the capacity of a compliant hand to passively conform to unknown object geometry to complete the task. Existing work in this area is also reviewed here. However, to date, realizations of body-brain control strategies for soft robots often exhibit a disproportionate reliance on the robot's body and this translates to significant human intervention and trial-and-error when performing novel tasks. We believe to fully realize the potential of embodied intelligence for soft elastomer systems, equal parts computation (brain) and body are required.

2.2.1 Models

Robot-Independent Kinematic Model

Despite the variability in continuum manipulator designs, their kinematics can often be represented using a piecewise constant curvature (PCC) model. This is the message within a 2010 review of continuum robots by Webster and Jones [2010]. That is, Webster and Jones review several seemingly distinct kinematic modeling approaches, but show that when using a PCC modeling assumption, these approaches yield identical robot-independent results. This assumption means each body segment of a multi-segment arm is assumed to deform with constant curvature. An early use of PCC modeling appears in Hannan and Walker [2003] where a bending robotic trunk is developed.

Again as Webster and Jones [2010] describe, the transformation from configuration space consisting of arc length $\mathbf{L}$, curvature $\boldsymbol{\kappa}$, and plane orientation $\boldsymbol{\gamma}$ (refer to Figure 2-3) to task space consisting of Cartesian position and orientation or pose can be derived in multiple ways. Differential geometry [Hannan and Walker, 2003] or the use of virtual links and a Denavit-Hartenberg (DH) approach [Jones and Walker, 2006a] are among a few of the derivations. The robot independent forward kinematic

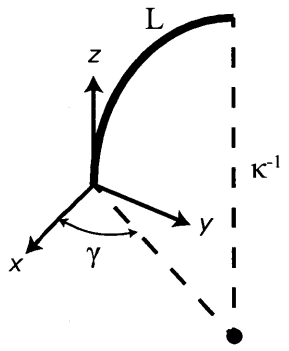


Figure 2-3: Arc parameters used to model the bending of a spatial continuum segment under the piecewise constant curvature assumption: Arc length $\mathbf{L}$, curvature $\boldsymbol{\kappa}$, and plane orientation $\boldsymbol{\gamma}$. This figure is adapted from Webster and Jones [2010].

transformation for a single continuum segment is

$$\mathbf{T} = \begin{bmatrix} \cos\gamma\cos\kappa s & -\sin\gamma & \cos\gamma\sin\kappa s & \frac{\cos\gamma(1-\cos\kappa s)}{\kappa} \\ \sin\gamma\cos\kappa s & \cos\gamma & \sin\gamma\sin\kappa s & \frac{\sin\gamma(1-\cos\kappa s)}{\kappa} \\ -\sin\kappa s & 0 & \cos\kappa s & \frac{\sin\kappa s}{\kappa} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

where $s = [0,\ L]$. Although the PCC modeling assumption is quite powerful, it does not always apply to continuum manipulators. The obvious limitation being that not *all* continuum robots deform according to segmented constant curvature. A common exception is when a continuum body segment assumes a spiral. Mahl et al. [2013] develop general variable curvature continuum kinematics for this reason. Giorelli et al. [2012] develop forward and Jacobian based inverse kinematics for a planar conical soft manipulator that again spirals. Accordingly, an important contribution of this work is that we validate this PCC modeling assumption in the context of a soft fluid-powered elastomer manipulator.

Robot-Dependent Kinematic Models

A robot specific mapping is required from actuation space (e.g. tendon lengths or pressures) to configuration space (i.e. $\mathbf{L}$, $\boldsymbol{\kappa}$, and $\boldsymbol{\gamma}$) as reviewed in Webster and Jones [2010]. Previous work on continuum robots has generally approached this robot dependent mapping or model in three ways.

First, Gravagne et al. [2003], Webster et al. [2009] and Trivedi et al. [2008] use Bernoulli-Euler beam mechanics to predict the deformation of continuum robots. When a beam is subject to a tip moment, the beam moment M_b along the length l is constant, $M_b(l) = c$, and it deforms under constant curvature, $\kappa = \frac{M_b}{EI}$. A common realization of this is an actuator parallel to the beam's neutral axis that applies a load to a fixture mounted orthogonal to the beam at the tip. This modeling approach does not apply to our device as it has a fundamentally different operating principle: the soft segment stores bending energy in its outermost actuated layer as opposed to its inner backbone. If the inner layer is assumed to store bending energy, and the Bernoulli-Euler beam model is applied, a non-constant moment along the backbone is predicted producing non-constant deformation.

Second, continuum robots with high pressure (50-100 psi max) [McMahan et al., 2006, Kang et al., 2013, Suzumori et al., 1991] and medium pressure (30 psi max) [Chen et al., 2006] PMAs as well as tendons and cables [Jones and Walker, 2006b, Hannan and Walker, 2003, McMahan et al., 2005, Wang et al., 2013] first make the assumption that segments are deforming under constant curvature and second make the assumption that actuators either lengthen or shorten as a function of input. Subsequently, a relationship between joint variables, i.e. 3 or 4 actuator lengths, and constant curvature arc parameters is developed. This approach does not generally apply to our device as the low durometer rubber composition and lack of kinematic constraints means our actuators are expanding spatially at varying rates, not only expanding axially.

Last and perhaps most relevant, are robot specific models developed to understand the deformation of extremely soft, low pressure fluidic elastomer actuators. For instance, Onal et al. [2011] develop a static analytical model under the conditions of: (i) longitudinal strain, (ii) uniform rectangular channels with square cross-sections orthogonal to the neutral axis, (iii) a rectangular actuator shape, and (iv) stresses that are based on initial channel geometry. Additionally, Shepherd et al. [2011] develop a model that assumes only circumferential strain in uniform round channels orthogonal to the neutral axis in a rectangular actuator and use a linear stress-strain relationship.

These models all assume a two-dimensional cross-section of infinite thickness.

Dynamic Models

Dynamic models of hard and semi-soft continuum robots have opened the door for a variety of control techniques. Chirikjian [1994] uses a continuum approach to model the dynamics of a hard hyper-redundant manipulator and uses this for computed torque control. Gravagne and Walker [2002] dynamically model the Clemson Tentacle Manipulator, a hard continuum robot, and show a PD plus feed-forward regulator is sufficient for stabilizing the system. They further develop a large deflection model and controller in Gravagne et al. [2003]. Snyder and Wilson [1990] and Wilson and Snyder [1988] dynamically model polymeric pneumatic tubes subject to tip loading using a bending beam model but do not use this for control. Using a Lagrangian approach Tatlicioglu et al. [2007] develop a dynamic model for and provide simulations of a planar extensible continuum manipulator. Braganza et al. [2007] develop a neural network controller for continuum robots such as OctArm [McMahan et al., 2005] based on a dynamic model.

2.2.2 Control

Most fluid powered soft elastomer robots use model-free open-loop valve sequencing to control body segment bending. That is, a control valve is turned on for a user-determined duration of time to pressurize an elastomer actuator and then off to either hold or deflate the actuator. For instance, there are soft rolling robots [Correll et al., 2010, Onal et al., 2011, Marchese et al., 2011] made of fluidic elastomer actuators that use this control approach. Also a soft snake-like robot developed by Onal and Rus [2013] uses this open-loop scheme to enable serpentine locomotion. Luo et al. [2014] develop and verify a planar dynamic model for this soft snake but do not use it for control. Again, Shepherd et al. [2011] use a model-free open-loop valve controller to drive body segment bending in an entirely soft multigait robot. Tolley et al. [2014b] extends this control approach to an untethered version of the multigait walker that is

five times larger. Passive control is demonstrated in an explosive, jumping robot in Shepherd et al. [2013a] and extended to use a valve controller in Tolley et al. [2014a]. Martinez et al. [2013] develop manually operated elastomer tentacles. There is also an example of controlling a soft pneumatic inchworm-like robot using servo-controlled pressure described in Lianzhi et al. [2010]. Recent work has aimed to create simpler passive valves [Napp et al., 2014], and this may enable other morphological control strategies.

Open-loop model-free control is also common for soft elastomer robots that do not use pneumatic actuation. For example, previous work on soft bioinspired octopus-like arms developed by Calisti et al. [2010] demonstrate open-loop capabilities like grasping and locomotion [Laschi et al., 2012] [Calisti et al., 2011]. Umedachi et al. [2013] developed a soft caterpillar robot that uses an open-loop SMA driver to control body bending.

To the best of our knowledge, highly compliant robots whose bodies are made from soft elastomer and distributed fluidic actuators have not used closed-loop curvature control nor dynamic model-based control. Prior work in this field uses model-free, open-loop control policies, but because this existing work does not derive control policies from nonlinear dynamic models nor control the bending of the robot using feedback, these approaches cannot efficiently plan motions for novel tasks without sufficient trial-and-error.

2.2.3 Planning

Previous research has shown that purely kinematic approaches to continuum robot planning work in simulation and on physical systems when the robot is sufficiently constrained by the rigidity of its actuators or backbone structure. For example, Hannan and Walker [2003] develop novel continuum kinematics for a hyper-redundant elephant trunk and demonstrate how these enable capabilities like obstacle avoidance. Also, Jones and Walker [2006b] and Jones and Walker [2006a] provide kinematic algorithms for controlling the shape of multi-segment continuum manipulators. Additionally, Chirikjian and Burdick [1995] use a continuous backbone model to plan optimal

hyper-redundant manipulator configurations using calculus of variations. Lastly, Xiao and Vatcha [2010] introduce a planar continuum arm planner that enables simulated grasping in uncertain, cluttered environment.

Solving the inverse kinematics (IK) problem is a significant component to planning for continuum robots, and existing literature has provided several approaches. For example, Neppalli et al. [2009] provide a closed-form IK solution for continuum arms, but this Jacobian-based solution only considers the endpoint of the final body segment and obstacle avoidance requires manual planning. Jones and Walker [2006a] control Air-OCTOR and OctArm using fast real-time Jacobian control over task-space, but rely on joystick control for whole arm tasks like manipulation and grasping [Csencsits et al., 2005]. A limitation of existing approaches to solving the IK problem for continuum arms is that currently the whole arm, in addition to the end effector's pose, is not considered in the solution. Autonomous obstacle avoidance and movement through a confined environment is difficult without a computational solution to the IK problem that is aware of the robot's whole body in space. Buckingham [2002] articulates a distinguishing advantage of a snake-like arm. This is, it can potentially achieve the primary task of tip control, while meeting the secondary task of shaping the whole arm. Local optimization has shown promising results for rigid-bodied redundant manipulators [Nenchev, 1989], but as far as we are aware such a technique has not been used to solve the whole body planning problem for a continuum robot. Furthermore, we are not aware of any existing fluidic elastomer systems [Correll et al., 2010, Onal et al., 2011, Marchese et al., 2011, Shepherd et al., 2011, Onal and Rus, 2013, Shepherd et al., 2013a, Martinez et al., 2013, Tolley et al., 2014b,a] that use model-based computations to position their bodies in task-space.

2.3 Robots: Systems and Applications

In recent years, roboticists have turned to soft-bodied systems in order to address challenges in robot locomotion and manipulation. In the following we provide a brief review of relevant systems having varying degrees of rigidity.

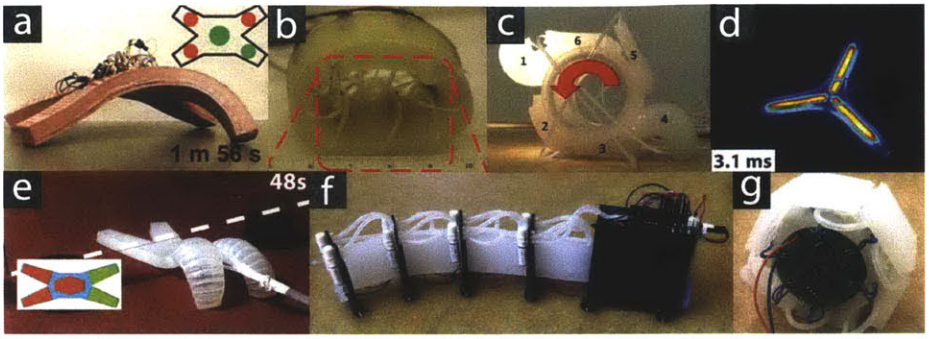


Figure 2-4: Various soft locomotory robots.

2.3.1 Locomotion

Soft-bodied robots are well-suited to emulate the locomotion of natural systems. Their compliant bodies deform continuously allowing peristaltic, crawling, undulatory, flapping, and serpentine locomotion. Further, soft materials and fabrication processes allow soft-bodied robots to readily realize biologically-inspired forms.

Land

In the past several years soft roboticists have made many notable low durometer rubber robots intended for land locomotion. For example, rolling belts have been produced by Correll et al. [2010] and Marchese et al. [2011]. Trimmer et al. [2006] and Umedachi et al. [2013] emulated the peristaltic locomotion of caterpillars. Shepherd et al. [2011] developed a multigait walking robot, and Shepherd et al. [2013a] developed a jumping robot powered by combustion. However, a limitation of the aforementioned locomotory robots is that they require an electrical and/or pneumatic tether. Soft actuation systems, especially fluidic actuation systems, typically require significant supporting hardware and often limit soft locomotory robots from being self-contained. That said, there are a few examples of untethered soft robots: Onal et al. [2011] created a rolling robot, Onal and Rus [2013] emulated the serpentine locomotion of snakes, and Tolley et al. [2014b] developed a quadrupedal walking robot; these are all soft-bodied fluidic elastomer systems. Seok et al. [2010] realize peristaltic locomotion with a self-contained SMA-based inchworm. However, a lim-

itation of all these untethered soft platforms is that performance is severely limited with respect to their rigid-bodied counterparts, and this limitation is due to the constraints imposed by bringing onboard all supporting hardware. More specifically, they all exhibit locomotory speeds of between 0.008 and 0.07 bodylengths per second.

Underwater

Soft under water robots exhibit continuum motion that conventional rigid-bodied robots cannot achieve. For instance, although many notable robotic fish exist [Liu and Hu, 2010, Barrett et al., 1999, Triantafyllou and Triantafyllou, 1995, Zhong et al., 2009, Wen et al., 2012] these prior robotic systems have bodies composed of rigid segments connected by fixed joints and are consequently incapable of reproducing the body kinematics observed during agile escape response maneuvers. Previous attempts to recreate an escape response used a body composed of four position-controlled, rigid links [Liu and Hu, 2005, Xu et al., 2012, Su et al., 2011]; however, such fully actuated, rigid-bodied systems inherently fail to capture the continuum motion of the escape response maneuver.

Prior work has aimed to create robotic fish using biologically inspired flexible posteriors. Self-propelling flexible foils driven by an external robotic actuator have been studied by Lauder et al. [2012] and Alben et al. [2012]. Valdivia y Alvarado and Youcef-Toumi [2006] use a compliant body in the design of a robotic fish to mimic the swimming kinematics of a natural fish. Similarly, the robot fish FILOSE developed by El Daou et al. [2012] and El Daou et al. [2011] has a compliant posterior and serves as a test bed for fish-like sensing and locomotion. Both of these systems are cable driven and actuated with an onboard servomotor but lack autonomy and require an external power supply. Recently researchers developed a cable driven, flexible spring-steel spine to model escape response behavior in Feng et al. [2011]; however, in this system the motor, control system, and power supply are external to the apparatus, and its motion is constrained. Long et al. [2011a] have developed a flexible biomimetic vertebral column used to propel an autonomous surface swimming robot. The vehicle can also perform an escape response [Long et al., 2011b]. Again, a single servomotor

is used to actuate the compliant spine. Although this system is autonomous, relative to the aforementioned work only a small portion of the body is flexible, namely its posterior tail, and because its large anterior is a surface vessel the system is limited to surface swimming. Notably, the above compliant bodied robotic fish operate on the principle of a passive, flexible mechanism driven by a traditional electromechanical actuator and the researchers' primary goals are to understand the hydrodynamics of the flexible body. However, in this thesis one of our goals is to develop a fluidic actuation system that is embedded within the flexible body, yielding a compliant and active body within a completely self-contained system.

Prior work has also introduced compliant active-bodied robotic swimmers. Shen et al. [2013] used an oscillating strip of ionic polymer-metal composite as the posterior trunk of a dolphin-like robot. This is a free-swimming robot, but again limited by an external power tether. Perhaps the closest precursor to our work is the Airacuda fish developed by Festo [Festo, 2006]. This robot has a flexible body and is driven by fluidic actuators. Similar to our system, the fluidic and electronic components are located in the fish's rigid anterior and its actuators extend along the length of its flexible trunk. However, this system differs considerably from ours in design: it is composed of a plastic skeleton covered by flexible skin with two actuators along the anteroposterior axis, whereas we have a body composed almost entirely of soft rubber with numerous actuators embedded in the dorsoventral orientation along the anteroposterior axis. Another difference is that Airacuda can do static diving and swimming using the on-board pneumatic actuation system, whereas the focus of the fish presented within this thesis is on forward swimming and planar escape response maneuvers.

2.3.2 Manipulation

Soft machines have many distinguishing design characteristics when compared to traditional hard robots. These differences are reviewed by Trivedi et al. [2008]. One salient characteristic is that soft robots have continuously deformable backbones that allow for large-scale deformations and provide theoretically infinite degrees of freedom

(DOF). This attribute classifies soft robotic manipulators as a subclass of continuum robotic manipulators, as reviewed by Robinson and Davies [1999]. However, not all continuum robotic manipulators are *soft*, and even continuum manipulators referred to as *soft* have varying degrees of rigidity. We provide a brief review of the field below where continuum manipulators are categorized by their degree of rigidity. Figure 2-5 details several relevant manipulation systems.

Discrete Hyper Redundant Manipulators

Before reviewing the spectrum of continuum manipulator designs, it is important to mention the discrete hyper-redundant robot configuration. In contrast to continuum robots, this configuration generally has a kinematically constraining backbone composed of multiple rigid links serially connected by revolute joints. Thus, this backbone has many actuatable DOF and approximates continuous body bending. Examples of discrete hyper-redundant robot manipulators include the 32 DOF elephant trunk manipulator developed in Hannan and Walker [2003] (Fig. 2-5a) and the 10 DOF snake arm robot developed in [Buckingham, 2002]. Another discrete hyper redundant morphology is the variable-geometry truss manipulator (VGTM) as initially explored in Miura et al. [1985] and Miura and Furuya [1988]. These designs also use discrete body links; however, here a link is composed of two parallel prismatic joints held at the top and bottom by rigid plates. Each of these body elements resembles a parallel manipulator. A notable VGTM is the planar prototype developed by Chirikjian and Burdick [1994] with 10 body modules and 30 DOF.

Hard Continuum Manipulators

As mentioned, continuum manipulators are realized through a spectrum of morphologies. Many continuum manipulators are composed of hard, but flexible material (e.g. materials having a Young's modulus of roughly 0.5 to 200 GPa). For example, the elastic elephant trunk manipulator developed in Cieslak and Morecki [1999] is composed of eight elastic elements having coil spring backbones and is controlled as 3 independent body segments yielding 6 DOF. Additionally, Gravagne et al. [2003]

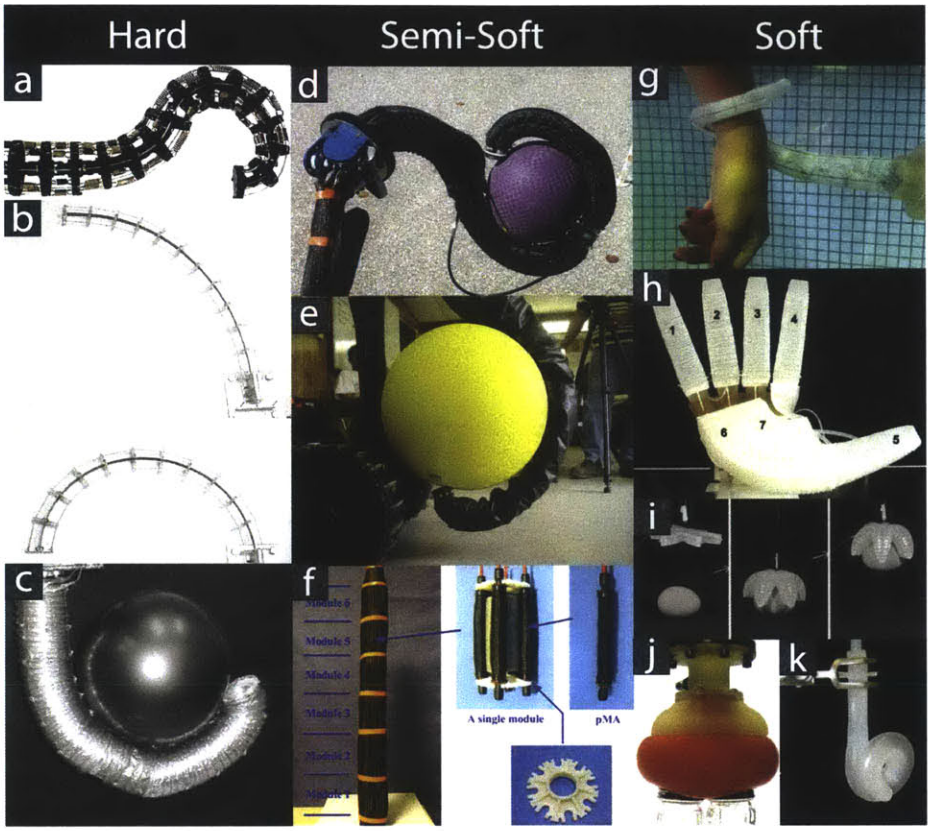


Figure 2-5: Various hard, semi-soft, and soft continuum manipulators. *Hard*: (**a**) Elephant trunk manipulator shown sideways developed by Hannan and Walker [2003], (**b**) Hard planar continuum robot developed by Gravagne et al. [2003], (**c**) Two-section Air-Octor manipulator developed by McMahan et al. [2005], *Semi-Soft*: (**d-e**) OctArm V developed by McMahan et al. [2006], (**f**) Biologically inspired continuum arm developed by Kang et al. [2013], *Soft*: (**g**) Octopus-bioinspired underwater arm developed by Calisti et al. [2010], (**h**) Anthropomorphic soft hand developed by Deimel and Brock [2014], (**i**) Star-fish like gripper developed by Ilievski et al. [2011], (**j**) Soft granular jamming gripper developed by Brown et al. [2010], and (**k**) Elastomer tentacle developed by Martinez et al. [2013].

(Fig. 2-5b) and Gravagne and Walker [2002] model and control the Clemson Tentacle Manipulator, a hard continuum manipulator. This platform has two planar sections, a total of 4 DOF, and uses a spring steel backbone. Air-Octor developed in McMahan et al. [2005] (Fig. 2-5c), the articulated catheter robot developed in Camarillo et al. [2009], and Tubot developed in Sanan et al. [2011] are also examples of continuum manipulators constructed of hard, but flexible materials. The aforementioned hard continuum manipulators use tendons to control bending. Typically, an array of servo-motors or linear actuators are used to pull cables that move connecting plates located between body segments.

Semi-Soft Continuum Manipulators

There are softer continuum manipulator morphologies that use a combination of soft and hard components. In these designs, distributed pneumatic muscle actuators (PMAs) are most prevalent, as the review by Trivedi et al. [2008] explains. OctArm IV developed by McMahan et al. [2006] (Fig. 2-5d and e) is a notable example. This robot uses 18 PMAs distributed throughout 4 arm segments yielding 12 DOF. It is capable of adaptively manipulating and grasping unknown objects that vary in both size and shape [McMahan et al., 2006]. Another example is the continuum manipulator developed by Pritts and Rahn [2004] that uses 14 PMAs within 2 body segments yielding 6 DOF. Also, the continuum manipulator developed by Kang et al. [2013] (Fig. 2-5f) uses 24 PMAs within 6 body segments yielding 18 DOF. Common features among all these designs are rigid end plates to which the PMAs are mounted and actuation pressures of between 50 and 100 psi. Accordingly, this manipulator morphology still relies on rigid components and undergoes significant strain stiffening, a phenomenon that adds rigidity to these robots.

Soft Continuum Manipulators

Recently, continuum manipulators composed from soft elastic material have been developed. These soft rubber manipulators can be categorized under two primary morphologies. First, there are tendon driven manipulators consisting of variable length

tendons, typically cables or shape memory alloy wire (see Section 2.1.1), embedded within and anchored to portions of a soft silicone arm. For example, previous work on soft bio-inspired octopus-like arms developed by Calisti et al. [2010] (Fig. 2-5g) used tendons and demonstrated capabilities like grasping and locomotion [Laschi et al., 2012, Calisti et al., 2011]. Also, Wang et al. [2013] developed a cable driven soft rubber arm consisting of one large actuated segment that bends bi-directionally. Lastly, McEvoy and Correll [2014a] and McEvoy and Correll [2014b] used a programmable stiffness spine in conjunction with tendons to achieve shape change in a soft rubber arm. The second morphology uses fluidic elastomer actuators (see Section 2.1.1) distributed among the manipulator's soft body segments. The primary advantages of using fluidic actuation for soft continuum manipulators is that this energy transmission system: (i) can be lightweight making for easy integration into distal locations of the body, (ii) conforms to the time varying shape of the manipulator, and (iii) does not require rigid components to implement. There are several examples of soft fluidic grippers described in recent literature. Deimel and Brock [2013] developed a pneumatically actuated three-fingered hand made of fiber reinforced silicone that is mounted to a hard industrial robot and capable of robust grasping. More recently, they have used similar fiber reinforced actuation technology to develop an anthropomorphic soft pneumatic hand capable of dexterous grasps [Deimel and Brock, 2014] (Fig. 2-5h). Ilievski et al. [2011] created a pneumatic starfish-like gripper composed of FEAs and demonstrated it grasping an egg (Fig. 2-5i). Stokes et al. [2014] used an FEA-based elastomer quadrupedal robot to grasp objects in a hard-soft hybrid robotic platform. A puncture resistant soft pneumatic gripper is developed in Shepherd et al. [2013b]. An alternative to positive pressure actuated soft grippers is the robotic gripper based on the jamming of granular material developed in Brown et al. [2010] (Fig. 2-5j). Another relevant piece of work is the manually operated 3D elastomer tentacles developed by Martinez et al. [2013] (Fig. 2-5k) containing 9 pneumatic crescent-shaped channels embedded within 3 body segments.

Chapter 3

Single-Segment Soft Robots[1].

3.1 System Overview

A defining characteristic of the soft-bodied robotic fish is the separation of actuation power from the rest of the system, specifically the utilization of mechanical energy in the form of pressurized fluid instead of electrical energy to power actuation. The body of the robotic fish (Fig. 3-1C-E, G-I, and Q) is entirely composed of Fluidic Elastomer Actuators or FEAs [Onal et al., 2011, Correll et al., 2010, Marchese et al., 2011], which are directly powered by pressurized fluid and accordingly no energy conversion takes place at the actuators. However, in order to control the fluidic system, supporting valve hardware is also incorporated into the architecture to electrically address and isolate the mechanical actuation system.

The soft robot has onboard all the subsystems of a conventional robot: an actuation system, power system, driving electronics, and computation and control systems. These systems (Fig. 3-1K-N), are stored in the fish's rigid anterior region, (A), a region with minimal contribution to body curvature during escape responses [Spierts and Leeuwen, 1999]. Technological advancements in these subsystems enable autonomous operation of a soft-bodied robot underwater.

[1]This chapter originally appeared in Marchese et al. [2014d] and is also based on results from Marchese et al. [2013] and Marchese et al. [2011]

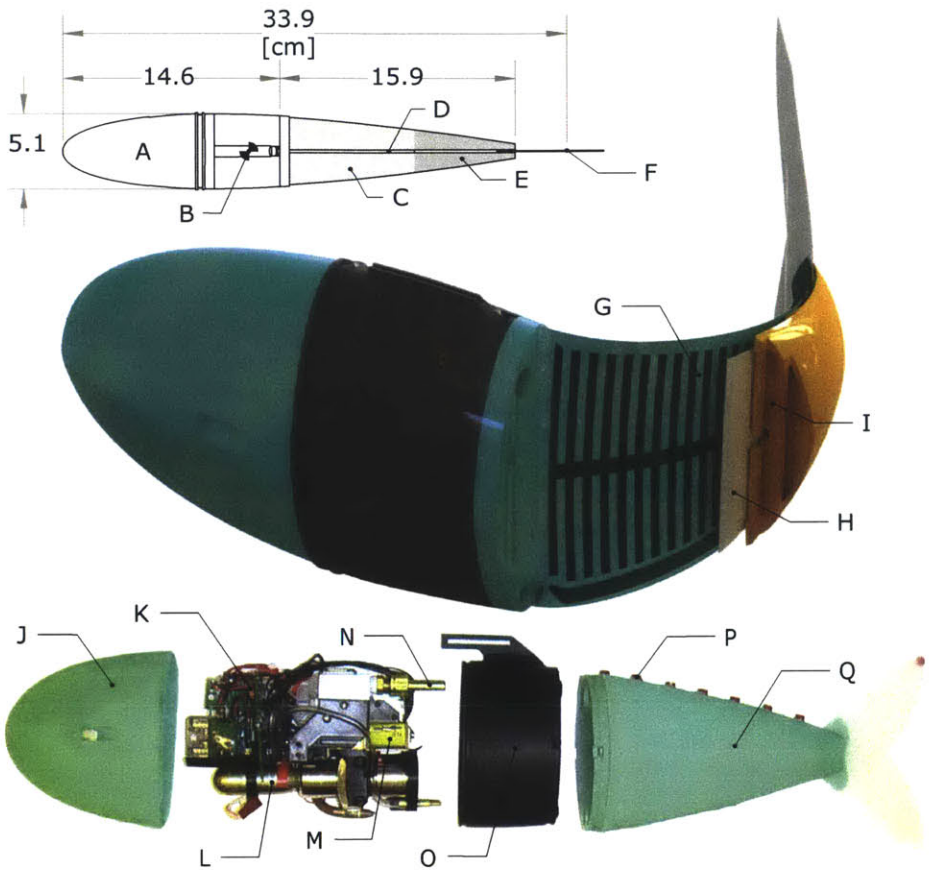


Figure 3-1: Details of a soft-bodied robotic fish. Top, a dorsal view of the fish showing (A) rigid anterior, (B) center of mass, (C) anterior trunk muscle-like actuator pair, (D) inextensible vertebrate-like constraint, (E) posterior trunk actuator pair, (F) passively actuated caudal fin. Center, a cross-sectional rendering of the mechanism showing, (G) fluidic elastomer channels grouped into antagonistic actuator, (H) flexible constraint layer, (I) pressurized elastomer channels in agonistic actuator. Bottom, an exploded view of the robot detailing (J) silicone skin, (K) communication and control electronics, (L) compressed gas cylinder and regulator, (M) flow control valves, (N) actuator access port, (O) plastic fuselage, (P) videography markers, (Q) silicone elastomer trunk.

3.2 Actuation

3.2.1 Fluidic Elastomer Actuator

Fluidic Elastomer Actuator (FEA) technology forms the core of the soft-bodied robotic fish. FEAs are elastomer modules that bend under fluid pressure. Bending is accomplished using a two layer bimorph structure. Pressurized gas expands fluidic channels embedded within an elastomer layer and a second inextensible but flexible layer functions to constrain the axial tension generated by the expanding channels along one side. This transforms lateral stress in the elastomer into a bending moment. Moreover, three layers can be used to form a bidirectional bending FEA: an inextensible constraining layer sandwiched between both an agonistic and antagonistic expanding layer, as is demonstrated in Onal and Rus [2013]. This bidirectional FEA structure is fundamental to the soft robotic fish (see Fig. 3-1G, agonistic layer; H, constraining layer; and I, antagonistic layer). However, in this work we advance FEA technology by abandoning a simple rectangular shape and creating FEAs that conform to the complex anatomical shape of a fish. The structure and operating principles of a tapered bidirectional FEA are illustrated schematically in Figure 3-2.

3.2.2 Modeling

Onal and colleagues present a model in Onal et al. [2011] describing the total bending angle θ of a rectangular FEA using both the physical properties of the channels and internal actuator pressure p_c,

$$\theta = 2n \arctan\left(\frac{\bar{w}\,\epsilon\,(\sigma)}{2\,\bar{h}}\right), \qquad \sigma = p_c \frac{\bar{h}}{\bar{t}}. \tag{3.1}$$

Here, n is the number of channels, $\bar{w}$ is channel width, ϵ is material strain and a nonlinear function of material stress σ, and lastly $\bar{t}$ is the wall thickness or the difference in heights of the actuator and channels. However, because our actuator

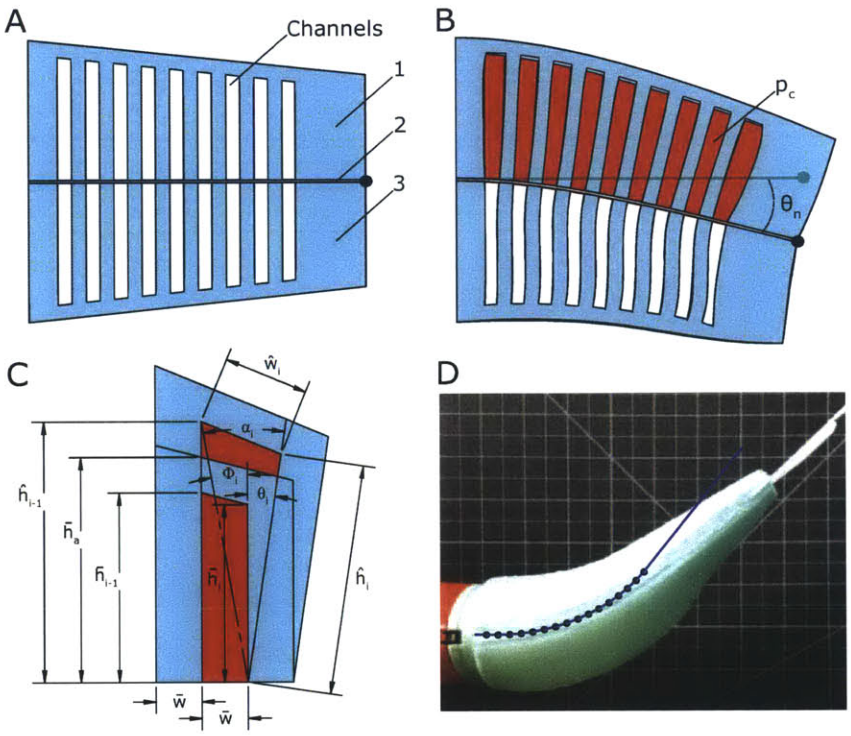


Figure 3-2: Schematic representation of a tapered bidirectional FEA in cross-section. Panel A shows the three layer structure: symmetric agonistic (1) and antagonistic (3) expanding layers sandwiching an inextensible but flexible constraining layer (2). Here, embedded channel groupings are in a depressurized state. Panel B depicts pressurized gas (red) expanding the agonistic channel group. Due to the constraining layer, fluid pressure induces a bending moment producing curvature. Panel C illustrates model parameters and D shows predicted curvature of the fish's anterior actuator overlaid atop the actuator's actual deformation.

differs considerably from a rectangular actuator with uniform channels we develop a new model. By extending the model presented in Onal et al. [2011] to include variable channel height as well as radial stress (i.e. normal to the inextensible constraint layer), we can statically model the non-uniform bending of a tapered FEA (Fig. 3-2). Specifically, the accumulated bend angle along the length of the actuator after the n^{th} embedded channel, represented as θ_n, can be estimated as a function of both the physical properties of the preceding channels and p_c,

$$\theta_n = \sum_{i=1}^{n} \alpha_i - \phi_i,$$

$$\theta_n = \sum_{i=1}^{n} \cos^{-1}\left(\frac{-\hat{w}_i^2 + \hat{h}_{i-1}^2 + \bar{w}^2 + \hat{h}_i^2}{2\hat{h}_i\sqrt{\hat{h}_{i-1}^2 + \bar{w}^2}}\right) - \tan^{-1}\left(\frac{\bar{w}}{\hat{h}_{i-1}}\right), \tag{3.2}$$

$$\hat{w}_i = \bar{w}\left[\epsilon\left(p_c\frac{\bar{h}_i}{2\bar{t}} + 1\right)\right], \quad \hat{h}_i = \bar{h}_i\left[\epsilon\left(p_c\right) + 1\right].$$

Here, i is the channel index, α and ϕ are construction angles for the indexed channel, $\bar{w}$ and $\hat{w}$ represent the initial and deformed widths of the indexed channel, and $\bar{h}$ and $\hat{h}$ represent the initial and deformed heights of the indexed channel. These parameters are illustrated in Figure 3-2C. It is important to note that this simplifying static model assumes channels deform purely by extending their side and top walls, and that these wall stresses are based on initial channel geometry. In reality, the wall stresses change as the channel surfaces deform. For this reason, this analytic model is most valid for small deformations, i.e. when pressure is low and the actual stresses approximate those calculated from initial channel geometry. The model also ignores external forces such as the compressive forces generated by the antagonistic half of the actuator. In Figure 3-2D we show a predicted bending of the fish's anterior actuator overlaid on top of the actuator's actual bending. Here, θ_n was predicted to be 52 degrees but measured approximately 45 degrees. Table 1 lists actuator specific parameter values for this experiment.

This static analysis suggests that by independently varying the height of embedded channels, net complex curvatures of the body can be achieved. A curvature profile can

be mechanically "programmed" into the body geometry of the fish. Such construction serves to simplify computational control inputs. For instance, a single binary control input can be used to drive the robot's body through a complex kinematic (Fig. 3-7).

The robotic fish used in this case study employs four silicone FEAs that are molded to replicate the slender anatomy of a natural fish, creating an actuated but continuously compliant body. The actuated body spans 43% to 100% of the robot's overall fork length (30.5 cm) (Fig. 3-1, top). Embedded fluidic channels are grouped into two independently actuated pairs: an agonistic and antagonistic anterior trunk pair ranging from 45% to 70% of fork length (Fig. 3-1C) and a posterior trunk pair ranging from 70% to 90% of fork length (Fig. 3-1E). Separating the agonistic and antagonistic channel groups is an inextensible but flexible constraining layer introduced along the fish's posterior midline (Fig. 3-1D and H). This layer enables channel stresses to generate body curvature, analogous to, though inverted from the process by which muscles generate a bending moment about the vertebrate column in a fish [Jayne and Lauder, 1993, Wakeling and Johnston, 1999].

3.2.3 Fabrication

A fabrication process was developed to first cast and then combine components of the fish's soft body. The process is illustrated in Figure 3-3. First, each halve of the body is cast from silicone rubber (Mold Star 15, Smooth-on, Easton PA) by a two-part mold. The top mold piece creates the embedded channels of both the anterior and posterior actuator grouping, and the bottom piece creates the anatomical fish shape (Fig. 3-3[1a]). In parallel, a connector piece is cast with holes to serve as access ports to each channel grouping and with an internal faceplate, allowing the connector to mate with the fish's rigid anterior (Fig. 3-3[1b]). In addition, a thin constraint layer is cast with an embedded 0.5 mm Acetal film that provides inextensibility and also forms the caudal fin (Fig. 3-3[1c]). Next, the four pieces undergo post-casting preparation and are sequentially bonded to each other using a thin layer of silicone (Fig. 3-3[2]). Once cured, the body is ready for attachment to the rigid anterior (Fig. 3-3[3]).

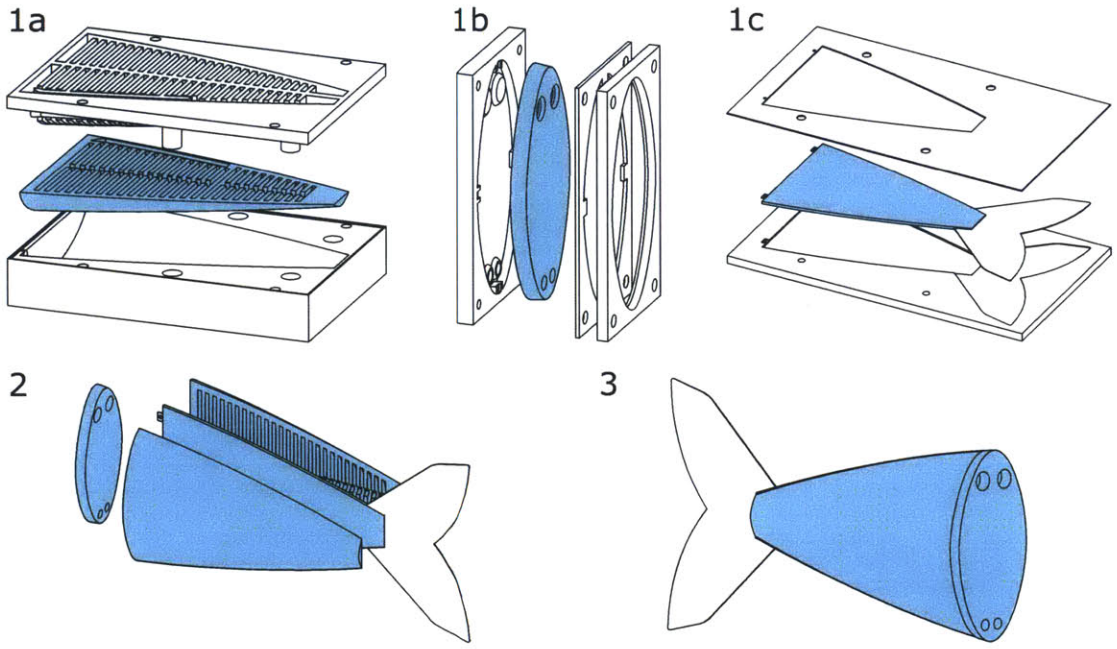


Figure 3-3: Illustration of the soft fish body fabrication process. First, two halves of the body (**1a**), a connector piece (**1b**), and a constraining layer (**1c**) are all cast from silicone using two-part molds. Next, these four pieces are sequentially bonded together using a thin layer of silicone (**2**). Lastly, once cured the fish body is ready for operation (**3**).

3.2.4 Energy

As mentioned, fluid energy is used to actuate the robot's body. Work is done on the body by the onboard power system. When the body is actuated, a portion of this work is stored as potential energy within the elastomer and compressible fluid, $V_{Elastic}$, and the remainder is dissipated due to friction inside and outside the body, $T_{Resistive}$. In order to characterize the relative amounts of work required to actuate the body at various rates, a pressure-volume analysis technique was used similar to that used to measure the work of breathing in Otis [1954]. While submerged in water, the agonistic anterior actuator was repeatedly filled to a target volume, $\mathbb{V}_c$, as flow rate into the actuator was varied, ranging from a baseline flow of 5 L/m to a maximum flow of 50 L/m. Both the change in internal actuator pressure, p_c, and change in volume, $\mathbb{V}$, were measured. We make the assumption that at the baseline flow rate, there are no resistive losses and all energy delivered to the actuator is stored elastically,

$$V_{Elastic} = \int_0^{\mathbb{V}_c} p_{c_{Baseline}}(\mathbb{V})\, \mathrm{d}\mathbb{V}. \tag{3.3}$$

At each flow rate above baseline, i, the resistive component of work done on the body can be approximated as

$$T_{Resistive} = \int_0^{\mathbb{V}_c} \left(p_{c_i}(\mathbb{V}) - p_{c_{Baseline}}(\mathbb{V})\right) \mathrm{d}\mathbb{V}. \tag{3.4}$$

Figure 3-4 shows the pressure-volume profiles at various flow rates as well as an illustration of the work calculations. Table 3.1 lists the total, elastic, and resistive components of work done on the soft body in water at various flow rates. Here, work is calculated according to Equations 3.3 and 3.4 not measured directly. The resistive component of work increases as the actuator is driven at higher rates; however, even at the highest rate of actuation nearly 78% of delivered energy is stored elastically.

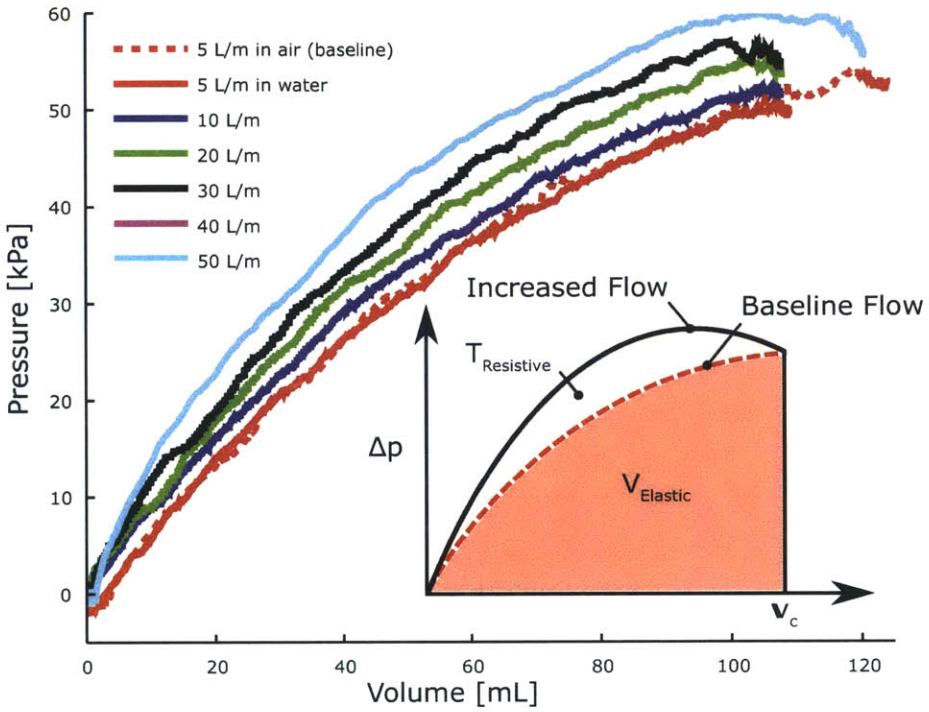


Figure 3-4: Pressure-volume profiles of fluid used to fill the anterior agonist actuator at various flow rates. By integrating the change in pressure as a function of displaced volume, the elastic and resistive components of work done on the actuator are determined.

Table 3.1: Elastic and Resistive Components of Work Done on the Soft Body in Water

Mass Flow (L/m)	Total Work (J)	Elastic Work (%)	Resistive Work (%)
5	2.89	100	0
10	3.11	93.5	6.5
20	3.36	87.1	12.9
30	3.60	81.9	18.1
40	3.69	80.6	19.4
50	3.88	77.7	22.3

3.3 Power

3.3.1 Power Supply

In order to drive locomotion there must be an apparatus onboard for supplying fluidic power. In an autonomous soft robot application, it is desirable to maximize the energy density $\left(\frac{\text{MJ}}{\text{L}}\right)$ of the power supply. As a solution, we expand on our approach in Marchese et al. [2013] and again use an 8 gram CO_2 gas cylinder (PN 80121, Leland Ltd, Inc., South Plainfield, NJ) housing fluid at high pressure and low volume (Fig. 3-5A). In this form, a relatively large amount of fluidic energy can be stored in a volume suitable for storage onboard the robot.

The total potential energy stored within, V_{Supply}, and energy density of, D, the high pressure container can be theorized by assuming the energy release is isothermal and by neglecting fluid phase changes,

$$D = \frac{V_{Supply}}{\forall} = \frac{\int_{\mathbb{V}_0}^{\mathbb{V}_f} p_s\left(\mathbb{V}\right) \mathrm{d}\mathbb{V}}{\forall}. \tag{3.5}$$

Here, the supply pressure p_s is a non-linear function of volume $\mathbb{V}$ and this relation is defined using Van der Waals gas law. The volume $\mathbb{V}_0$ is the fluid's volume corresponding to the vapor pressure of the gas, $p_s\left(\mathbb{V}_0\right) = p_{vapor}$. The volume $\mathbb{V}_f$ is the fluid's volume at standard temperature and pressure. The volume occupied by the container is $\forall$. Table 3.2 contains application specific parameters for the power supply. According to Equation 3.5 the estimated energy density of this power supply is 0.1 $\left(\frac{\text{MJ}}{\text{L}}\right)$ and the estimated total energy stored is 1.86 kJ. Although a typical lithium polymer battery has an energy density of between 0.9-1.5 $\left(\frac{\text{MJ}}{\text{L}}\right)$, electrical energy is not suitable for directly powering FEAs and in such a case the energy must first be converted to the fluid domain using supporting hardware, occupying additional volume and imposing energy losses. This motivates our use of a fluidic power supply.

3.3.2 Gas Delivery

Both an onboard gas regulation mechanism [2] shown in Figure 3-5B and delivery system, (Fig. 3-5C and D), are required to distribute the stored fluid to the robot's actuators. First, fluid is regulated to a suitable driving pressure, p_d, an order of magnitude below supply pressure. Subsequently, gas flows through a series of control valves and into the body actuators. The network of valves and fluid pathways between the regulator and actuators resist fluid flow. Consequently, inlet flow v to the actuators is driven by Δp, the pressure gradient between p_d and the actuator's internal pressure p_c. To ensure p_d is of sufficient magnitude to drive the body through its target kinematic profile, a relationship between v and Δp is theorized.

From our analysis in Figure 3-4, inlet flows of between 30 and 50 L/m are required for several hundred milliseconds. Within the gas delivery system Reynolds numbers are in excess of $3{\times}10^4$, above the critical Reynolds number of $2.3{\times}10^3$, indicating turbulent flow. Assuming fully developed turbulent flow as well as constant fluid density, viscosity, and temperature, the Δp required to drive v can be estimated follows:

$$\Delta p = p_d - p_c = \rho\, g\, h_T \tag{3.6}$$

where ρ is the fluid's density, g gravity, and h_T the total head loss of the fluid delivery system, defined as,

$$h_T = \sum_{i=1}^{n} \frac{8\, v^2}{\pi^2\, d_i^4\, g} \left(\frac{f_i\, L_i}{d_i} + \sum_{j=1}^{m} K_{i,j} \right) \tag{3.7}$$

where i indexes the n serial pathways within the fluid delivery system, L and d are the length and diameter of the pathway, K_j the loss coefficients of a pathway containing m losses, and f the friction factor as defined by the Haaland equation [Haaland, 1983].

Specifically, we calibrate Equation 3.7 to characterize the case of maximum excursion. Here, the anterior agonistic control valve is fully open, the antagonistic valve is fully occluded, and for simplicity we ignore the posterior actuator pair. The fluid pathway has three main parts: (1) the section between the regulator and valve ori-

[2] 0.29 MPa regulator (PN 50044; Leland Ltd., Inc.).

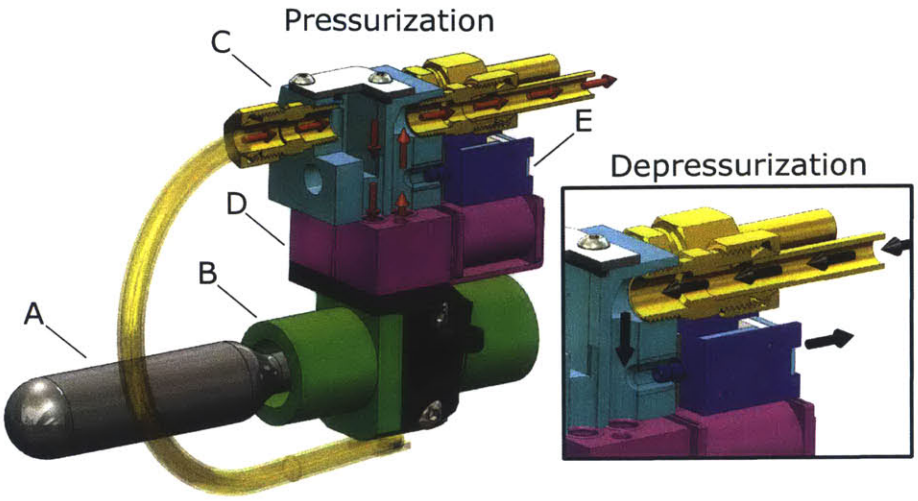


Figure 3-5: Details of gas storage, regulation, and release mechanism. The mechanism consists of a high pressure CO2 gas cylinder (A), a passive mechanical regulator (B), an interface manifold (C), proportional control valves (D), and exhaust valves (E). On the left, red arrows illustrate gas flow during actuator pressurization, and on the right, black arrow illustrate gas flow during depressurization.

fice, (2) the valve and, (3) from the valve to the actuator inlet. Pathways 1 and 3 are designed to minimize resistance and accordingly contribute little to h_T. However, constraints on valve performance and size force pathway 2 to be of high resistance. Finite Element Analysis was used to estimate losses ($\sum K$). See Table 3.2 for parameter estimates. As a result of this analysis, Δp is required to between 0.06 and 0.10 MPa and considering the range of p_c in Figure 3-4 and commercial availability, a regulator providing a p_d of 0.29 MPa was chosen.

The above analysis suggests that the gas delivery system is responsible for considerable resistive energy losses. Given an anterior actuator frequency of ω we can estimate the losses of this subsystem over a single bidirectional actuation cycle: $T_{Resistive} = (p_d - p_c)\, v\, \frac{2\pi}{\omega}$. Based on experimental data detailed in Figure 3-4, we can expect losses of around 50 J per cycle while driving the anterior actuators during forward swimming.

Table 3.2: Robot Parameters Used in Modeling Fluid Energy Supply and Fluid Delivery System

Parameter	Value
$\mathbb{V}_0$	42.7 mL
$\mathbb{V}_f$	4.37 L
$p_s(\mathbb{V}_0)$	6.08 MPa
$\vee$	18.6 mL
ρ	1.83 kg/m^3
d_1	4.72 mm
d_2	2.75 mm
d_3	4.72 mm
L_1	61 mm
L_2	3.5 mm
L_3	83 mm
f_1	4.7×10^{-3}
f_2	4.4×10^{-3}
f_3	4.7×10^{-3}
$\sum K_1$	2.35 mm
$\sum K_2$	4.79 mm
$\sum K_3$	4.76 mm

3.4 Processing and Control

The robot contains an onboard microprocessor[3] and wireless communication module[4] that enable it to both process external inputs and execute control policies. This computational system interfaces directly with onboard hardware such as control valves and any available sensors. Two proportional valves[5] are used to control the pressurization of the agonistic and antagonistic anterior actuators and two exhaust valves[6] control depressurization. Two solenoid valves[7] control pressurization and depressurization of the posterior actuator pair. Depressurization of the body actuators is primarily passive. The exhaust and solenoid valve orifices resist fluid outflow and the actuators act as fluid energy storage devices, analogous to a capacitor in the electrical domain. If we ignore the fluid's negligible inertia, we can represent actuator

[3]Atmega 644P (Atmel Corporation, San Jose, CA).
[4]XBee-PRO 900 (Digi International Inc., Minnetonka, MN).
[5]PN 921-111051-000 (Parker Precision Fluidics).
[6]PN PND-05A-12 (Parker Precision Fluidics).
[7]PN 914-232123-000 (Parker Precision Fluidics).

depressurization as an undriven first order system, analogous to a capacitor discharging through a resistor. The instant the exhaust valve is open, the pressure gradient driving exhaust flow is equal to p_c, and flow is at a maximum. Subsequently, the flow exponentially drops to zero. Although the computational system can provide onboard autonomy, the robot wirelessly communicates with a host PC to receive user commands. The fish is negatively buoyant and held at a fixed submergence by means of a floating support.

The control policy of fully actuated fish locomotion can be described using seven parameters, t_1, M_1, t_2, M_2, ϕ, t_3, and t_4. The anterior control valves are driven using a square wave input and agonistic and antagonistic valves are energized sequentially ensuring while one actuator is pressurizing the other is exhausting. Here, t_1 and t_2 are the open periods of the anterior agonistic and antagonistic actuator proportional control valves, respectively. The corresponding orifice magnitudes M_1 and M_2 of these control valves are defined as a percent of maximum available flow. The parameter ϕ is the phase delay between the anterior and posterior actuator pairs, and t_3 and t_4 are the open periods of the posterior agonistic and antagonistic actuator control valves respectively and their magnitudes are fixed. However, the body can also be driven as an underactuated system where fewer than the total number of available actuators are used. For forward locomotion this abbreviated policy can be parameterized using only t_1, M_1, t_2, and M_2. For escape responses this policy can be further reduced to using only t_1 and M_1. During escape responses the posterior agonistic actuator is also used, but t_3 is set equal to t_1.

3.5 Capabilities

3.5.1 Swimming

As mentioned, the fish's body can function as an underactuated system, where a single control input can excite multiple modes of motion. Using just the anterior actuator pair as opposed to all four available actuators for forward swimming allows the fish

to conserve the limited fluid energy stored onboard. Similar to the work of Valdivia y Alvarado and Youcef-Toumi [2006], our fish can use just anterior trunk actuators to excite movement in the entire compliant body, producing fast forward swimming. During the experiment detailed in Figure 3-6 the fish operated using an open-loop controller and its objective was to swim in a straight line. Here, the posterior trunk actuator pair was passive, the anterior trunk actuators were periodically driven at 1.67 Hz, and linear velocities of 150 mm/s, or 0.44 body lengths/s, were attained. Approximately 30 tail beats were available from the 8 gram CO_2 cylinder under these experimental conditions. Forward locomotion performance parameters were measured using a high speed video camera and a mirror mounted at 45 degrees underneath the robot. Video was recorded at 300 fps and 9 colored points along the robot's dorsal midline were digitized during post-processing.

These forward swimming results show that the self-contained actuation system is capable of sustained operation, exemplified by forward fish-like locomotion. However, the swimming gait we present here is certainly sub-optimal. A comprehensive sweep over driving pressures, tail frequencies, and body stiffness would allow optimization of forward locomotion.

3.5.2 Escape Response

A critical behavior exhibited among fish is an escape response, which involves rapidly bending the fish's body to large angles in order to accelerate away from adverse stimuli. Experiments were conducted to investigate the escape response of our soft-bodied robotic fish and compare its maneuvers to that of natural fish. A total of 28 escape-response experiments are reported and were carried out in a 76 cm long by 76 cm wide by 61 cm tall water tank filled to 41 cm. The robot was completely submerged and held at an initial fixed submergence by means of a 172 cm long nylon string. Escape responses were generated using an open-loop controller. Escape response performance parameters were measured using the same high speed videography techniques as listed for the forward locomotion experiment; however, here the camera was positioned overhead. *Heading angle* is defined in degrees as the rotational

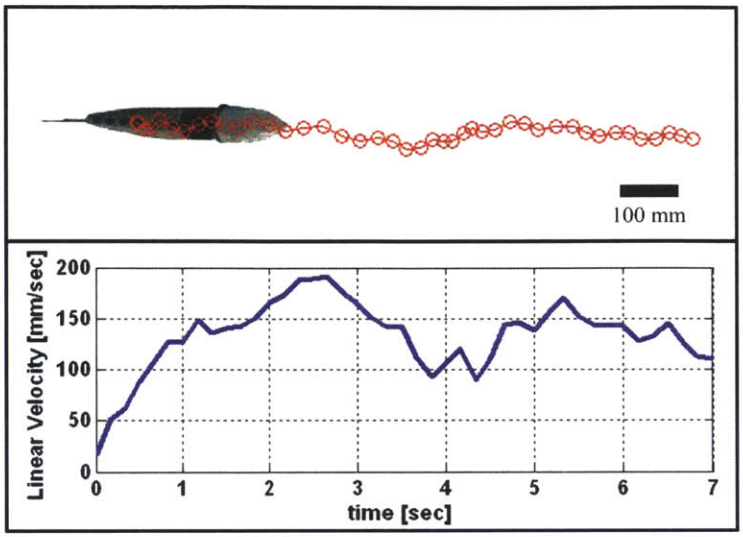


Figure 3-6: Experimental results of the robotic fish during forward swimming. The top panel shows the digitized average body midline position moving as a function of time. The bottom panel details the corresponding linear velocity of the center of mass as a function of time. During this experiment tail stroke frequency is 1.67 Hz and a velocity of approximately 150 mm/s is attained.

displacement of the rigid anterior midline. *Escape angle* is defined as the resulting heading angle after the maneuver when the angular rate drops to zero. *Linear escape velocity* is defined as the Cartesian velocity of the center of mass 500 milliseconds after the onset of the maneuver.

Fish 30-40 cm in length (rainbow trout and northern pike) have been shown to perform type I, or single-bend, escape responses where the body initially bends into an "S" shape and subsequently into a "C" shape with heading angles greater than 90 degrees [Harper and Blake, 1990]. Our robot exhibits fishlike mechanisms to accelerate (see the exemplary single-bend escape response in Figure 3-7A-D and Figure 3-8A). We create single-bend responses by using just the anterior and posterior agonistic body actuators. Here, from 0 to 150 ms the body's midline assumes an "S" shape with the caudal fin curving in the opposite direction to the trunk. From 150-500 ms the body's midline assumes a "C" shape as the caudal fin, peduncle region, and trunk exhibit the same direction of curvature. The robot reaches a maximum heading angle of 100 degrees at approximately 520 ms. The aforementioned natural fish also

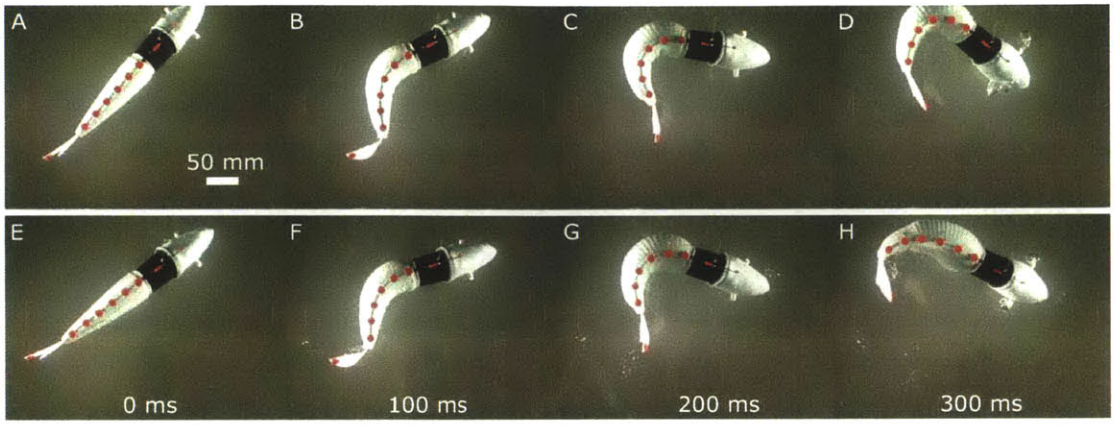


Figure 3-7: Sequences depicting the soft robotic fish performing both a single-bend (**A-D**) and double-bend (**E-H**) escape response. The single-bend response requires only agonistic actuator effort. The double-bend response requires sequential agonistic and antagonistic actuator efforts causing a significant decrease in heading angle and ultimately resulting in lower escape angles than single-bend responses. Actuator effort durations of 160 ms were used in both escape responses.

exhibit type II, or double-bend, escape responses characterized by the fish's anterior region bending away from the posterior midline's primary direction of curvature as the body assumes a "C" shape [Harper and Blake, 1990]. In Figure 3-7E-H and Figure 3-8B, the robot performs a double-bend escape response using both anterior and posterior agonistic and antagonistic body actuators. Here, the "C" shape is assumed at 170 ms and is quickly preceded by the body straightening. A maximum heading angle of 72 degrees is obtained at 410 ms. In general, the robotic fish exhibits higher escape angles in single-bend than in double-bend escape responses. Figure 3-9 portrays both a single and double-bend escape response of an angelfish (body length of 7.3 cm) exhibiting a similar kinematic pattern to our robot; that is, the angelfish is also shown to have higher escape angles in single-bend than in double-bend responses [Domenici and Blake, 1991].

Foreman and Eaton [1993] document that single-bend maneuvers result in higher escape angles and slower center of mass motion, while double-bend maneuvers result in lower escape angles and faster center of mass motion. Consequently, although antagonistic effort is not required for center of mass motion, it is responsible for amplifying the center of mass motion and a period of negative angular rate in natural

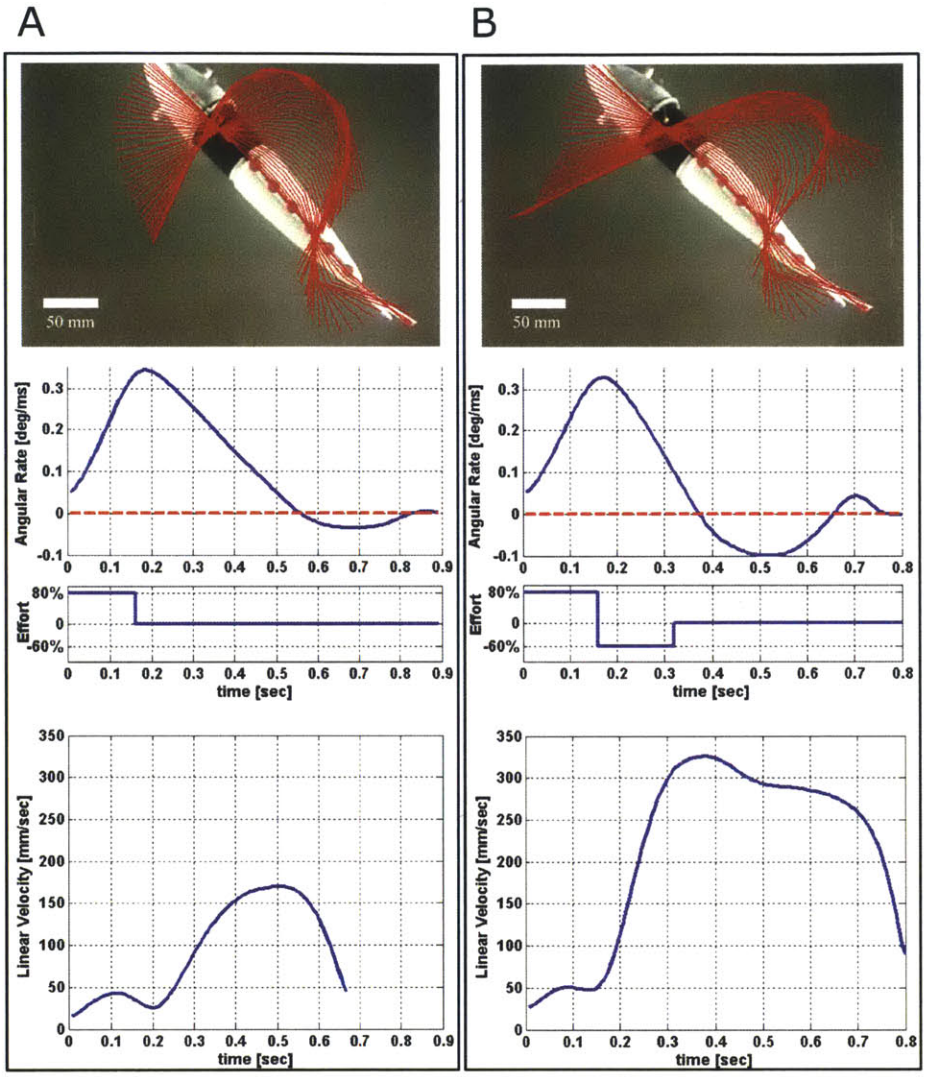


Figure 3-8: Escape response kinematics of the soft-bodied robotic fish. Panel (**A**) details kinematics of a typical single-bend escape response for the robotic fish; similarly, panel (**B**) details a double-bend escape response. The top portions of the panels show the digitized body midline (red) overlaid every 10 ms from the first detectable motion to the end of maneuver. The middle portions show the corresponding angular velocity of the head along with actuator effort (agonistic is positive, antagonistic is negative). And at the bottom is the resulting center of mass velocity for each maneuver.

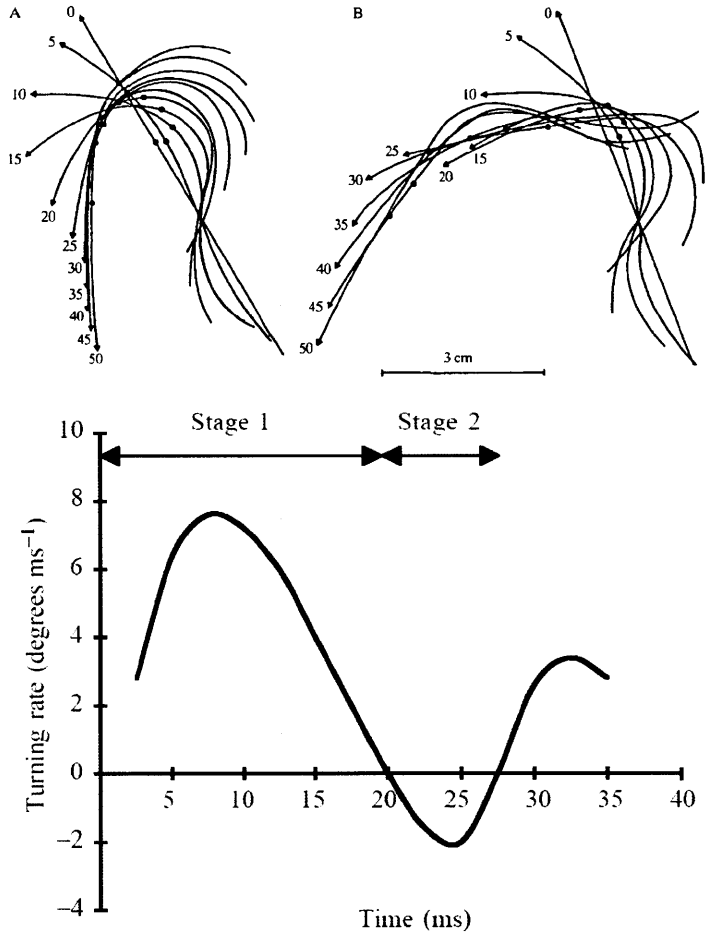


Figure 3-9: Fast-start kinematics of an angelfish. At the top is the body midline plotted for a single-bend (**A**) and a double-bend (**B**) fast-start. At the bottom is the corresponding angular velocity profile for the double-bend fast-start. These figures are reproduced with permission from Domenici and Blake [1991] and Domenici and Blake [1997], respectively.

fish. An important result is that our robot exhibits similar behavior: In double-bend responses escape angle is lower and linear escape velocity higher than in the single-bend responses. In double-bend escape responses having antagonistic actuator activity (Fig. 3-8B), the escape angle is reduced, as there is significant angular rate in the negative (antagonistic) direction starting at approximately 380 ms (60 ms after the completion of antagonistic activity). Onset of significant center of mass motion occurred at 160 ms, approximately synchronized with the onset of antagonistic activity.

A point of contrast is that in natural fish the head and tail both move towards each other in the first stage of the escape response (Fig. 3-9). However, because the center of mass of our robot is in the anterior head region, head movement is greatest during the second stage of the escape response.

Eaton, Lee, and Foreman's direction change hypothesis in natural fish [Eaton et al., 2001] is consistent with the behavior of our robotic fish. Specifically, a combination of agonistic and antagonistic actuator inputs can independently influence both escape angle and linear escape velocity during the robot's escape response. A series of experiments were carried out to investigate the effect of two parameters, t_1[8] and M_2, on the escape angle and linear escape velocity of the robotic fish (Fig. 3-10). As a result we found t_1 has marked influence on escape angle. Mean escape angle at t_1 equal to 100 ms was 26.6 degrees, whereas mean escape angle was 81.4 degrees at t_1 equal to 160 ms. Also, M_2 provides control over escape angle. Increasing M_2 decreased escape angle (Fig. 3-10). Unlike escape angle, t_1 has minimal influence on linear escape velocity, mean values of 193.1 mm/s and 207.2 mm/s for 100 and 160 ms, respectively. However, M_2 provided control over linear escape velocity in that increasing effort level exponentially increased velocity (Fig. 3-10B). These findings indicate that through a combination of agonistic and antagonistic actuator efforts, escape angle and linear escape velocity can be independently altered. Given a fixed maximal agonistic effort level, altering t_1 and M_2 can independently influence both the resulting escape angle and escape velocity.

[8] $t_1 = t_2$ in all tests.

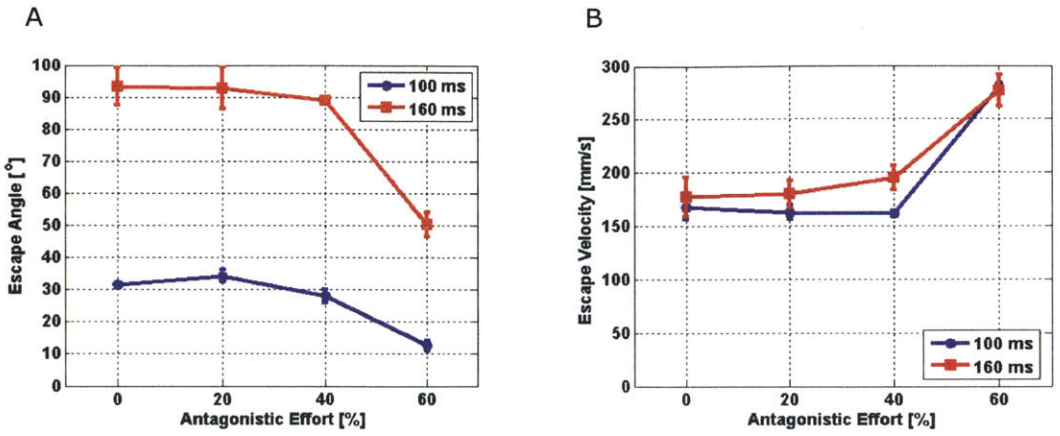


Figure 3-10: Input-output relationship of escape response maneuvers in the robotic fish. (**A**) Escape angle as a function of antagonistic actuator effort. (**B**) Escape velocity as a function of antagonistic actuator effort. In both cases equal duration agonistic and antagonistic efforts of 100 and 160 ms were used, blue and red lines respectively. Data points represent mean values (n=4 and n=3 for 100 and 160 ms scenarios respectively for a total of 28 tests) and error bars represent S.D.

3.6 Discussion

The robotic fish provides an instantiation of our approach to creating autonomous soft-bodied robots capable of rapidly achieving continuum-body motion. In this system, soft muscle-like actuators generate curvature in a continuously deformable, vertebrate-like body. Novel, form-independent actuator technology as well as miniaturization of supporting hardware enable the robot to take on the fundamental anatomical structure of a fish while being self-contained and unconstrained.

The programmability of our system allows repeatable evaluation of the robotŠs escape response maneuvers. By directly controlling the duration and magnitude of agonistic and antagonistic actuator efforts and measuring the resulting escape response performance, we conclude that agonistic duration has strong authority over escape angle and minimal authority over linear escape velocity. Antagonistic magnitude has nonlinear control authority over both escape angle and escape velocity. Increasing agonistic effort duration allows for greater angular displacement of the head to be reached during the first stage of the response. Soon after this duration of time, the robotŠs head begins to decelerate. The longer the agonistic effort is applied, the

greater the escape angle. When antagonistic effort magnitude increases, there is more energy to decelerate the turn and consequently escape angle lowers. The antagonistic effort also provides the propulsive stroke, so greater antagonistic effort yields higher escape velocity.

Evidence suggests that a similar inputŨoutput relationship holds in biological escape response behavior. Consistent with our robotic system, Foreman and Eaton [1993] presented the direction change concept, where they show that escape response heading angle is a function of the relative magnitudes and timings of agonist and antagonist muscle contractions. Wöhl and Schuster [2007] investigated the predictive start of hunting archer fish and showed that in the underlying C-start behavior, escape angle and escape velocity need to be decoupled. Also in line with our robotic system, Tytell and Lauder [2002] found that, among investigated variables, stage-one duration correlated most strongly with escape angle and that antagonistic effort magnitude correlated most strongly with escape velocity in biological fish. Our findings also suggest that despite the apparent complexity of the maneuver, it is feasible that a robotic system with limited onboard computational power could determine required escape response control parameters in real time and with no a priori planning. Because the robotŠs escape response performance outputs, angle and velocity, are inherently decoupled by the physical form of the body and structure of the maneuver, control parameters could potentially be computed onboard. For instance, the maneuver may start in immediate response to a perceived external stimulus with a predetermined maximal agonistic effort. As this effort occurs, the desired escape angle and velocity may be computed relative to the perceived stimuli. Accordingly, the control variables t_1 and M_2 can be determined given a mapping similar to that provided in Figure 3-10.

Chapter 4

Planar Multi-Segment Soft Robots[1]

4.1 System Overview

Our desired capabilities for the planar manipulation system include autonomous trajectory-following in free space, moving dexterously through confined spaces, and/or grasping and placing objects, all without human intervention. Figure 4-1 depicts the aggregate physical system and highlights the major subsystems.

Three multi-segment planar soft manipulator morphologies were developed. Each design consists of two distributed fluidic elastomer actuators per segment. These manipulators move with minimal friction on a level surface (Fig. 4-1d). A localization system (Fig. 4-1b) provides real-time measurements of segment end-point positions along the neutral-axis of the manipulator. This localization data can be provided by an exteroceptive system (e.g., motion capture) or a proprioceptive system (e.g., embedded sensors). Our current implementation uses an external motion capture system (Opti Track, NaturalPoint) and this supports our experimental evaluation by providing accurate ground truth measurements.

Low-level controllers continuously adjust the fluid volume and pressure within the manipulator using an array of fluidic drive cylinders (Fig. 4-1c) to control the curvature of each segment. Every low-level curvature controller runs on its own

[1]This chapter is based on Marchese et al. [2015a], Marchese et al. [2014c], Marchese et al. [2014a], and Katzschmann et al. [2015]

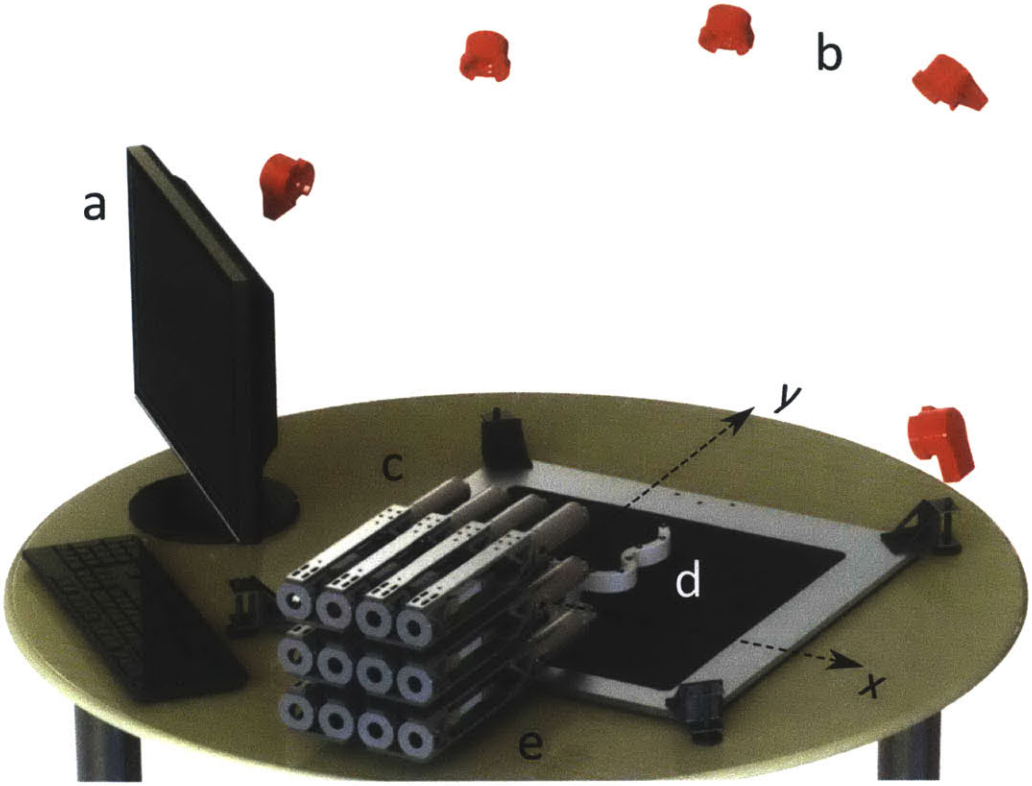


Figure 4-1: An overview of the major components comprising the soft robotic manipulation system. These are: a soft and highly compliant manipulator (d), a localization system (b), an array of fluidic drive cylinders (c), arm segment curvature controllers (e), and a computer for multi-segment kinematics computations (a).

embedded system (Fig. 4-1e), which is mounted within the housing of each fluidic drive cylinder. A nearby computer (Fig. 4-1a) takes the aforementioned tracking information as input and then runs a suite of motion planning algorithms. A rigid frame, not shown in Figure 4-1, holds all the subsystems together providing reliable and consistent hardware experiments.

4.2 Actuation

In this section we detail the design and fabrication of three soft fluidic elastomer manipulators. The primary design constraint is that the manipulators should be com-

posed almost entirely from soft materials. Each of the three actuation morphologies detailed in this section can be integrated within a planar soft manipulation system. The morphology of the actuator affects the curvatures achievable by the actuator and its scope.

4.2.1 Operating Principles

Despite the variability in fluidic elastomer manipulator morphologies, their fundamental operating principles are universal. This section provides an overview of these operating principles. Generally, each segment of a fluidic elastomer manipulator bends and this bending is due to material strain. Figure 4-2 illustrates how unidirectional bending arises from material strain. Consider a block of elastomer where the edges of the top and bottom surfaces have equal lengths, L_0. If the top surface is strained such that its new edge length is $L_0 + \Delta L$, but the bottom of this block remains unextended, then the elastomer will bend. Bending is the basic motion primitive of the fluidic elastomer manipulator.

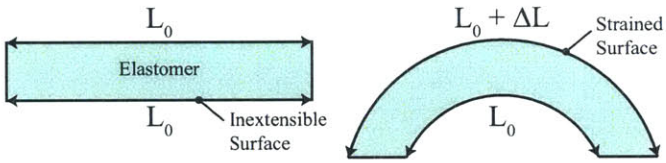


Figure 4-2: Operating principle of a bending elastomer segment. One surface of the elastomer is strained while the opposite side remains unextended. The difference in length produces bending.

In order to generate strain within the elastomer, this class of manipulators uses pressurized fluids. Essentially, expandable, fluid-filled chambers are embedded within the elastomer. When these chambers are pressurized, the entrapped fluid generates stress in the material causing the material to strain. This concept is illustrated in Figure 4-3A and Figure 4-3B. Here, the entrapped fluid is shown in yellow and its pressure is p_c. In order to express the relationship between fluid pressure and elastomer deformation, we can use a one-dimensional simplification of an iterative model presented in Marchese and Rus [2015]. Let $\bar{h}$ and $\bar{t}$ be the initial undeformed diam-

eter and wall thickness of a cylindrical elastomer channel, and let $\hat{h}$ and $\hat{t}$ represent the deformed diameter and wall thickness. Algorithm 1 expresses how the channel's diameter grows as a function of pressure. Stresses are successively updated based on deformed channel dimensions. Here, $\Delta \mathbf{p}_c$ is a vector of all consecutive incremental pressure increases until the maximum channel pressure $p_c^{\max}$ is reached. The stress and strain in the elastomer are represented by σ and ϵ, respectively. The procedure `strainLookUp()` provides a nonlinear mapping from stress to strain.

Algorithm 1: Iterative Channel Deformation

Input: $\bar{t}$, $\bar{h}$, $\Delta \mathbf{p}_c$, $p_c^{\max}$

$\hat{h} \leftarrow \bar{h}$.

$\hat{t} \leftarrow \bar{t}$.

$\bar{c} \leftarrow \pi \left(\frac{\bar{t}}{2} + \bar{h} + \frac{\bar{t}}{2} \right)$.

$p_c \leftarrow p_{\text{atm}}$.

$i \leftarrow 0$.

repeat

$\sigma \leftarrow p_c \frac{\hat{h}}{2\hat{t}}$.

$\epsilon \leftarrow$ `strainLookUp`(σ).

$\hat{c} \leftarrow \bar{c}(1+\epsilon)$.

$\hat{h}, \hat{t} \leftarrow$ `solve` $\left\{ \begin{array}{l} \textbf{Circumferential Strain:} \\ \hat{h} = \frac{\hat{c}}{\pi} - \hat{t} \\ \textbf{Conservation of Material Volume:} \\ \pi \left[\left(\frac{\hat{h}}{2} + \hat{t} \right)^2 - \frac{\hat{h}^2}{4} \right] = \pi \left[\left(\frac{\bar{h}}{2} + \bar{t} \right)^2 - \frac{\bar{h}^2}{4} \right] \end{array} \right\}$.

$p_c \leftarrow p_c + \Delta p_{c,i}$.

$i++$

until $p_c \geq p_c^{max}$

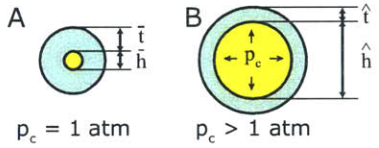


Figure 4-3: Operative principle of producing material strain through fluidic power. (**A**) Fluid, shown in yellow, is entrapped in an elastomer channel. (**B**) When the fluid is pressurized, stress and therefore strain are generated in the material.

4.2.2 Actuator Morphologies

This section provides an in-depth look at three separate soft elastomer body segments actuated using pressurized fluids. We use a defining structural feature to refer to each

of the presented segment morphologies, those are (i) ribbed, (ii) cylindrical, and (iii) pleated. In Section 4.2.3, these segments are combined serially to form multi-body manipulators. Although similar in material composition and function, differences in internal and external structure and form lead to several distinct difference between the three presented morphologies. First, we present each morphology, examining the structural differences. Then, we provide a comparative characterization of the segments, highlighting salient performance characteristics.

Ribbed Segment

The ribbed fluidic elastomer actuator with its multiple rectangular channels was first implemented and characterized in Correll et al. [2010] followed by Onal et al. [2011], Onal and Rus [2013]. Joining two fluidic elastomer actuators in an agonist-antagonist pairing provides bidirectional bending. This actuator type provided the fundamental segment-level structure of the manipulator developed in Marchese et al. [2014c]. We refer to this three layer composite here as a ribbed segment. That is, two actuator layers are combined in a pair, but separated by an inextensible constraint layer. An implementation of this segment morphology is shown in both a neutral (Fig. 4-4A) and bent state (Fig. 4-4B). Bending is produced through the pressurization of agonist fluidic channels (Fig. 4-4b) that are embedded within the actuated layers (Fig. 4-4, layers 1 and 3). The structure of the actuated layers is cast from soft elastomer (Fig. 4-4a). When pressurized, the agonist fluidic channels expand and strain the elastomer. This deformation is transferred into bending by means of an inextensible but flexible constraint (Fig. 4-4c) embedded within the center layer (Fig. 4-4, layer 2). Ribs located between channels (Fig. 4-4e) mitigate strain normal to the inextensible neutral axis. At the segment level, Marchese et al. [2014c] extended the ribbed segment design to make it suitable for inclusion in a multi-segment manipulator. Specifically, fluidic supply channels (Fig. 4-4d) were introduced on either side of the inextensible constraint and embedded within the center layer. Each segment accommodates multiple, parallel supply channels, two for each body segment within the manipulator. For a detailed model of how a ribbed segment deforms under flu-

idic pressure input, please refer to Marchese et al. [2014d]. It is important to note that this simplifying static model assumes that ribbed channels deform purely by extending their side and top walls, and that these wall stresses are based on initial channel geometry. In reality, as is shown here in Algorithm 1, wall stresses change as a function of the deformed geometry. If needed, Algorithm 1 can be used to augment the ribbed model in Chapter 3, Section 3.2.

Pros: The primary benefits of this morphology in relation to alternatives presented in this section are: (1) Ribs between channels mitigate strain normal to the neutral axis. (2) For a fixed fluid energy input, this segment exhibits greater bending than the cylindrical segment.

Cons: The primary disadvantages of this morphology in relation to alternatives presented in this section are: (1) The three layer structure is prone to delamination and rupture under high strain. (2) Manufacturing this rectangular, layered structure is challenging because all transmission lines must be embedded within the thin constraint layer.

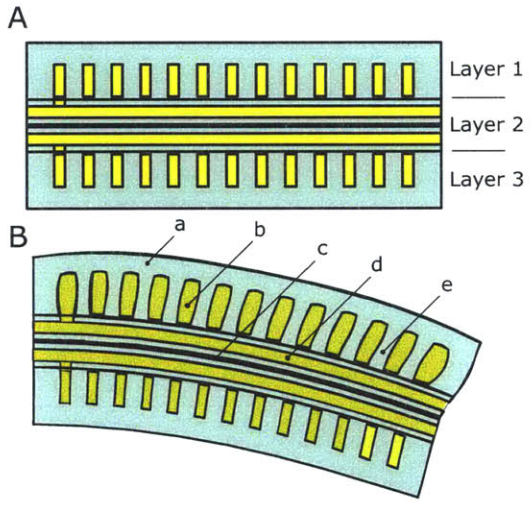


Figure 4-4: A conceptual representation of the ribbed segment morphology. The segment is composed of three layers produced from soft elastomer (a), embedded fluidic channels (b), inextensible, but flexible constraint (c), embedded fluid transmission lines (d), and ribbed structures (e). (**A**) The segment in an unactuated, or neutral state. (**B**) The segment in an actuated state where fluid within the agonist channel group is pressurized producing bending about the inextensible axis.

Cylindrical Segment

The cylindrical fluidic elastomer segment is an alternative to the ribbed design. This design was first presented by the authors in Marchese et al. [2014b]. Design inspiration was drawn from the soft rubber tentacles developed by Martinez et al. [2013] which use embedded crescent-shaped channels in a similar two-layer rubber construction. Although the cylindrical segment morphology is notably different from the ribbed segment, the fundamental operating principles are the same. In the cylindrical morphology (Fig. 4-5A and B), we transition from a rectangular, planar-layered composite to a cylindrical, concentric-layered composite. Specifically, the segment consists of three concentric layers: (i) an outer soft layer (Fig. 4-5b, *transparent*), (ii) a slightly stiffer inner layer (Fig. 4-5d, *green*), and (iii) a hollow core that accommodates a bundle of fluid transmission lines (Fig. 4-5f, *white*). Two fluid-filled, and cylindrically-shaped channels are embedded laterally within the outermost layer (Fig. 4-5c). These channels interface with the transmission lines by means of a stiffer rubber inlet piece (Fig. 4-5a, *brown*). When pressurized, the entrapped fluid deforms the embedded channel both circumferentially and longitudinally (Fig. 4-5B). Specific to this morphology, the inner tube-like layer composed of slightly stiffer rubber serves as an inextensible constraint, transforming channel deformation into segment bending.

Pros: The primary benefits of this morphology in relation to alternatives presented in this section are: (1) Entirely composed of rubber, the resiliency and the durability of the actuator are increased. (2) The two cylindrical channels make this segment the simplest to fabricate. (3) Embedded fluidic channels are not at the interface between fabricated layers, making this morphology robust against delamination under high pressures.

Cons: The primary disadvantages of this morphology in relation to alternatives presented in this section are: (1) The simple channel design exhibits high circumferential strain. Compared to the ribbed and pleated morphologies, more fluid energy is required to produce bending. (2) When the segment bends, an increased volume

of rubber on the antagonist side of the actuator has to be compressed. This inhibits a high maximum curvature.

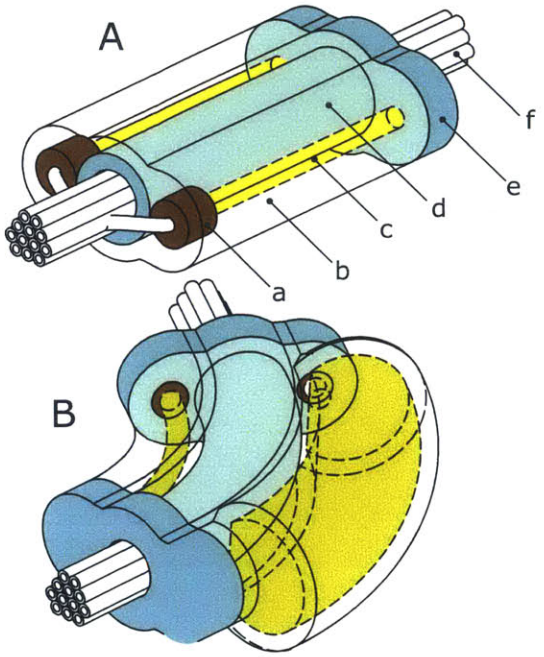


Figure 4-5: A conceptual representation of the cylindrical segment morphology. The segment consists of a soft silicone rubber outer layer (b, *transparent*), a slightly stiffer silicone inner layer (d, *cyan*), crush resistant silicone inlets (a, brown), expanding embedded fluidic channels (c, *yellow*), and an internal tubing bundle (f, *white*). The segment terminates in soft endplates (e). (**A**) A depiction of the segment in an unactuated state. (**B**) A depiction of the body segment in an actuated state where the expansion of the pressurized fluidic channel is schematically represented.

Pleated Segment

The pleated channel design is detailed in Figure 4-6 and consists of evenly spaced, discrete elastomer sections (Fig. 4-6d), which are separated by gaps (Fig. 4-6c). Embedded within each elastomer section is a hollow channel (Fig. 4-6e). Cut views of the un-actuated and actuated states are shown in Figure 4-6A and Figure 4-6B, respectively. This design approach draws inspiration for its pleats from the soft pneumatic gloves developed by Polygerinos et al. [2013] and its homogeneous body design is inspired from the tail design of a soft robotic fish developed by Katzschmann et al. [2014]. The hollow channels within each pleat are connected via a center channel and

are accessible through a front inlet (Fig. 4-6a). When fluid within these channels is pressurized (Fig. 4-6, *yellow*), an individual pleat undergoes a balloon-like expansion of the thin exterior skin both normal and parallel to the neutral axis. Similar to the cylindrical actuator design, a stiffer silicone layer (Fig. 4-6, *blue*) serves as an almost inextensible constraint layer. The sum of the balloon-like expanding motions leads to bending of the less extensible center constraint layer.

Pros: The primary benefits of this morphology in relation to alternatives presented in this section are: (1) A unidirectional pleated actuator is capable of bending to higher curvatures than the ribbed or cylindrical morphology. (2) A bidirectional pleated segment is capable of exerting higher maximum forces because of its ability to accommodate the largest energy input. (3) Using a lost-wax casting approach, the *cyan* portion of this segment can be cured in a single step, avoiding seams that are prone to delamination.

Cons: The primary disadvantages of this morphology in relation to alternatives presented in this section are: (1) The morphology is more complex to manufacture because it requires a lost-wax casting procedure detailed in Section 4.2.4. (2) The implementation of this morphology requires the most fluid energy to actuate it to appreciable tip forces. This might very well be due to the fact that, when compared to the other implementations, this implementation is larger in size and uses a higher shore hardness elastomer.

Comparative Characterization

To characterize the actuated segments, we first perform bending tests to experimentally determine the relationship between the segment's neutral axis bend angle θ, internal channel pressure p_c, and supplied volume $\mathbb{V}_s$ for each morphology. In these experiments, the base of each segment is grounded securely in a fixture and the segment's tip is supported vertically with a ball transfer. Then, the segment's agonist channel is incrementally filled under closed-loop volume control via the displacement of a fluidic drive cylinder; please refer to Section 4.3. After each incremental fill, we allow pressure within the cylinder and within the actuated channel to equalize be-

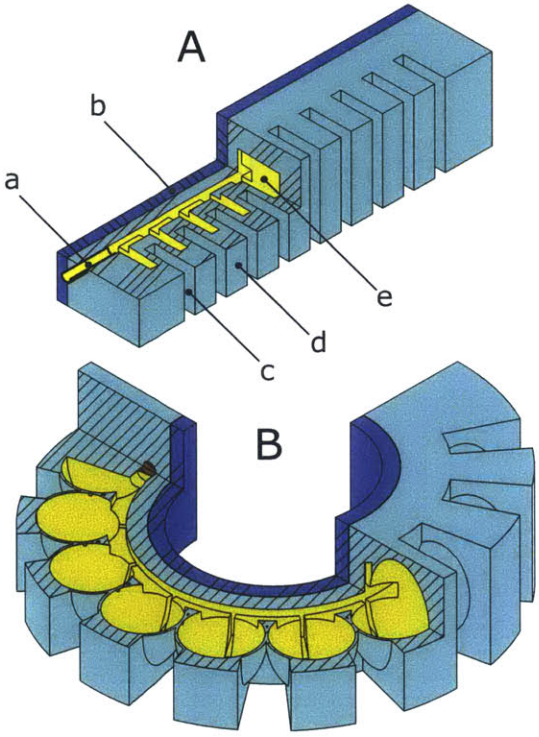


Figure 4-6: A conceptual representation of the pleated segment morphology. The design consists of a channel inlet (a), an almost inextensible constraint layer (b), uniform pleats (d) separated by even gaps (c), and internal channels within each pleat (e). (**A**) depicts the segment in an unactuated state and (**B**) shows the segment in an actuated and therefore bent state. The expansion of the pressurized channels are schematically represented.

fore measurements of the channel's pressure and the segment's curvature are taken. Curvature is assumed to be constant along the length of the segment and is uniquely defined by measuring the cartesian locations of the base and the tip of the segment; refer to Section 4.4.2. From this curvature we compute the segment's bend angle. Since this is a quasi-static process, fluid pressure and supply volume measurements can be used to determine the elastic potential fluid energy input into the actuation system, which consists of the elastomer segment and the internal compressible transmission fluid. The potential energy is calculated by

$$V_{Elastic} = \int_0^{\mathbb{V}_c} p_c\left(\mathbb{V}\right) \mathrm{d}\mathbb{V}. \tag{4.1}$$

Additionally, a blocking force test is performed in order to understand the variability in tip force output between segment morphologies. Again, a similar experimental procedure is used as for the bending characterization; however, during blocking force experiments a plate attached via a force transducer to ground is mounted in contact with the segment's tip, orthogonal to the bending plane. This effectively measures the force required to block the actuator from bending.

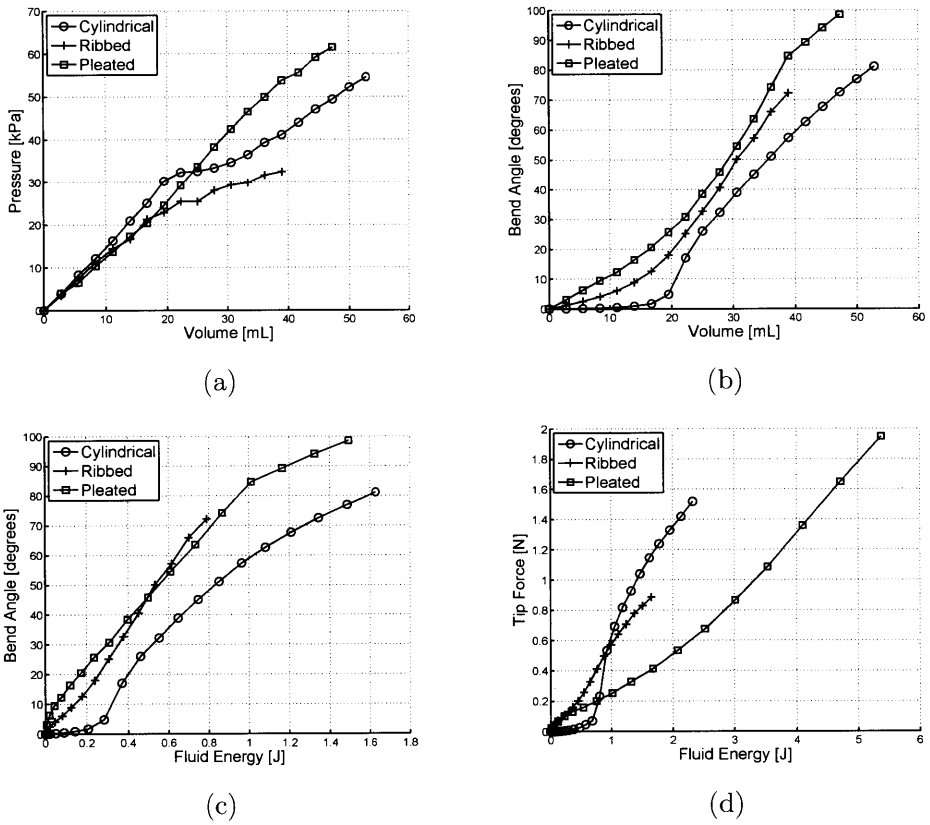


Figure 4-7: Experimental characterizations of three actuated segment morphologies performed by filling each actuator by means of controlled volumetric displacements and measuring internal pressure, neutral axis bend angle under a constant-curvature assumption, and blocking force.

Figure 4-7 details the results of these characterization experiments from which we can make several observations. First, the pleated morphology is generally the stiffest, followed by the cylindrical, and then the ribbed, where stiffness is defined as $\frac{\partial p_c}{\partial \mathbb{V}_c}$ (Fig. 4-7a). Second, the cylindrical morphology has a salient bend angle

nonlinearity (Fig. 4-7b). More specifically, small volumetric fluid changes of less than 15 mL provide little control authority over curvature; however, above 25 mL displacements, the control authority is strong and the curvature-volume relationship is approximately linear. Third, the cylindrical morphology requires the most amount of fluid energy to produce a given bend angle and the ribbed and pleated segments require approximately the same amount of fluid energy to generate equivalent bending (Fig. 4-7c). Lastly, the pleated segment generally requires more fluid energy than both the ribbed and cylindrical morphologies to produce a given tip force. However, the pleated segment can accommodate significantly higher input energies and therefore can reach the highest maximum tip force.

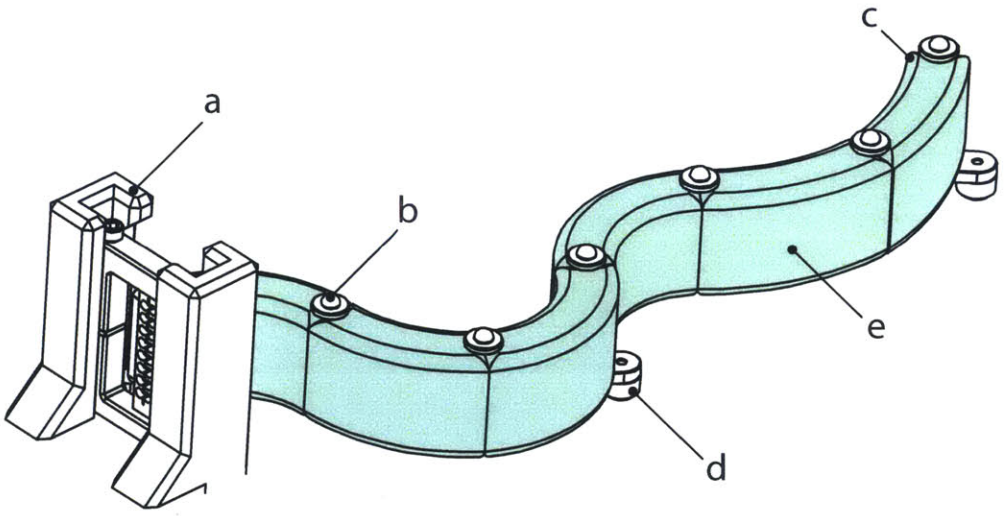
4.2.3 Multi-Segment Manipulators

Soft and continuously deformable manipulators can be assembled from bending fluidic elastomer segments. Specifically, in this section we detail how multi-segment manipulators can be composed by serially concatenating the actuated segments that were presented in Section 4.2.2.

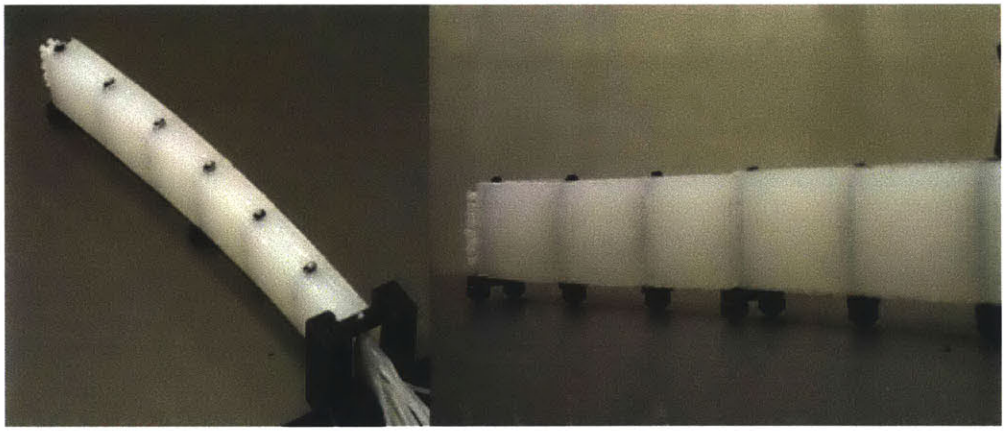
Ribbed

Structurally, a ribbed arm is composed of serially concatenated, homogeneous ribbed body segments. In this work, the manipulator's reachable envelope is constrained to the x-y plane, as shown in Section 4.1. By volume, over ninety-seven percent of the ribbed manipulator is soft silicone rubber, excluding the feet. This manipulator is depicted in Figure 4-8a and can theoretically be composed of any number of the aforementioned ribbed segments (Fig. 4-8ae). Practically, we have constructed a six-segment prototype (Fig. 4-8b). All twelve fluidic transmission lines as well as channel-to-supply interfaces are embedded within the manipulator's center layer. Markers are located at the interface between segments (Fig. 4-8ab), making segment endpoints identifiable to an external localization system. The starting point of the arm's first segment (Fig. 4-8aa) is grounded to the platform on which the arm moves and we

refer to this as the base. Ball transfers (Fig. 4-8ad) are also located at each segment endpoint to allow the arm to move on a two-dimensional plane with minimal friction. In many experiments conducted throughout this work, the pose of the arm's end-effector (Fig. 4-8ac) is controlled.



(a)



(b)

Figure 4-8: A ribbed soft manipulator prototype. (**a**) The arm is composed of homogeneous and independently actuated ribbed segments (e). The base of the arm's first segment is fixed (a) and the end of its last segment is the end-effector (c). Markers (b) identify the endpoints of each segment and ball transfers (d) mitigate friction. (**b**) Photographs of the ribbed manipulator prototype.

Cylindrical

We can also compose a manipulator from cylindrical fluidic elastomer segments, as shown in Figure 4-9. Just as in the ribbed composition, cylindrical segments are joined end-to-end. Here, fluid transmission lines are passed through the manipulator's hollow center. This feature not only facilitates segment concatenation, but also allows for modular composition of a manipulator, because transmission lines are not permanently embedded within the elastomer. Additionally, this manipulator type is only composed of soft silicone rubber as there is no inextensible constraint. No other materials are used, except for the attached ball transfers to mitigate ground friction.

Pleated

A manipulator can also be composed from pleated fluidic elastomer segments, as shown in Figure 4-10. Just as in the ribbed and cylindrical composition, pleated segments are joined end-to-end. The fluid transmission lines are passed through along the central axis of the segments. A supportive hollow profile can be added to combine two segments. This pleated design allows for modular composition of a manipulator, because transmission lines are not permanently embedded within the elastomer. Additionally, this type of manipulator is, like the cylindrical manipulator, composed entirely of soft silicone rubber.

4.2.4 Fabrication

The fabrication processes for the previously described multi-segment arm designs are described in the following. Table 4.1 contains the superscript references to machine tools and materials used.

Ribbed

The ribbed manipulator is composed of six segments and is fabricated from various soft and semi-soft materials. Seven constraint supports (Fig. 4-11d) are 3D printed[1]

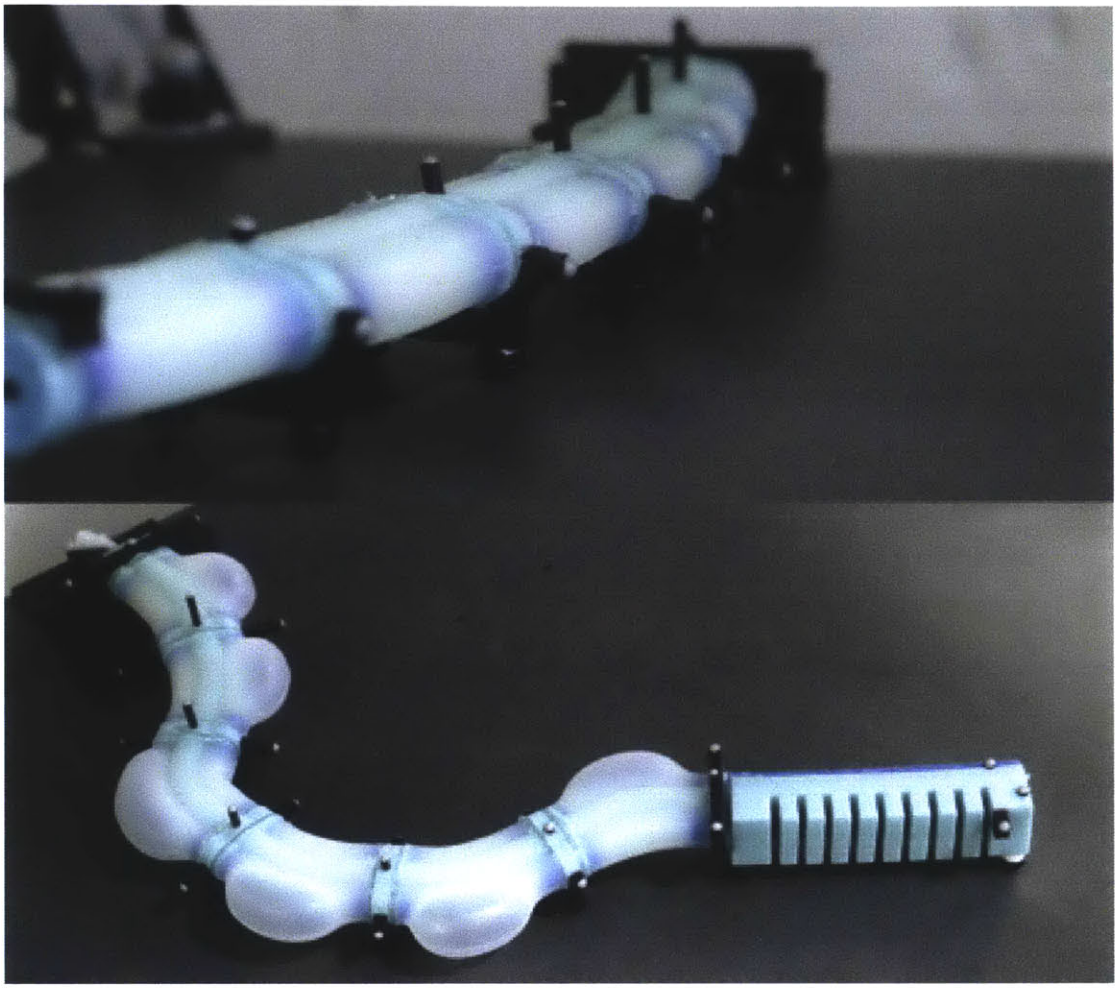


Figure 4-9: A cylindrical soft manipulator prototype with and without a pleated finger-like gripper.

and placed into a constraint layer mold (Fig. 4-11f), which is also 3D printed. The constraint film (Fig. 4-11c) is cut from a thin acetal sheet[8] using a laser[2] and inserted through the aforementioned supports. Above and below the constraint film, eight pieces of silicone tubing (Fig. 4-11a) are threaded through the supports. Silicone rubber[3] is then mixed and poured into the constraint layer mold, immersing tubing, film, and supports in a layer of elastomer to create the composite constraint layer (Fig. 4-11g). The uncured rubber inside the mold is then immediately degassed using a vacuum chamber[4]. Once cured, small holes are created in the constraint layer to pierce the embedded tubing at specific locations, allowing each line to independently

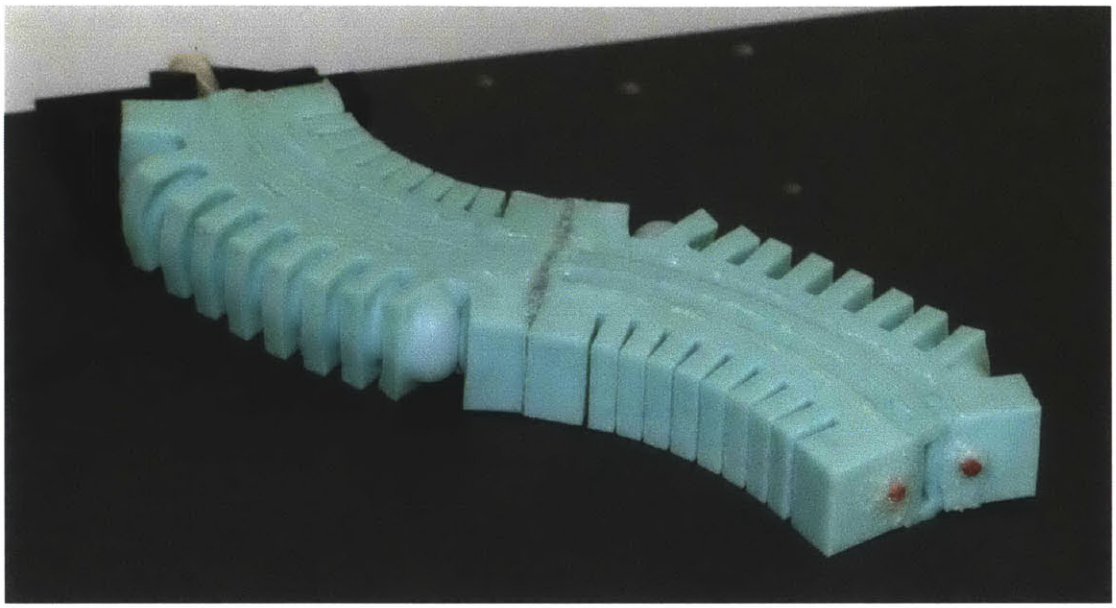


Figure 4-10: A pleated soft manipulator prototype, composed of two segments with two degrees of freedom each.

address a group of fluidic channels. Elastomer pieces containing channels (Fig. 4-11b) are casted and cured separately using a similar molding technique. Those cured elastomer pieces (Fig. 4-11b) are then carefully attached to both faces of the constraint layer using a thin layer of silicone rubber. Lastly, the printed feet (Fig. 4-11e) are attached to the constraint supports (Fig. 4-11d) to create an attachment point for ball transfers (Fig. 4-8ad). These mechanisms prevent the arm from tipping and help constrain the arm's motion to the X-Y plane.

Cylindrical

The cylindrical manipulator is fabricated through a casting process that uses pourable silicone rubber[3,5] and 3D printed molds[1]. Figure 4-12 details this process.

First, each body segment is independently fabricated in steps 1-3 and later these segments are joined serially to form the arm in steps 4 and 5. To start, a four piece mold is printed. The mold is then poured in two steps. In step 1, a low elastic modulus rubber[3] is mixed, degassed in a vacuum[4], and poured to form the body segment's soft outer layer shown in *white*. The mold's outer piece, one half of it is

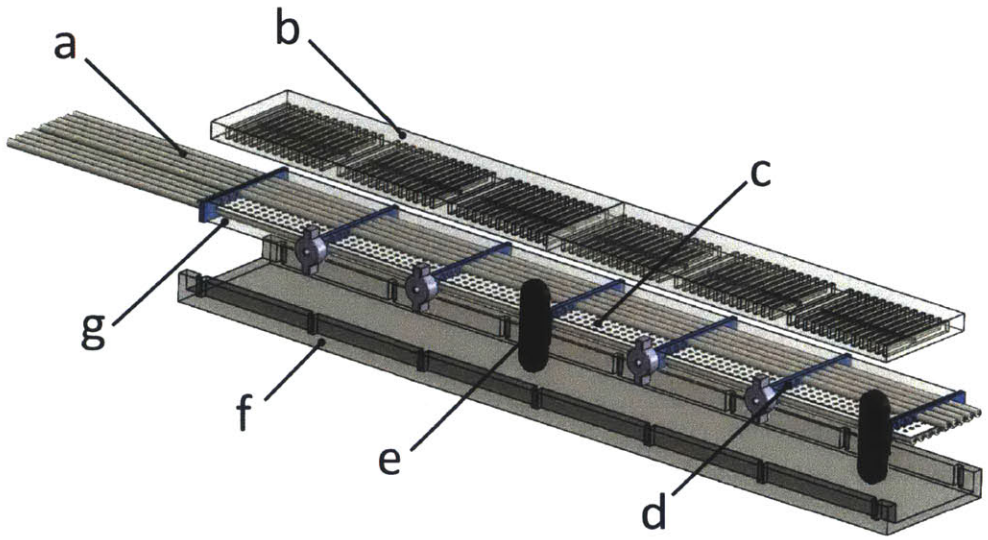


Figure 4-11: Fabrication process for a ribbed manipulator morphology: silicone tubing (a), elastomer pieces containing channels (b), constraint film (c), constraint supports (d), feet (e), constraint layer mold (f), and composite constraint layer (g).

shown in *green*, functions to form the segment's exterior. Metal rods shown in *pink* are inserted into the mold and are held in place by the *orange* bottom piece of the mold. These rods will form the cavities for the segment's two lateral fluidic actuation channels.

After the outer layer has cured, the *red* rigid sleeve is removed in step 2 from the extruded feature of the *orange* bottom piece of the mold. This produces a cavity into which a slightly stiffer rubber[5] is poured, forming the segment's partially constraining inner layer shown in *cyan*. The extruded feature of the *orange* bottom piece, shown by its *orange* end tip, functions to produce the segment's hollow interior core. In step 3, the body segments are removed from their molds and joined to rubber[5] endplates shown in *cyan* using silicone adhesive[6]. The small *yellow* channel inlets were added on one side of the *pink* metal pins during step 1. In step 4, soft silicone tubes[7] are joined to each embedded channel's inlet. The resulting bundle of tubes is passed through each segment's hollow interior. Lastly, in step 5 multiple body segments are attached at their endplates using the same adhesive[6].

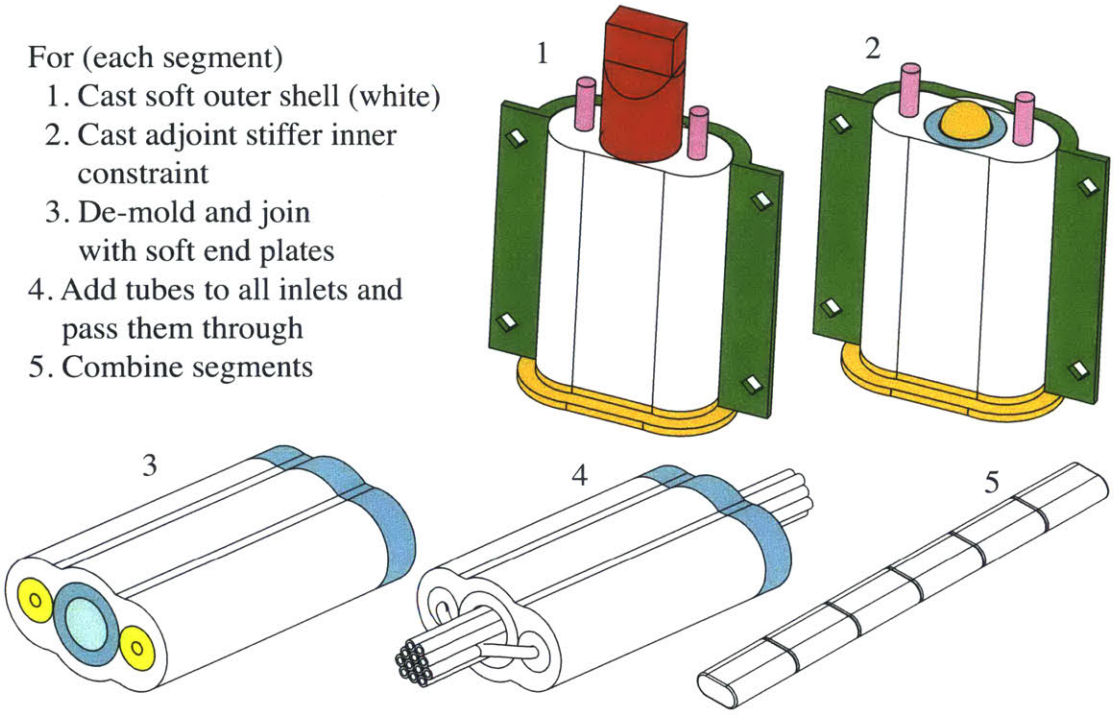


Figure 4-12: Fabrication process for the cylindrical manipulator morphology: Each body segment is casted using a two step process where the outer soft layer (**1**) and inner stiffer layer (**2**) are poured. Once cured, the segments are joined to endplates using silicone adhesive (**3**). Next, silicone tubing is connected to each embedded channel and the resulting tubing bundle is run inside each segment's hollow interior (**4**). Lastly, the body segments are serially connected using adhesive to form the manipulator (**5**).

Pleated

Existing soft robots are typically produced through a multi-step soft lithography process, which produces seams that are prone to delamination. This limits their range of applications and lifetime. By abandoning the need for a lamination-based process, arbitrarily shaped internal channels can be achieved to enable a wider range of applications. For these reasons, we introduce lost-wax casting as part of the fabrication process for soft actuators. As an example, we fabricate a pleated actuator using the lost-wax approach. The complete fabrication process for a pleated manipulator consists of eight steps that are depicted in Figure 4-13 and a final concatenation step.

In step (A), harder silicone rubber[10] is poured into a mold, which contains a 3D

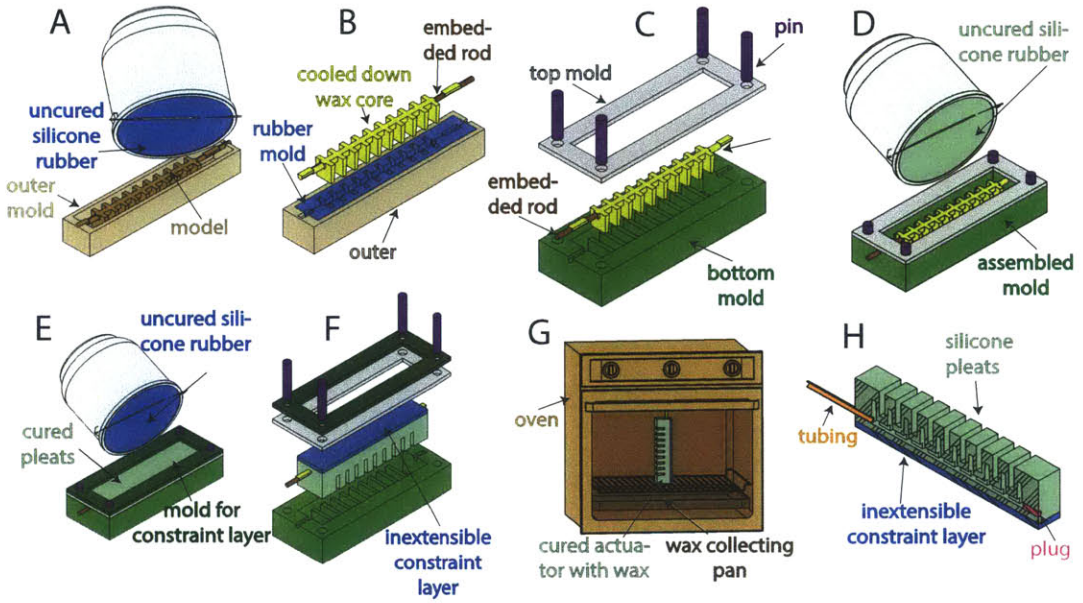


Figure 4-13: Fabrication process for the pleated actuator morphology: (**A**) Pour and cure a rubber mold, (**B**) pour wax core with embedded supportive rod, (**C**) combine bottom mold, top mold and wax core using pins, (**D**) pour rubber into assembled mold, (**E**) pour stiffer rubber on top of the cured actuator to form a constraint layer, (**F**) remove cured actuator from mold, (**G**) melt out wax core from the actuator using an oven, and (**H**) add silicone tubing and plug using silicone sealant.

printed model of the wax core. In preparation for step (B), the model is removed and the rubber mold is left inside the outer mold. Next, a rigid rod or tube, for example made of carbon fiber[12], is used as a supportive inlay for the wax core. The rod is laid into the cavity of the rubber mold, supported on both ends by the outer mold. This ensures that the wax core does not break when removed from the rubber mold. Mold release spray is applied to the silicone rubber mold to ease the wax core removal process. The wax[11] is heated up until it becomes fully liquefied. The assembly of the rubber mold and the outer mold is heated up for a few minutes to the same temperature as the wax. Using a syringe, the liquid wax is injected into the assembly. Within a few minutes, the injected wax will start to solidify and significantly shrink in volume; this is counteracted by injecting more hot wax into the solidifying wax core during the cool down period. In step (B), the wax core is first allowed to completely cool down, then it is released from the mold. In step (C), the cooled down wax

core is assembled together with the bottom mold, which defines the pleated structure of the actuator. The mold assembly is aligned with a top mold using pins. This top mold provides additional volume to cover the wax core. In step (D), low elastic modulus rubber[3] is mixed, degassed in a vacuum[4], and poured to form the pleats and allowed to cure. In step (E), stiffer rubber is poured on top of the cured pleats to form a constraint layer. In step (F), the cured actuator is removed from the mold. In step (G), most of the wax core is melted out by placing the cured actuator into an oven in an upright position. After this, remaining wax residues are cooked out in a boiling water bath. Finally, in step (H) a silicone tube[9] and a piece of silicone cord[13] get covered with silicone adhesive[6] and are inserted into the front and back holes, respectively.

Lastly, the actuator can be used as a unidirectional gripper or as one agonist side segment of a multiple body manipulator, where each segment can be actuated bidirectionally; see Section 4.2.3.

Table 4.1: Commercially Available Tools and Equipment

#	Product Name	Company
1	Fortus 400mc	Stratasys
2	VLS3.50	Universal Laser Systems
3	Ecoflex 0030	Smooth-On
4	AL Cube	Abbess Instr. & Systems
5	Mold Star 15	Smooth-On
6	Silicone Sealant 732	Dow Corning
7	PN 51845K52	McMaster
8	PN 5742T51	McMaster
9	PN 51845K53	McMaster
10	Mold Star 30	Smooth-On
11	Beeswax	Jacquard
12	PN 2153T31	McMaster
13	PN 9808K21	McMaster
14	L16-50-35-12-P	Firgelli Technologies
15	122-D	Bimba
16	1394	Pololu Corp.
17	Opti Track	NaturalPoint

4.3 Power

The objective of the power system is to provide real-time adjustment of fluid energy input to the manipulators. The power system is used in conjunction with the curvature controllers developed in Section 4.5 and the kinematic algorithms developed in Section 4.4 to achieve closed-loop configuration control of the manipulators. In general, fluid power can be achieved by controlling either pressure or volume. We choose volume control because it offers the ability to: (i) measure the control variable via linear displacement as opposed to pressure transducers, (ii) set a maximum safe displacement limit, and (iii) vary pre-pressurization of segments to accommodate for differences in actuator compliance.

4.3.1 Fluidic Drive Cylinder

In order to independently and bidirectionally actuate arm segments, an array of custom fluidic drive cylinders were developed. These cylinders provide fluidic power to the arm by producing volumetric changes within a segment's embedded channels. Linear actuators are directly coupled to and control the positional displacements of the pistons within fluidic cylinders. Accordingly, these linear actuators govern the volumetric displacement of fluid out of the cylinders and into a segment's embedded channels, and vice versa. Figure 4-14 illustrates the components of a fluidic drive cylinder.

4.3.2 Fluidic Drive Cylinder Model

In the following, a model of the fluidic drive cylinder is developed. Although this plant model is not used in the control of the cylinder, it serves to identify the impact design decisions have on the system's input-output relationships. Figure 4-15 shows a simplified schematic representation of the closed-loop fluidic power delivery system. A linear time-invariant dynamic model can be created to *approximate* the behavior of a fluidic drive cylinder connect to an elastomer channel, and we refer to this as the plant in Figure 4-15.

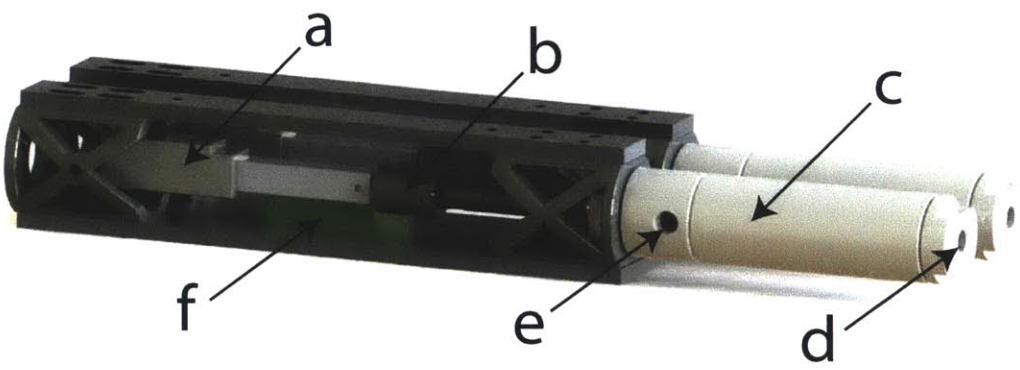


Figure 4-14: An overview of two fluidic drive cylinders used to drive the curvature of a bi-directional arm segment. (a) An electric linear actuator[5] is directly attached to the piston of a fluidic cylinder[6] (c) via a 3D-printed coupler (b). Fluid is displaced through the inlet (e) and outlet (d) of the cylinder. A motor controller[7] (f) allows digital command signals to govern fluid movement.

In order to develop the plant model, we make the following assumptions:

1. The motor has no inductance. We justify this by observing that the dynamics of the electrical system are considerably faster than the dynamics of the fluidic system.

2. The piston moves without friction inside the cylinder. We justify this by observing that most of the plant's energy is removed via the highly resistive fluid transmission line between the cylinder and channel.

3. The channel's compliance C_c, which is the ratio of its change in volume to change in pressure, is piecewise constant with respect to channel pressure.

4. The fluid's compliance C_f, which is also the ratio of its change in volume to change in pressure, is constant with respect to fluid pressure. This is, because linearizing the ideal gas law around the operating point p_0 at constant temperature T_0 leads to $\mathbb{V} = -\frac{mRT_0}{p_0^2}\, p = C_f\, p$.

5. The fluid's mass is negligible, because we are currently only working with gases, specifically air.

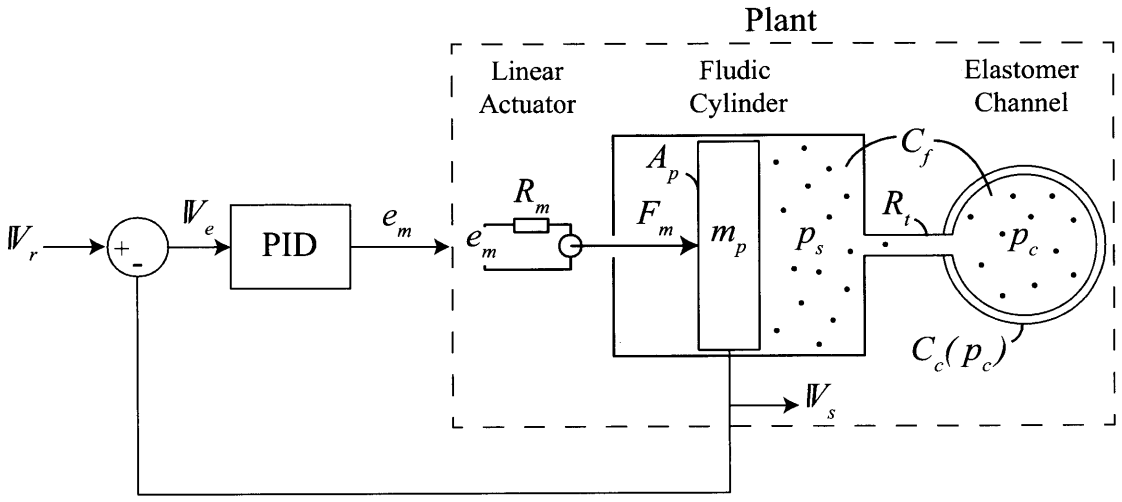


Figure 4-15: Parameters used in developing a simplified fluidic drive cylinder model. A schematic representation of the plant shows a linear electric actuator (left), a piston and fluidic cylinder (middle), and an elastomeric channel grouping within an arm segment (right). A PID controller runs the plant with the volumetric displacement as its reference value, the motor voltage as its control effort and the sensed volumetric displacement as its feedback.

The equations of motion for the drive cylinder are expressed in the following. First, the piston's linear motion can be approximated as

$$F_m - p_s A_p \approx \frac{m_p}{A_p} \ddot{\mathbb{V}}_s, \tag{4.2}$$

where F_m is the force exerted by the linear motor on the piston, p_s is the gauge pressure of the fluid inside the cylinder, A_p is the cross sectional area of the piston, m_p is the mass of the piston, and $\mathbb{V}_s$ is the volumetric displacement of the cylinder. We can derive the next equation of motion by observing that the total volume of fluid within the plant (represented as the dotted area in Fig. 4-15), does not change as this is a closed-circuit drive system. It follows that $\Delta \mathbb{V} \approx 0$ and therefore

$$\dot{\mathbb{V}}_s + \dot{p}_s C_f + \dot{p}_c \left(C_f - C_c\right) \approx 0. \tag{4.3}$$

Here, p_c is the gauge pressure of the fluid within the elastomer channel and C_f and C_c are the compliances of the transmission fluid and elastomer channel, respectively.

The volumetric fluid flow into the cylinder is approximated as

$$\dot{\mathbb{V}}_s + \frac{p_c - p_s}{R_t} \approx -C_f \dot{p}_s, \tag{4.4}$$

where R_t is the resistance of the fluid transmission line connecting the cylinder and channel. Lastly, the force output of the linear actuator can be approximated as

$$F_m \approx -\frac{\beta\,\lambda}{R_m\,A_p}\,\dot{\mathbb{V}}_s + \frac{\beta}{R_m}\,e_m. \tag{4.5}$$

Above, β is the motor constant relating motor current to the force of the linear actuator. The motor constant λ relates linear velocity to the counter EMF voltage, R_m is the motor's resistance, and e_m is the input motor voltage.

Using these equations of motion, a SISO LTI open-loop plant model can be expressed in the form $\dot{\mathbf{x}} = \mathbf{A}_{\mathrm{OL}}\,\mathbf{x} + \mathbf{B}_{\mathrm{OL}}\,u$ and $y = \mathbf{C}_{\mathrm{OL}}\,\mathbf{x}$, and is explicitly written in Equation 4.6. Combining Equation 4.2 and 4.5 yields the first row. The second row is found by combining Equation 4.3 and 4.4, and the third row is Equation 4.4.

$$\begin{bmatrix} \ddot{\mathbb{V}}_s \\ \dot{p}_c \\ \dot{p}_s \end{bmatrix} = \begin{bmatrix} -\frac{\beta\lambda}{m_p R_m} & 0 & -\frac{A_p^2}{m_p} \\ 0 & \frac{1}{(C_f - C_c)R_t} & -\frac{1}{(C_f - C_c)R_t} \\ -\frac{1}{C_f} & -\frac{1}{C_f R_t} & \frac{1}{C_f R_t} \end{bmatrix} \begin{bmatrix} \dot{\mathbb{V}}_s \\ p_c \\ p_s \end{bmatrix} + \begin{bmatrix} \frac{\beta A_p}{m_p R_m} \\ 0 \\ 0 \end{bmatrix} e_m$$

$$y = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \dot{\mathbb{V}}_s \\ p_c \\ p_s \end{bmatrix} \tag{4.6}$$

For bounded-input, bounded-output stability, it is sufficient to use a proportional control law taking the form

$$e_m(t) = K_p\,\left(\mathbb{V}_r(t) - \mathbb{V}_s(t)\right), \tag{4.7}$$

where $\mathbb{V}_r$ is a reference volume and K_p is a proportional gain constant. However, to ensure zero steady-state error in tracking the reference volume and for a suitable

dynamic response, we implement a traditional PID control law

$$e_m(t) = K_p \mathbb{V}_e(t) + K_i \int_0^t \mathbb{V}_e(\tau)\mathrm{d}\tau + K_d \frac{\mathrm{d}\mathbb{V}_e(t)}{\mathrm{d}t} \tag{4.8}$$

$$\mathbb{V}_e(t) = \mathbb{V}_r(t) - \mathbb{V}_s(t). \tag{4.9}$$

Approximations of the model's parameters are listed in Table 4.2. We approximate the transmission line resistance as

$$R_t \approx \frac{128\,\mu\,L_t}{\pi\,d_t^4}, \tag{4.10}$$

where μ is the fluid's viscosity, L_t is the transmission line's length, and d_t is the transmission line's inside diameter. Figure 4-16 details the experimentally determined pressure-volume relationship of both the transmission fluid and the compliant elastomer channel. Channel compliance was determined by actuating a fluidic elastomer actuator with an incompressible fluid and measuring the resulting internal pressuring while incrementally increasing channel volume. The incompressible fluid enables channel compliance to be decoupled from transmission fluid compliance. By linearizing this experimental data, the fluid compliance C_f and the piecewise defined channel compliance C_{c1} and C_{c2} can be determined. Also the linearization offset $\mathbb{V}_{\text{off}}$ can be found. To verify the plant model, model predicted transmission line flow $\frac{p_s - p_c}{R_t}$ was compared to experimentally measured flow (Fig. 4-17). To obtain flow data, a flow transducer was placed inline with the transmission tubing.

β	λ	m_p	R_m	A_p	C_f	C_{c1}	C_{c2}	V_{off}	R_t
232	475	0.19	18.5	7.92	-2.0e-10	1.25e-10	4.15e-9	-9.3e-05	3.4e8
$\frac{\text{N}}{\text{A}}$	$\frac{\text{V s}}{\text{m}}$	kg	Ω	cm^2	$\frac{\text{m}^3}{\text{Pa}}$	$\frac{\text{m}^3}{\text{Pa}}$	$\frac{\text{m}^3}{\text{Pa}}$	m^3	$\frac{\text{Pa s}}{\text{m}^3}$

Table 4.2: Approximations of Fluidic Drive Cylinder Parameters

4.3.3 Fluidic Drive Cylinder Implementation

Fluidic drive cylinders are mechanisms, which interface the computational and algorithmic aspects of the manipulation system with the soft arm. Specifically, they input

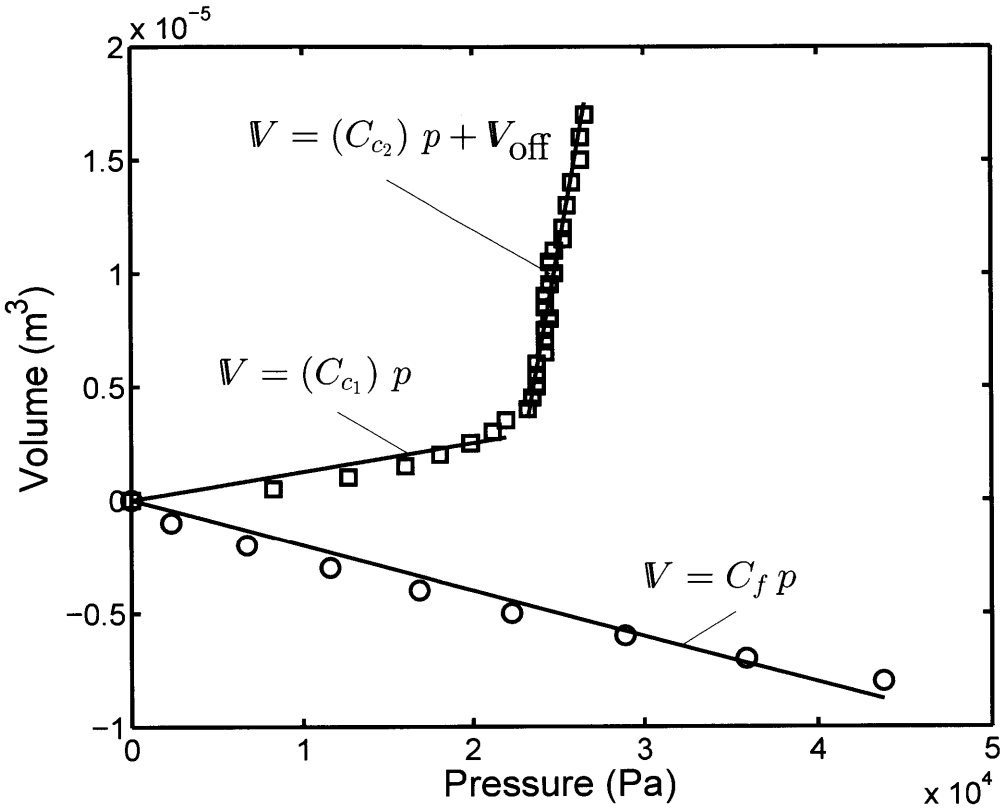


Figure 4-16: Experimentally measured pressure-to-volume relationship of air (C_f) and of air and elastomer channel combined ($C_f + C_c$).

digital command signals from a control algorithm and generate fluidic pressure that drives the curvature of the arm segments.

Two drive cylinders are used to control a single bidirectional segment. Although the mapping of either the agonistic or antagonistic channel deformation is monotonically related to a single piston's displacement, when considering bidirectional segment movement as well as positive and negative curvatures, the two drive cylinders must be controlled synchronously. One piston is held still and the other piston is moved in either forward or reverse to increase or decrease curvature. There are four distinct states of a pair of fluidic drive cylinders in this arrangement as detailed in Figure 4-18.

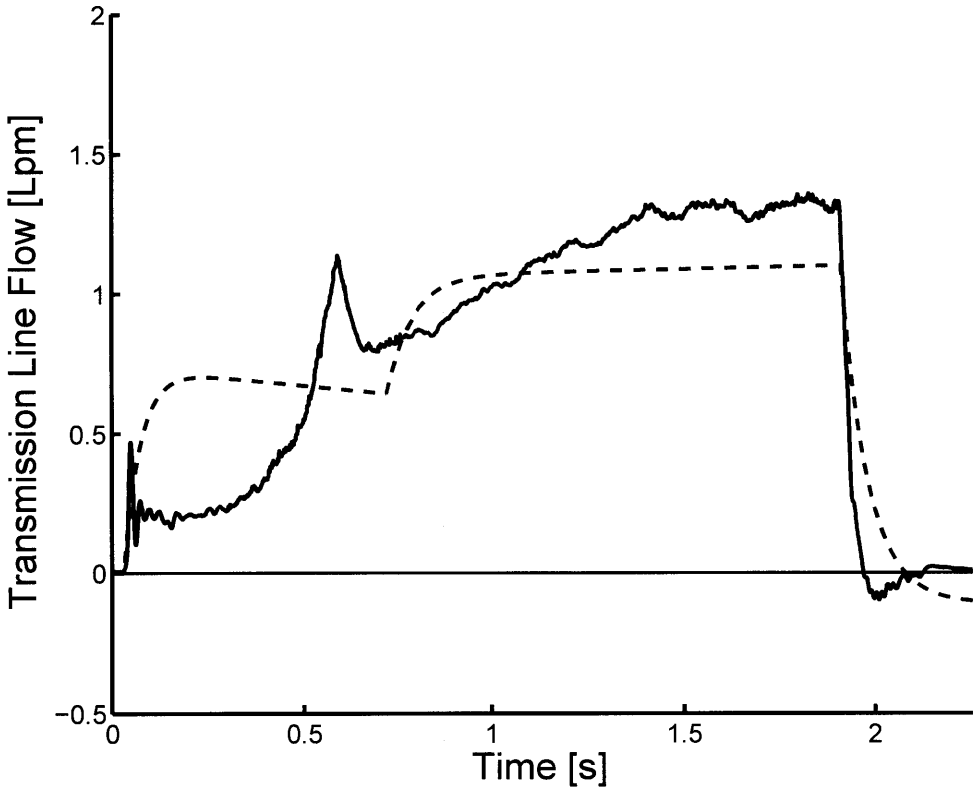


Figure 4-17: Experimental verification of the fluidic drive cylinder plant model. The solid line represents measured transmission line flow, and the dotted line represents model predicted flow.

4.4 Kinematic Modeling

In this section, we develop a model sufficient for describing the kinematics of the soft manipulators described in Section 4.2. Specifically, we first discuss the application of a piecewise constant curvature modeling assumption [Webster and Jones, 2010] to describe these soft multi-segment arms. Next, we provide a procedure for measuring a configuration-space representation of each segment by means of external localization data. Then, we show how these measurements can be used to compute the multi-segment manipulator's task-space representation by means of a recursive, integral-based forward kinematics algorithm. Lastly, we provide two methods for solving the inverse kinematics problem for this manipulator type, that is a Jacobian-based

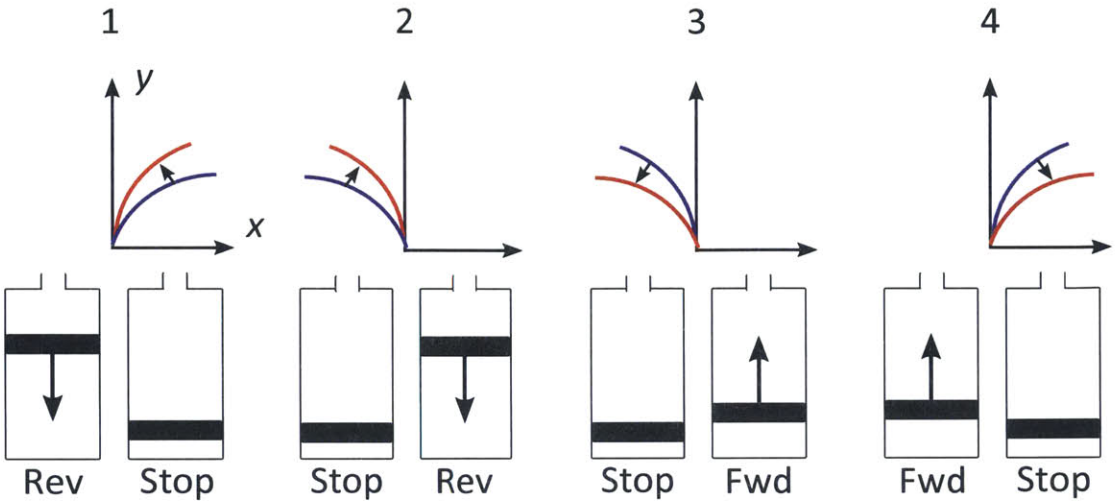


Figure 4-18: Diagram depicting the four driving states of the two fluidic drive cylinders used to control an arm segment. These states depend on the error in curvature (measured state shown as blue arcs minus target state shown as red arcs) as well as the sign of the curvature (right hand rule). (**1**) The curvature is negative and the error is positive. (**2**) The curvature is positive and the error is negative. (**3**) The curvature is positive and the error is positive. (**4**) The curvature is negative and the error is negative.

approach and a nonlinear optimization-based approach.

4.4.1 Piecewise Constant Curvature

Building on previous results describing the deformation of a fluidic elastomer actuator (see Section 4.2), we assume the state of an arm segment can be represented by a signed curvature κ and arc length L as well as the knowledge of its starting orientation, θ_0. Accordingly, within the two-dimensional x-y plane, the kinematic transformation from the base of an arm segment to the tip can be represented as

$$\mathbf{S}_{\text{tip}}^{\text{base}} = \mathbf{R}_z\left(\frac{\theta}{2}\right)\mathbf{T}_y\left(d\right)\mathbf{R}_z\left(\frac{\theta}{2}\right), \tag{4.11}$$

$$\kappa = \frac{\theta}{L}, \tag{4.12}$$

$$d = \frac{2\,L}{\theta}\sin\left(\frac{\theta}{2}\right). \tag{4.13}$$

Above, $\mathbf{T}$ and $\mathbf{R}$ represent coordinate translations and rotations along and about the subscript axes and θ is the segment's bend angle. This representation is consistent with the literature on continuum manipulation [Webster and Jones, 2010]. The multi-segment kinematic model of the planar soft manipulator can be defined by cascading N single segment transformations together, where N is the number of individual body segments

$$\mathbf{M}_{\text{tip,N}}^{\text{base,1}} = \mathbf{S}_{\text{tip,1}}^{\text{base,1}}(\theta_1)\ \mathbf{S}_{\text{tip,2}}^{\text{base,2}}(\theta_2)\ \ldots\ \mathbf{S}_{\text{tip,N}}^{\text{base,N}}(\theta_N)\,. \tag{4.14}$$

4.4.2 Single-segment Inverse Kinematics

In order to control the pose of arbitrary points along the soft robot arm in task space, it is first necessary to estimate an arm segment's state in configuration space using available localization data. To do this, we develop a procedure referred to as `singleSegInvKin()` detailed in Algorithm 2 and in Figure 4-19. This procedure fits the n^{th} segment's configuration-space representation, that is curvature κ_n and arc length L_n, to its measured Cartesian end point data $\mathbf{E}_n \subset \mathbb{R}^2\ \forall n = 1..N$. End point data for a segment $\mathbf{E}_n$ consists of two x-y points, a base point $\mathbf{A}$ and a tip point $\mathbf{B}$. Although we provide our approach to the single segment inverse kinematics problem, relating an initial angle and the positions of end points to curvature has been previously addressed by Hannan and Walker [2003] and Jones and Walker [2006a].

Algorithm 2: Single Segment Inverse Kinematics (Refer to Fig. 4-19)

Procedure `singleSegInvKin(` $\mathbf{A}$, $\mathbf{B}$, θ_0`)`

1. A line aa, orthogonal to the starting tangent vector $\mathbf{u}$ and passing through $\mathbf{A}$, is constructed.
2. A perpendicular bisector pp to the line segment $\overline{\mathbf{AB}}$ is constructed.
3. The intersection of aa and pp forms the center point $\mathbf{O}$ of a constant radius arc that connects $\mathbf{A}$ and $\mathbf{B}$. Line segment $\overline{\mathbf{AO}}$ is the radius r of this arc.
4. The arm segment's curvature $\kappa_n \leftarrow \frac{1}{r}$.
5. The bend angle $\theta \leftarrow \angle\mathbf{AOB}$.
6. The arc length $L_n \leftarrow \frac{\theta}{\kappa_n}$.

return κ_n *and* L_n

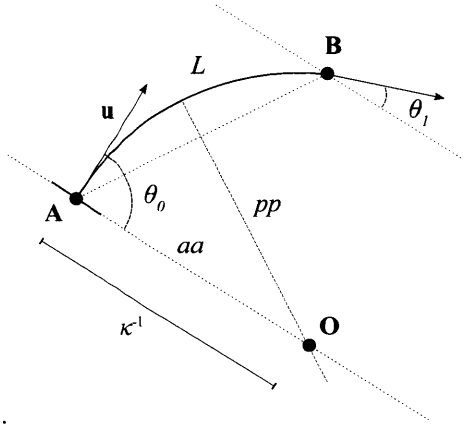


Figure 4-19: Visualization of Algorithm 2 used to determine the configuration of a soft arm segment at a given point in time from end point data. The n^{th} segment's base point, **A**, and tip point, **B**, are measured and the initial orientation, θ_0, of the segment is provided. The n^{th} segment's curvature κ_n and length L_n is determined. Again, we assume the deformation of each segment at a point in time can be adequately represented by a signed curvature value.

4.4.3 Forward Kinematics

Having identified independent segment curvatures and lengths, we can now write the forward kinematics of serially connected bidirectional bending segments in a manner consistent with Onal and Rus [2013]. The orientation at any point $s \in [0,\, L_n]$ along the arc representing segment n within a chain of N segments composing the arm can be expressed as

$$\theta_n(s) = k_n\, s + \theta_n(0) \tag{4.15}$$

Because these segments are serially connected and continuous we assume $\theta_n(0) = \theta_{n-1}(L_{n-1})$. In the case of our soft manipulators, this allows us to define a procedure `forwardKin()` described in Algorithm 3, which uniquely identifies the state of the entire arm by starting at the grounded base segment ($n = 1$) with *a priori* known orientation. Subsequently, each consecutive segment's state is determined.

The position of any point along the arm can be expressed as

$$x_n(s) = x_{n-1}(L_{n-1}) + \int_0^s \cos\left[\theta_n(s')\right] \, \mathrm{d}s' \tag{4.16}$$

$$y_n(s) = y_{n-1}(L_{n-1}) + \int_0^s \sin\left[\theta_n(s')\right] \, \mathrm{d}s' \tag{4.17}$$

$$\forall n = 1 \,..\, N.$$

Frequently throughout this work, we will refer to the manipulator's end effector, which is defined as $\mathbf{w}_N = [x_N(L_N), y_N(L_N), \theta_N(L_N)]$. Algorithm 3 recursively determines a point on the arm located at s on segment n, given curvature and length are provided.

Algorithm 3: Forward Kinematics Algorithm

Procedure `forwardKin(` $\boldsymbol{\kappa}$, $\mathbf{L}$, i, s`)`

Input: The inputs $\boldsymbol{\kappa}$ and $\mathbf{L}$ represent the current manipulator configuration typically determined by running Algorithm 2 $\forall n = 1 \,..\, N$ prior. The input i is the index of the segment of interest and s is the arc length along the indexed segment.

if $i = 1$ **then**

1 $\quad x_i(0) \leftarrow 0.$

2 $\quad y_i(0) \leftarrow 0.$

3 $\quad \theta_i(0) \leftarrow \theta_0.$ $\quad \Leftarrow$ known *a priori*

else

4 $\quad [x_i(0), y_i(0), \theta_i(0)] \leftarrow$ `forwardKin` $(\boldsymbol{\kappa}, \mathbf{L}, i-1, L_{i-1})$.

end

5 $x \leftarrow x_i(0) + \frac{\sin\theta}{k_i} - \frac{\sin\theta_i(0)}{k_i}.$

6 $y \leftarrow y_i(0) - \frac{\cos\theta}{k_i} + \frac{\cos\theta_i(0)}{k_i}.$

7 $\theta \leftarrow \theta_i(0) + k_i s.$

return $[x, y, \theta]^{\mathrm{T}}$ *or* $[x, y]^{\mathrm{T}}$

4.4.4 Multi-Segment Inverse Kinematics

Besides the procedures `singleSegInvKin()` (Algorithm 2) and `forwardKin()` (Algorithm 3), a critical component to multi-segment manipulation is the arm's inverse kinematics procedure. Given the pose of a point or points along the arm's backbone, this procedure returns the manipulator's arc-space configuration. In the following, we develop two distinct approaches to the inverse kinematics (IK) problem. First, we extend an iterative Jacobian-transpose IK approach [Balestrino et al., 1984, Wolovich and Elliott, 1984] to our soft continuum manipulator. Second, we develop a nonlinear

optimization-based approach that allows easy encoding of primary task-space and secondary arc-space manipulation objectives in the form of nonlinear constraints and cost functions. The optimization-based approach seems particularly well-suited for soft manipulators as is demonstrated in Sections 4.6.2 and 4.6.3.

Jacobian-based Approach

In the Jacobian-based approach we run an iterative algorithm each control step. The procedure is explicitly defined in Algorithm 4. We effectively determine a small update to segment curvatures that will move a controlled point or points towards their desired pose $\mathbf{w}_d = [x_d,\ y_d,\ \theta_d]$. Each iteration, the algorithm calculates the incremental curvature update $\Delta\boldsymbol{\kappa}$ based on the manipulator's Jacobian $\mathbf{J} = \frac{\partial \mathbf{w}}{\partial \boldsymbol{\kappa}}$. After sufficient iteration, the curvatures required to attain the desired task-space pose are returned. In other words, given the start and end points of each arm segment and θ_0 as well as the desired pose(s), the algorithm determines a curvature discrepancy for each arm segment. We are able to write the Jacobian in closed-form using Equation 4.14, which is fundamental to line 3 of the inverse kinematics Algorithm 4.

Algorithm 4: Jacobian-based Inverse Kinematics Algorithm

Procedure `inverseKinJac(` $\boldsymbol{\kappa}$, $\mathbf{L}$, $\mathbf{w}_d$, i, s`)`

Input: The inputs $\boldsymbol{\kappa}$ and $\mathbf{L}$ represent the current manipulator configuration typically determined by running Algorithm 2 $\forall n = 1..N$ prior. The input $\mathbf{w}_d$ is the desired pose of a point along the arm defined by a segment i and the arc length s along the indexed segment. ϵ is the maximal pose error.

for $i = 0,\ i \leq$ *maxIter*, i++ **do**

1. $\mathbf{w}_c \leftarrow$ `forwardKin(` $\boldsymbol{\kappa}$, $\mathbf{L}$, i, s `)`. $\Leftarrow$ current pose
2. $\mathbf{w}_e \leftarrow [\mathbf{w}_d - \mathbf{w}_c]$. $\Leftarrow$ pose error

   if $\|w_e\| < \epsilon$ **then**

3. **break**.

   end

4. $\mathbf{J} \leftarrow \left.\frac{\partial \mathbf{w}_c}{\partial \boldsymbol{\kappa}}\right|_{\boldsymbol{\kappa},\,\mathbf{L},\,\theta_0}$.
5. $\Delta\boldsymbol{\kappa} \leftarrow \alpha\,\mathbf{J}^{\mathrm{T}}\mathbf{w}_e$. $\Leftarrow \alpha$ is a step size correction
6. $\boldsymbol{\kappa} \leftarrow \boldsymbol{\kappa} + \Delta\boldsymbol{\kappa}$.

end

return $\boldsymbol{\kappa}$

Optimization-based Approach

We can pose the IK problem as a constrained nonlinear optimization.

Algorithm 5: Optimization-based Inverse Kinematics Algorithm

Procedure `inverseKinOpt(` $\boldsymbol{\kappa}$, $\mathbf{L}$, $\mathbf{w}_d$, i, s`)`

$\mathbf{R}$ is a weighting vector.

1 $\boldsymbol{\kappa}^* \leftarrow \min_{\boldsymbol{\kappa}} \sum_{n=1}^{N} R_n \kappa_n^2$.

2 subject to $\mathbf{f} \leftarrow$ `forwardKin` $(\boldsymbol{\kappa}, \mathbf{L}, i, s)$.

3 $\begin{bmatrix} x \\ y \\ \theta \end{bmatrix} - \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} = \mathbf{0}$. $\Leftarrow$ linear kinematic constraint

4 $h(\boldsymbol{\kappa}) = \mathbf{0}$. $\Leftarrow$ nonlinear constraints

5 $\kappa_n^{\min} \leq \kappa_n \leq \kappa_n^{\max} \quad \forall n = 1 \,..\, N$.

return $\boldsymbol{\kappa}^*$

That is, we can find an optimal manipulator configuration $\boldsymbol{\kappa}^*$ such that a point or point(s) along the arm have a desired posed while simultaneously minimizing an objective function and considering other nonlinear constraints in the form $h(\boldsymbol{\kappa}) = \mathbf{0}$. This IK approach is well-suited for soft continuum manipulation where there is typically a primary manipulation task, for example end-effector positioning, as well as a secondary task, for example avoiding collisions, minimizing material strain, or exploiting environmental contact.

Algorithm 5 details a simple instantiation of this approach, where the objective is to find a minimal-strain configuration that simultaneously satisfies a pose constraint imposed on a back-bone point and that is within allowable curvature limits.

4.5 Control

This section provides the design of a control system used to govern the manipulator's position and orientation in real-time. That is, we build on the power system developed in Section 4.3 and the kinematic algorithms developed in Section 4.4 to develop a controller for the soft manipulator. In subsequent sections, this controller in combination with task-space planning algorithms is used to provide autonomous

capabilities for the developed soft machines.

4.5.1 Main Controller

The main controller determines in real-time the curvature adjustments required to move the end-effector or points along the arm through their requested pose trajectory. A state flow diagram describing at a high-level the main controller's sensing, task planning, and execution states is given in Figure 4-20.

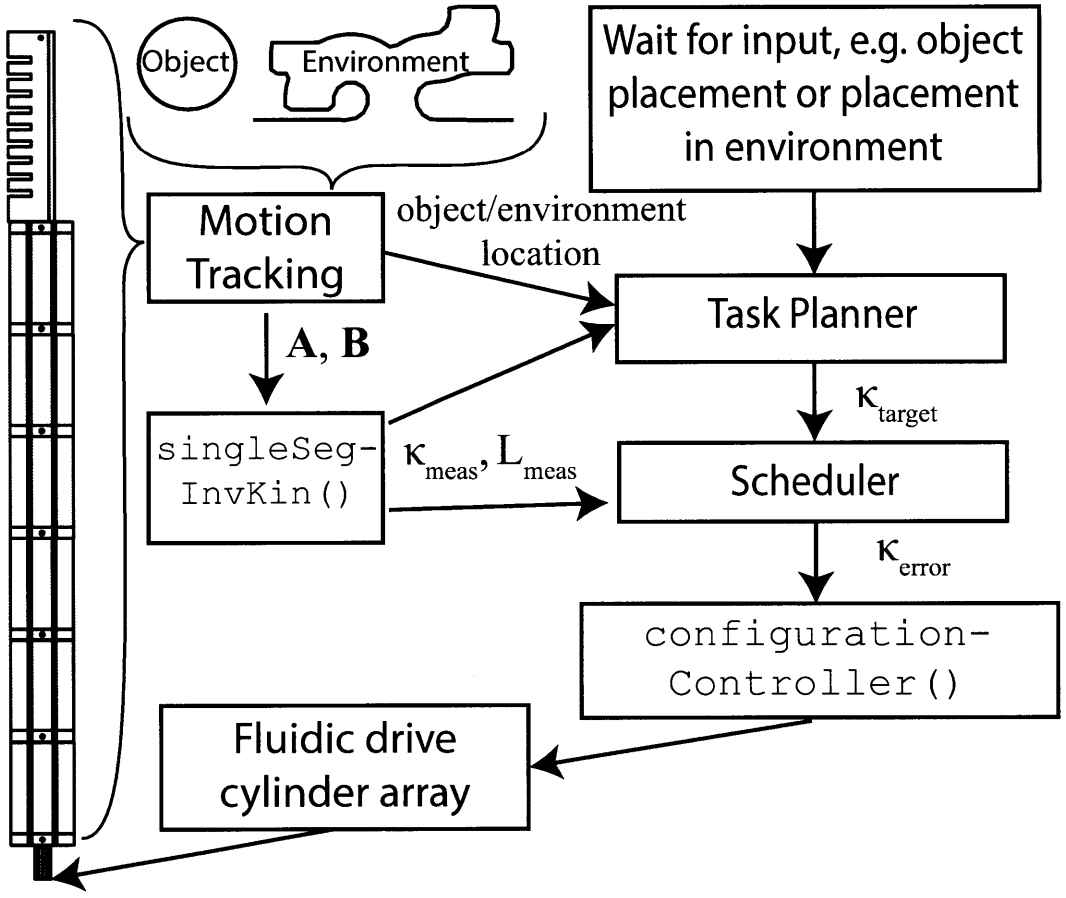


Figure 4-20: State flow diagram of the main controller, describing the flow and processing of information from the motion tracking system to the actuators.

A task for the soft manipulator could be, for example, free space motion, whole arm planning, or grasping-and-placing an object, see also Section 4.6. While the task planner waits for an input that triggers it, the localization system is constantly capturing segment endpoint locations along the soft manipulator, any objects in the

workspace, and the environment boundaries at a constant rate. Our experimental implementation uses an external motion capture system to provide this positional data. The localization system recursively calls the `singleSegInvKin()` procedure using the manipulator's segment endpoint locations as input (see Section 4.4.2). The procedure outputs curvatures $\boldsymbol{\kappa}_{\text{meas}}$ and lengths $\boldsymbol{L}_{\text{meas}}$ at an update rate of 20 Hz. After the task planner has been triggered, it uses the current location of any objects in the workspace and the environment together with the current curvature values of the manipulator to solve the task at hand. The task planner then passes the target configuration $\boldsymbol{\kappa}_{\text{target}}$ of the manipulator to the scheduler for execution in real-time.

The scheduler has access to a trajectory generation procedure `trajGen()`, needed for some of the possible tasks. It generates in real-time for each individual degree-of-freedom velocity profiles with acceleration and velocity constraints. These profiles allow real-time interpolation between the intermediate target configurations of the manipulator and avoid overshoot at the next target configuration. The scheduler also constantly receives the manipulator's measured configuration representation $\boldsymbol{\kappa}_{\text{meas}}$, $\boldsymbol{L}_{\text{meas}}$ from `singSegInvKin()` and uses this information in combination with intermediate target configurations determined by `trajGen()` to determine the curvature error $\boldsymbol{\kappa}_{\text{error}}$ at a given point in time.

The `configurationController()`, described in Section 4.5.2, receives $\boldsymbol{\kappa}_{\text{error}}$ as an input and accordingly controls the fluidic drive cylinder array to achieve the intermediate target configuration.

4.5.2 Configuration Controller

In order to control in real-time the manipulator's configuration, a closed-loop control algorithm `configurationController()` was developed. A block diagram representation of this controller is shown in Figure 4-21. An instance of this controller is instantiated for each segment of the manipulator. This algorithm periodically receives discrepancies $\boldsymbol{\kappa}_{\text{error}}$ between the soft arm's measured and reference curvatures and uses a cascaded control structure to effectively adjust the reference volumetric displacement of the fluidic drive cylinders to resolve manipulator's configuration er-

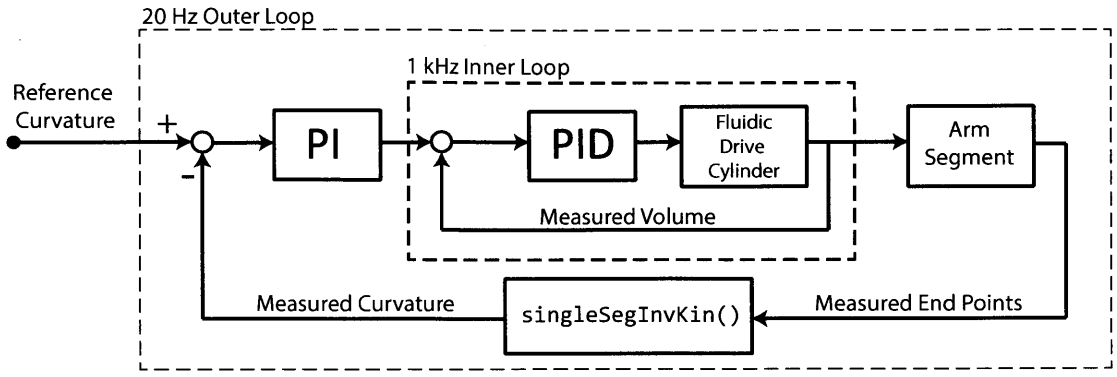


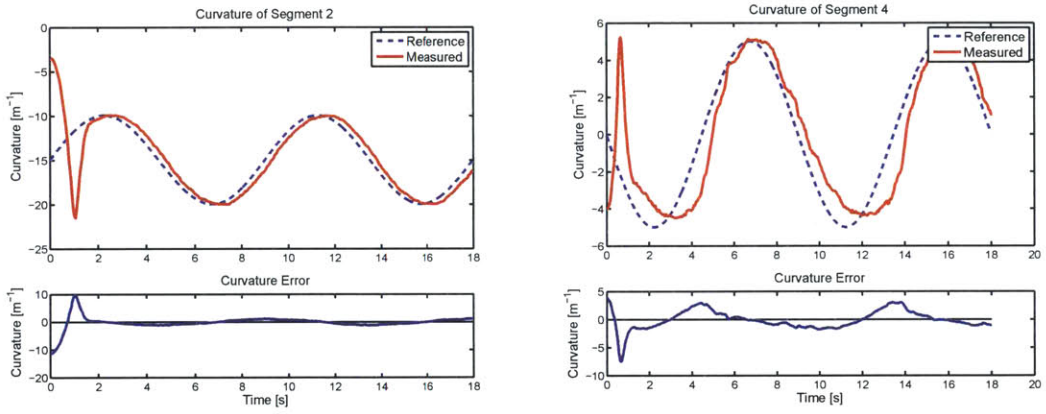
Figure 4-21: A block diagram of the manipulator's configuration controller. A cascaded curvature-volume control structure is used to achieve closed-loop configuration control. The inner volume control loop runs at approximately 1 kHz, which is substantially faster than the outer curvature control loop, which runs at approximately 20 Hz.

ror. The controller achieves this by running a PI computation on the curvature error in order to generate a set-point for the positional control of the drive cylinder's linear actuator. The outer curvature control loop is triggered by the availability of sensor measurements from the external localization system and runs at approximately 20 Hz. The inner positional control loop runs at 1 kHz and ensures the cylinder's piston displacement matches the reference displacement determined by the outer control loop. The cylinder's piston displacement is the primary manipulated variable as input fluid volume is monotonically related to the curvature of a manipulator segment (see Section 4.2).

4.5.3 Configuration Tracking

The capability of an individual segment to track a curvature profile varying over time is a fundamental requirement for free-space motion and more advanced manipulator tasks. In order to find an initial set of gains for the configuration controller, the Ziegler-Nichols tuning method was used. Figure 4-22a details both the target and measured curvature over time as well as the error of the arm's second segment. Here, the target profile is a sine wave of amplitude $5\,\mathrm{m}^{-1}$ with a period of 9 seconds, centered about -$15\,\mathrm{m}^{-1}$. A challenge for the controller is transitioning from driving either of

the two fluidic cylinders to the other and this occurs when the segment's curvature passes zero. The drive cylinder's control authority over the ribbed segment around zero makes tracking reference trajectories around zero possible. Figure 4-22b details curvature tracking of a ribbed segment when a similar target sinusoidal profile is centered about $0\,\mathrm{m}^{-1}$. However, because the drive cylinder's control authority over the cylindrical segment is limited around zero, a pre-pressurization strategy is used to achieve better tracking performance. That is, during the initialization of the main controller, `configurationController()` performs a pre-pressurization of both lateral channels. This is only done for the segments closest to the root of the manipulator in order to increase their impedance and shorten their response time constant.



(a) (b)

Figure 4-22: Closed-loop curvature tracking of an arm segment. The sinusoidal trajectory in (**a**) is entirely negative, only one drive cylinder is actuated. The sinusoidal trajectory in (**b**) is centered about zero, both drive cylinders are actuated. At the top of each subfigure, the reference (*blue dotted*) and measured (*red*) curvatures over time for an individual arm segment are shown. At the bottom of each subfigure, the error in curvature, which is reference minus measurement, is shown.

4.6 Capabilities

This section details several capabilities of these new soft fluidic elastomer manipulators including tracking a linear path in free-space, maneuvering in a confined pipe-like

environment, and grasping-and-placing objects. We provide task-space planning algorithms that enable these capabilities.

4.6.1 Free Space Motion

The robotic manipulation system is able to accurately and precisely control the pose of points along the soft arm in real-time. Specifically, the manipulation system's task-space planner can move the manipulator's end-effector or any intermediate point along it to a user specified pose using Algorithm 4, the Jacobian-based inverse kinematics procedure. We refer to this motion capability as point-to-point movements and we experimentally characterize them in Section 4.7.1.

The arm's end effector can also track Cartesian trajectories in real-time. At every time step, Algorithm 4 is used to identify a feasible target configuration. We refer to this capability as path tracking and it is validated experimentally in Section 4.7.2.

4.6.2 Whole Arm Planning

We develop a task-space planner to autonomously advance the soft planar multi-segment manipulator through a confined environment. The task-space of the arm consists of the position and orientation of a point $\mathbf{w} = [x,\ y,\ \theta]$ or points along the arm (see Section 4.4.3). The goal of this algorithm is to find locally optimal manipulator configurations (curvatures $\boldsymbol{\kappa}$) that (1) position the end-effector to points along a path through the confined environment and (2) consider the proximity of the points on the robot's envelope $\mathbf{h} \in \mathbb{H} \subseteq \mathbb{R}^2$ in relation to points on the surrounding environment $\mathbf{e} \in \mathbb{E} \subseteq \mathbb{R}^2$. The envelope $\mathbb{H}$ is modeled by trapezoids scaled linearly according to the measured curvature values. The environment $\mathbb{E}$ is represented as a set of boundary points and is known *a priori*. We assume that the arm is initially always placed in a free space outside of the bounded environment we want to enter. Also, a series of task-space waypoints $\mathbf{w}_t$, $\forall\, t = 1\,..\,t_f$, is predefined by the user, where the first waypoint is placed at the inlet to the environment. All following waypoints describe a path that would lead to a final goal position inside the environment.

The optimization-based whole arm configuration planning is outlined in Algorithm 6, visually supported by Figure 4-23, and explained in further detail in the following. At the start time $t = 1$, we set the arm segments to a locally optimal $\boldsymbol{\kappa}$

Algorithm 6: Whole Arm Planner Algorithm Outline

Input:
$\mathbb{E} \subseteq \mathbb{R}^2 \leftarrow$ environment
$\mathbf{w}_t, \quad \forall t = 1..t_f \leftarrow$ path waypoints
$t \leftarrow 1$ ▷ set first time step to 1
$\boldsymbol{\kappa}_{\text{meas}}, \mathbf{L}_{\text{meas}} \leftarrow$ `singleSegInvKin()`
$\boldsymbol{\kappa}_t \leftarrow$ `inverseKinOpt(` $\boldsymbol{\kappa}_{\text{meas}}, \mathbf{L}_{\text{meas}}, \mathbf{w}_t, N, L_N$)
while $t \leq t_f$ **do**
▷ loop until last waypoint index t_f is reached
1 $\mathbb{E}_1 \leftarrow \mathbb{E} \mid \sin(\alpha) > 0$ ▷ environment to the left of the arm hull
2 $\mathbf{e}_{1,t} \leftarrow \underset{\mathbf{e}\in\mathbb{E}_1}{\text{argmin}}\{\|\mathbf{e} - \mathbf{w}_t\|\}$
3 $\mathbb{E}_2 \leftarrow \mathbb{E} \setminus \mathbb{E}_1$ ▷ environment to the right of the arm hull
4 $\mathbf{e}_{2,t} \leftarrow \underset{\mathbf{e}\in\mathbb{E}_2}{\text{argmin}}\{\|\mathbf{e} - \mathbf{w}_t\|\}$
5 $(\mathbb{H}_1, \mathbb{H}_2) \leftarrow$ `armHull`$(\boldsymbol{\kappa}_t, \mathbf{L}_{\text{meas}})$ ▷ left and right hull points
for $j = 1 \ldots t$ **do**
▷ for all times until t
for $i = 1 \ldots 2$ **do**
▷ for left and right
6 $\mathbf{h}_{i,j} \leftarrow \underset{\mathbf{h}\in\mathbb{H}_k}{\text{argmin}}\{\|\mathbf{e}_{i,j} - \mathbf{h}\|\}$ ▷ closest hull points
7 $\mathbf{b}_{i,j} \leftarrow \mathbf{e}_{i,j} - \mathbf{h}_{i,j}$ ▷ vector betw. hull and environment
8 $\mathbf{a}_{i,j} \leftarrow$ `envNormal`$(\mathbf{e}_{i,j})$ ▷ normal to wall
end
end
9 $\boldsymbol{\kappa}_t \leftarrow \max_{\boldsymbol{\kappa}} \left\{ \sum_{j=1}^{t} \sum_{i=1}^{2} \|\vec{b}_{i,j}\| \ln\left(\vec{a}_{i,j} \cdot \vec{b}_{i,j} + 1\right) \right\}.$
10 subject to $\mathbf{f} \leftarrow$ `forwardKin`$(\boldsymbol{\kappa}, \mathbf{L}, N, L_N)$.
11 $\mathbf{w}_t - \mathbf{f} = \mathbf{0}.$
12 $h(\boldsymbol{\kappa}) = \mathbf{0}.$
13 $\kappa_n^{\min} \leq \kappa_n \leq \kappa_n^{\max} \quad \forall n = 1..N.$
$t \leftarrow t + 1$
end

that places the manipulator's end-effector at the first waypoint $\mathbf{w}_1$ according to the forward kinematics procedure `forwardKin()` (see Algorithm 3). The solution also ensures that each segment does not exceed its curvature limits κ^{min} and κ^{max}. The extreme curvatures are experimentally determined to be either the maximum curvatures achievable given the fixed volumetric displacements of the fluidic drive cylinders

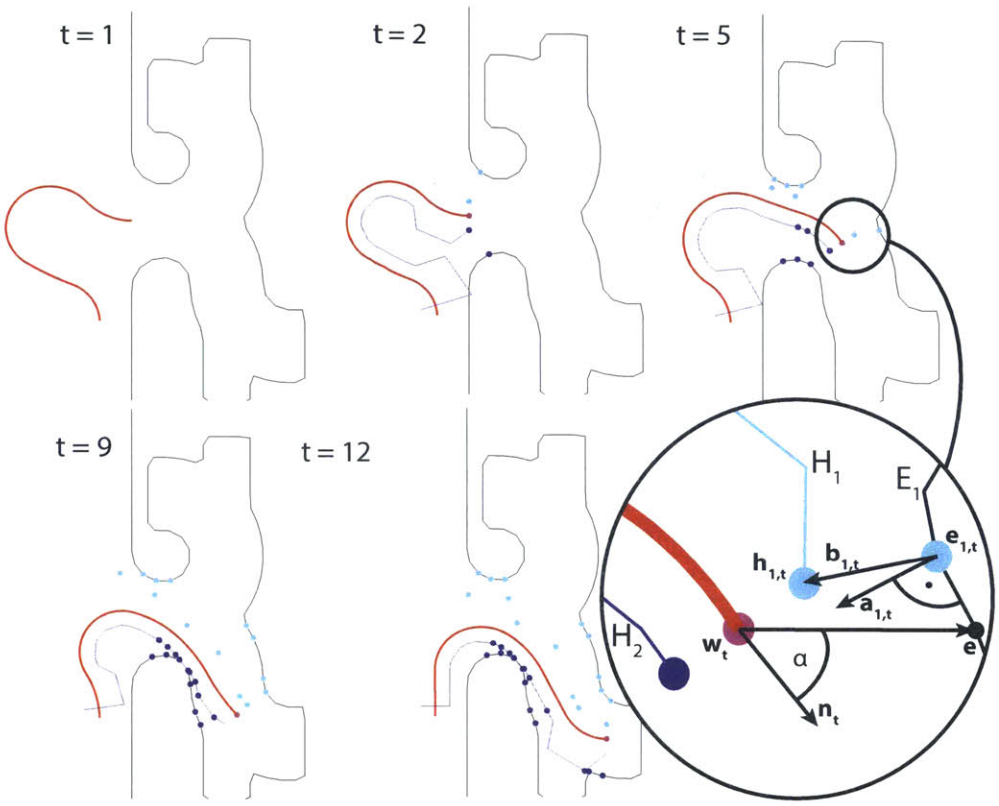


Figure 4-23: Whole Arm Planner: The soft robot is depicted for several time steps t by a *red* body center line, a *cyan* line for the left body shape and a *blue* line for the right body shape. The *black* line depicts the bounded environment. The *purple* dot at the end of the *red* center line represents the end-effector of the manipulator. At every time step t, the end-effector reaches the planned waypoint by using the controller developed in Section 4.5.1. At timestep $t = t_f = 12$, the last point inside the bounded environment is reached. A close-up view shows the vectors relevant for the whole body planner described in Subsection 4.6.2 and Algorithm 6.

or the maximum curvatures which would not yet damage the segments due to over inflation, whichever of the two is lower. After the start, we enter a loop until the last waypoint at time step t_f is reached. At the start of the loop, we find the closest environment points left $\mathbf{e}_{1,t}$ and right $\mathbf{e}_{2,t}$ of the arm's body shape. To define the two sides of the arm, we use the arm's approach vector $\mathbf{n}_t$ and the angle $\alpha = \angle(\mathbf{e} - \mathbf{w}_t, \mathbf{n}_t)$. The origin of the end-effector's approach vector $\mathbf{n}_t$ is defined by the positions x and y and its direction by the angle θ. Positive values for α result in the closest point $\mathbf{e}_{1,t}$ left of $\mathbf{w}_t$ and negative values result in $\mathbf{e}_{2,t}$ to the right of $\mathbf{w}_t$. Next, we calculate the arm hulls $\texttt{armHull}(\boldsymbol{\kappa}_t, \mathbf{L}_{\text{meas}})$ of both sides by modeling each segments expanded side with

a trapezoid that is linearly scaled in height by the segment's curvature value. The concave side is not vacuumed during the actuation process, so the hull is described by a constant offset from the curved body centerline. For every stored environment point $\mathbf{e}_{i,j}$, $\forall\, i = 1 \ldots 2$ and $\forall\, j = 1 \ldots t$, we find the closest hull points, calculate their connection vector and calculate the normals of the environment $\texttt{envNormal}(\mathbf{e}_{i,j})$. We then perform an optimization to find the curvatures that best fit the hull into the environment with the constraint that the end-effector pose must match the current target waypoint. Finally, the time step is incremented and the loop repeats until the last waypoint is reached.

The optimization algorithm was implemented using Matlab's Optimization Toolbox with the function call *fmincon*, which finds the minimum of a constrained nonlinear multivariable function. Sequential Quadratic Programming was used as the solver with a relative upper bound of 2×10^{-3} on the magnitude of the constraint functions. The maximal function evaluations were bounded by 2000 and the lower bound on the size of a step was given by 1×10^{-6}. The function *dsearch* was used to find the minimal distant points on the environment and on the hull.

4.6.3 Grasp-and-Place

The robotic manipulation system is also capable of autonomously performing grasp-and-place operations. A state flow diagram describing the grasp-and-place task-space planner is shown in Figure 4-24. As mentioned in Section 4.5.1, a motion tracker constantly captures the position of the object and passes it along to a routine, which checks for the object to settle. After it has settled, the grasp object planner receives the coordinates and radius of the settled object and together with the current curvature values of the arm and gripper, it solves a series of constrained nonlinear optimization problems to generate end-effector poses approaching the object. This is described in more detail in Section 4.6.3.

The grasp object planner passes the locally-optimal approach configurations $\boldsymbol{\kappa}_i^*$ to the scheduler as $\boldsymbol{\kappa}_{\text{target}}$ for execution in real-time (see Section 4.5.1). To allow for smoother transitions between each approach configuration, the scheduler uses a

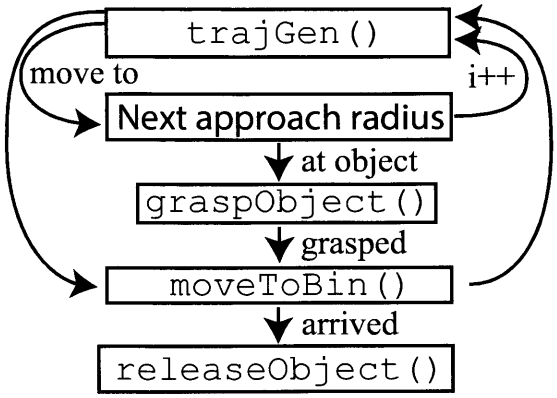


Figure 4-24: State flow diagram of the grasp-and-place planner developed for the autonomous grasp-and-place operation of the manipulator. This diagram describes the flow of information from the motion tracking system to the discrete hardware.

trajectory generation procedure `trajGen()` detailed in Chapter 3 to generate velocity profiles with acceleration and velocity constraints for each independent DOF. These velocity profiles allow the scheduler to calculate intermediate manipulator configurations between the given approach configurations by performing a time-based integration. These reference configurations are used to determine $\boldsymbol{\kappa}_{\text{error}}$, which is sent to the configuration controller. When the arm has arrived at the object, the planner initiates `graspObject()`. After encapsulating the object, `moveToBin()` is initiated, which also uses `trajGen()` to generate a trajectory from the current pose to a pre-defined bin location. After the manipulator has arrived at the bin location, the procedure `releaseObject()` causes the gripper to open and release the object.

Grasp Planner

A fundamental component of the grasp-and-place task planner is the system's ability to plan a feasible approach to the object. That is, given the location (x_o, y_o) and radius r_o of a round object as well as the manipulator's current configuration $\boldsymbol{\kappa}_{\text{meas}}$, we determine a series of locally optimal manipulator configurations $\boldsymbol{\kappa}_i^*$, $\forall i = 1 \ldots numMoves$, that will, if sequentially achieved, bring the manipulator gradually closer to the object while not touching it. We refer to these as approach configurations. The process for determining these approach configurations is detailed in

the `planGraspApproach()` procedure (see Algorithm 7). The planner is visualized

Algorithm 7: Plan Grasp Approach

Procedure `planGraspApproach()`

Input: $\boldsymbol{\kappa}_{\text{meas}}$, $\mathbf{L}_{\text{meas}} \leftarrow$ measured arm configuration
$\boldsymbol{\kappa}_{\text{off}} \leftarrow$ bias configuration offset at start
$g_{\text{off}} \leftarrow$ gripper offset normal to end-effector
x_o, y_o, $r_o \leftarrow$ object center coordinates and radius
$N \leftarrow$ number of manipulator segments
$\mathbf{w}_t \leftarrow$ `forwKin` $(\boldsymbol{\kappa}_{\text{meas}}, \mathbf{L}_{\text{meas}}, N, L_N)$.
$d_1 \leftarrow \| [x_o, y_o]^{\mathrm{T}} - \mathbf{w}_t \|$.
$d_2 \leftarrow d_1 - r_o - g_{\text{off}}$.
$numMoves \leftarrow \lfloor \frac{d_2}{\Delta d} \rfloor$.
$i = 0$.
repeat
  $i = i + 1$
  $r_{a_i} \leftarrow d_1 - i \frac{d_2}{numMoves}$.
  $\boldsymbol{\kappa}_i^* \leftarrow$ `findOptimalConfig`(r_{a_i})
until $i = numMoves$
return $\boldsymbol{\kappa}_i^* \quad \forall i = 1..numMoves$

Procedure `findOptimalConfig`(r_a)

$$\boldsymbol{\kappa}^* \leftarrow \min_{\phi, \boldsymbol{\kappa}} \ \mathbf{R} \left(\boldsymbol{\kappa} - \boldsymbol{\kappa}_{\text{off}}\right)^2.$$

subject to
$$\mathbf{w}_t \leftarrow \begin{bmatrix} x_o + r\cos\phi \\ y_o + r\sin\phi \\ \phi + \frac{\pi}{2} \end{bmatrix}.$$
$$\mathbf{f} \leftarrow \text{`forwardKin`}\,(\boldsymbol{\kappa}, \mathbf{L}, N, L_N).$$
$$\mathbf{w}_t - \mathbf{w}_{\text{off}}\,(r_o, \phi) - \mathbf{f} = \mathbf{0}.$$
$$\kappa_n^{\min} \leq \kappa_n \leq \kappa_n^{\max} \quad \forall n = 1..N.$$

return $\boldsymbol{\kappa}^*$

in Figure 4-25.

In short, we define a series of approach radii r_{a_i}, $\forall i = 1 \ldots numMoves$, that define concentric circles shrinking from the manipulator's starting tip pose towards the center of the object. Given actuator limits, we then search for a series of feasible manipulator configurations $\boldsymbol{\kappa}_i^*$ that will place the robot's end-effector on these circles, parameterized by r_a and ϕ, while minimizing manipulator deformation.

The procedure first determines the manipulator's current tip pose $\mathbf{w}_t$ and the Euclidean distance d_1 between the tip and the object's center. Next, considering the object's radius r_o and the gripper normal offset $\mathbf{g}_{\text{off}}$, the minimal tip transit distance d_2 is calculated. The symbol $\mathbf{g}_{\text{off}}$ represents the component of the end effector offset $\mathbf{w}_{\text{off}}$, which is normal to end-effector orientation. Also, the number of

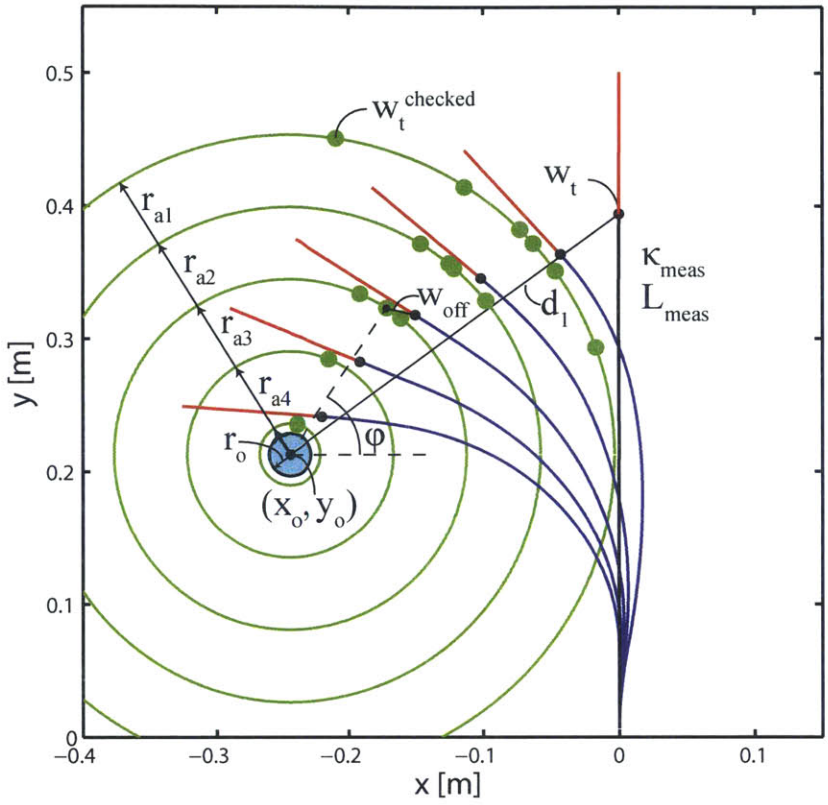


Figure 4-25: Grasp approach planner visualization. Concentric approach circles are shown in *green* and are centered about the object shown in *cyan*. The locally optimal approach configurations are shown in *blue* and the gripper is shown in *red*. The initially measured manipulator configuration is shown in black.

approach configurations $numMoves$ is determined as $\lfloor \frac{d_2}{\Delta d} \rfloor$, where Δd is an allowable incremental distance. Using these parameters, approach radii r_a are iteratively calculated and their corresponding locally optimal configurations are found using the `findOptimalConfig()` procedure. This process is posed as a nonlinear optimization. Here, the objective function represents the summation of independently weighted manipulator curvatures and the constraints force the manipulator's tip to lie on and be tangent to the approach circle. This process leverages the arm's forward kinematics `forwardKin()`, or Algorithm 3, defined in Section 4.4.3.

4.7 Experimental Results

In this section, we experimentally validate all of the proposed algorithms and planners outlined in this work. Please refer to Extension 1 of this paper for videos of the experiments described in the following. Experiments are performed with multiple fluidic elastomer manipulator prototypes within the aggregate experimental setup (see Section 4.1).

4.7.1 Point-To-Point

In order to verify the system's ability to accurately and precisely control the pose of a point on the soft manipulator, point-to-point movement experiments were conducted. A six DOF ribbed manipulator prototype was used. During the experiment, the manipulator's end effector was commanded to move to four reachable poses. Figure 4-26 visualizes the manipulator's pose for each trial as well as the target poses (Fig. 4-26A-D). Before moving to one of the commanded poses, the manipulator was initialized to a resting state where all manipulator segments were depressurized. The time history of each marker position as well as the manipulator's state ($\boldsymbol{\kappa}$, $\mathbf{L}$, θ_0), as determined by Algorithm 2, were logged. The manipulator was moved to each target pose ten consecutive times. After a settling period, the end effector's error in pose was measured. The mean and standard deviation of the positional and rotational error over all ten trials are reported for the four target poses in Table 4.3.

Point	Mean $\pm$ S.D. of Pos. Err. (cm)	Rot. Err. (deg)
A	0.95 ± 0.42	1.75 ± 1.26
B	0.45 ± 0.17	0.58 ± 0.58
C	0.61 ± 0.13	0.91 ± 0.58
D	0.84 ± 0.22	1.28 ± 0.60

Table 4.3: Mean errors and S.D. for point-to-point movements.

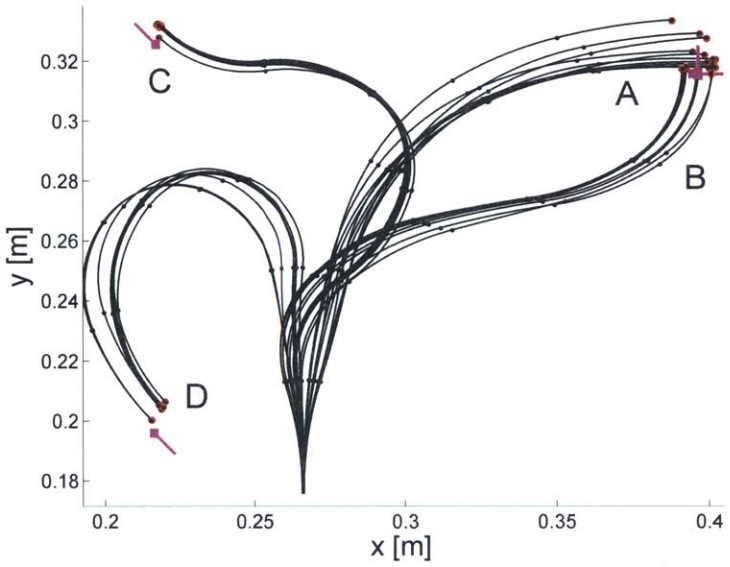


Figure 4-26: Point-to-point movement results. The magenta squares and lines represent the end effector's four target poses (A-D). The red circles, black curves, and black points represent the manipulator's measured end effector positions, configurations, and segment end points respectively for each independent trial.

4.7.2 Path Tracking

In order to verify the system's ability to accurately and precisely follow a trajectory in real-time, path tracking experiments were conducted. Again, the six DOF ribbed manipulator's end effector was controlled. Target poses were updated every iteration of the main controller (see Section 4.5.1). Before each experimental trial, the manipulator was initialized to a resting state where all segments were depressurized. During the experiment, the same data as in the point-to-point experiments were collected. A total of ten trials where conducted. Figure 4-27 details the L-shaped path the manipulator was commanded to follow in *magenta*. The manipulator's measured configuration representation is shown in *black* at the experiment's start at $t = 0$, the path's starting point A, and the path's end point B for an exemplary trial. The end effector's measured position and orientation is shown at each time step along the path in *red* and *blue*, respectively. Figure 4-28 shows the compiled results for all ten trials. The mean positional and rotational errors are calculated at each moment in time and

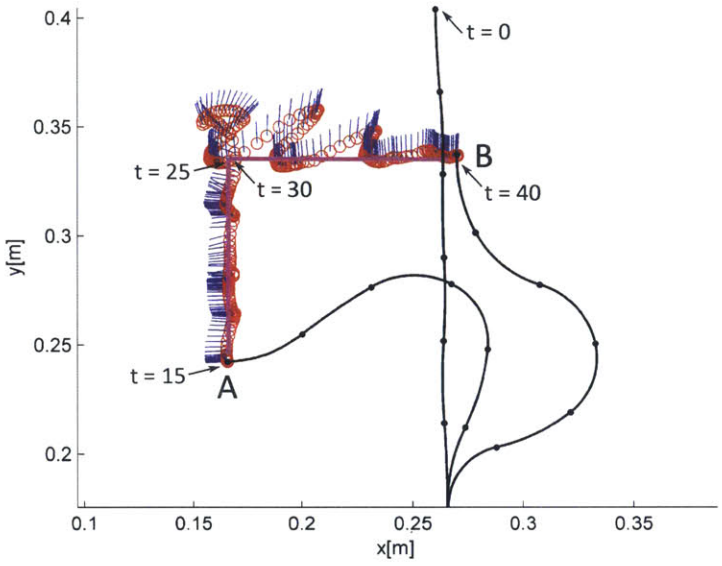


Figure 4-27: A path tracking experimental trial. The L-shaped target path of the manipulator is shown in magenta. The vertical line has a target orientation of π and the horizontal line has a target orientation of $\frac{\pi}{2}$. The manipulator's measured configuration representation at the path's start (A) and end (B) is shown in black. The end effector's measured pose is shown at each time step. The end effector's orientation is shown in blue and its position is represented by red circles. Important path times are noted.

shown in *red*. These are overlaid for individual trial errors shown in *blue*.

4.7.3 Confined Environment

To experimentally validate the soft robot's ability to successfully advance through a confined environment, we carry out a series of experiments using a six DOF cylindrical manipulator prototype. The primary goal of these experiments is to see if the whole body planner outlined in Section 4.6.2 can incrementally advance the robot's end effector pose according to a series of user-specified waypoints, ultimately terminating at the end of one of four distinct pipe-like sections. To achieve this primary goal, the planner also has to successfully accomplish the intermediate task of conforming the whole manipulator's shape to the pipe-like environment. The purpose of these experiments is to show that through a combination of computation (i.e., Algorithm 6)

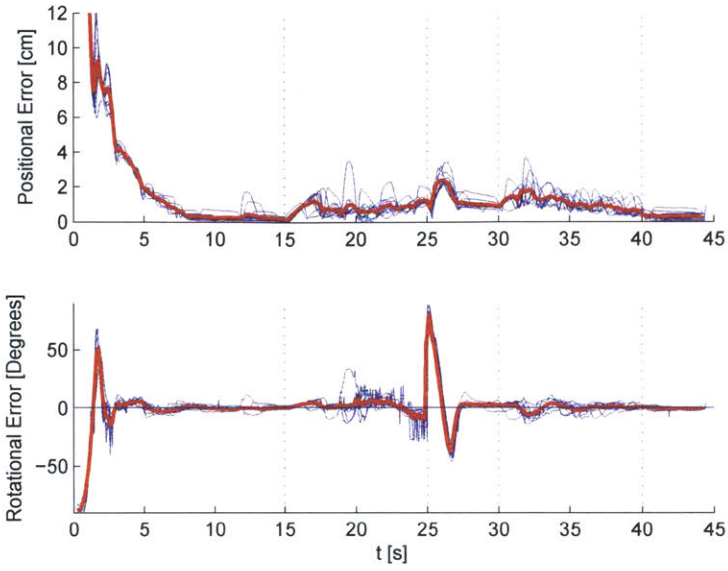


Figure 4-28: Line tracking results for all ten trials. The positional and rotational error of the manipulator's end effector are reported as a function of time. Mean errors are shown in red and calculated at each moment in time. These are overlaid on individual trial errors shown in blue. Vertical lines represent important timing events (see Fig. 4-27)

and design (i.e., the inherent softness and compliance of the manipulator), we can successfully accomplish the primary task. If we ignore the manipulator's softness and high compliance, the planar manipulator provides only six DOF and the primary task constrains three degrees of freedom (X-Y position and orientation). In the non-compliant case, this would leave only three DOF for the whole body planner to conform the manipulator's whole body to the environment's interior at any point along the waypoint path. Fortunately, the compliance of the manipulator provides many passive DOF and we leverage this by making occasional, almost frictionless contacts with the rigid environment. Again, the computational solution does not prevent collisions, but simply minimizes their likelihood and magnitude.

Specifically, a rigid pipe-like environment with 3D printed walls is used in these experiments. To create frictionless collisions, we coat the exterior of the robot with a thin layer of talc powder, reducing the friction between the robot and the wall. In order to constrain the robot's motion to the X-Y plane, reduce friction between the

robot and the ground, and prevent segment torsion about the hollow core, small ball transfers are placed underneath each segment's soft endplate.

Before the robot autonomously snakes through the environment, it is reset to an initial near-zero curvature outside the confined environment (Fig. 4-29A). Independent of the experiment's targeted final waypoint, the planner finds the same solution to maneuver the manipulator to the pipe-like environment's entrance (Fig. 4-29B). As the planner autonomously advances the manipulator's end effector through the environment's center and considers the proximity of the changing envelope to the boundaries, collisions inevitably happen and are highlighted in red in Figure 4-29C. The manipulator successfully reaches one of four goal states depicted in Figure 4-29D-F. Table 4.4 contains data for several successful demonstrations of the manipulator advancing through the environment to the end goal poses. Approximately twelve waypoints were used along each path. t_{solve} is the average time the planner took to compute a solution at a given path waypoint in simulation. The position and orientation errors between the manipulator's end effector and the goal pose are also listed. t_{task} is the time the manipulator took to advance from the starting location to the goal pose. We have also indicated the number of collisions between the manipulator and the environment for several of these trials. Of thirty path attempts, twenty-four resulted in successful advances to goal locations and six were unsuccessful. The typical failure scenario was that the manipulator became lodged in the environment due to excessive contact friction.

The speed of the manipulator advancing along the path is limited by how quickly the target curvatures determined by the algorithm are realized by the physical system. In the ideal case, the system would realize target curvatures as fast as the algorithm computes them; however, because of the increased compliance of the manipulator and the control approach outlined in Section 4.5, the algorithm must halt until the physical system can catch up. In order to improve the speed of the given physical system, the control policy could be optimized by also considering the manipulator's dynamics as constraints.

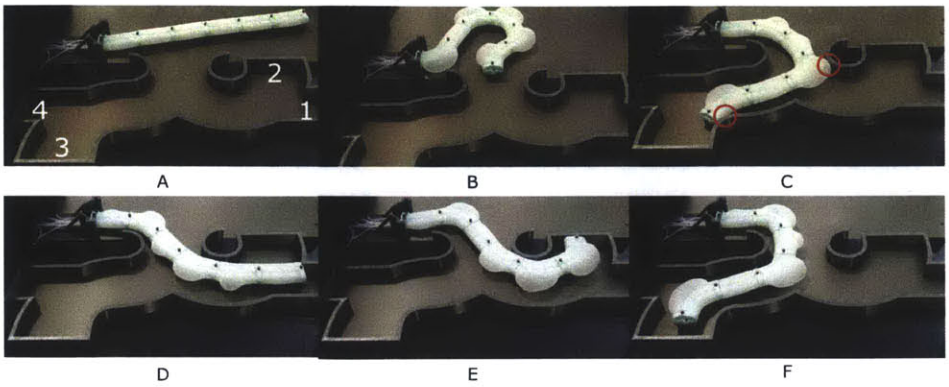


Figure 4-29: Experimental validation of the soft robot's ability to autonomously navigate through a confined environment. (**A**) The starting point of the robot and the positions of the goal locations, 1-4, are shown. (**B**) The manipulator advances to the entrance of the environment. (**C**) Example of the soft robot harmlessly colliding at two points with the boundaries of the environment. (**D-F**) The final goal points 1-3 respectively are shown.

4.7.4 Grasp-and-Place

To verify the fluidic elastomer manipulator's ability to grasp-and-place objects, we perform twenty-five experimental trials at various positions to demonstrate the repeatability of our system. The results of all the trials are shown in Figure 4-30. One representative approach, grasp, and retract move is shown in Figure 4-31. In twenty-three of twenty-five experimental trials, the manipulator successfully achieved the task of grasping an object and bringing it to a bin location. The object was placed on one of five points marked on the board, so that the user places the object in roughly the same point in every repetition. These five points were chosen to roughly represent the major portion of the manipulator's reachable workspace. The system was able to autonomously locate the object, plan its whole body motion, and perform the task of grasping and placing the object. Further individual trials have shown that the system can successfully grasp and place a randomly positioned object within the workspace. The two unsuccessful trials can be explained by stick-slip friction and motion tracking problems. Our modeling does not account for the stick-slip friction between the roller bearings and surface. Accordingly, this highly non-linear actuation behavior can lead to instabilities in the control, as observed in one of the trials. The incapability of

Goal #	t_{solve} (sec)	Trial #	Pos. Err. (cm)	Rot. Err. (deg)	t_{task} (sec)	Collisions #
1	0.9	1	0.9	10.4	44	
		2	0.5	0.1	54	
		3	1.3	1.6	55	6
		4	0.7	1.3	50	4
		5	0.4	0.3	50	2
2	0.9	1	0.7	2.7	44	
		2	0.3	0.6	53	
		3	0.4	0.6	53	4
		4	1.1	5.3	53	2
		5	1.2	4.9	55	3
3	1.0	1	3.1	2.1	42	
		2	3.5	2.5	44	
		3	2.8	0.6	50	8
		4	2.4	4.5	49	6
		5	3.6	3.0	48	12
		6	1.3	1.7	50	
		7	3.5	8.7	52	7
		8	3.4	6.0	53	8
		9	3.4	4.9	50	8
4	1.1	1	4.1	10.7	60	
		2	5.1	17.5	48	4
		3	8.0	25.3	47	12
		4	6.1	21.4	48	7
		5	5.6	17.1	50	6
		6	4.4	14.7	53	13

Table 4.4: Experimental Validation

the motion tracking system to distinguish between two adjacent markers lead to the second failure.

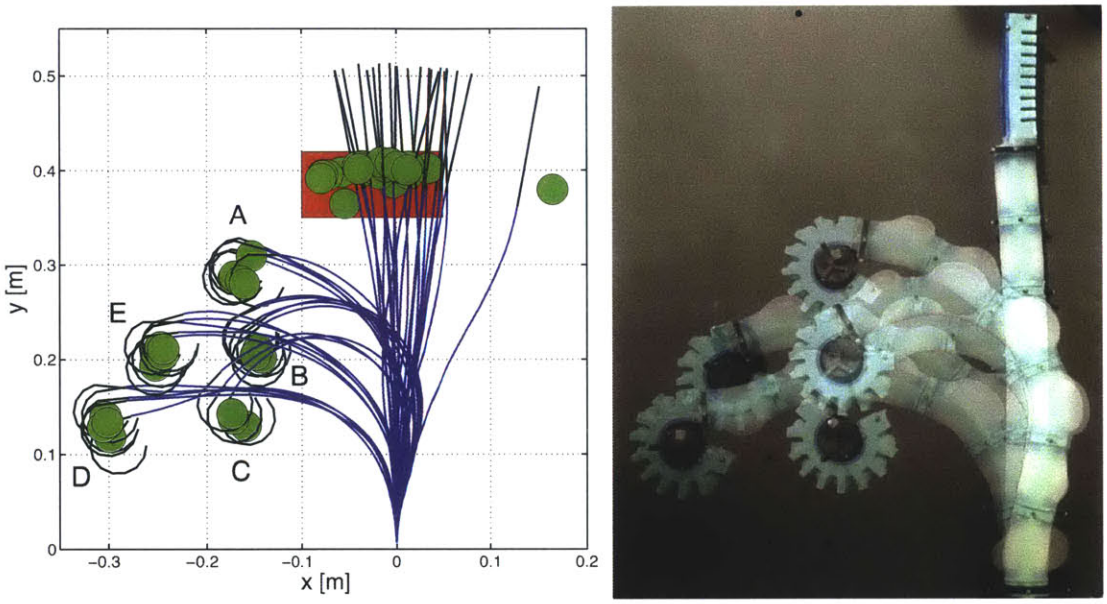


Figure 4-30: Complete set of experimental grasp-and-place trials. In these experiments, the manipulator moves from an initial straightened configuration to grasp a round object placed in one of five locations (A-E). The manipulator then returns the object to a bin location shown in red. For each trial, a seven degrees of freedom manipulator representation is generated at both the *grasped* and *released* state using experimental data and is shown in blue. The corresponding 1 degree-of-freedom end-effector representation is shown in black. The round object's measured position at each state is shown in green. Right: Overlaid photographs of the manipulator grasping an object placed at each of the five locations.

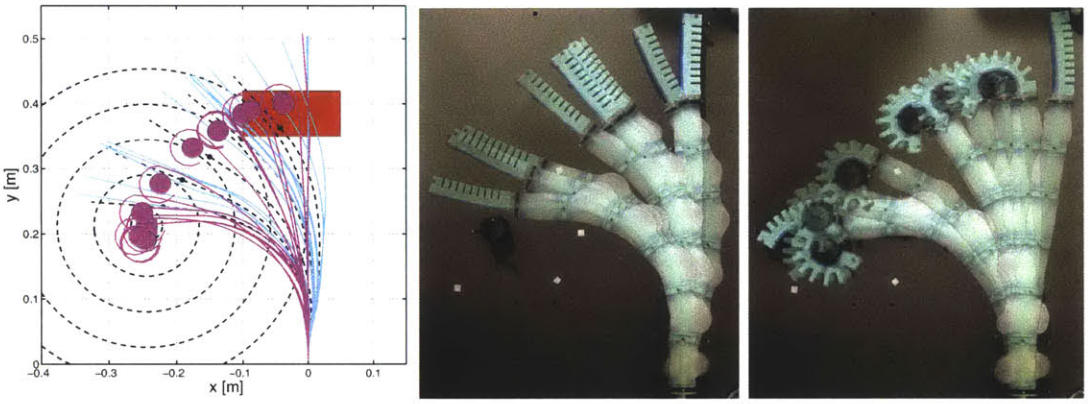


Figure 4-31: Left: A time series representation of an experimental grasp-and-place trial for an object located at point E. Here, the locally-optimal planned manipulator configurations as well as planned sequential approach circles are shown as black dotted curves. The manipulator and gripper are shown in their experimentally determined configuration representations at 1 second intervals. The cyan configurations represent the manipulator prior to grasping the object, that is moving from its initial configuration to the object's location. The magenta configurations represent the manipulator after grasping the object, that is moving from the object's location back to the bin location shown in red. Right: A photograph of the manipulator moving from its initial pose to the object and from the object to the release location, respectively.

Chapter 5

Spatial Multi-Segment Soft Robots[1]

5.1 System Overview

The soft fluidic elastomer manipulation system can move in three dimensions and is subject to gravity. The aggregate system is depicted in Figure 5-1 and contains actuation, perception, power, and computation subsystems. A fluidic elastomer arm composed entirely from soft silicone rubber is suspended subject to gravity (Fig. 5-1A). Each of the manipulator's four body segments (Fig. 5-1, 1-4) are independently actuatable with two bending DOF and is fabricated through a multi-step casting process. The arm attaches to a grounded frame by means of a rotary stage at its base (Fig. 5-1B). A bundle of transmission lines (Fig. 5-1D) carries pressurized fluid from an array of fluidic drive cylinders (Fig. 5-1E), or pumps, to actuated channels that are distributed throughout the arm. A system of cameras (Fig. 5-1C) is used to localize markers on the soft connectors between the arm's body segments. The computational subsystem (Fig. 5-1F) uses kinematic algorithms and cascaded controllers to provide closed-loop curvature control of the manipulator. The subsystem determines the arm's measured representation and target configurations as well as sends control signals to the fluidic drive cylinders.

[1]This chapter is based on Marchese and Rus [2015] and Marchese et al. [2015b]

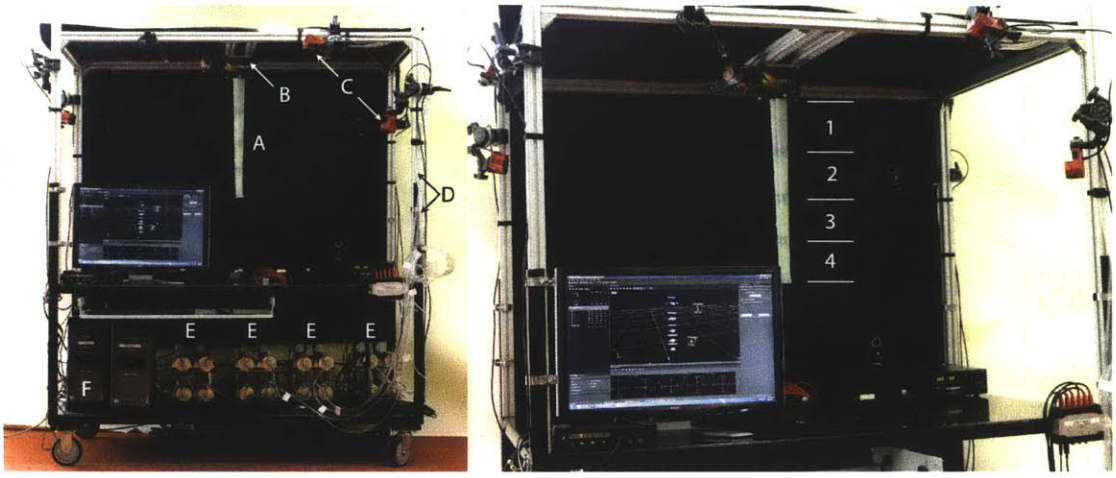


Figure 5-1: Overview of the spatial fluidic elastomer manipulation system. The system consists of (A) soft spatial manipulator composed of four body segments, (B) rotary stage, (C) a system of cameras for external perception, (D) fluid transmission lines, (E) fluidic drive cylinders, and (F) processing and control algorithms.

5.2 Actuation

5.2.1 Soft Manipulator Design

The soft spatial manipulator is composed entirely from low durometer silicone rubbers and is actuated by pressurizing fluid channels embedded within the arm. The 50 cm long 600 g arm, shown in Figure 5-2A, is composed of independently casted and serially concatenated modular segments that each move in three spatial dimensions with two DOF. The arm is held horizontally from its midpoint subject to gravity emphasizing the manipulator's high body compliance in Figure 5-2B. The durability of such an *entirely* soft design is shown in Figure 5-2C-E as well as in Media Extension 1. These demonstrations are inspired by the resiliency tests that Tolley et al. [2014b] use in the evaluation of a soft walking robot. The arm remains functional in extreme heat and in underwater environments while the actuation system is placed external to the extreme condition. Additionally, after being subjected to large compressive loads, the arm resumes normal operation. The same robot prototype used in these durability tests was used to perform all 42 reported experimental evaluations in the following sections. In total, this prototype withstood several hundred experimental

evaluations.

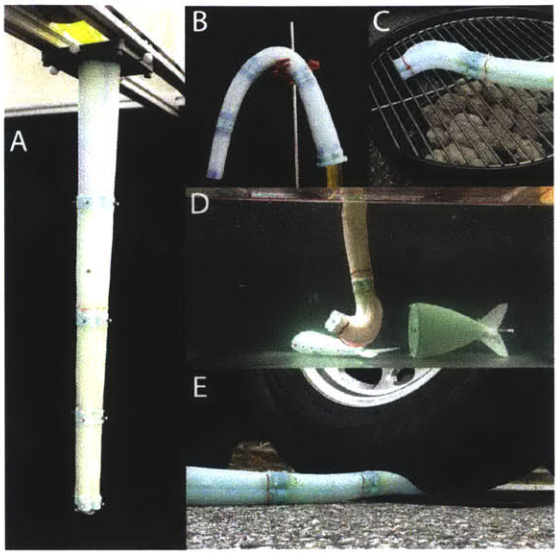


Figure 5-2: (**A**) The soft spatial manipulator. (**B**) The arm is suspended horizontally from its midpoint subject to gravity illustrating its high body compliance. (**C**) The arm is operated on the surface of a grill measuring 230°C. (**D**) The arm is operated under water. (**E**) The arm is run over by a 4,000 kg truck at approximately 5 mph and afterwards resumes normal operation.

The morphology of our spatial manipulator draws inspiration from previous work. The soft rubber tentacles developed by Martinez et al. [2013] use embedded pneumatic actuators in a similar two layer rubber construction. Additionally, inspiration is drawn from a planar manipulator developed by the authors in Marchese et al. [2014b]. However, the manipulator presented here differs from these works in several ways: First, similar to our planar arm, this manipulator is modular in both design and fabrication; however, the spatial morphology differs in that each module or arm segment has four distributed fluidic actuators allowing it to move in three spatial dimensions. Second, similar to our planar design, but different from the design in Martinez et al. [2013], the manipulator presented here has features such as a hollow cylindrical interior and soft segment endplates. The hollow interior accommodates fluid transmission lines and allows modular assembly. The soft endplates allow the arm to be localized externally. Both features are essential to an autonomous manipulation system. Third, the arm has a longitudinally tapered and clover-shaped

exterior and this has a significant impact on the manipulator's dynamics [Marchese et al., 2015b] because of the reduced mass of the distal segments.

In Figure 5-3B, a schematic of the spatial manipulator is shown and as mentioned before, this manipulator is composed of four tapering segments separated by soft segment connectors (Fig. 5-3g). Figure 5-3A depicts the two layer concentric structure of a segment. The outer layer (Fig. 5-3e) is made from a very soft silicone rubber and the inner layer (Fig. 5-3d) is made from a slightly stiffer silicone rubber. Within the outer layer of the segment, four cylindrical fluidic channels (Fig. 5-3c) are embedded along the z-axis and are spaced at 90 degree intervals. These channels enable the arm to bend in two orthogonal planes. When a channel is pressurized with fluid, the outer layer undergoes both longitudinal and circumferential strain; however, the inner layer is designed to be inextensible under the induced stresses and this difference in material strain results in bending about the inextensible z-axis. Bending motion primitives are shown in Figure 5-3C. Note that a composite bend defined by the bend angle γ and segment curvature κ is generally achievable by actuating adjacent channels within a segment. Alternatively, a traditional rotational DOF can be introduced at the segment's base to define γ and a single channel can be actuated to define κ. To achieve three-dimensional positioning but to prevent the segments from twisting about their neutral axis (an unactuated DOF), in this work we actuate the manipulator in a sagittal plane defined by the rotation of a stage to which the base connector (Fig. 5-3b) is mounted. Each segment's center is hollow and used to accommodate internal fluid transmission lines (Fig. 5-3a) that function to deliver pressurized fluid to each actuated channel as well as facilitate the concatenation of multiple segments. On one end, each transmission line is individually connected to each actuated channel in the multi-segment arm by means of a crush resistant silicone insert (Fig. 5-3f) shown in red. On the other end, each line connects to the outlet of a fluidic drive cylinder, a device originally developed by the authors in Marchese et al. [2014c].

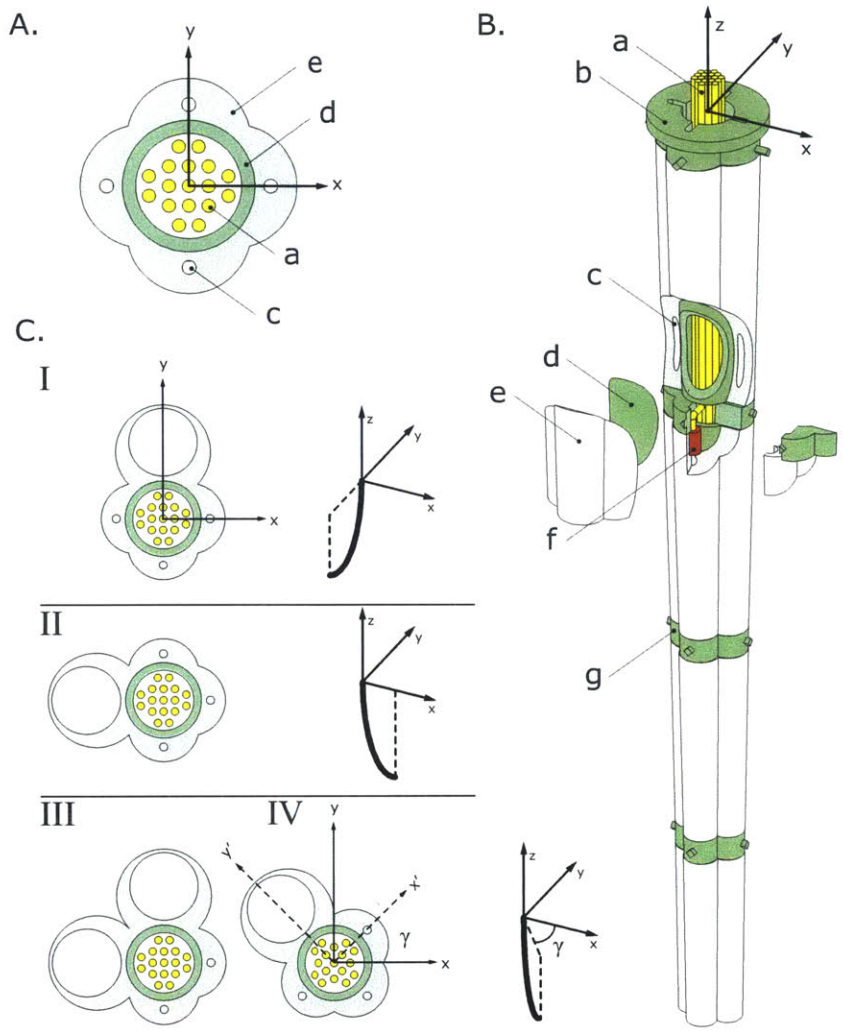


Figure 5-3: (**A**) A cross-section in the x-y plane details the concentric layered structure of the manipulator's soft segments. (**B**) A schematic of the four segment spatial manipulator implemented in this work with the following components: (a) fluid transmission line bundle, (b) base connector, (c) expansion channels, (d) inner layer, (e) outer layer, (f) crush resistant channel inlet, and (g) segment connector. (**C**) The actuation motion primitives of a single soft segment: A segment bends through pressurization of the embedded fluidic channels.

5.2.2 Alternative Designs Considered

To arrive at the current entirely soft manipulator design, we evaluated many alternative designs with varying degrees of rigidity. Through an iterative process, we were able to design away all non-rubber components. For example, Figure 5-4 illustrates several of these past design alternatives. In Figure 5-4A, a semi-soft manipulator is

shown and it is composed of several arm segments each consisting of a plastic backbone and metal ribs surrounded by casted rubber (Fig. 5-4B). Figure 5-4C shows one of these kinematically constrained soft segments under actuation. In this design iteration, the manipulator's endplates were rigid, and without a hollow interior, fluid transmission lines were run external to the manipulator. These lines interfered with localization and mobility. Additionally, the arm segments shown here are cylindrical and uniform along the length of the manipulator producing a larger suspended mass than the current design. Figure 5-4D shows a second alternative design where a plastic helix was introduced around the segment's exterior. Noticeably, although this kinematic constraint reduced circumferential strain, it produced torsion about the neutral axis under actuation (Fig. 5-4E).

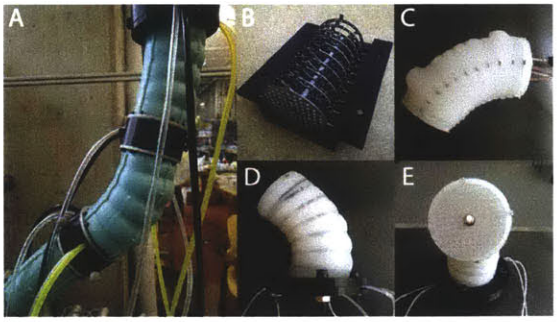


Figure 5-4: Example design alternatives having varying degrees of rigidity. Through an iterative design process, all non-rubber components were removed.

5.2.3 Kinematic Modeling

In order to autonomously perform tasks with the manipulator developed in Section 5.2.1, we must develop a multi-segment kinematic model. The objectives of this section are to: (i) predict how an individual body segment will deform under fluidic input, (ii) determine what simplifying assumptions can be made in a multi-segment kinematic model, and (iii) experimentally evaluate the proposed model. An important result is that we use and validate a PCC modeling assumption [Webster and Jones, 2010], explained in Chapter 2 Section 2.2.1, to describe the multi-segment manipulator under self-weight. The kinematic model developed here will be used by

a controller, detailed in Section 5.4.1, to control the manipulator's configuration in real-time.

Segment Deformation

We extend an analytical modeling approach presented by the authors in Marchese et al. [2014d] to understand the deformation of the manipulator's soft arm segments. Figure 5-5 provides a schematic representation of a soft arm segment in both a neutral state (Fig. 5-5A) and a deformed state (Fig. 5-5B). Additionally, Fig. 5-5C illustrates an isometric view of a thin slice of the segment's elastically deformed outer layer.

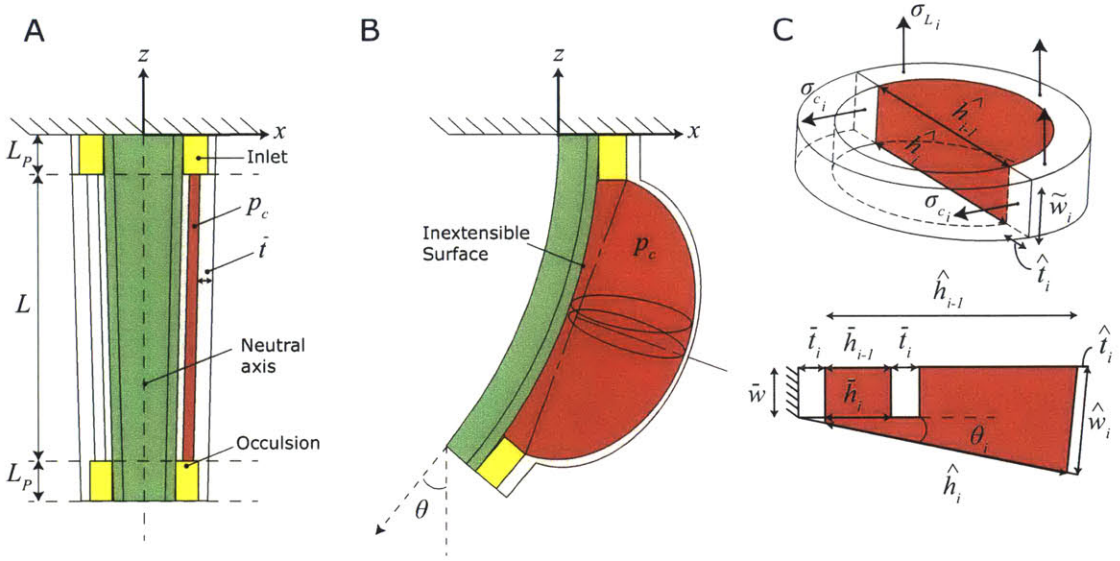


Figure 5-5: Schematic representation of a soft arm segment under deformation.

The segment bends due to the relative difference in stiffness between the inner and outer rubber layers. The inner layer is cast from a rubber chosen to have an elastic modulus approximately 5.5 times greater than that of the outer layer, allowing us to make the assumption that the inner layer is inextensible relative to the outer layer. By pressurizing the fluidic channel embedded within the outer layer, stress is induced both circumferentially (normal to the neutral z-axis) and longitudinally (parallel to the neutral z-axis) in the outer layer, resulting in circumferential expansion and longitudinal elongation respectively. Due to the difference in lengths between the

inner and outer layer, the segment bends.

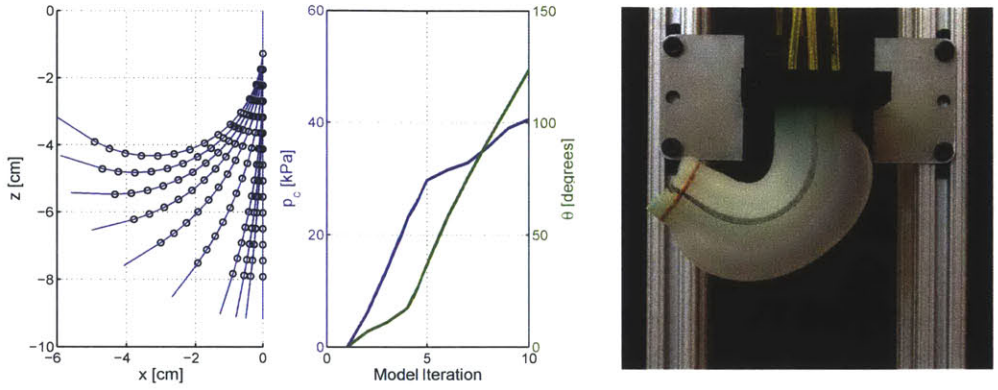
In order to understand how a segment bends, we consider the case of one of the segment's channels deforming under pressure. Our approach is to first model the deformation of this channel as a function of the channel's fluid pressure p_c. The model is based on dividing the channel into J cross-sectional slices of uniform z-axis length cut parallel to the x-y plane. Calculations are independently performed on each slice. In general, a bar is used to described an initial, undeformed channel dimension, whereas a hat is used to describe a deformed channel dimension. Then, to simulate segment bending, we iteratively apply this model each time increasing p_c. More specifically, at each iteration the channel's circumferential and longitudinal deformed dimensions are recalculated based on material stresses that are derived from the previous iteration's deformed geometry. Figure 5-6a depicts the bending of the inextensible layer predicted by the simulation as well as apriori channel pressure data that was successively input into the simulation at each iteration. Also, the resulting accumulated bending angle θ is shown at each iteration. Figure 5-6b is a photograph of the actual segment that the model is based on at the maximum pressure used in the simulation. Table 5.1 lists important model and simulation parameters.

Table 5.1: Segment Parameters Used in Simulation

Parameter	Symbol	Value
Segment Length	$L + 2\,L_P$	9.7 cm
Channel Length	L	7.1 cm
Number of Slices	J	15
Slice Wall Length	$\bar{w} = \frac{L}{J}$	0.47 cm
Channel Diameter,	$\bar{h}$	0.32 cm
Channel Wall Thickness	$\bar{t}$	0.32 cm
Simulation Iterations		10

Stresses and Strains

Stresses and strains in the elastomer can be approximated at a simulation iteration given the channel's current deformed geometry as well as the pressure input. After these stresses and strains are computed, they can be used to update the channel's deformed geometry for the next simulation iteration. First, we find the material



(a) (b)

Figure 5-6: (**a**) The neutral axis bending of the soft arm segment whose physical parameters are listed in Table 5.1 is simulated. The right panel details apriori channel pressure data successively input into the simulation each iteration as well as the resulting accumulated bending angle θ. (**b**) A photograph of the modeled segment at the maximum pressure used in the simulation.

stresses. Due to the fact that each of the segment's embedded channels are cylindrical, we can approximate the circumferential stress σ_c in a thin cross-sectional channel slice by balancing static forces. That is, the pressure force acting to circumferentially expand the channel should balance with the product of circumferential wall stress and wall area (please refer to Fig. 5-5C for visualization),

$$2\,\sigma_{c_i}\,\hat{t}_i\,\tilde{w}_i \;\approx\; p_c\,\frac{\hat{h}_{i-1}+\hat{h}_i}{2}\,\tilde{w}_i \quad \forall\, i \;=\; 1\,..\,J, \tag{5.1}$$

$$\sigma_{c_i} \;\approx\; p_c\,\frac{\hat{h}_{i-1}+\hat{h}_i}{4\,\hat{t}_i} \quad \forall\, i. \tag{5.2}$$

When the condition $\hat{h}_{i-1} = \hat{h}_i$ is met,

$$\sigma_{c_i} \approx p_c\,\frac{\hat{h}_{i-1}}{2\,\hat{t}_i} \quad \forall\, i. \tag{5.3}$$

In the above approximations, i is the slice index, J is the total number of slices, p_c is the channel's fluid pressure, and $\hat{h}$, $\hat{w}$, and $\hat{t}$ are a slice's deformed channel diameter, distal wall length, and wall thickness, respectively. The dimension $\tilde{w}$ represents the

slice's deformed wall length midway through the slice (refer to Fig. 5-5C).

Using a similar approach to determining the circumferential stress, we can approximate the longitudinal stress σ_L in a slice as

$$\sigma_{L_i} \approx \frac{p_c \hat{h}_{i-1}^2}{16\,\hat{t}_i\left(\hat{t}_i + \hat{h}_{i-1}\right)} \quad \forall i = 1..J. \tag{5.4}$$

To arrive at the above approximation, only one quarter of the apparent longitudinal pressure force is balanced with the longitudinal stress-area product. This is motivated first by the fact that the inner most portion of the channel wall is laminated to an inextensible surface, counteracting a significant portion of the applied pressure force. Second, the opposing, unactuated channel (i.e. on the other side of the neutral axis) is under compression while the segment bends and this too counteracts the applied pressure force. We determined the one-quarter scaling experimentally. An important insight from this is that the longitudinal stress is predicted to be approximately one eighth the circumferential stress.

Next, the circumferential and longitudinal strain, ϵ_c and ϵ_L respectively, can be determined using a look-up table based on the material's measured true stress true strain relationship, shown in Figure 5-7.

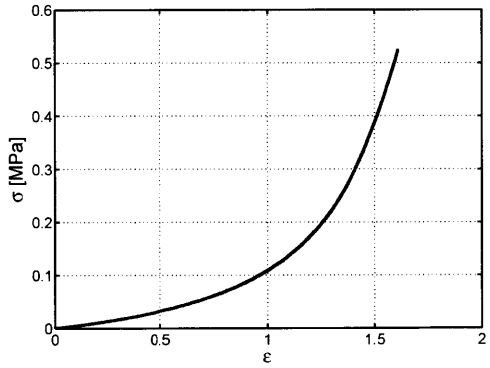


Figure 5-7: True stress true strain relationship of the rubber material composing a segment's outer layer.

Updating Channel Geometry

Having determined the material's strain, we can now update the deformed dimensions

of a channel slice under the increased pressure. That is, the elongated wall $\hat{w}$, the expanded diameter $\hat{h}$, and compressed thickness $\hat{t}$ can be expressed using the calculated material strain, the initial channel geometry (represented by $\bar{h}$, $\bar{w}$, and $\bar{t}$), and the conservation of material volume. The dimension $\hat{w}$ is updated according to

$$\hat{w}_i \approx \bar{w}_i \left(\epsilon_{L_i} + 1\right) \quad \forall i. \tag{5.5}$$

The updates for $\hat{h}$ and $\hat{t}$ can be found by solving a system of two equations. The first equation, Equation 5.6, is found using the previously determined circumferential strain,

$$\hat{h}_i \approx \frac{\hat{c}_i}{\pi} - \hat{t} \quad \forall i, \tag{5.6}$$

where

$$\hat{c}_i \approx \bar{c}_i \left(\epsilon_{c_i} + 1\right), \tag{5.7}$$

$$\bar{c}_i = \pi \left(\bar{h}_i + \bar{t}_i\right). \tag{5.8}$$

The dimensions $\bar{c}$ and $\hat{c}$ represent the undeformed and deformed channel circumference measured half-way through the wall's thickness. The second equation, Equation 5.9, is found using the conservation of material volume. That is, the initial volume of elastic material in a slice is equal to the volume of material in the slice when it is deformed

$$\bar{A}_i \, \bar{w}_i \approx \hat{A}_i \, \bar{w}_i + \hat{A}_i \, \frac{\hat{w}_i - \bar{w}_i}{2} \quad \forall i, \tag{5.9}$$

where $\bar{A}$ and $\hat{A}$ are the initial and deformed cross-sectional area of a slice, respectively. More specifically,

$$\bar{A}_i = \pi \left(\frac{\bar{h}_i}{2} + \bar{t}_i\right)^2 - \pi \left(\frac{\bar{h}_i}{2}\right)^2 \quad \forall i, \tag{5.10}$$

$$\hat{A}_i = \pi \left(\frac{\hat{h}_i}{2} + \hat{t}_i\right)^2 - \pi \left(\frac{\hat{h}_i}{2}\right)^2 \quad \forall i. \tag{5.11}$$

According to the technique outlined in the actuation section of Marchese et al. [2014d],

the accumulated angle θ along the neutral axis at this simulation iteration is

$$\theta = \sum_{i=1}^{J} \cos^{-1}\left(\frac{-\hat{w}_i^2 + \hat{h}_{i-1}^2 + \bar{w}^2 + \hat{h}_i^2}{2\hat{h}_i\sqrt{\hat{h}_{i-1}^2 + \bar{w}^2}}\right) - \tan^{-1}\left(\frac{\bar{w}}{\hat{h}_{i-1}}\right). \tag{5.12}$$

Conditions for Constant Curvature

From this model we can make the following observation: if the initial undeformed channel dimensions are constant along the length of the channel, i.e.

$$\bar{w}_i = C_1 \quad \forall\, i = 1\,..\,J, \tag{5.13}$$

$$\bar{h}_i = C_2 \quad \forall\, i, \tag{5.14}$$

$$\bar{t}_i = C_3 \quad \forall\, i, \tag{5.15}$$

then the deformed dimensions $\hat{w}$, $\hat{h}$, and $\hat{t}$ and accordingly the bending angle θ are constant $\forall\, i = 2\,..\,J-1$, along the length of the channel.

Segment Transformation

Given that a channel deforms under approximately constant curvature, we can write the following transformation from the base of a segment to the tip

$$\mathbf{S}_{tip}^{base} = \mathbf{R}_z(\gamma)\ \mathbf{T}_z(L_P)\ \mathbf{R}_y\left(\frac{\theta}{2}\right)\mathbf{T}_z(d(\theta))\ \mathbf{R}_y\left(\frac{\theta}{2}\right)\ \mathbf{T}_z(L_P). \tag{5.16}$$

Here, $\mathbf{T}$ and $\mathbf{R}$ represent coordinate translations and rotations along and about the subscript axes; refer to Figure 5-5 for coordinate definitions. The center three transformations are consistent with kinematic representations of continuum manipulators, and are used when the constant curvature assumption is valid [Webster and Jones, 2010]. In the context of this work we need to add additional translations along z before and after the bent channel to account for the relative rigidity of the inlet pieces. Additionally, we redefine $d(\theta)$ with an approximation that is singularity free for the

reason of numerical stability

$$d(\theta) = \frac{2L}{\theta} \sin\left(\frac{\theta}{2}\right), \tag{5.17}$$

$$d(\theta) \approx L\left(a\theta^2 - 1\right), \tag{5.18}$$

where a is a fitting coefficient. An important observation is that the segment transformation $\mathbf{S}_{tip}^{base}$ has two DOF. That is, if we know the sagittal plane along which the segment is bending, specified by γ, and are given one point along the segment's neutral axis in reference to the segment's base we can uniquely define θ. Additionally, in order to specify a point along the neutral axis of the curved region $s = [0, L]$ in reference to the base of the channel we use the following transformation

$$\mathbf{S}_{s}^{base} = \mathbf{R}_z(\gamma)\, \mathbf{T}_z(L_P)\, \mathbf{R}_y\left(\frac{\kappa s}{2}\right) \mathbf{T}_z(d(\kappa s))\, \mathbf{R}_y\left(\frac{\kappa s}{2}\right), \tag{5.19}$$

$$\kappa = \frac{\theta}{L}.$$

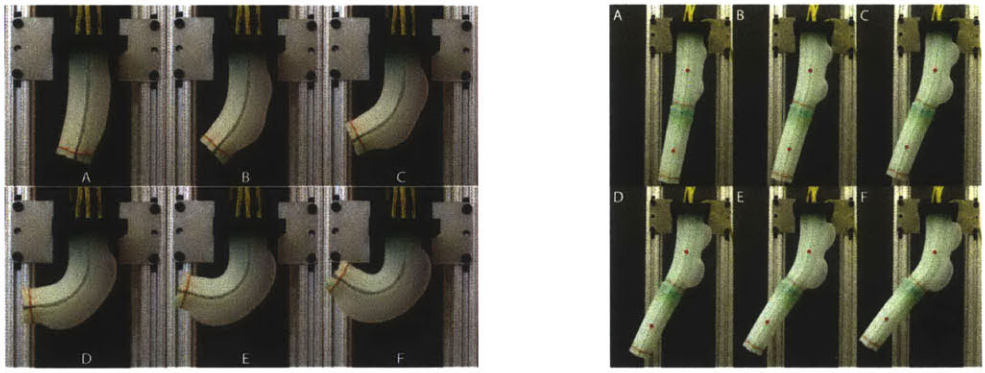
The kinematics of a multi-segment soft arm composed of N segments can be represented by cascading single segment transformations together

$$\mathbf{M}_{tip_N}^{base} = \mathbf{S}_{tip}^{base}(\gamma_1, \theta_1)\, \mathbf{S}_{tip}^{base}(\gamma_2, \theta_2) \cdots \mathbf{S}_{tip}^{base}(\gamma_N, \theta_N). \tag{5.20}$$

Model Evaluations

In order to validate this kinematic model we fabricated and actuated the segment described physically in Table 5.1 subject to gravity. The segment weighs 65 grams. The primary concept we are evaluating in this experiment is: if we localize a single point on the arm segment, i.e. the tip, then we can uniquely define the entire segment's neutral axis. The experiment consists of incrementally filling a single channel with between 0 and 70 mL of air and photographing the segment's deformation. As a metric of goodness of fit, we compare our model's predicted bending angle θ^P with the measured bending angle θ^M at each incremental fill level. Additionally, we compare our model's predicted center of gravity G^P with the measured center of gravity G^M.

Measurements were taken along the neutral axis of the segment using image analysis. θ^M was measured from the unbent tip region and G^M was taken to be at $s = \frac{L}{2}$, the center point of the neutral axis. The model is defined using a single point, in this case the segment's tip. Figure 5-8a shows the test segment being incrementally filled from 32 to 72 mL of air. Table 5.2 contains a comparison between measured and model predicted deformation kinematics.



(a) (b)

Figure 5-8: Experimental validation of the proposed segment transformation. (**a**) A single segment is incrementally filled with air in the sequence of images (A-F). (**b**) A larger soft segment is actuated in the image sequence (A-F) with the suspended weight of a second distally attached segment.

Table 5.2: Comparison between measured and model predicted deformation kinematics

Fig 5-8a	Volume (mL)	Pressure (kPa)	θ^P (°)	θ^M (°)	$\lvert G^P - G^M \rvert$ (cm)
(**F**)	72	40.6	121	122.5	0.54
(**E**)	64	39.0	104	105.6	0.45
(**D**)	56	35.7	83	83.6	0.48
(**C**)	48	32.8	61	61.4	0.30
(**B**)	40	31.5	39	40.2	0.20
(**A**)	32	29.7	16	13.8	0.24

Additionally, we performed the same experiment on a segment weighing 173 grams under the external load of an attached second segment weighing 85 grams. The intent of this experiment was to quantify the error in the proposed kinematic segment model when the segment is under significant external load. For the same volumetric input,

the maximum measured bending angle 40° is significantly smaller than in the unloaded case, 123°. As expected, the error in the model predicted bending angle is higher in the loaded case than in the unloaded case, but within the range $\pm 40°$ does not exceed an error of 7.0°, with a mean error of 2.7°. Figure 5-9 depicts the percent error in the prediction as a function of measured bending angle. Generally, as bending angle increases the error in predicted bending angle grows. Figure 5-8b shows the loaded test segment being incrementally filled from 32 to 72 mL of air.

Table 5.3: Comparison between measured and model predicted deformation kinematics for segment under external load

Fig 5-8b	Volume (mL)	Pressure (kPa)	θ^P (°)	θ^M (°)	$\lvert G^P - G^M \rvert$ (cm)
(**F**)	72	40.6	33.1	39.9	0.30
(**E**)	64	39.0	28.3	34.5	0.18
(**D**)	56	35.7	24.4	28.1	0.12
(**C**)	48	32.8	19.2	20.5	0.03
(**B**)	40	31.5	14.6	15.3	0.09
(**A**)	32	29.7	9.9	10.0	0.08
-	24	23.0	4.3	3.7	0.09

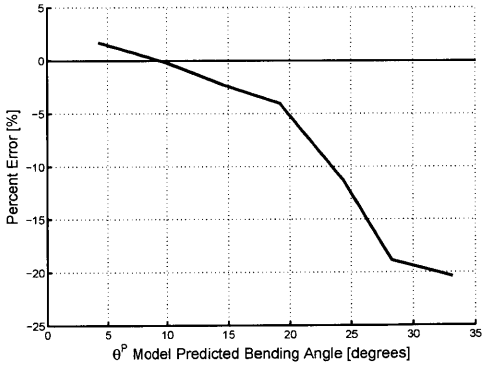


Figure 5-9: The percent error in model predicted bend angle as a function of model predicted bending angle θ^P

5.2.4 Dynamic Model

Energetics

Our objective is to write the equations of motion for this soft fluidic elastomer manipulator. To do this we can first find the potential, kinetic, and input components of energy for a single arm segment and then use a Lagrangian approach to derive the equations of motion with respect to the segment's generalized coordinate. A fundamental difference between soft and hard robot manipulators is in the way energy is stored. In a soft fluidic elastomer manipulator, input fluid energy is delivered from a power supply and stored as both strain energy along its continuum segments V_ϵ and gravitational potential energy V_g. Both forms of stored energy serve to deform the manipulator and are transferred to kinetic energy T.

Potential Energy of a Segment

Consider a single spatial arm segment deforming in a sagittal plane defined by a fixed γ. By approximating the center of mass to be located half-way along the segment's neutral axis, we can use S_s^{base} to express the center of mass position in $\mathbb{R}^3$ as $(x_m(\theta),\, y_m(\theta),\, z_m(\theta))$. Bend angle θ is understood to be time dependent. The gravitational potential energy of the segment is

$$V_g(\theta) = m\, g\, z_m\,(\theta)$$

where m is the segment's mass and g is the gravitational constant. For a fluidic soft manipulator made of deformable elastomer, a significant component of potential energy is strain energy. For strain below 60%, we can approximate the stress strain relationship of the arm segment's outer layer with a constant elastic modulus E. With this, the strain energy developed in an actuated channel is

$$V_\epsilon = \frac{1}{2} \vee E\, \epsilon^2 \quad \rightarrow \quad V_\epsilon = \frac{1}{2}\, \pi\, \bar{t}\, (\bar{h} + \bar{t})\, L\, E\, \epsilon^2$$

where ϵ is material strain, $\vee$ is the material volume incurring strain, and $\bar{t}$ and $\bar{h}$ are the wall thickness and diameter of the actuated channel. In a segment subject to circumferential and longitudinal strain that deforms under constant curvature, material strain ϵ and bend angle θ can be related by decomposing the actuated region into J cross-sectional x-y slices of z-axis length w as outlined in Section 5.2.3 and the law of cosines

$$\epsilon_j = \frac{\bar{h}_j}{w_j}\sqrt{2 - 2\cos\theta_j} \quad \forall j = 0..J \quad \rightarrow \quad \epsilon = \frac{\bar{h}}{w}\theta.$$

There are several important observations that allow us to express this relationship between ϵ and θ: First, the dimensions of each slice are uniform under the aforementioned constant curvature assumption. Second, in general $\bar{h}$ is not constant, but rather changes as a function of strain $\bar{h}(\epsilon)$ and this is consistent with the analysis contained in Shepherd et al. [2011] where pneumatic channels similar to the type described here increase in stiffness and potential energy when pressurized. However, we observe that after undergoing initial circumferential expansion, the diameter of the actuated channels here changes little. Approximating the diameter $\bar{h}$ to be constant is valid to describe the regime of operation after the initial circumferential change. Lastly, using the small angle approximation $\cos\theta \approx 1 - \frac{\theta^2}{2}$ for the argument $\frac{\theta}{J}$ where J is chosen such that the approximation is valid, we can linearize the relationship between ϵ and θ in order to arrive at a constant stiffness coefficient and help reduce the complexity of the model.

Now, we can write strain energy in the segment as a function of bend angle

$$V_\epsilon(\theta) = \frac{1}{2}\left(\frac{\pi\,\bar{t}\,(\bar{h}+\bar{t})\,L\,E\,\bar{h}^2}{w^2}\right)\theta^2 \quad \rightarrow \quad V_\epsilon(\theta) = \frac{1}{2}\,k\,\theta^2,$$

where k is an effective stiffness for the generalized coordinate θ. The total potential energy of the arm segment in the sagittal plane defined by γ is $V(\theta) = V_g + V_\epsilon$.

Input to a Segment

We develop an independent generalized force τ that acts on an arm segment by differentiating the total potential energy with respect to the generalized coordinate, i.e. $\tau = \frac{\partial}{\partial\theta} V$

$$\tau = k\,\theta + a\,g\,Lm\,\cos\left(\frac{\theta}{2}\right)\theta - \frac{1}{4}\,g\,L\,m\,\sin\left(\frac{\theta}{2}\right)\,\left(-1 + a\,\theta^2\right)$$

We can substitute in the approximations $\sin\left(\frac{\theta}{2}\right) \approx \theta$ and $\cos\left(\frac{\theta}{2}\right) \approx 1 - \frac{1}{8}\theta^2$ with less than 5% error at θ equal to 50° and 100° respectively

$$\tau = k\,\theta + \frac{1}{8}\,(1\,+\,8\,a)\,g\,L\,m\,\theta\,-\,\frac{1}{4}\,a\,g\,L\,m\,\theta^3.$$

This approximation will help simplify the identification process in Section 5.4.2. Next, we can express the change in channel volume $\mathbb{V}_c$ as a function of material strain and, because of our aforementioned strain assumption, a function of θ

$$\mathbb{V}_c = \frac{1}{2}\,\frac{\pi\,\bar{h}^2}{4}\,L\,\epsilon \quad \rightarrow \quad \mathbb{V}_c = \frac{\pi\,\bar{h}^3\,L}{8\,w}\,\theta.$$

Substituting this into the generalized force yields:

$$\tau = -\frac{128\,a\,g\,m\,w^3}{L^2\,\pi^3\,\bar{h}^9}\,\mathbb{V}_c^3 + \left(\frac{8\,k\,w}{\pi\,\bar{h}^3\,L} + \frac{(1\,+\,8\,a)\,g\,m\,w}{\pi\,\bar{h}^3}\right)\,\mathbb{V}_c,$$

revealing that there is a cubic relationship between the generalized force and the change in channel volume.

Multi-Segment Equations of Motion

We can write the equations of motion for a multi-segment soft manipulator using multiple generalized coordinates as follows. The center of mass position of the n^{th} soft segment is represented by $\mathbf{P}_n$ and can be expressed as

$$\mathbf{P}_n = M^{base}_{tip_{n-1}}\,S^{base}_{\frac{Ln}{2}}\,\mathbf{0} \quad \forall\, n = 1\,..\,N.$$

The total kinetic energy of a manipulator with N segments is

$$T = \sum_{n=1}^{N} \frac{1}{2} m_n \frac{d}{dt}\mathbf{P}_n \cdot \frac{d}{dt}\mathbf{P}_n.$$

And the total potential energy is

$$V = \sum_{n=1}^{N} \frac{1}{2} k_n \theta_n^2 + g \sum_{n=1}^{N} m_n \mathbf{P}_n \cdot \hat{\mathbf{k}}.$$

Using the Lagrangian $L = T - V$, N independent nonlinear equations of motion can be written, one for each generalized coordinate

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}_n} - \frac{\partial L}{\partial \theta_n} = \tau_n - b_n \dot{\theta}_n \quad \forall n = 1 \,..\, N.$$

where b is a damping term used to account for the non-conservative nature of the generalized forces. The soft robot dynamics can now be written in traditional manipulator equation form

$$\mathbf{H}(\boldsymbol{\theta})\,\ddot{\boldsymbol{\theta}} + \mathbf{C}\left(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}\right)\,\dot{\boldsymbol{\theta}} + \mathbf{G}(\boldsymbol{\theta}) = \mathbf{B}\,\boldsymbol{\tau}.$$

It is important to note that the nonlinear equations of motion are complex enough that for the 4-segment case we were unsuccessful in symbolically determining $\mathbf{H^{-1}}$ using both Mathematica mat [2012] and Maple map [2011] on an Intel i7-2820QM 2.3GHz processor with 20 GB of RAM. Figure 5-10 provides an illustration of this model for a soft manipulator composed of four segments. The sagittal plane is defined by a traditional rotational degree of freedom γ located at the manipulator's base. In the most general case, the dynamic model is parameterized by four generalized coordinates $\theta_1 \ldots \theta_4$ and four corresponding segment masses $\mathbf{m}$, generalized stiffnesses $\mathbf{k}$, and damping coefficients $\mathbf{b}$. Additionally there are three generalized input forces $\boldsymbol{\tau}$.

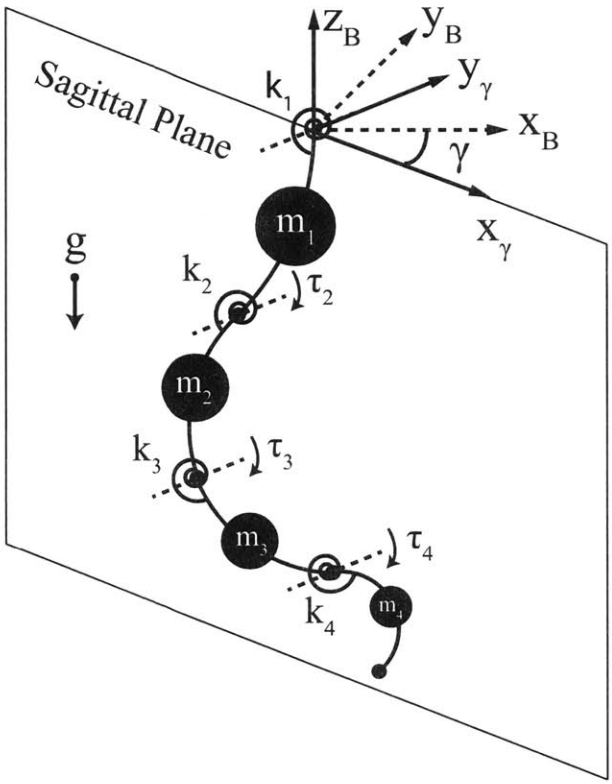


Figure 5-10: Visualization of the multi-segment soft manipulator model. The base frame is rotated by γ by a traditional rotational degree of freedom and defines the sagittal plane within which the manipulator moves. The first soft segment is unactuated.

5.2.5 Manipulator Fabrication

The fabrication process for this spatial soft arm is a three-dimensional extension of a fabrication process the authors developed to construct a planar soft manipulator presented in Marchese et al. [2014b]. The tools and equipment used for fabrication are listed in Table 5.4. Generally, we fabricate the arm through a casting process that uses pourable silicone rubber[2,4] and 3D printed molds[1]. Figure 5-11 and Algorithm 8 detail the fabrication process for the spatial manipulator. The general approach is to first independently cast each segment and connector in steps 1-5 and subsequently join these segments serially to form the manipulator. Prior to the process, a four piece mold was designed and printed. The mold consists of: (1) a two-part outer shell (Fig. 5-11 *orange*) that functions to form the segment's clover-like exterior, (2)

the mold's interior piece (Fig. 5-11 *blue*) that functions to produce the body's hollow interior, (3) metal rods (Fig. 5-11 *grey*) inserted into the interior piece to form cavities for the segment's four embedded pneumatic channels, and lastly (4) a removable sleeve (Fig. 5-11 *fuchsia*).

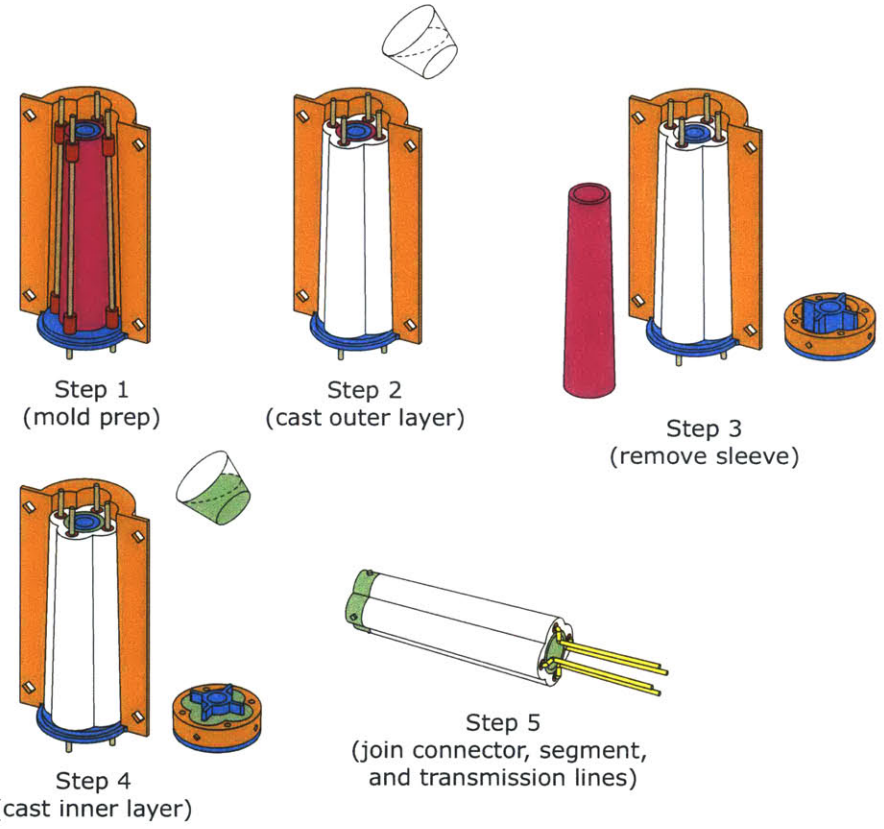


Figure 5-11: A sequence of pictures depicting the 5 step casting process used to fabricate a soft spatial arm segment. A mold is prepared in step 1 and poured in steps 2-4. First, in step 2 a low elastic modulus rubber is poured to form the segment's outer layer. In step 3, a sleeve is removed from the mold after the outer layer has cured and functions to produce a cavity into which a second stiffer rubber is poured during step 4, forming the segment's constraining inner layer. In step 5 the segment is demolded and transmission lines as well as a connector piece are attached.

In **step 1**, the four mold parts are assembled and eight crush resistant silicone pieces (Fig. 5-11 *red*) are slid over the metal rods. These relatively stiff pieces will eventually form inlets to the embedded channels. All inlet pieces are cut to length and cleaned with rubbing alcohol to ensure a good bond between their surface and the later introduced silicone rubber. To prepare four molds for the parallel casting

Algorithm 8: Fabrication Process (See Fig. 5-11)

Result: A multi-segment soft manipulator

repeat

1. Assemble and prepare mold (Step 1).
2. Pour soft rubber[2] to cast segment's outer layer (Step 2).
3. Remove sleeve to create inner cavity (Step 3).
4. Pour stiffer rubber[4] to cast segment's inner layer and connector piece (Step 4).
5. Demold and attach[5] transmission lines and connector piece to segment (Step 5).

until *All N segments are complete*

6. Join N segments together using adhesive[5] to form manipulator.

of four segments, this step takes approximately one hour. In **step 2**, a low elastic modulus rubber[2] is mixed, degassed in a vacuum chamber[3], and poured to form the body segment's soft outer layer (Fig. 5-11 *white*). The rubber is poured slowly into the top of the mold. Because the material gradually thickens as a function of time (i.e., a pot life of 45 minutes), we often mix and pour two sequential batches of rubber to ensure the material remains at a sufficiently low viscosity to enter the mold. When pouring four molds in parallel, this stage of the process can take between 1 and 2 hours depending on experience. Once the outer layer has fully cured (4 hours), the mold is disassembled and the rigid *fuchsia* sleeve is removed in **step 3**. When the mold and outer layer are reassembled, this produces a cavity between the outer layer (Fig. 5-11 *white*) and interior mold piece (Fig. 5-11 *blue*). Also, we prepare a smaller separate mold for the segment's connector. This step takes approximately 1 to 1.5 hours while preparing four segments. In **step 4**, we pour a slightly stiffer rubber[4] into this newly formed cavity and this forms the segment's partially constraining inner layer (Fig. 5-11 *green*). Simultaneously, we pour the segment's connector, also shown in *green*. The mixing and pouring process is identical to that in step 2 and this stage can also take between 1 and 2 hours depending on experience. During **step 5** the pieces are removed from their molds, cleaned, and the segment is joined to the rubber connector using adhesive[5]. Also, soft rubber tubes[6] (Fig. 5-11 *yellow*) are used for transmission lines and are joined to the channel's inlets. To make four segments, this step takes approximately 2.5 hours. Lastly, to assemble the manipulator in **step 6** the transmission line bundle is passed through each segment's hollow interior and the segments are joined together using the same silicone adhesive.

This fabrication process was used to make four complete, functional soft manipulators (Fig. 5-12) and numerous individual segments not shown. For each prototype, minor variations in the manufacturing process were evaluated, but fundamentally each fabrication cycle is represented by the 6 step process outlined above. One variation to this process included casting the inner layer in a separate mold *apriori* and then using this in place of the removable sleeve to eliminate intermediate demolding.

Table 5.4: Fabrication Tools and Materials

1	Fortus 400mc, Stratasys Ltd., Eden Prairie, MN
2	Ecoflex 0030, Smooth-On, Easton, PA
3	AL Cube, Abbess Instruments and Systems, Inc., Holliston, MA
4	Mold Star 15, Smooth-On, Easton, PA
5	Silicone Sealant 732, Dow Corning Corp, Midland, MI
6	PN 51845K52, McMaster, Princeton, NJ

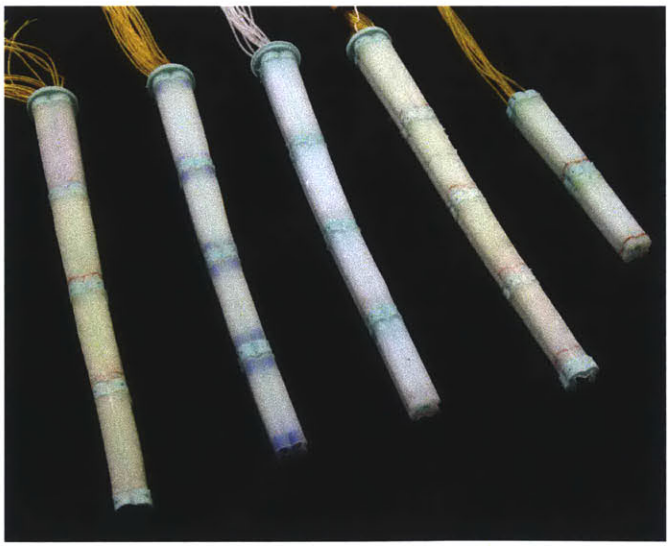


Figure 5-12: Functional soft fluidic elastomer manipulators fabricated using the process outlined in Figure 5-11 and Algorithm 8.

5.3 Power

The soft manipulator is powered by pressurized fluids. Specifically we use air because of its low viscosity. This feature presents the unique challenges of (1) needing to supply each of the manipulator's actuators with a continuous source of fluid energy for prolonged operation and (2) needing the ability to continuously adjust input energy for smooth curvature control. To address these challenges we constructed an array of 16 fluidic drive cylinders, detailed in Figure 5-13, and these serve to individually pressurize each of the manipulator's embedded channels. We conceived and implemented a smaller version of the fluidic drive cylinder array in previous work [Marchese et al., 2014c]; however, both the maximum achievable flow rate and maximum volumetric capacity of the original array were insufficient for the functional requirements of the larger spatial manipulation system, necessitating the array to be redesigned. Each cylinder maps one-to-one with an actuated channel, and agonist-antagonist pairs of actuators are pressurized by a pair of cylinders driven in sequence.

The fluidic drive cylinder has several advantages when compared to other fluidic energy supplies (e.g., rotary pumps [Katzschmann et al., 2014] and compressed gas [Marchese et al., 2014d]): (1) segment curvature is monotonically related to cylinder displacement, (2) the drive system is closed-circuit meaning a constant volume of transmission fluid is moved around the system and none is intentionally exhausted to the environment, and (3) flow can be continuously adjusted. These advantages enable real-time curvature control, as initially demonstrated in Marchese et al. [2014c], as well as prolonged operation of the manipulation system.

Specifically, each fluidic drive cylinder consists of several components: a nose-mounted, 63.5 mm bore, 76.2 mm stroke air cylinder (PN 503-D; Bimba Manufacturing Company, University Park, IL) shown in Figure 5-13a and b; a 12 volt, 100 mm stroke, 43 mm/sec electric linear actuator (PN 2319; Pololu Corporation, Las Vegas, NV) shown in Figure 5-13c and d, as well as a positional controller for these linear actuators (PN 1393; Pololu Corporation).

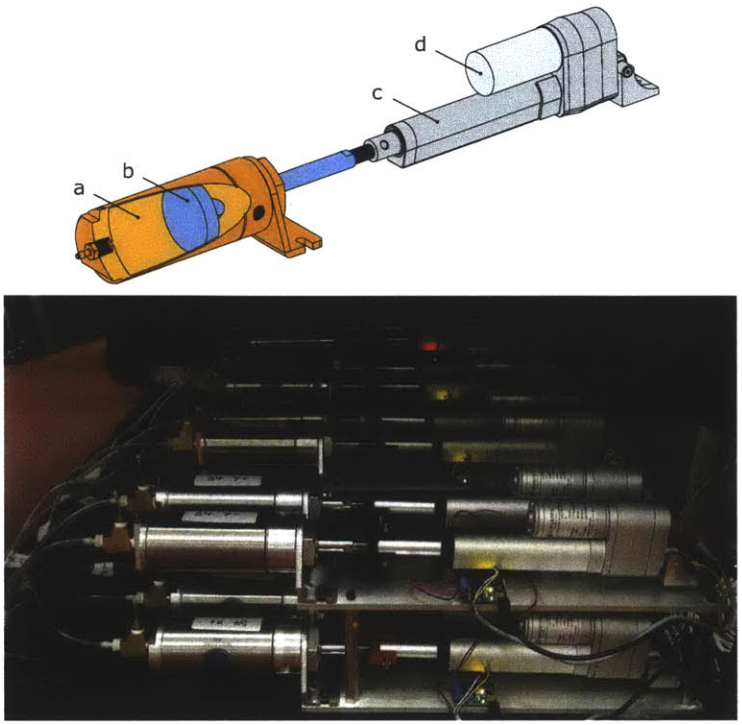


Figure 5-13: Top: schematic representation of a high capacity fluidic drive cylinder. The device consists of a (a) fluidic cylinder, (b) piston, (c) linear drive, and (d) electric motor. Bottom: An array of 16 fluidic drive cylinders used to power the spatial soft arm.

5.4 Processing and Control

5.4.1 Kinematic Controller

In order to autonomously control the spatial manipulator's configuration in real-time, we develop a kinematic controller. The controller (see Algorithm 9) performs several tasks: First, the procedure `generateTraj()` generates a realizable curvature velocity trajectory $\dot{\boldsymbol{\kappa}}(t)$ that takes the arm's segments from their initial state $\boldsymbol{\kappa}_0$ at $t = t_0$ to a desired state $\boldsymbol{\kappa}_f$ at $t = t_f$ (see Fig. 5-14b). This velocity trajectory is trapezoidal and computed from empirically determined constraints on acceleration and velocity, $\ddot{\boldsymbol{\kappa}}^{max}$ and $\dot{\boldsymbol{\kappa}}^{max}$ respectively. Second, the procedure `driveSegments()` physically advances the arm segments along the aforementioned trajectory under closed-loop control. More specifically, when a new measurement is available from the external

localization system[2] segment endpoints are identified in reference to a base frame (i.e., $\mathbf{E}_n^{\text{base}} \subset \mathbb{R}^3 \quad \forall n = 1..N$) and subsequently transformed into the manipulator's sagittal plane (i.e., $\mathbf{E}_n^{\gamma} \subset \mathbb{R}^2 \quad \forall n = 1..N$) specified by the rotation γ. Next, these endpoints are used to determine the manipulator's measured curvature configuration representation $\boldsymbol{\kappa}_{\text{meas}}$ at the current instant in time (see Fig. 5-14a). This stage is accomplished using a geometric algorithm referred to here is `singleSegInvKin()` and developed by the authors in Marchese et al. [2014c]. This procedure computes the inverse kinematics of a single planar segment. That is, given the start and end points of a segment as well as its starting orientation we can uniquely determine its signed curvature. Next, a reference configuration $\boldsymbol{\kappa}_{\text{target}}$ is computed from the realizable reference trajectory $\dot{\boldsymbol{\kappa}}(t)$ at the current instant in time. The computed error then becomes the input to an array of cascaded PI-PID (curvature and cylinder displacement, respectively) controllers, a control strategy developed by the authors in Marchese et al. [2014c].

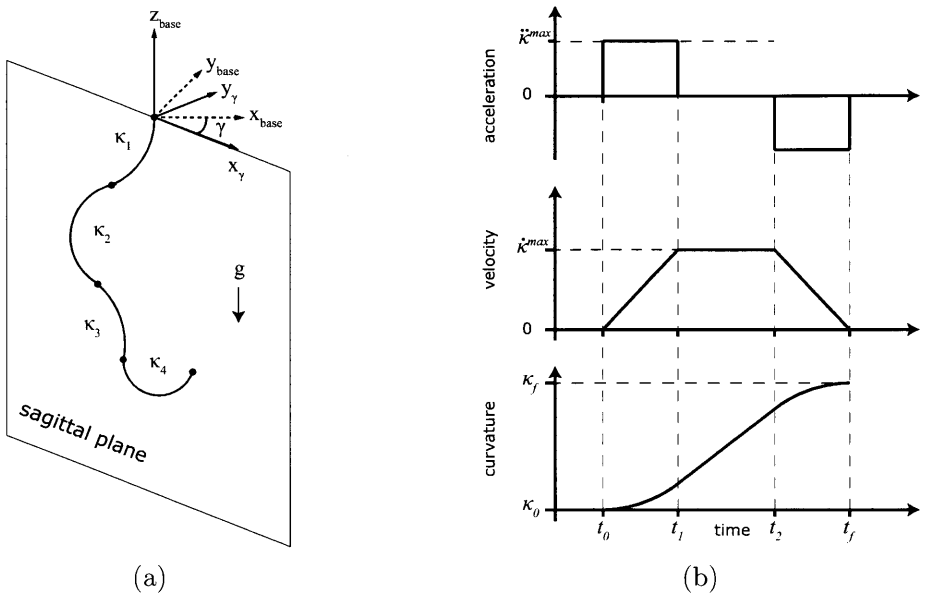
5.4.2 System Identification

In order to use the dynamic model developed in Section 5.2.4 for automated control we must first develop a strategy for identifying the model's unknown physical parameters. In addition to this, we must also define an approach for identifying an accurate model for the manipulator's soft actuators as well as its drive mechanisms. In this section we first present a high-level algorithm used to identify the aggregate manipulation system composed of three distinct subsystems: fluidic drive cylinders, distributed soft actuators, and the soft manipulator. Then, we look specifically at how these unknown model parameters arise from each subsystem.

Approach Overview

Identification of the aggregate dynamical manipulation system *arm* is performed by iteratively adjusting a parameter set $\mathbf{p}$ such that a model instantiated from $\mathbf{p}$ fol-

[2]OptiTrack; NaturalPoint, Inc., Corvallis, OR



(a) (b)

Figure 5-14: (**a**) A simplified state representation of the soft fluidic elastomer manipulator using PCC modeling [Webster and Jones, 2010] rationalized and experimentally validated in the context of this arm in Section 5.2.3. The simplifying assumptions of this model include: (1) the N arm segments bend according to constant curvature within a sagittal plane defined by the rotation $\mathbf{R}_z(\gamma)$ of a base frame and (2) the lengths of the passive portions of each segment are negligible, i.e. $L_P \rightarrow 0$. (**b**) A realizable reference curvature trajectory generated by the controller (Algorithm 9) is shown for a single segment. The trajectory is computed under the constraints of (1) a trapezoidal velocity profile as well as (2) empirical constraints on acceleration and velocity, $\ddot{\boldsymbol{\kappa}}^{max}$ and $\dot{\boldsymbol{\kappa}}^{max}$ respectively.

lows the same N-segment endpoint Cartesian trajectory as measured on the physical system. Specifically, we do this by solving the nonlinear optimization within Algorithm 10 for a locally optimal parameter set $\mathbf{p}^*$. Here, $\mathbf{E}_{n,i}$ is a discrete trajectory of the *measured* cartesian endpoint coordinates of the n^{th} arm segment. The manipulator state trajectory $\mathbf{x}(t)$ is composed of segment bend angles $\boldsymbol{\theta}$ and corresponding velocities $\dot{\boldsymbol{\theta}}$. The function FORWARDKIN$_n$ uses the multi-segment transformation to return the cartesian endpoint coordinates of the n^{th} arm segment. The function UPDATEMODEL instantiates *arm* according to the parameter set $\mathbf{p}$ and the function SIMULATE forward simulates the response of the dynamic model to input trajectory $\mathbf{u}(t)$ over the time interval $t = [0, t_f]$ from initial conditions $\mathbf{x}_0$.

The aggregate manipulation system *arm* consists of four fluidic drive cylinder pairs

Algorithm 9: Configuration Controller

Input: $\boldsymbol{\kappa}_0$, $\boldsymbol{\kappa}_f$, $\dot{\boldsymbol{\kappa}}^{max} > \mathbf{0}$, $\ddot{\boldsymbol{\kappa}}^{max} > \mathbf{0}$, γ
Result: Manipulator moves along configuration trajectory from $\boldsymbol{\kappa}_0$ to $\boldsymbol{\kappa}_f$.

1 $\dot{\boldsymbol{\kappa}}(t) \leftarrow$ `generateTraj` ($\boldsymbol{\kappa}_0$, $\boldsymbol{\kappa}_f$, $\dot{\boldsymbol{\kappa}}^{max}$, $\ddot{\boldsymbol{\kappa}}^{max}$)
2 `driveSegments` ($\dot{\boldsymbol{\kappa}}(t)$, γ)

Procedure `generateTraj`($\boldsymbol{\kappa}_0$, $\boldsymbol{\kappa}_f$, $\dot{\boldsymbol{\kappa}}^{max}$, $\ddot{\boldsymbol{\kappa}}^{max}$)
Result: $\dot{\boldsymbol{\kappa}}(t)$ that takes segments from $\boldsymbol{\kappa}_0$ at $t = t_0$ to $\boldsymbol{\kappa}_f$ at $t = t_f$
for $n = 1..N$. **do**
All κ in this loop are indexed by i but omitted for convenience
if $|\kappa_f - \kappa_0| < 2\frac{(\dot{\kappa}^{max})^2}{\ddot{\kappa}^{max}}$ **then**
$\dot{\kappa}^{max} \leftarrow \text{SGN}(\kappa_f - \kappa_0)\sqrt{|\kappa_f - \kappa_0|\,\ddot{\kappa}^{max}}$.
$t_1 \leftarrow \frac{|\kappa_f - \kappa_0|}{|\dot{\kappa}^{max}|}$.
$t_2 \leftarrow t_1$.
else
$\dot{\kappa}^{max} \leftarrow \text{SGN}(\kappa_f - \kappa_0)\,\dot{\kappa}^{max}$.
$t_1 \leftarrow \frac{|\dot{\kappa}^{max}|}{\ddot{\kappa}^{max}}$.
$t_2 \leftarrow \frac{|\kappa_f - \kappa_0|}{|\dot{\kappa}^{max}|}$.
end
$t_f \leftarrow t_1 + t_2$.
$\mathbf{t} \leftarrow [0,\, t_1,\, t_2,\, t_f,\, t_f + \epsilon]$.
$\mathbf{v} \leftarrow [0,\, \dot{\kappa}^{max},\, \dot{\kappa}^{max},\, 0,\, 0]$.
$\dot{\kappa}(t) \leftarrow$ `firstOrderHold`($\mathbf{t}$, $\mathbf{v}$).
end
return $\dot{\boldsymbol{\kappa}}(t)$

Procedure `driveSegments`($\dot{\boldsymbol{\kappa}}(t)$, γ)
Result: Drive segments to desired configuration
$t \leftarrow t_0$.
while $t <$ `max`$(\mathbf{t}_f)$ **do**
if *A new sensor measurement* $\mathbf{E}^{base}$ *is available* **then**
$\mathbf{E}_n^{\gamma} \leftarrow \left[\mathbf{T}_{\gamma}^{\text{base}}(\gamma)\right]^{-1}\mathbf{E}_n^{\text{base}} \quad \forall\, n = 1..N.$
$\boldsymbol{\kappa}_{\text{meas}} \leftarrow$ `singleSegInvKin` $(\mathbf{E}^{\gamma})$ [Marchese et al., 2014c].
$\boldsymbol{\kappa}_{\text{target}} \leftarrow \int_{t_0}^{t} \dot{\kappa}_n(t)\,\mathrm{d}t \quad \forall\, n.$
Send error $(\boldsymbol{\kappa}_{\text{target}} - \boldsymbol{\kappa}_{\text{meas}})$ to low level cascaded PI-PID controllers [Marchese et al., 2014c].
end
$t \leftarrow$ elapsed time.
end
return

connected to eight fluidic elastomer actuators distributed within the soft manipulator. We break this aggregate system into three distinct subsystems with the following input $\rightarrow$ output relationships:

1. Fluidic Drive Cylinders:

   reference inputs $\mathbf{u} \rightarrow$ cylinder displacements $\mathbb{V}_s$

Algorithm 10: System Identification

$$\min_{\mathbf{p}} \quad \sum_{i}\sum_{n=1}^{N} \|arm.\text{FORWARDKIN}_n(\mathbf{x}_i) - \mathbf{E}_{n,i}\| \quad \Leftarrow \textbf{Match Endpoint Data}$$

$$\text{subject to} \quad arm \leftarrow \text{UPDATEMODEL}(\mathbf{p})$$

$$\mathbf{x}(t) \leftarrow \text{SIMULATE}\left(\mathbf{u}(t),\, arm,\, [0,\, t_f],\, \mathbf{x}_0\right),$$

$$i = \lfloor \frac{t}{\mathrm{d}t} \rfloor \quad \forall t = 0\,..\,t_f.$$

And initial conditions $\mathbf{x}_0$ are found according to

$$\mathbf{x}_0 = \min_{\mathbf{x}} \quad \sum_{n=1}^{N} \|arm.\text{FORWARDKIN}_n(\mathbf{x}) - E_{n,0}\|$$

$$\text{subject to} \quad x_n^{min} \leq x_n \leq x_n^{max} \quad \forall n = 1\,..\,N.$$

2. Fluidic Elastomer Actuators:

   cylinder displacements $\mathbb{V}_s \rightarrow$ generalized torques $\boldsymbol{\tau}$

3. Soft Manipulator:

   generalized torques $\boldsymbol{\tau} \rightarrow$ manipulator states $\mathbf{x}$

Both the dynamic manipulator model and system identification algorithm were implemented using Drake Tedrake [2014], which is an open-source planning, control, and analysis toolbox for nonlinear dynamical systems.

Fluidic Drive Cylinders

Volumetric fluid changes to each agonistic pair of embedded channels within a soft arm segment are controlled by a pair of position-controlled fluidic drive cylinders, a device developed by the authors in Marchese et al. [2014c]. In this work we further develop and identify the device's dynamic model. Each pair is identified as an independent subsystem, and under the sagittal plane assumption N of these subsystems are required.

The input to each subsystem is u, a reference differential volumetric displacement to the position controlled cylinder pair and the output of each subsystem is $\mathbb{V}_s$, the differential volumetric displacement of the cylinders. One of two identical cylinders in the pair is driven at a time and pressurizes either half of the attached bending

segment.

To experimentally identify this subsystem we conduct several trials of the same experiment. The experiment consists of exciting the system with a reference wave $u(t)$ that is the summation of W sinusoidal waves with randomized phase delays ϕ, frequencies ω, and amplitudes a_w randomly sampled from a Bretschneider wave spectrum $S^+(\omega)$ with peak frequency ω_p of $2\,\pi$ and significant wave height ζ equal to twice the maximum displacement $\mathbb{V}_{max}$.

$$\begin{aligned} u(t) &= \sum_{i=1}^{W} a_{w_i} \sin\left(\omega_i\, t + \phi_i\right), \\ S^+(\omega) &= \frac{1.25}{4}\frac{\omega_p^4}{\omega^5}\zeta^2 \exp\left(-1.25\left(\frac{\omega_p}{\omega}\right)^4\right). \end{aligned}$$

To experimentally identify this subsystem we conduct several trials of the same experiment. The experiment consists of exciting the system with a reference wave $w(t)$. We fit a second order state space model to measured input-output data from one of five trials and then validated the model prediction against the remaining four trials. An example verification is shown in Figure 5-15. The identification and verification process was repeated for each of the 4 cylinder pairs used in later experiments.

Fluidic Elastomer Actuators

To identify the dynamics of the arm's soft actuators, we rely on the predicted cubic relationship between internal channel volume $\mathbb{V}_c$ and generalized torque τ as developed in Section 5.2.4. Also, the relationship between piston pressure p_s and channel volume $\mathbb{V}_c$ indicates a delay due to the impedance of the transmission line R_t

$$\begin{aligned} \dot{p}_s &= \frac{1}{C}\dot{\mathbb{V}}_s(t) - \frac{1}{C\,R_t}p_s + \frac{1}{C_a\,C\,R_t}\mathbb{V}_c, \\ \dot{\mathbb{V}}_c &= -\frac{1}{(C+C_a)\,R_t}\mathbb{V}_c + \frac{C_a}{(C+C_a)\,R_t}p_s \end{aligned}$$

where $\mathbb{V}_s(t)$ is considered the subsystem input and C_a and C is a first order approximation of the actuator's channel compliance and air respectively. Combining these

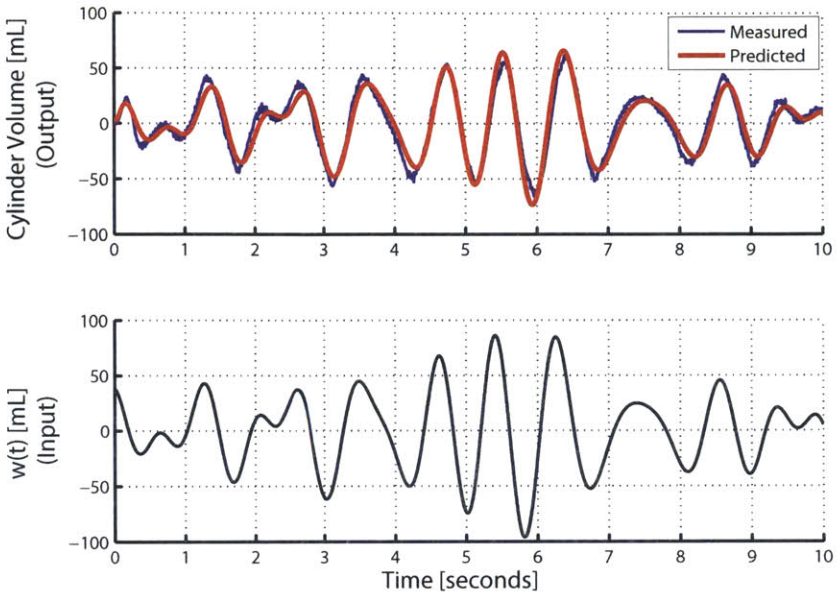


Figure 5-15: Example experimental identification of a position controlled fluidic drive cylinder subsystem. The identification process consists of exciting each independent subsystem with several randomized wave profiles and fitting and verifying a two state LTI black-box model to measured input-output data. Top: model predicted and measured output in blue and red respectively. Bottom: subsystem input.

effects we define a simplified identifiable model in the form

$$\tau(t) = c\,\mathbb{V}_s^3\,(t - t_d)\,.$$

The model constants $\mathbf{c}$ for each actuator pair and a single t_d are added to the main algorithm's parameter set $\mathbf{p}$ for identification, as the soft actuators are subject to dynamic fatigue and their performance is susceptible to change over time.

To validate this input output relationship, we again perform several trials of the aforementioned experiment, this time deriving actuator torque through a custom apparatus that measures the blocked tip force exerted by a segment fixed at its base. Figure 5-16 shows an example input-output identification for this subsystem.

Soft Manipulator

The manipulator's dynamic model is symbolically parameterized by N masses $\mathbf{m}$, stiffnesses $\mathbf{k}$, and damping coefficients $\mathbf{b}$. In the actuated case, there are also N

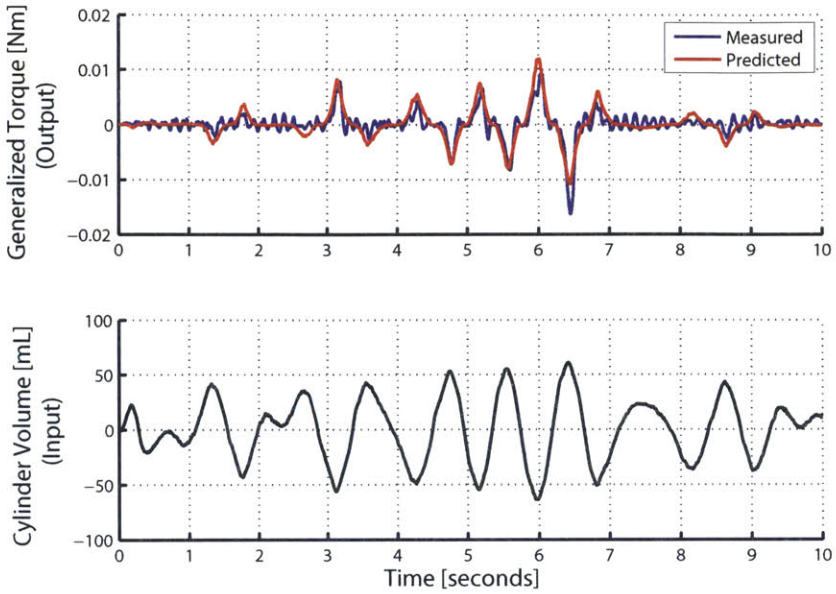


Figure 5-16: Example experimental identification of a soft actuator subsystem. Again the identification process consists of exciting each independent subsystem with several randomized wave profiles, but here we fit and verify a two parameter nonlinear model to measured input-output data. Top: model predicted and measured output in blue and red respectively. Bottom: subsystem input.

additional actuator parameters, $N-1$ unknown coefficients $\mathbf{c}$ and a single time delay t_d. To reduce the parameter set $\mathbf{p}$ from $4\,N$ parameters to $2\,N\,+\,2$ parameters we make the following observations: according to the expression for V_ϵ in Section 5.2.4 stiffness changes linearly with channel length L and therefore we can replace $\mathbf{k}$ with $\frac{L_i}{L_1}\,k$ where i is the segment index and k is a single unknown stiffness. Furthermore, we hypothesize the non-conservative components of force $\mathbf{b}\,\dot{\theta}$ are similar along the length of the arm, therefore we approximate the coefficients b_i to be equal $\forall\, i$.

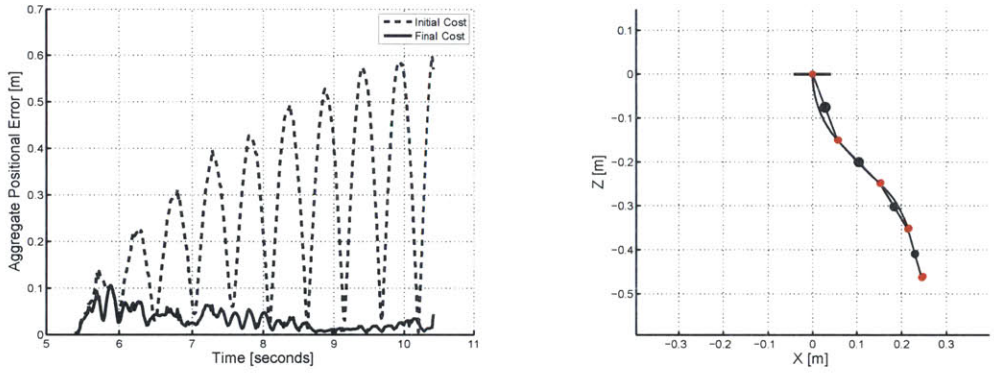
Evaluations

Measurements provided a coarse estimate of each parameter in $\mathbf{p}$. The identification algorithm, Algorithm 10, then freely adjusts these parameters. Initial mass and stiffness parameters were bound by a $\pm 38\%$ and $\pm 23\%$ change respectively, and the damping coefficient was adjusted on the interval $[1,\,5]\cdot 10^{-3}$. Multiple identifications were performed using random perturbations in the initial parameter set. Table 5.5

summarizes the results 4 trials and Figure 5-17a shows an example initial and final aggregate positional error between measured and simulated segment endpoints over time. When summed over time this is the algorithm's objective function.

Table 5.5: Identification of Passive Arm

	p						cost
	m_1	m_2	m_3	m_4	k	b	$\sum_i \sum_n$
			(kg)		(N m)	(N m s)	(m)
Initial	0.21	0.17	0.085	0.065	0.12	$2.0{\cdot}10^{-3}$	10
Final	0.190 $\pm$0.012	0.146 $\pm$0.001	0.090 $\pm$0.002	0.090 $\pm$0.003	0.108 $\pm$0.003	$4.2{\cdot}10^{-3}$ $\pm 0.1{\cdot}10^{-3}$	0.969 $\pm$0.004



(a) (b)

Figure 5-17: At (a) the positional error between measured and simulated endpoints summed over segments both for the initial parameter set (dashed line) and final parameter set (solid line) over time. At (b) the initial pose of the arm is shown at $\dot{\boldsymbol{\theta}} = 0$. Measured segment endpoints are shown in red and the modeled neutral axis of the arm is shown in black. The black circles indicate the approximated center of mass locations.

5.5 Capabilities

5.5.1 Confined Environment

The kinematics and extreme compliance of an entirely soft manipulator enable it to fit within and advance through confined environments. Consider the task of conform-

ing to an environment whose boundaries can be parameterized by curved cylinders, or tubes. For a manipulator operating in such an environment, we define the minimum confining space $\hat{V}$ as the volume of the smallest curved environment through which a given robot can pass. Figure 5-18 illustrates this concept. We assume a rigid robot link is cylindrical with diameter w_r and length L_r. Figure 5-18A depicts a rigid link advancing through the link's minimum confining space, parameterized by the environment's radius r_e and its curvature κ_e. The minimum confining space is dependent on both the link's geometry and the environment's curvature. A segment of a continuum manipulator with the same initial cylindrical dimensions as the rigid link is advanced through the segment's minimum confining space given the same environmental curvature in Figure 5-18B. Noticeably, the robot and environment radius only differ by ϵ to account for circumferential expansion under actuation. Lastly, in Figure 5-18C it is shown that the curvature of the environment is only limited by the curvature bounds of the continuum segment.

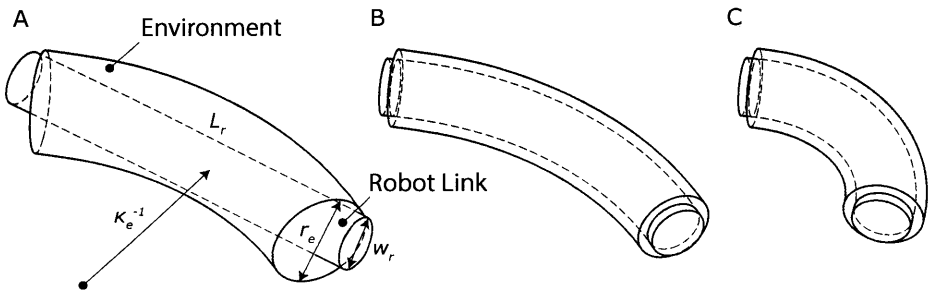


Figure 5-18: (**A**) A rigid robot link advancing through its minimum confining space at a given environmental curvature κ_e. (**B**) A soft robot link advancing through its minimum confining space given the same environmental curvature. (**C**) A soft robot link advancing through its minimum confining space given an increased environmental curvature.

More concretely, we can write $\hat{V}$ as a function of robot link geometry, L_r and w_r. By using the geometric construction outlined in Figure 5-19a we can express $\hat{V}_H$, or the minimum confining space for a rigid link. We start by finding the environment

radius r_e via the triangle composed of sides s_1, s_2, and s_3 and the law of cosines

$$s_1 = \frac{1}{\kappa_e} - r_e + w_r, \tag{5.21}$$

$$s_2 = \frac{L_r}{2}, \tag{5.22}$$

$$s_3 = \frac{1}{\kappa_e} + r_e, \tag{5.23}$$

$$r_e = \frac{1}{8} \frac{8\, w_r + 4\, w_r^2\, \kappa_e + L_r^2\, \kappa_e}{2 + w_r\, \kappa_e}, \tag{5.24}$$

$$\frac{\partial r_e}{\partial \kappa_e} = \frac{1}{4} \frac{{L_r}^2}{(2 + w_r \kappa_e)^2}. \tag{5.25}$$

An important observation is that under these assumptions the environment radius always increases with curvature. Next, the environment's bending angle θ_e can be found using the law of sines

$$\theta_e = 2\,\mathbb{R}\left(\arcsin\left(\frac{1}{2}\frac{L_r\,\kappa_e}{1 + r_e\,\kappa_e}\right)\right), \tag{5.26}$$

and using the volume of a torus scaled by the bending angle we can express $\hat{V}_H$

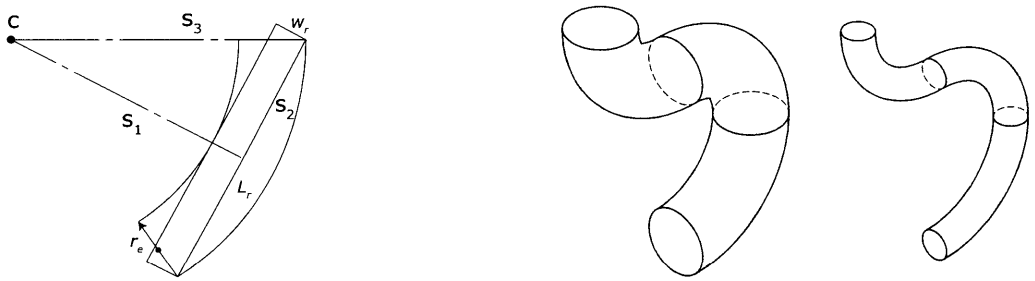
$$\hat{V}_H\,(w_r,\, L_r,\, \kappa_e) = 2\,\pi^2\,\frac{1}{\kappa_e}\,r_e^2\,\left(\frac{\theta_e}{2\,\pi}\right). \tag{5.27}$$

For a continuum manipulator the expression for the minimum confining space $\hat{V}_S$ is similar but with the difference that r_e is not necessarily a function of κ_e. In fact, in the ideal case r_e is constant and equal to $\frac{w_r}{2} + \epsilon$ where ϵ accounts for circumferential strain and/or clearance. Then,

$$\hat{V}_S\,(w_r,\, L_r,\, \kappa_e) = 2\,\pi^2\,\frac{1}{\kappa_e}\,\left(\frac{w_r}{2}\right)^2\,\left(\frac{\theta_e}{2\,\pi}\right). \tag{5.28}$$

This implies the minimum confining space for a continuum manipulator segment can be significantly smaller than a rigid manipulator link in this class of environments. To illustrate, Figure 5-20 shows the ratio $\frac{\hat{V}_H}{\hat{V}_S}$ as a function of κ_e at various manipulator link aspect ratios, $L_r : w_r$. Under the aforementioned assumptions, as

$\epsilon \to 0$ a continuum robot will always be able to advance through a smaller confining environment than a rigid linked robot when $\kappa_e > 0$. This volume difference increases with aspect ratio and environmental curvature. Figure 5-19b visualizes an example of this volume difference for a robot link aspect ratio of 6:1 and environment bend angle of $\frac{\pi}{2}$.



(a) (b)

Figure 5-19: (**a**) The triangle with sides s_1, s_2, and s_3 is constructed and used in determining the relationship between environmental radius r_e and environment curvature κ_e and robot link diameter w_r and length L_r. (**b**) A visualization of the volume difference between hard and soft minimum confining spaces for a robot link aspect ratio of 6:1 and environment bend angle of $\frac{\pi}{2}$.

Confined Environment Evaluations

For an entirely soft robot, where continuum kinematics and high compliance are combined, this result implies that such a manipulator can passively conform to the aforementioned class of confined environments. When the portion of the arm within the confined environment is passive $\epsilon \to 0$ and when the environment is composed of PCC cylinders, the above assumptions are met. In order to experimentally validate the soft fluidic elastomer manipulator's ability to advance through a confined environment we autonomously advance a passive manipulator with an actuated base through a pipe. Then, we actively perform a task once the manipulator is fully inserted. The experiment is shown in Figure 5-21 as well as Extension 1. The goal of the experiment is to repeatably advance the soft arm (Fig. 5-21b) through a tightly bending pipe (Fig. 5-21d) using only a constant linear and rotational motion of the

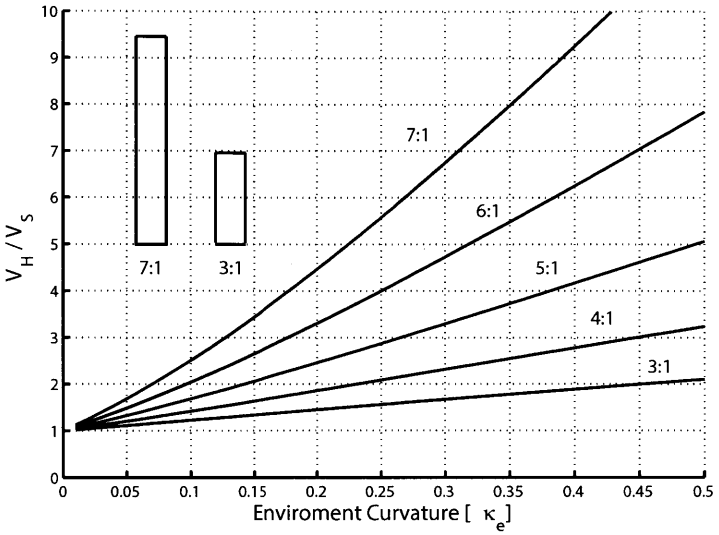


Figure 5-20: The ratio of a hard to soft robot's minimum confining volume $\frac{\hat{V_H}}{\hat{V_S}}$ as a function of the curvature of the surrounding environment κ_e. The robot's minimum confining volume is the volume of the smallest curved environment through which a given robot can pass. The ratio is expressed for various robot link aspect ratios $L_r : w_r$. Both the hard and soft links have the same initial geometry.

manipulator's base. This insertion motion is provided by a motorized rotary and linear stage (Fig. 5-21h and i, respectively). A camera located at the arm's end effector (Fig. 5-21c) can be used to localize the pipe entrance, inspect the pipe, and view objects of interest (Fig. 5-21f) once the arm has exited the pipe. Video from this end effector camera is shown in the supplementary video submission. Because the arm's fluidic transmission system can deliver power independent of the arm's configuration, a tether (Fig. 5-21e) allows the user to actuate the most distal segment while the proximal portion of the arm has passively conformed to the pipe environment.

Ten consecutive trials of the confined environment experiment were performed, and the manipulator successfully advanced through the pipe every trial and was able to perform an actuated task after insertion every trial. The confined environment consists of three 5 cm insider diameter curved pipe segments, motivated by the fact that the manipulator's maximum outside diameter is 5 cm with a 2 degree taper from tip to base. In all tests the manipulator's base was moved at a constant linear

velocity and rotational velocity of 12 mm/sec and 0.11 rev/sec, respectively. The base was moved until a fixed insertion depth of 40 cm was reached. A thin layer of Talc powder was applied to the surface of the silicone arm to reduce the coefficient of friction between the robot's surface and plastic pipe. A load cell (Fig. 5-21g) was used to measure insertion force as a function of time. Figure 5-22 shows this required resulting insertion force and insertion depth as a function of time for all ten trials as well as the applied base linear and rotational velocities as a function of time. Across the ten trials, mean peak insertion force was 11.8 $\pm$ 1.2 Newtons.

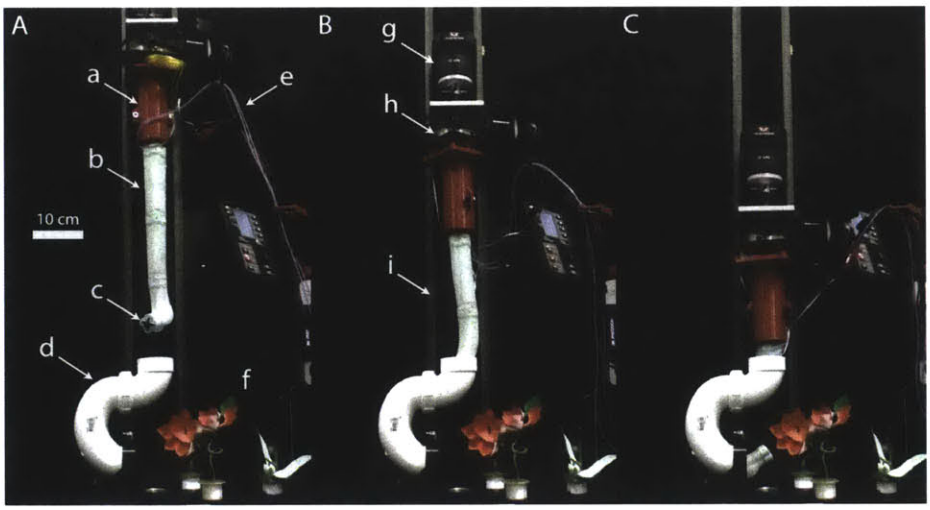


Figure 5-21: Pipe insertion experiment. The experimental setup consists of serval components: a (a) plastic sheath, (b) soft manipulator prototype of maximum O.D. 5 cm, (c) end effector camera, (d) confining pipe of I.D. 5 cm, (e) power tether, (f) objects of interest, (g) load cell, (h) automated rotary stage, and (i) automated linear stage. (**A**) Depiction of the manipulator prior to pipe insertion. Here, the distal most arm segment is actuated to simulate localization of the pipe entrance. (**B**) Depiction of the passive arm being autonomously inserted through the pipe by means of actuation at its base. (**C**) Depiction of the manipulator fully advanced through the pipe and performing the actuated task of inspecting flowers with its end effector camera.

5.5.2 Shape Fitting

The continuum kinematics of a soft robot manipulator enable a high degree of dexterity. Specifically, in an environment where a collision-free path is parameterized by

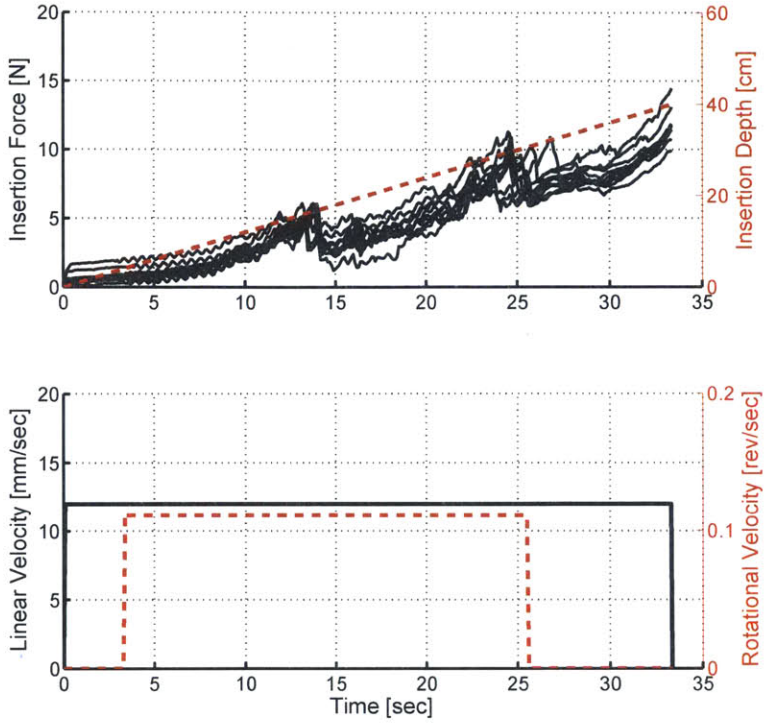


Figure 5-22: Results of pipe insertion experiment. At the top, measured insertion force as well as insertion depth is reported for all 10 experimental trials. Generally, insertion force increased with insertion depth. At the bottom, the commanded linear and rotational velocity of the manipulator's base are shown during the insertion phase of the experiment. The insertion phase has positional end criteria, that is linear velocity is constant at 12 mm/sec until a fixed insertion depth of 40 centimeters is reached upon which motion is stopped. Note that base rotation begins several seconds after linear motion, approximately when the manipulator enters the pipe, and base rotation has a fixed velocity of 0.11 rev/sec until 2.43 revolutions are reached.

a curved shape, a continuum manipulator can generally fit the curvature of the path better than a rigid link manipulator with discrete joints. Consider a path in planar cartesian coordinates $P\left(x(t),\, y(t)\right)$ parameterized by time t or equivalently arc length s. The curvature at t can be expressed as:

$$\kappa_p(t) = \frac{\frac{\partial x}{\partial t}\frac{\partial^2 y}{\partial t^2} - \frac{\partial y}{\partial t}\frac{\partial^2 x}{\partial t^2}}{\left(\frac{\partial x}{\partial t}^2 + \frac{\partial y}{\partial t}^2\right)^{3/2}} \tag{5.29}$$

Because the first order Taylor series expansion $TS_1(\kappa_p(t))$ at t is constant, a soft continuum robot has the capacity to locally approximate the path's curvature to the first order by adjusting the curvature of its neutral axis over a continuum. In the case of no external forces, each of the soft continuum N segments can be fit to approximate the local curvature of the path under actuation:

$$\kappa_i = \frac{1}{L_i} \int_{-\frac{L_i}{2}}^{\frac{L_i}{2}} TS_1(\kappa_p(s))\, \mathrm{d}s \quad \forall i = 1..N, \tag{5.30}$$

Where L is the soft segment length. There are several scenarios that cause error between the continuum robot's curvature and the path's curvature. These scenarios are detailed in Figure 5-23. Figure 5-23A depicts a scenario where the path $P(s)$, shown by the dotted line, has locally constant curvature on the interval $s = [a, b]$; however, the arm segment's neutral axis of approximately equal length, shown as the solid line, has a maximum curvature κ^{max} less than $\kappa_p(s)$ and this produces an error in shape fitting. Figure 5-23B depicts a scenario where the path's local curvature on the interval $s = [a, b]$ is non-constant and this too causes an error in the goodness of fit between the path and arm segment of approximately equal length. Lastly, Figure 5-23C illustrates the scenario where arm segment length L is greater than the total path length. Although a quality shape fit may be possible, this results in positional error either normal to the path (left) or tangent to the path (right).

In the case of a rigid link manipulator with discrete joints, curvature along the neutral axis is either 0 or $\pm\infty$, as by definition this structure is not a continuum. In this case the best way to approximate the curvature of a path is to increase the number of links and decrease the link length relative to the path length.

Shape Fitting Algorithm

To incrementally advance a soft continuum manipulator along a curved path we propose the path following strategy described in Algorithm 11. The fundamental idea of this path following approach is that each segment is sequentially advanced onto the path. For each segment on the path, two points along the segment's neutral axis,

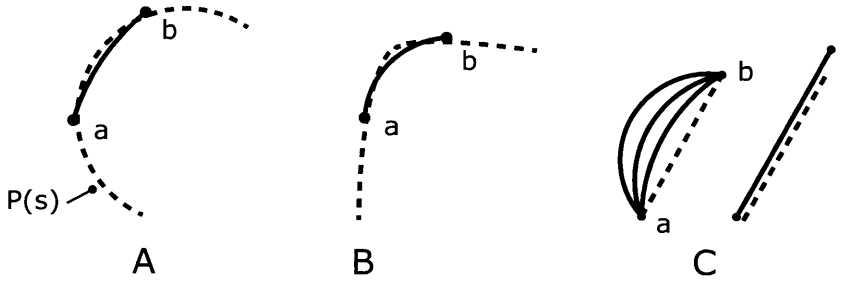


Figure 5-23: Several common scenarios that cause an error between a continuum robot's curvature and a path's curvature. (**A**) The maximum curvature of the robot is less than the local curvature of the path on the interval $s = [a, b]$. (**B**) The local curvature of the path is non-constant and the robot's deformation is parameterized by PCC arcs. (**C**) Description of a scenario where the robot's segment length is greater than the total path length.

the mid and endpoint, are fit to the path by choosing the segment's curvature κ to minimize the Euclidean norm between the two points and their projection onto the path. By adding two points for each segment advanced onto the path plus the origin pose constraint, this approach effectively samples the path's curvature $\kappa_p(s)$ as two points and an initial orientation uniquely define a constant curvature arc. This approach assumes the manipulator has the capacity to position and orient a point along its neutral axis that is a distance s from the manipulator's tip, to the path's origin.

Algorithm 11: Path Fitting Algorithm

for $t = 0 : dt : t_f$ **do**

1 $s \leftarrow t \frac{\Delta s}{\mathrm{dt}}$.

2 N, $\mathbf{L} \leftarrow$ Determine the number of segments N that can be projected onto P and their corresponding lengths $\mathbf{L}$ as a function of s and $\mathbf{L}_0$.

3 $\boldsymbol{\kappa}^* \leftarrow$

$$\begin{cases} \min\limits_{\boldsymbol{\kappa}} & \sum\limits_{i=1}^{N} \left\| \texttt{forwardKin}(\boldsymbol{\kappa}, \mathbf{L}, i, L_i) - P\left(\sum\limits_{j=1}^{i} L_j\right) \right\| + \sum\limits_{i=1}^{N} \left\| \texttt{forwardKin}\left(\boldsymbol{\kappa}, \mathbf{L}, i, \frac{L_i}{2}\right) - P\left(\sum\limits_{j=1}^{i} L_j - \right. \right. \\ \text{subject to} & \kappa_i^{min} \leq \kappa_i \leq \kappa_i^{max} \quad \forall i = 1..N. \end{cases}$$

4 Servo continuum segments to $\boldsymbol{\kappa}^*$

end

Shape Fitting Evaluations

Simulations

To demonstrate the difference between fitting a soft continuum and rigid-linked robot to a curved path, we perform several simulations of advancing both a soft and rigid manipulator backbone through a curved path. The results of the simulations are shown in Figure 5-24. The first column describes a given collision free path through an environment. Columns 2 through 4 show the backbone of a soft robot (blue) and hard robot (red) advancing along and conforming to the path with different link lengths. Their neutral axis is overlaid at every time step. Both the soft and hard manipulators have the same link length, and this is varied from 25% of the total path length in column 2 to 41% of path length in column 4. In general, as the ratio of link length to path length increases the number of error scenarios outlined in Figure 5-23 increases, and the quality of fit decreases; However, in these examples the soft continuum robot approximated the curved path better than the rigid link robot in all cases.

Experiments

The purpose of the experiments performed in this section is to evaluate the aggregate soft manipulation system's ability to autonomously control the configuration of the soft arm. That is, in real-time with the closed-loop controller developed in Section 5.4, we drive the arm's measured curvature representation $\boldsymbol{\kappa}_{\text{measured}}$ to a desired reference shape parameterized by $\boldsymbol{\kappa}_{\text{target}}$. We demonstrate that the arm is able to conform to a user specified configuration with a sagittal plane index by γ subject to the self-loading effects of gravity. To the best of our knowledge, this is the first demonstration of closed-loop curvature control for a fluidic elastomer robot subject to gravity.

The goal of the experiments detailed in Figures 5-25 and 5-26 as well as Extension 1 is for the soft manipulator to autonomously transition from a neutral, hanging state (i.e. $\theta_0 = \frac{3\pi}{2}$ and $\boldsymbol{\kappa}_0 \approx \mathbf{0}$) to either an S shape (see Fig. 5-25) or a C shape (see Fig. 5-26). Ten and twelve trials of the S and C shape experiments were performed,

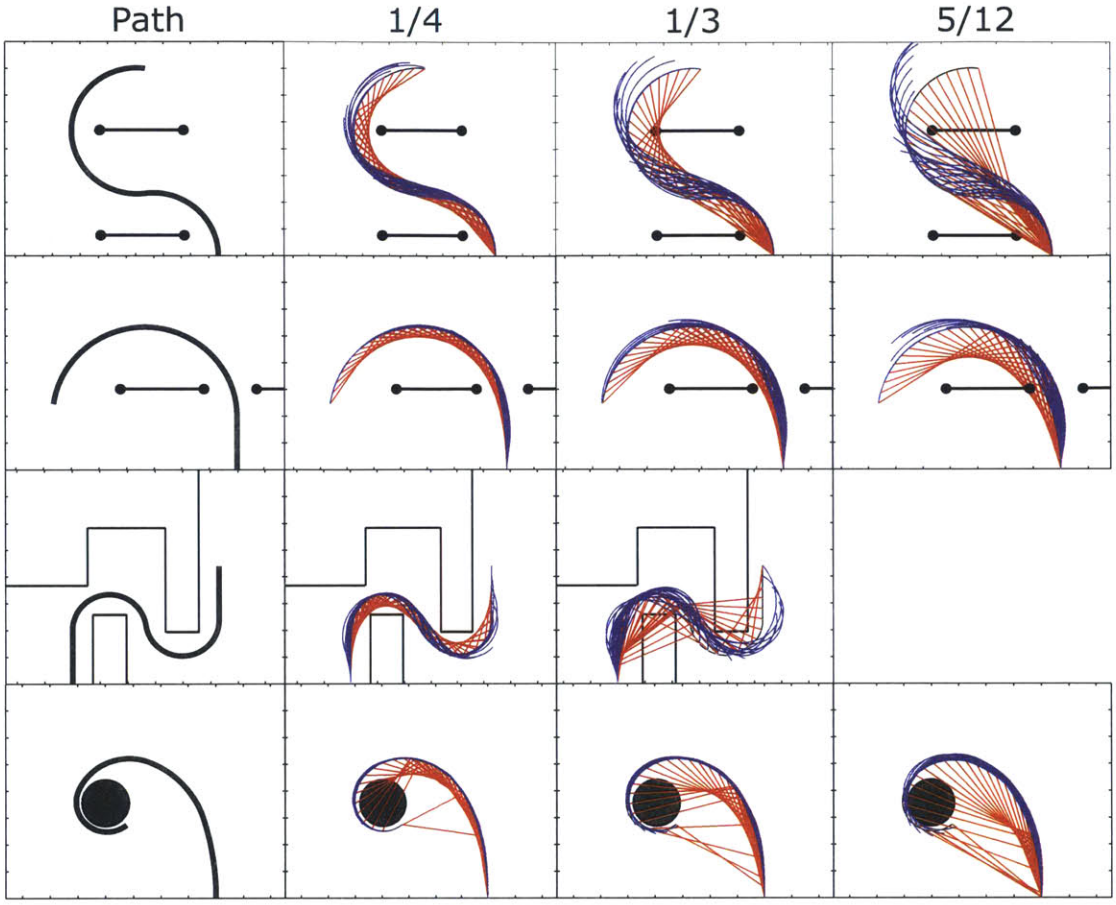


Figure 5-24: Shape fitting simulations. In the first column, a collision free curved path through several environments is given. In the second, third, and fourth columns a soft robot with a PCC backbone [Webster and Jones, 2010], shown in blue, and a hard robot with fixed joints, shown in red, are sequentially advanced along and fit to the path in the first column using Algorithm 11. Manipulator link length is varied from 25% of total path length in column 2 to 41% of path length in column 4.

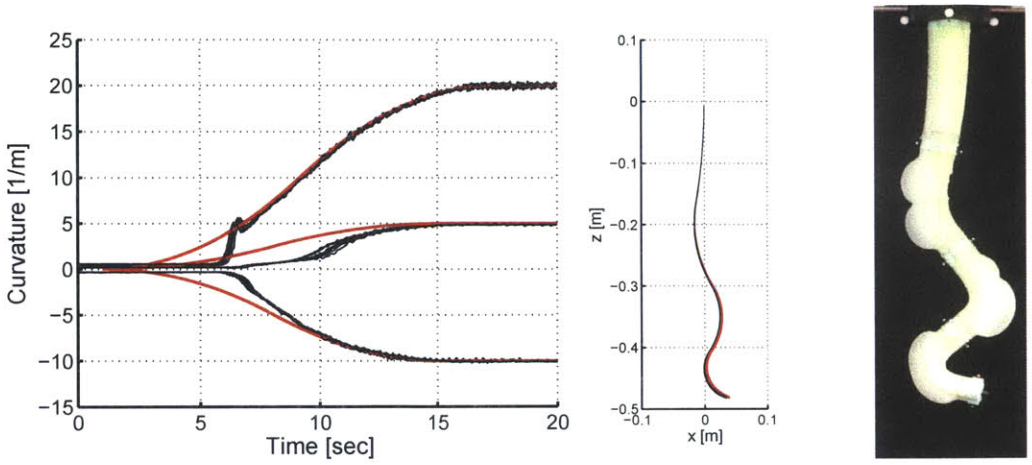
respectively, and the results for all trials are shown in Figures 5-25 and 5-26, respectively. In the S-shaped configuration experiment, segments were less prone to twisting about their neutral axis (an unactuated DOF), and accordingly it was sufficient to use a single pair of agonist actuators collocated with the sagittal plane to achieve the required bending (schematically represented Fig. 5-3C IV). However, during the C-shaped configuration experiment, segments were more prone to twisting about their neutral axis. To counter this phenomenon, two pairs of agonist actuators were collocated with the sagittal plane, i.e. each pair was offset 45° from the sagittal plane

(schematically represented Fig. 5-3C III). This actuation strategy provides equal and opposite components of force that stabilize the segment's bending along the sagittal plane.

Panel (a) of Figures 5-25 and 5-26 depicts the reference curvature trajectory $\boldsymbol{\kappa}_{\text{target}}(t)$ in red and measured curvature trajectory $\boldsymbol{\kappa}_{\text{measured}}$ in black as a function of time, as determined by Algorithm 9 developed in Section 5.4. The initial lag between reference and measured curvatures can be explained by the compressibility of the transmission fluid, in this case air. The residual volume in the fluidic drive cylinder must be compressed prior to embedded channel deformation. This phenomenon can be corrected for using a feed-forward control component, but in this work we do not correct for this lag. Note that, three out of the four segments are actively controlled, and the base segment is left passive in order to dampen oscillations arising from distal segment movement. Panel (b) shows the reference (red) and measured (black) manipulator configuration at $t = 20$. Panel (c) contains photographs of the soft fluidic elastomer arm prototype during the experiments. Mean $\pm$ one standard deviation steady state curvature errors across all trials for the C shape were 0.05 $\pm$ 0.09, -0.04 $\pm$ 0.07, and 0.02 $\pm$ 0.09 m^{-1}, respectively for segments 2 through 4. And steady state curvature errors across all trials for the S shape were -0.11 $\pm$ 0.03, -0.02 $\pm$ 0.07, and 0.03 $\pm$ 0.12 m^{-1}, respectively.

5.5.3 Positioning

Additionally the soft arm is capable of Cartesian positioning in three dimensions. Similar to the approach in Marchese et al. [2014b], we pose the inverse kinematics problem (i.e., the transformation from the manipulator's task space to its arc space) as a constrained nonlinear optimization, allowing phenomena arising from the arm's softness to be expressed as numerical constraints. More specifically, we can calculate γ as well as find a locally-optimal reference configuration $\boldsymbol{\kappa}^*$ that position the arm's end-effector at a goal point within a statically reachable envelope. For more information on finding points within a statically reachable envelope please refer to Marchese et al. [2015b]. Once a suitable reference configuration is determined, we can again use the



(a) (b) (c)

Figure 5-25: Experimental evaluations of closed-loop real-time configuration control. In 10 consecutive trials, the aggregate manipulation system is provided an S shaped goal configuration to reach, shown in red in (**b**). The goal configuration state is parameterized by three target curvatures $\boldsymbol{\kappa}_f = [5, -10, 20]$ for segments 2-4 respectively. Algorithm 9 plans a reference configuration trajectory $\boldsymbol{\kappa}(t)$, shown in red in (**a**), and drives the arm along this configuration trajectory by measuring the arm's current configuration, shown in black in (**a**), and adjusting the volumetric displacement of the fluidic drive cylinder array. (**c**) The prototype fluidic elastomer manipulator in the target configuration.

procedures outline in Algorithm 9 to both plan an execute a curvature trajectory trajectory $\dot{\boldsymbol{\kappa}}(t)$.

Positioning Algorithm

First, the procedure `moveToObject` in Algorithm 12 localizes an object's center in $\mathbb{R}^3$ using the external camera system, a process represented by `localizeObject()`. To obtain a target end-effector position, the object's position is translated a predetermined offset based on the object's geometry using the transformation $\mathbf{T}^{\text{object}}_{\text{offset}}$.

The strategy for moving the arm to this goal point is represented in the procedure `moveToPoint()`. First, the arm is rotated into a sagittal plane defined by the angle γ by means of the base rotary stage. The vertical plane is coincident with both the base frame's origin and the goal point. The end-effector is then iteratively moved in this plane towards the goal point by finding locally-optimal reference con-

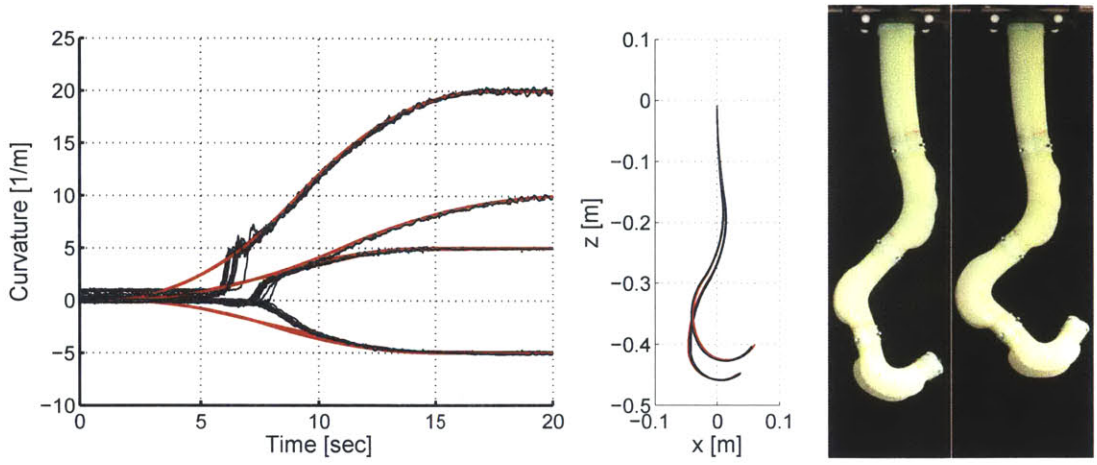


Figure 5-26: Experimental evaluations of closed-loop real-time configuration control. In 10 consecutive trials, the aggregate manipulation system is provided a C shaped goal configuration to reach, shown in red in (**b**). The goal configuration state is parameterized by three target curvatures and 10 trials used the goal $\boldsymbol{\kappa}_f = [-5, 5, 20]$ and 2 trials used the goal $\boldsymbol{\kappa}_f = [-5, 10, 20]$ for segments 2-4 respectively. Algorithm 9 plans a reference configuration trajectory $\boldsymbol{\kappa}(t)$, shown in red in (**a**), and drives the arm along this configuration trajectory by measuring the arm's current configuration, shown in black in (**a**), and adjusting the volumetric displacement of the fluidic drive cylinder array. (**c**) The prototype fluidic elastomer manipulator in the target configuration.

figurations. This iterative search for manipulator configurations starts, as previously mentioned, by taking a set of real-time endpoint measurements $\mathbf{E}_n^{\text{base}}$, projecting them into the sagittal plane using the simple transformation $\mathbf{T}_\gamma^{\text{base}}(\gamma)$ to arrive at the set $\mathbf{E}_n^\gamma$. This data is passed to the procedure `singleSegInvKin()` in order to uniquely fit a PCC model to the data, providing a measured manipulator configuration representation $\boldsymbol{\kappa}_{\text{meas}}$. This provides the current manipulator configuration from which the trajectory $\dot{\boldsymbol{\kappa}}(t)$ is started. To determine the trajectory's end configuration $\boldsymbol{\kappa}_f$, the optimization-base inverse kinematics procedure, represented by `inverseKin()`, is invoked. This process finds a weighted minimal-strain solution subject to (i) the end-effector being positioned at the goal point projected into the sagittal plane, (ii) the state of the passive base segment, κ_{meas_1}, appropriately limiting the configuration's DOF, (iii) end-effector orientation being constrained, in combination with curvature limitations, in order to prevent configurations where the center of mass of segment

n is above connector piece $n-1$ for static stability purposes, and lastly (iv) empirically determined curvature limits not being exceeded. After defining $\boldsymbol{\kappa}_{\text{meas}}$ and $\boldsymbol{\kappa}_f$ the previously defined procedure `generateTraj()` is used to generate a realizable configuration trajectory and the procedure `driveSegments()` is used to move the manipulator along this trajectory. The process is repeated until the end-effector is within ϵ of the goal location.

Algorithm 12: Algorithm for 3D positioning

Procedure `moveToObject()`

$[\, x_{\text{object}},\, y_{\text{object}},\, z_{\text{object}} \,] \leftarrow$ `localizeObject()`.

$[\, x_{\text{offset}},\, y_{\text{offset}},\, z_{\text{offset}} \,]^{\text{T}} \leftarrow \mathbf{T}^{\text{object}}_{\text{offset}} [\, x_{\text{object}},\, y_{\text{object}},\, z_{\text{object}} \,]^{\text{T}}$.

`moveToPoint`$(\, x_{\text{object}},\, y_{\text{object}},\, z_{\text{object}} \,)$.

return

Procedure `moveToPoint`$(x^{base},\, y^{base},\, z^{base})$

$\gamma \leftarrow \arctan\left(\frac{y^{\text{base}}}{x^{\text{base}}}\right)$.

$[\, x^{\gamma},\, y^{\gamma},\, z^{\gamma} \,]^{\text{T}} \leftarrow \left[\mathbf{T}^{\text{base}}_{\gamma}(\gamma)\right]^{-1} [\, x^{\text{base}},\, y^{\text{base}},\, z^{\text{base}} \,]^{\text{T}}$.

Move rotary stage to γ.

while $\left| \textit{forwardKin}(\, \boldsymbol{\kappa}_{meas},\, \mathbf{L}_{meas},\, N,\, L_{meas_N} \,) - \begin{bmatrix} x^{\gamma} \\ z^{\gamma} \end{bmatrix} \right| \geq \epsilon$ **do**

Wait until a new sensor measurement $\mathbf{E}^{\text{base}}$ is available.

$\mathbf{E}^{\gamma}_{n} \leftarrow \left[\mathbf{T}^{\text{base}}_{\gamma}(\gamma)\right]^{-1} \mathbf{E}^{\text{base}}_{n} \quad \forall n = 1 \,..\, N.$

$\boldsymbol{\kappa}_{\text{meas}},\, \mathbf{L}_{\text{meas}} \leftarrow$ `singleSegInvKin`$(\mathbf{E}^{\gamma})$ [Marchese et al., 2014c].

$\boldsymbol{\kappa}_f \leftarrow$ `inverseKin`$(x^{\gamma},\, z^{\gamma},\, \kappa_{\text{meas}_1})$.

$\dot{\boldsymbol{\kappa}}(t) \leftarrow$`generateTraj`$(\, \boldsymbol{\kappa}_{\text{meas}},\, \boldsymbol{\kappa}_f,\, \dot{\boldsymbol{\kappa}}^{\max},\, \ddot{\boldsymbol{\kappa}}^{\max} \,)$.

`driveSegments`$(\, \dot{\boldsymbol{\kappa}}(t),\, \gamma)$

end

return

Procedure `inverseKin`$(x,\, y,\, \kappa_{meas_1})$

$\mathbf{R}$ a weighting vector,

$\theta^{\min}$ and $\theta^{\max}$ constraints on end-effector approach angle.

$$\boldsymbol{\kappa}^{*} \leftarrow \min_{\boldsymbol{\kappa}} \quad \sum_{i=1}^{N} R_i\, \kappa_i.$$

subject to $\quad \mathbf{f} \leftarrow$ `forwardKin`$(\boldsymbol{\kappa},\, \mathbf{L}_{\text{meas}},\, N,\, L_{\text{meas}_N})$.

$$\begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \mathbf{0}.$$

$$\kappa_{\text{meas}_1} - \kappa_1 = 0.$$

$$\theta^{\min} \leq f_3 \leq \theta^{\max}.$$

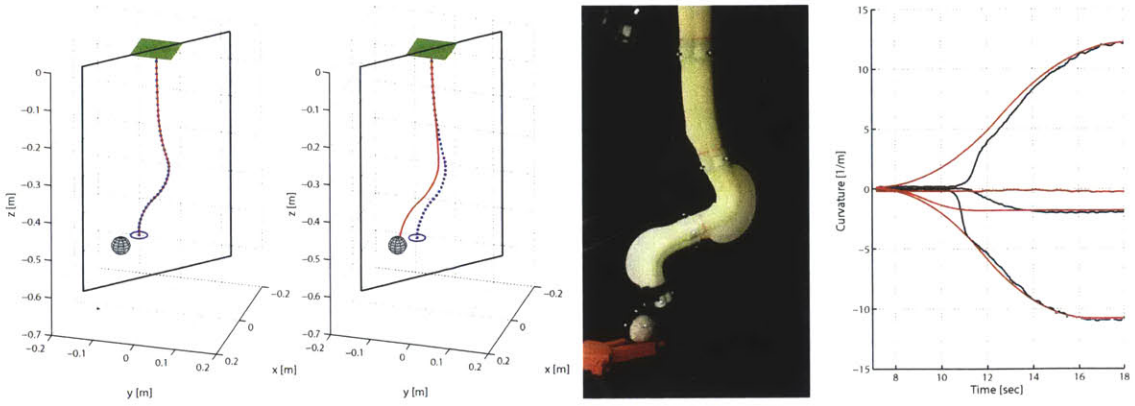
$$\kappa^{\min}_{n} \leq \kappa_n \leq \kappa^{\max}_{n} \quad \forall n = 1 \,..\, N.$$

return $\boldsymbol{\kappa}^{*}$

Positioning Evaluations

In order to experimentally evaluate the three dimensional positioning capabilities of the soft arm we ran Algorithm 12 on the manipulation system. The goal of the experiment was for the soft arm to move its end-effector into contact with a 4cm diameter table tennis ball placed within the arm's statically reachable envelope. The purpose of the experiment was to evaluate the arm's positioning capabilities, not to robustly pick-up the ball. Accordingly, velcro was simply fixed to both the ball and end-effector to perform the pick-up.

In 10 consecutive attempts, the manipulation system was able to successfully position the arm's end effector to make contact with the ball. The object offset translation $\mathbf{T}_{\text{offset}}^{\text{object}}$ underwent minor adjustments throughout the tests to continually improve the contact between the velcro pieces. Figure 5-27a-d as well as Extension 1 detail one of these trials. Figure 5-27a shows the first reference configuration $\boldsymbol{\kappa}_f$ planned by Algorithm 12 in red as well as the configuration representation $\boldsymbol{\kappa}_{\text{meas}}$ fit to measured endpoint data at $\mathbf{t}_f$ in blue. The discrepancy between $\boldsymbol{\kappa}_f$ and $\boldsymbol{\kappa}_{\text{meas}}$ is almost imperceptible. At this point the end-effector and object made contact (shown in Fig. 5-27c). Figure 5-27b shows the second reference configuration planned by the algorithm, considering the passive bending of the first segment, again shown in red. Noticeably, the manipulator's measured configuration representation, again shown in blue, deviates significantly from this plan as the manipulator's end-effector is already in contact with the object inhibiting the manipulator's motion in the desired direction. Lastly, Figure 9d shows intermediate target curvatures $\boldsymbol{\kappa}_{\text{target}}$ (red) as well as intermediate measured configuration representations $\boldsymbol{\kappa}_{\text{meas}}$ (black) from t_0 to t_f for each segment. The compressibility of the transmission medium as well as the relatively large residual volume of the pistons explain the several second delay in tracking the reference configuration. We have recently shown that by using techniques such a pre-pressurization and tandem inflation this performance characteristic can be improved [Katzschmann et al., 2015].



(a) (b) (c) (d)

Figure 5-27: Experimental evaluations of three-dimensional end-effector positioning with the soft manipulator prototype. (**a**) Algorithm 12 plans a locally-optimal, minimal-strain reference configuration (red) to which the arm's measured configuration representation is driven (blue). Here, contact is made between the end-effector and object. (**b**) The algorithm's next plan is shown that would theoretically bring the manipulator's end-effector even closer to the intended target but physical contact with the object prevents the new target configuration from being reached. (**c**) A photograph of the manipulator's end effector making contact with the object of interest. (**d**) The first intermediate reference configuration trajectory $\boldsymbol{\kappa}_{\text{target}}(t)$ is shown in red and the measured configuration representation of the manipulator is shown in black $\boldsymbol{\kappa}_{\text{meas}}(t)$.

5.5.4 Grabbing

A primitive enabled by the developments in Sections 5.2.4 and 5.4.2 is grabbing. Grabbing is defined as bringing the arm's end effector to a user specified, statically unreachable goal point with near zero tip velocity. Grabbing is an advantageous strategy to employ during manipulation as it enables the soft arm to reach areas that are statically unreachable due to gravity.

There are several major challenges that arise when trying to autonomously move the soft manipulator. First, we leave the top segment unactuated to accommodate external loads acting on the distal segments. Second, the system is tightly constrained by generalized torque limits. That is, the low operating pressures of the fluidic actuators in combination with their very low durometer rubber composition equate to constraints on input forces. To exemplify this problem consider the following search for feasible solutions that statically position the arm's end effector to a goal point in

task space

$$
\begin{aligned}
&\text{find} \quad \text{s.t.} \quad \mathbf{C} - \mathbf{B}\boldsymbol{\tau} = \mathbf{0}, \\
&\boldsymbol{\tau},\, \boldsymbol{\theta} \qquad \| arm.\text{FORWARDKIN}_N(\boldsymbol{\theta}) - \text{Goal} \| - \epsilon = 0, \\
&\qquad\qquad \tau_m^{\min} \leq \tau_m \leq \tau_m^{\max} \quad \forall\, m = 1\,..\,M, \\
&\qquad\qquad \theta_n^{\min} \leq \theta_n \leq \theta_n^{\max} \quad \text{and} \quad \dot{\theta}_n = 0 \quad \forall\, n = 1\,..\,N.
\end{aligned}
$$

By looking for solutions to goal points in the vicinity of the end effector, we quickly bring to light the limitations of a purely kinematic approach to motion planning for this class of manipulators subject to gravity. Table 5.7 depicts feasible static solutions in green for identified arms under estimated torque limits.

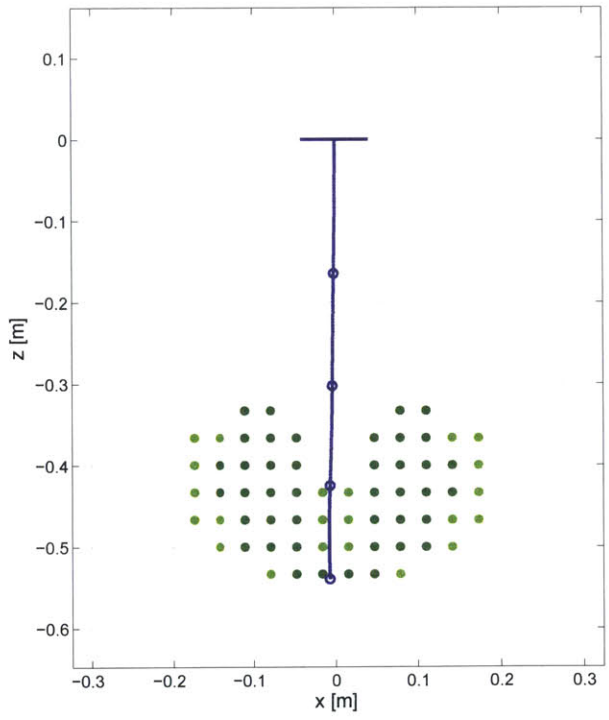


Figure 5-28: Feasible static solutions for an identified soft manipulator under estimated torque limits. The solid blue lines represent the initial state of the manipulator. Dark and light green circles indicate points that were statically reachable under the torque limits of $|\boldsymbol{\tau}| = [0.13,\, 0.13,\, 0.13,\, 0.13]^{\mathrm{T}}$ and $|\boldsymbol{\tau}| = [0,\, 0.12,\, 0.13,\, 0.18]^{\mathrm{T}}$ respectively.

Grabbing Algorithms

We develop an algorithm, Algorithm 13, that can plan and execute a grab maneuver. The algorithm uses trajectory optimization to both plan a locally-optimal policy in generalized torque space as well as to determine an optimal input trajectory to the aggregate manipulation system to realize this policy. The trajectory optimizations were implemented using Drake [Tedrake, 2014]. Algorithm 13 can be interpreted as an iterative learning control, which after a couple grabbing attempts is able to successfully perform the desired maneuver.

Algorithm 13: Iterative Learning Control

1 $arm_0 \leftarrow$ SYSTEMID $(\mathbf{x}_m(t), \mathbf{u}(t))$.
2 $i = 0$.
3 **while** *Goal is not met* **do**
4 $\quad \mathbf{\Pi} \leftarrow$ TRAJOPT $(arm_i$, Goal$)$.
5 $\quad \mathbf{u}(t) \leftarrow$ INVERTACTUATORS $(arm_i, \mathbf{\Pi})$.
6 $\quad \mathbf{x}_m(t) \leftarrow$ RUNPOLICY $(\mathbf{u}(t))$.
7 $\quad arm_{i+1} \leftarrow$ SYSTEMID $(arm_i, \mathbf{x}_m(t), \mathbf{u}(t))$.
8 $\quad i++$.
9 **end**

Here, $\mathbf{x}_m(t)$ represents a measured state trajectory of the soft manipulator over the time interval $t = [0, t_f]$, $\mathbf{u}(t)$ is the reference input trajectory to the manipulation system, and $\mathbf{\Pi}$ represents a matrix of locally-optimal generalized torque and state trajectories. The function SYSTEMID describes the identification process in Section 5.4.2, the functions TRAJOPT and INVERTACTUATORS embody processes described in Subsections 5.5.4 and 5.5.4, and RUNPOLICY represents executing the reference input policy $\mathbf{u}(t)$ on the physical manipulation system.

Trajectory Optimization

We use a direct collocation approach to trajectory optimization von Stryk [1993] in line 4 of Algorithm 13. In short, this is a model-based open-loop policy search that finds a feasible input trajectory that moves the manipulator from an initial state to a goal state given both input and state constraints. The policy $\mathbf{\Pi}$ can generally be

a function of both state and time, but in this case is parameterized by $M \times \frac{t_f}{\mathrm{d}t}$ free parameters α where M is the number of inputs and $\mathrm{d}t$ is a discrete time step

$$\begin{aligned} \boldsymbol{\Pi}_\alpha(\mathbf{x}, t) &= \alpha_{m,i} \quad \forall m = 1..M, \\ i &= \lfloor \frac{t}{\mathrm{d}t} \rfloor \quad \forall t = 0..t_f. \end{aligned}$$

In the case of the soft manipulator each α is a generalized torque τ for each actuated segment augmented with the manipulator's state vector at each time step

$$\boldsymbol{\Pi}_\alpha = \begin{bmatrix} \boldsymbol{\tau}_0 & \boldsymbol{\tau}_1 & \boldsymbol{\tau}_2 & \dots & \boldsymbol{\tau}_{\frac{t_f}{\mathrm{d}t}} \\ \mathbf{x}_0 & \mathbf{x}_1 & \mathbf{x}_2 & \dots & \mathbf{x}_{\frac{t_f}{\mathrm{d}t}} \end{bmatrix}.$$

The following trajectory optimization is performed to identify a locally-optimal policy $\boldsymbol{\Pi}_\alpha^*$

$$\begin{aligned} \boldsymbol{\Pi}_\alpha^* = \min_\alpha \quad & \sum_i g(\mathbf{x}_i, \boldsymbol{\tau}_i) \quad \Leftarrow \textbf{Objective Function} \\ \text{subject to} \quad & 0 = \mathbf{x}_i - f(\mathbf{x}_{i-1}, \boldsymbol{\tau}_{i-1})\,\mathrm{d}t - \mathbf{x}_0 \quad \forall i = 1..\frac{t_f}{\mathrm{d}t}, \\ & 0 = h(\mathbf{x}_{\frac{t_f}{\mathrm{d}t}}), \quad \Leftarrow \textbf{Enforce Tip Motion} \\ & \tau_m^{\min} \le \tau_{m,i} \le \tau_m^{\max} \quad \text{and} \quad \tau_{m,0} = 0 \quad \forall m, \forall i, \\ & \theta_n^{\min} \le \theta_{n,i} \le \theta_n^{\max} \quad \forall n, \forall i, \\ & \theta_{n,0} \leftarrow \text{measured} \quad \text{and} \quad \dot{\theta}_{n,0} = 0 \quad \forall n. \end{aligned}$$

The first line of constraints forces the policy to obey the manipulator's dynamics and leverages a sequential quadratic program's ability to handle constraints. The second line consists of general nonlinear constraints enforced at the last point in the trajectory $t = t_f$. In the specific case of performing a *grab* we formulate h as follows:

$$\begin{aligned} h_p &= \|arm.\text{FORWARDKIN}_N(\boldsymbol{\theta}) - \text{Goal}\| - \varepsilon_p, \\ h_v &= \|arm.\text{FORWARDVEL}_N\left(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}}\right)\| - \varepsilon_v, \end{aligned}$$

where h_p constrains end effector position to the goal point and h_v constrains end effector velocity to be near zero at the point in time the goal is reached. In both constraints ε represents a definable error tolerance.

For the task of grabbing, the objective function $g()$ can be used to minimize end effector velocity at t_f, i.e. taking the form $g\left(\mathbf{x}_{\frac{t_f}{dt}}\right) = \|arm.\text{FORWARDVEL}_N\left(\mathbf{x}_{\frac{t_f}{dt}}\right)\|$. Alternatively, $g()$ can be used to find a minimal effort policy and take the form $g\left(\boldsymbol{\tau}_i\right) = \boldsymbol{\tau}_i^{\mathrm{T}} R \boldsymbol{\tau}_i$, where R is a scalar weight.

Inverting Actuators

The manipulator's motion is planned in reference to its generalized torques. Using the soft actuator model developed in Section 5.4.2, this motion plan can be expressed in reference to cylinder displacements $\mathbb{V}_s^m$, where superscript m denotes an individual cylinder model for each input

$$\mathbb{V}_s^m(t) = \begin{cases} -1^{2/3} \frac{\tau_m^{1/3}(t)}{a_m^{1/3}} & : \tau_m(t) \leq 0 \\ \frac{\tau_m^{1/3}(t)}{a_m^{1/3}} & : \tau_m(t) > 0 \end{cases}$$

Since the target motion plan $\mathbb{V}_s^*(t)$ is a volume profile, many alternative drive systems can be used to realize the manipulator's trajectory, e.g. high pressure gas and valves [Marchese et al., 2014d], rotary pumps [Onal and Rus, 2013, Katzschmann et al., 2014], or fluidic drive cylinders [Marchese et al., 2014c,b]. In this work we use fluidic drive cylinders and this approach allows us to closely match the prescribed volume profile. To effectively invert the LTI fluidic drive cylinder model, developed in the Section 5.4.2, we use M direct collocation trajectory optimizations. In these problems

$$\mathbf{\Pi}_\alpha^m = \begin{bmatrix} u_0^m & u_1^m & u_2^m & \dots & u_{\frac{t_f}{\mathrm{dt}}}^m \\ \mathbf{x}_0^m & \mathbf{x}_1^m & \mathbf{x}_2^m & \dots & \mathbf{x}_{\frac{t_f}{\mathrm{dt}}}^m \end{bmatrix}.$$

And the following optimization, performed for each cylinder model, identifies a locally-optimal reference input $\mathbf{u}^*(t)$. The superscript m has been omitted for convenience

$$\begin{aligned}
\boldsymbol{\Pi}_\alpha{}^* = \min_\alpha \quad & \sum_i \|\mathbb{V}_s(i) - \mathbf{C}\,\mathbf{x}_i + \mathbf{D}\,u_i\| \quad \Leftarrow \textbf{Track } \mathbb{V} \textbf{ Profile} \\
\text{subject to} \quad & 0 = \mathbf{x}_i - (\mathbf{A}\,\mathbf{x}_{i-1} + \mathbf{B}\,u_{i-1})\,\mathrm{d}t - \mathbf{x}_0 \quad \forall\, i = 1 \,..\, \frac{t_f}{\mathrm{d}t}, \\
& u^{\min} \leq u_i \leq u^{\max} \quad \forall\, i \quad \text{and} \quad \mathbf{x}_0 = 0.
\end{aligned}$$

It is important to note that the locally-optimal input trajectories $\mathbf{u}^*(t)$ returned by the above optimization represent the best realization of a given volume profile subject to the dynamic limitations of the drive mechanism. For example, areas of high-frequency oscillation within $\boldsymbol{\tau}^*(t)$ can result in significant localized tracking errors. As a solution, if the discrepancy between simulated model output and volume profile, i.e. $\|\mathbb{V}_s(t) - \mathbf{C}\,\mathbf{x}(t) + \mathbf{D}\,\mathbf{u}(t)\|$, exceeds an experimentally determined threshold for some span of time, we simply rerun the policy search procedure with a randomized $\boldsymbol{\tau}(t)$ until a suitable realization is found. Alternative solutions may include planning directly in $\mathbf{u}$ space; however, this requires a single optimization to handle a dynamic model of the entire manipulation system, i.e. manipulator, actuator models, and cylinder models.

Grabbing Evaluations

Simulations

To validate this approach to dynamic motion planning for the soft arm, we run direct collocation trajectory optimization on an experimentally identified model of the arm. We find a minimal tip velocity open-loop policy that executes a *grab*. Figure 5-29 depicts four different grab states (A-D) and Figure 5-30a-5-30d shows corresponding locally optimal policies generated by the planning approach. Table 5.6 lists the goal points and corresponding positional errors, or the error between the manipulator's simulated end effector position at t_f and the goal point, as well as simulated end effector velocity at t_f for seven trials per goal point. Positional errors and velocities

that exceed ε_p and ε_v are explained by the fact that the trajectory optimization only enforces dynamic constraints every $\mathrm{d}t$, initialized at 80 ms, which is orders of magnitude greater than the time step used to integrate the manipulator's equations of motion in the approximately continuous time simulation.

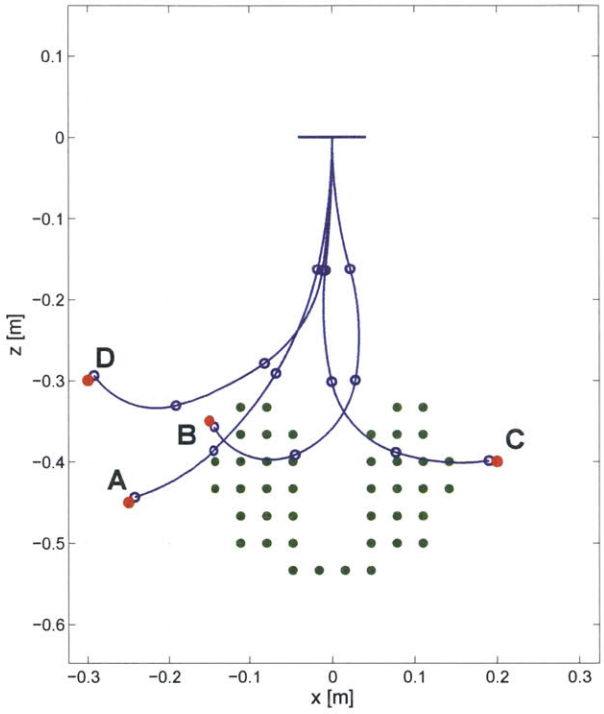


Figure 5-29: The neutral axis of an experimentally identified model of a four segment soft manipulator is shown in blue at four different grab states (A-D), where the goal position of the grab is shown in red. Green points represent goal positions that are statically feasible under the estimated torque limits of $|\boldsymbol{\tau}| = [0,\ 0.12,\ 0.13,\ 0.18]^{\mathrm{T}}$.

Experiments

In order to experimentally validate the outlined approach for grabbing with a soft and highly-compliant arm, we conduct multiple trials of four experiments, summarized in Table 5.7. The goal of these experiments is to have the aggregate manipulation system autonomously perform a grab maneuver. A successful grab is defined as attaching to and removing a 4 cm diameter table tennis ball from a holder at the goal

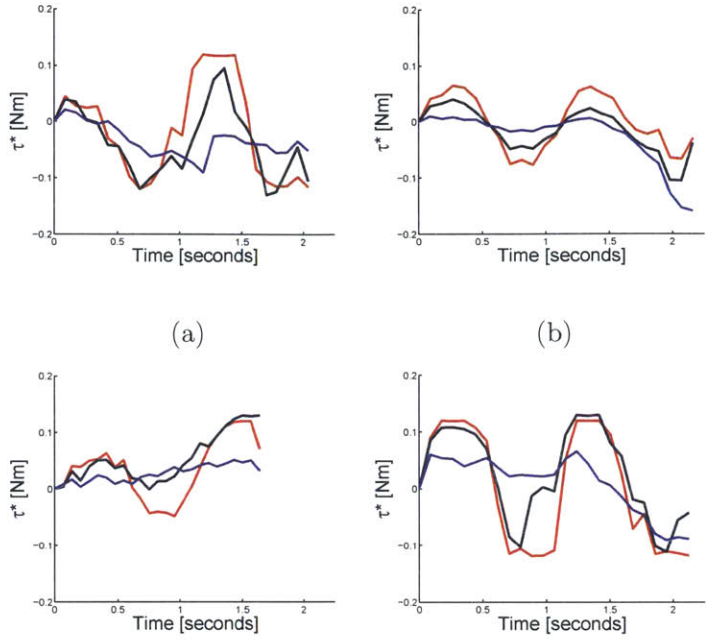


Figure 5-30: The corresponding locally-optimal generalized torque trajectories (a-d) for each of the grab states shown in Fig. 5-29 (A-D), respectively. The input trajectory to segment 2 is shown in red, segment 3 in black, and segment 4 in blue.

position; refer to Figure 5-32. Locally-optimal input trajectories $\mathbf{u}^*(t)$, as determined in Section 5.5.4, are executed on the aggregate manipulation system. Trials reported in Table 5.7 and Figure 5-31 occurred after successful completion of Algorithm 13. The arm's torque limits are controlled and varied between experiments, i.e. experiments one and two to three and four. Among these groups goal location is also controlled for and varied, i.e. one to two and three to four. In experiments one and two the ball, represented as the black circle in Table 5.7, is fixed at the user specified goal location around which the plan is derived. In experiments three and four the ball location underwent an initial one-time adjustment by 2 cm to ensure it corresponded to the simulated realization of the plan, which considers the dynamic limitations of the fluidic drive system. Important simplifications: In these evaluations the unactuated regions between segments $\mathbf{L}_p$ were assumed zero. Additionally, for model stability

Table 5.6: Dynamic motion planning with direct collocation

Goal	Coordinates (cm)	$R = 0.1$ Error (cm)	$R = 0.1$ Velocity ($\text{cm}\cdot\text{s}^{-1}$)	$R = 0.01$ Error (cm)	$R = 0.01$ Velocity ($\text{cm}\cdot\text{s}^{-1}$)
A	(-25, -45)	1.1	1.4	1.1 ± 0.2	1.5 ± 1.2
B	(15, -35)	1.1	2.4	0.8 ± 0.4	2.7 ± 1.0
C	(20, -40)	0.9	0.1	1.0 ± 0.1	0.8 ± 0.4
D	(-30, -30)	1.7*	7.6*	0.9	3.7

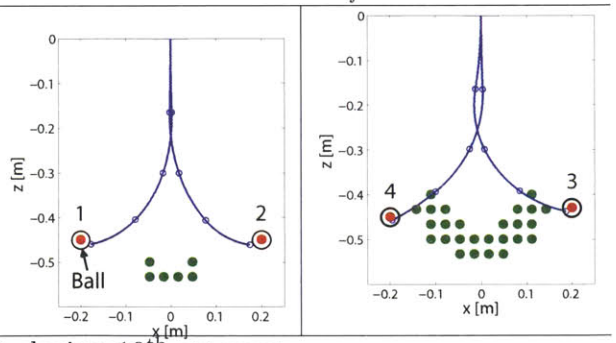
$\varepsilon_p = 1$ cm and $\varepsilon_v = 2$ $\text{cm}\cdot\text{s}^{-1}$ in all cases.
*Solver terminated after numerical difficulties.

purposes, the center of mass locations were redefined as

$$\mathbf{P}_n = M_{tip_{n-1}}^{base}\, \mathbf{R}_z(\gamma)\, \mathbf{T}_z(L_P)\, \mathbf{R}_y\left(\frac{\kappa\, s}{2}\right) \mathbf{T}_z\left(\frac{d(\kappa\, s)}{2}\right) \mathbf{0} \quad \forall\, n.$$

This adjustment effectively amplifies center of mass motion as segment curvature increases; However, for segment curvatures achieved during these experiments, this model assumption captures the dynamics of interest.

Table 5.7: Summary of Grabbing Experiments

Exp. #	Sys IDs	Consec. Attempts	Success Grabs	Plan Realization at $t = t_f$
1	2	10	10	
2	0	10	9†	
3	2	5	4⋆	
4	1	12	11⋆	

† Actuator burst during 10^{th} attempt.
⋆ A successful grab occurred after the failed attempt.

The aggregate system was able to successfully grab the ball in approximately 92% of trials. Experiments one and two were performed consecutively. Although 2 iterations of system identification were performed on the actuator model parameter set during experiment one, no additional identifications were performed during experi-

ment two. Similarly, experiments three and four were performed consecutively and two identifications were required during experiment three and one during experiment four.

Figure 5-31 shows the cartesian state trajectories of the manipulator's end effector for each experiment. The left and right figures show x and y velocity versus position, respectively. Multiple trial trajectories are overlaid on each figure and these trajectories originate from the origin and terminate at red markers. Trials for which motion capture data was lost for a significant portion of time were omitted. This occurred when the end-effector endpoint was misinterpreted as the ball center-point and is a limitation of the experimental setup. Raw end effector velocity measurements were filtered using a 5-point moving average, removing jitter from numerical differencing.

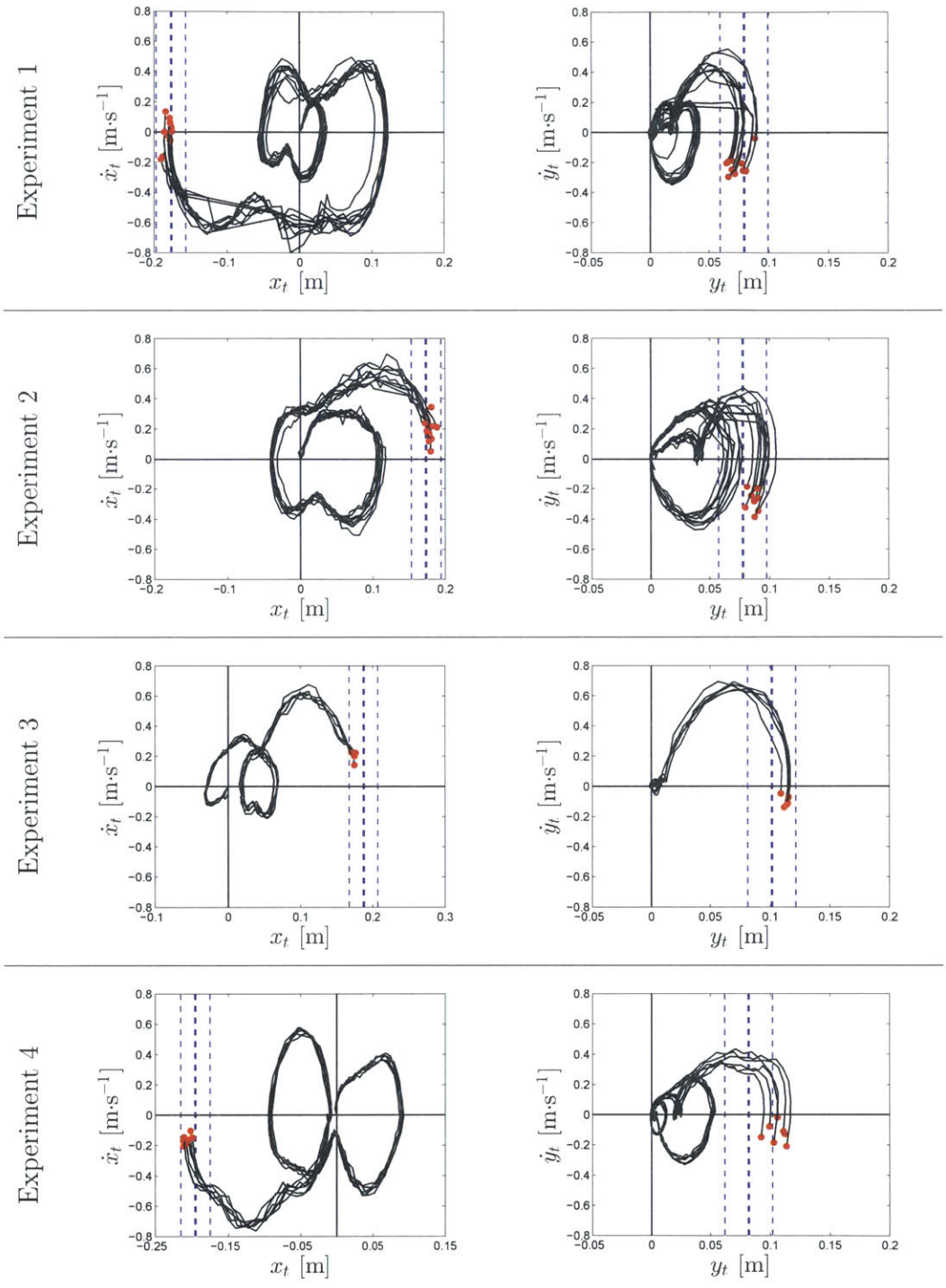
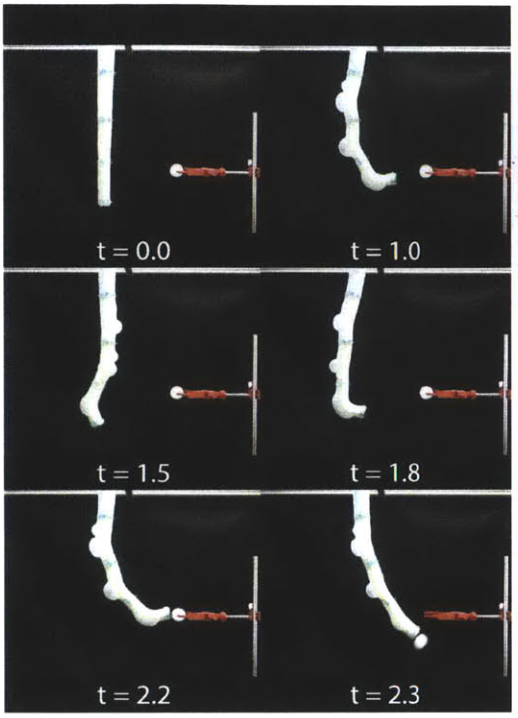
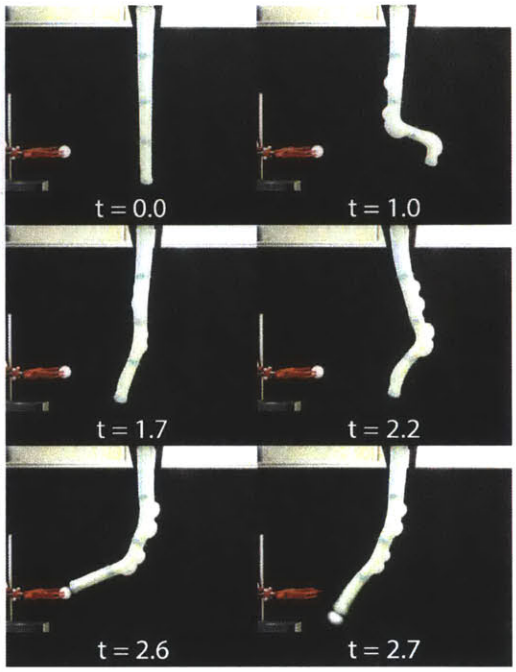


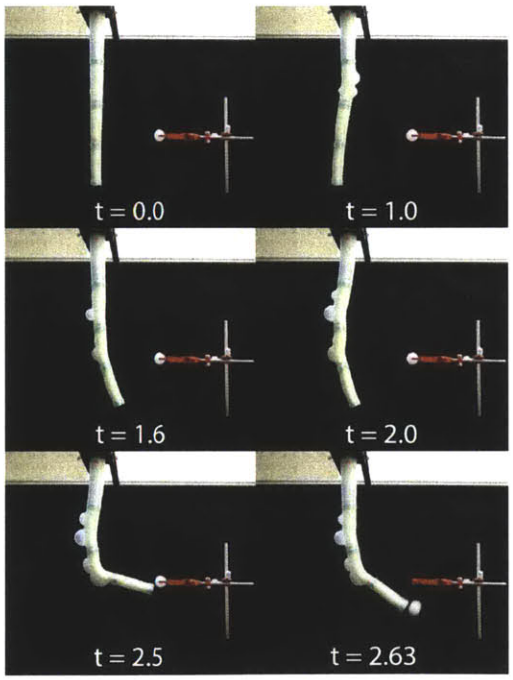
Figure 5-31: Cartesian state trajectories of the manipulator's end effector for each experiment. The left and right figures show x and y tip velocity versus position, respectively. The trajectories of independent trials for each experiment are overlaid in black. These trajectories originate from the origin and terminate at red markers indicating $t = t_f$. The vertical blue lines represent planned end-effector realizations $\pm$ 2 cm.



(a) Experiment 2

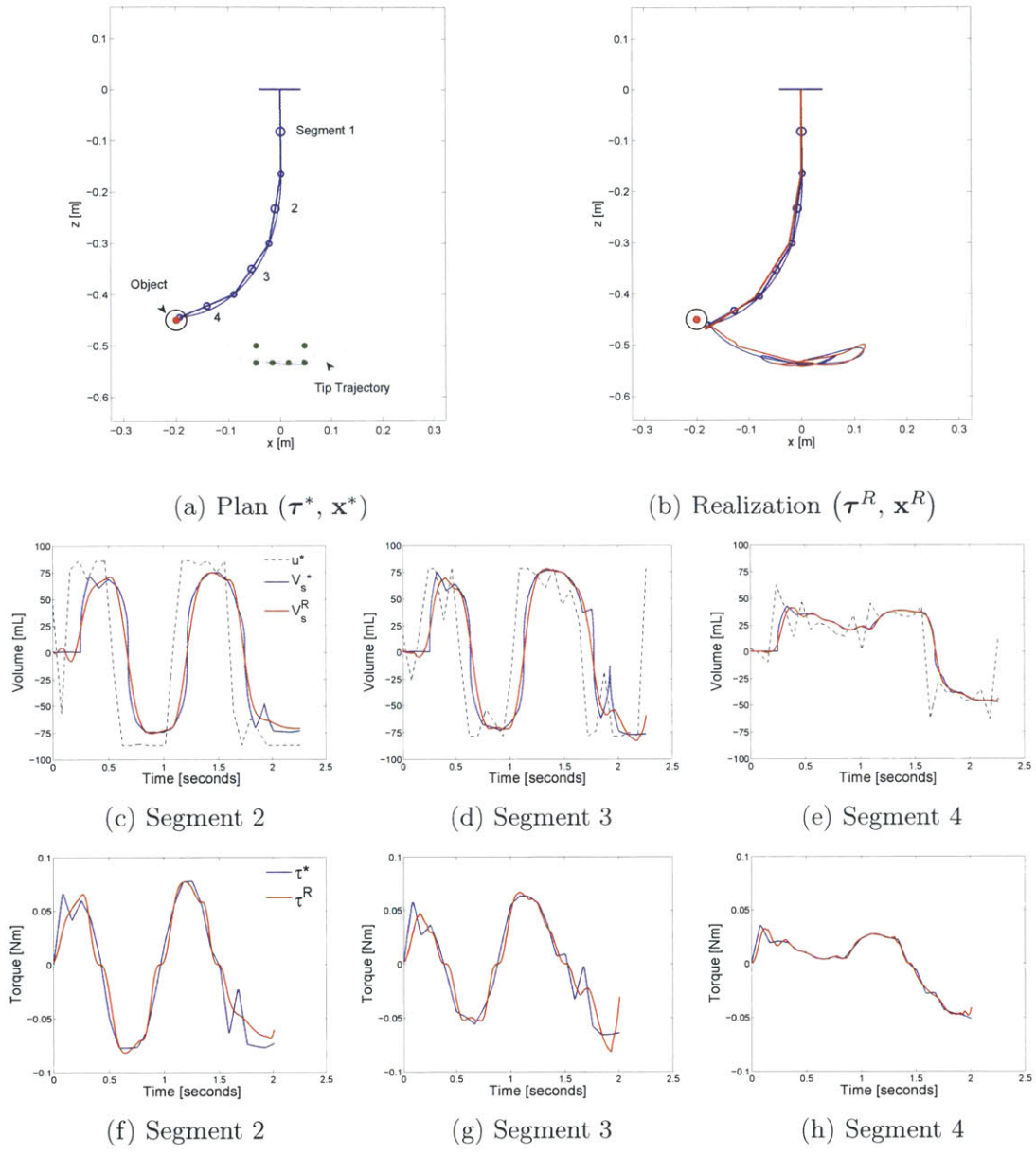


(b) Experiment 3



(c) Experiment 4

Figure 5-32: Sequenced photographs from experiments two, three, and four.



(a) Plan $(\boldsymbol{\tau}^*, \mathbf{x}^*)$ (b) Realization $(\boldsymbol{\tau}^R, \mathbf{x}^R)$

(c) Segment 2 (d) Segment 3 (e) Segment 4

(f) Segment 2 (g) Segment 3 (h) Segment 4

Figure 5-33: Experimental characterization of a dynamic *grab* maneuver performed with a four segment soft manipulator. Panels (a) and (b) depict the planned and realized manipulator motion in cartesian space respectively. In panel (a) the manipulator's predicted neutral axis is shown in blue and blue circles represent modeled center of mass locations. Here, green points represent a set of statically reachable points under estimated torque limits $|\boldsymbol{\tau}| = [0.08, 0.07, 0.09, 0.13]^{\mathrm{T}}$ and the red point represents the goal point of the maneuver. In (b) blue and red represent simulated and experimentally measured realizations of the ideal motion plan presented in (a). In panels (c)-(e) the locally optimal reference input trajectories $\mathbf{u}^*$ (dotted line), the target piston displacements $\mathbb{V}_s^*$ (blue line), and the realized piston displacements $\mathbb{V}_s^R$ (red line) are shown for segments 2, 3, and 4 respectively. Similarly, in panels (f)-(h) the locally optimal torque trajectories $\boldsymbol{\tau}^*$ (blue) and realizations $\boldsymbol{\tau}^R$ (red) are again shown for each actuated segment.

Chapter 6

Conclusion

Although this thesis presents a paradigm shift for robotics, it comes at a time when our culture is seemingly ready to adopt the idea. That is, the notion that robots of the future might look more like a balloon animal or teddy bear and less like C-3PO or R2D2[1] is gaining momentum, especially where robots cohabitate with humans. As I draft this chapter, Disney's animated film *Big Hero 6* has just opened[2] in American theaters and stars an inflatable, soft healthcare robot named Baymax (See Fig. 6-1).

Figure 6-1: Baymax from Walt Disney's *Big Hero 6* [Hall and Williams, 2014].

Although there are still numerous technical challenges to overcome before a system like Baymax is viable, this thesis has made significant progress towards the vision

[1] Characters from the film *Star Wars Episode IV: A New Hope*
[2] November 7th, 2014 (USA)

that machines can be created from soft materials matching the inherent elasticity of natural systems and also be capable of reliably executing autonomous tasks. The goal of this thesis was to explore how autonomous robotic systems can be designed to incorporate and leverage *softness*. To do this, we developed extremely soft robot morphologies that are radically different from today's mainstream rigid-body robotic platforms. Our efforts can be viewed as an attempt to break the mold on how we think about designing, fabricating, and controlling robotic systems.

We have demonstrated new approaches to constructing both the brains and bodies of soft robots. Specifically, an approach to creating autonomous, untethered soft-bodied robots capable of rapid continuum body motion is developed in the context of a fish-like robot in Chapter 3. It was experimentally demonstrated that this robot can travel with relative speeds 1 to 2 orders of magnitude faster than the current state-of-the-art untethered soft robot. In Chapter 4, these concepts are extended to demonstrate, for the first time, that manipulation is possible with a soft, multi-body robot made entirely from low durometer rubber and powered by fluids. New autonomous capabilities such as maneuvering through confined environments and encapsulating and placing objects were experimentally demonstrated. By using a drive cylinder's volumetric displacement and each soft segment's endpoints as measured feedback, we achieved closed-loop configuration control of the presented soft manipulators despite their low structural impedance. This control strategy is an advantageous alternative to open-loop or morphological control in that we can achieve repeatable and precise task-space positioning for novel tasks without trial-and-error. We then extended this work in Chapter 5 to design, fabricate, and control multi-body fluidic elastomer robots that can move in three spatial dimensions subject to gravity. For the first time, we utilized dynamic control algorithms to combat the underactuated nature of fluidic elastomer robots. Here, actuation limits, the effects of gravity, and the high compliance of the manipulator, all of which are physical phenomena common among soft robots, are represented as constraints within a nonlinear optimization.

6.1 Summary of Contributions

This thesis makes a number of contributions in the areas of device design, hardware processes, modeling, and algorithms:

6.1.1 Devices

- **Device 1**: A soft fluidic actuation system capable of rapidly achieving continuum-body motion.
- **Device 2**: A self-contained soft-bodied fluidic robot with 3 DOF and 2 bidirectional actuators that embodies our approach to mobile soft robotics and emulates planar forward swimming and escape maneuvers of biological fish.
- Experimental evaluations of Devices 1 and 2 that demonstrate the energy consumption, motion, and speed of the combined devices and a comparison to biological fish.
- **Device 3**: A soft multi-segment planar arm made of 6 distributed bidirectional rectangular fluidic elastomer actuators with 6 DOF.
- **Device 4**: A mechanism for the closed-circuit continuous driving of soft fluidic robots, the fluidic drive cylinder.
- Experimental evaluations of Devices 3 and 4 that demonstrate curvature control as well as autonomous point-to-point positioning and line following precision and accuracy.
- **Device 5**: An entirely soft multi-segment planar arm composed of 6 distributed bidirectional cylindrical fluidic elastomer actuators with 6 DOF.
- Experimental evaluations of Devices 5 and 4 that demonstrate autonomous navigation through a confining environment while minimizing collisions and primary task accuracy.

- **Device 6**: An actuated, soft finger-like gripper capable of passively adapting to variations in object geometry.

- Experimental evaluations of Devices 4, 5, and 6 that demonstrate repeatable and autonomous grasping and placing of randomly placed round objects.

- **Device 7**: An entirely soft multi-segment spatial arm composed of 8 distributed bidirectional cylindrical fluidic elastomer actuators with 8 DOF.

- Experimental evaluations of one and two segments of Device 7 that quantify the accuracy of bending angle and center of mass model estimations.

- Experimental evaluations of Device 7 that demonstrate repeatable maneuvering through a confined pipe-like environment, active configuration control, and kinematic positioning.

- Experimental evaluations of Device 7 that demonstrate repeatable positioning of the aforementioned manipulator to points outside of the statically reachable workspace.

6.1.2 Hardware Processes

- A method for fabricating soft-bodied fluidic robots that emulate the anatomical form and function of biological fish.

- A lamination-based casting process for fabricating a soft multi-segment planar fluidic elastomer manipulator made of uniform rectangular actuators.

- A retractable-pin-based casting process for the modular fabrication of an entirely soft planar fluidic elastomer manipulator composed of cylindrical actuators.

- A lost-wax-based casting process for the modular fabrication of an entirely soft planar fluidic elastomer manipulator composed of pleated actuators.

- A retractable-pin-based casting process for the modular fabrication of an entirely soft spatial fluidic elastomer manipulator composed of cylindrical actuators.

6.1.3 Models

- A static model for the non-constant bending deformation of a ribbed fluidic elastomer actuator.
- A dynamic model for a fluidic drive cylinder.
- An iterative model for the deformation of a spatial fluidic elastomer actuator with nonlinear material properties subject to both circumferential and longitudinal strain but no external load.
- A kinematic model based on piece-wise constant curvature to describe an entirely soft spatial manipulator.
- A kinematic model describing the expanding envelope of a soft planar multi-segment fluidic elastomer manipulator.
- A dynamic model to describe a soft spatial fluidic elastomer manipulator subject to gravity.
- An iterative approach to system identification for the aforementioned dynamic model under the sagittal plane assumption.

6.1.4 Algorithms

- An algorithm for generating open-loop forward swimming and escape response policies for Device 2.
- An algorithm for the general closed-loop curvature control for a fluidic elastomer actuator.

- Task-space planning algorithms that solve the IK problem and enable soft planar fluidic elastomer manipulators to autonomously (i) position their end-effector in free-space, (ii) maneuver in confined environments, and (iii) grasp-and-place objects.

- Processing and control systems that enable the first autonomous capabilities for a spatial fluidic elastomer manipulator, e.g. (i) advancing through a confined environment, (ii) following configuration trajectories within a sagittal plane, and (iii) positioning in 3D.

- Dynamic control algorithms that allow a spatial fluidic elastomer manipulator operating under gravity to be precisely positioned

- A manipulation primitive built on these dynamic control algorithms, *grabbing*.

6.2 Limitations and Near-Term Improvements

There are several notable limitations to the soft robotic systems presented in this thesis. Here, we discuss some of these limitations and offer possible near-term improvements, some of which we have already begun to execute.

6.2.1 Single-Segment Soft Robots

In Chapter 3 it was demonstrated that single segment robots can be both self-contained and capable of rapid body motion. However, there are several shortcomings of this implementation: (1) The fish uses an open-circulation pneumatic actuation system making prolonged operation difficult; after energy is delivered to the distributed body actuators, it is exhausted to the environment. This limitation is addressed within our recent work Katzschmann et al. [2014] by designing a closed-circulation pump-based drive system. (2) The implementation is pneumatically actuated, and without an air bladder the center of buoyancy is uncontrollably changing throughout the actuation cycle. In Katzschmann et al. [2014], the actuation system is redesigned

to use water as a transmission fluid to address this issue. (3) The implementation is constrained to move on a pre-defined trajectory in two dimensions. This is addressed in Katzschmann et al. [2014] by introducing dive planes to allow for dynamic diving and motion in three spatial dimensions. Obviously, there are always design tradeoffs. By introducing a closed-circuit pump-based hydraulic actuation system in place of the open-circuit pneumatic actuation system detailed in Section 3.3.1, high speed and agility were sacrificed for prolonged operation at a slower swimming speed. In the future, a synthesis of these two approaches within a single hybrid actuation system may allow the robot to transition between task-space performance regimes on the fly.

6.2.2 Multi-Segment Planar Soft Robots

In Chapter 4 it was demonstrated that planar manipulation with fluidic elastomer manipulators is viable. However, there are several notable limitations to the present systems: (1) The manipulator's dynamics are not modeled, and therefore not used in the manipulator's control. Although sufficient performance is achieved for the investigated manipulation tasks, it is reasonable to expect that the manipulator's speed could be increased and its settling time decreased if a dynamics-based control strategy were derived. (2) Further, in conducting these experiments it was observed that the soft manipulator can harmlessly collide with its environment, and this contact can be potentially leveraged to either increase primary task precision or its likelihood of success. However, a method for detecting these collisions is not provided, and it is unclear to what extent the piece-wise constant curvature modeling assumption used here is valid under environmental contact. Accordingly, this is strictly a passive feature but future work could provide strategies for incorporating and/or exploiting collisions into the task-space planner. Likely, more advanced kinematic models are needed to understand the deformation of soft manipulators constrained by environmental contact. (3) Lastly, this work brings to attention the need for proprioceptive localization systems for soft robots. Feedback for our controller comes currently from an exteroceptive localization system. This is a reasonable method for indoor, laboratory or factory environments where there is sufficient line of sight to the manipulator. How-

ever, this sensor approach is prohibitive in that the environment must be equipped with cameras and the tasks cannot occlude the view of these cameras onto the manipulator. Proprioceptive localization for soft robots is an important avenue for future research.

6.2.3 Multi-Segment Spatial Soft Robots

In these initial experiments we found it feasible to compute a sufficiently accurate dynamic model to make planning viable for a soft elastomer manipulator. However, to obtain the required performance for executing specific tasks, like grabbing, we found it necessary to use iterative learning control. In future work, these trajectories may be stabilized using linear time-varying linear quadratic regulators (LTV LQRs) [Tedrake, 2009] making them robust to uncertainty in initial conditions and tolerant of modeling inaccuracies. Although this class of robot is well-suited for environmental contact (e.g. whole arm grasping and bracing), the modeling assumptions used here may not suffice under these conditions. This work suggests dynamic model-based planning and control may be an appropriate approach for soft robotics.

6.3 Lessons Learned

- **No free lunch**: A fish was chosen as a case study because it naturally exhibits: continuum body curvature, rapid motion during an escape response, a compliant posterior that bends under hydrodynamic resistance, and an anterior suitable for housing rigid supporting hardware. Although this morphology offered many advantages, it offered just as many disadvantages. We spent a considerable amount of time and effort combating challenges unique to marine robots. For instance, creating water-tight chambers to house the novel electromechanical actuation system was difficult primarily because the robot's unique geometry and soft exterior prohibited the use of commercially available solutions for water-tight sealing. In hindsight, it may have been advantageous to relax the design constraints on emulating the exact size and form of a fish in

order to accommodate commercially available sealing solutions, therefore reducing time spent on engineering challenges outside the direct focus of the research. Furthermore, underwater wireless communication is extremely challenging but critical for an autonomous marine robot. We found it suitable to use a relatively low frequency, 900 MHz, radio transmitter for the shallow-water experiments presented in this thesis. However, at greater depths these signals would almost certainly be attenuated by water and either an acoustic or visible light modem would need to be employed.

- **Mission critical**: While working on some of the hardware platforms in this thesis, we uncovered seemingly advantageous design changes. An extremely important lesson learned was differentiating between incremental design changes with minimal return on time invested and important design changes that aid in achieving the functional specifications of the project, that is changes that are mission critical. Hardware can always be iteratively improved, but at some point engineering judgement must be used to deem the system reliable and robust enough to carry on with the next phases of the project. For example, about mid way through the autonomous fish robot project we discovered that scaling the fish up by approximately 20% would allow us to accommodate a larger compressed gas cylinder and equated to an increased operating life. However, redesigning the various components of the platform would have taken considerable time and the intent of the project was to demonstrate rapid continuum body motion and high forward swimming speeds, not necessarily prolonged operation.

- **Know when to fold 'em**: Throughout this work, we often faced the difficult decision of abandoning a design alternative after significant time and effort was invested into its development. Although a thorough analysis of alternatives was done prior to each project, often the technology being applied was so new or was being used in such a new application that we were forced to learn of prohibitive limitations first hand. For example, because of their small size and energy effi-

ciency we originally decided electro-permanent magnet (EPM) valves [Marchese et al., 2011] would be well-suited for mobile soft robot applications with high locomotory speeds. However, after empirically determining the actuation frequency of our fish-like robot (1-2 Hz), we observed that EPM valves generated significant heat when attempting to realize the control policy, primarily because of the frequency of current pulses sent through the coils. Additionally, the small orifice size of the valves produced high impedance to fluid flow requiring fluid driving pressures that were unnecessarily high. After learning these lessons, we opted for commercially available proportional control valves and reworked our design to accommodate their increased size.

It is also important to mention that prior to the development of the fluidic drive cylinder, we attempted to control the planar fluidic elastomer manipulators with multiple control valves. After having great success with control valves on the fish platform, we decided to scale this control strategy up to the multi-segment case. However, we failed to anticipate the complexity of transforming between the discrete states offered by valves and the continuous control authority required for curvature tracking. Using an arrangement of valves and one positive fluid source per bidirectional body segment offers only the ability to actively pressurize fluidic chambers. Depressurization is achieved by exhausting to a lower-than-supply pressure outlet. A key feature of the fluidic drive cylinder is that it offers control authority during depressurization of fluidic chambers, effectively vacuuming rather than exhausting.

- **The right tool for the job**: An important lesson learned is to use the right tool for the task at hand. In a literal interpretation, we learned that fused deposition 3D printing machines were certainly the proper tool to use for creating molds during the fabrication processes outlined in this work. They enable alternative designs to be rapidly prototyped and can realize complex spatial forms. However, we also learned this lesson on a smaller scale. Simply taking the time to find the right sized screw driver or the proper length USB cable can

prevent you from unnecessarily damaging a fragile prototype or from spending twice the time fiddling with a tool not quite intended for what you're trying to accomplish.

- **Time is money**: Knowing when to utilize commercially available solutions to technical challenges was a valuable less learned. As roboticists, we often have the ability to create in-house solutions to problems that arise during our research, but suppressing the temptation to tinker can be beneficial in some situations. For example, our original attempt at externally localizing the soft manipulator endpoints was a downward facing webcam and a sophisticated image processing algorithm. After spending roughly a month trying to robustly localize colored markers on the manipulator with the aforementioned system, we realized we were wasting our time when there existed many commercially available external, portable localization systems, namely OptiTrack from Natural point. By making the investment in this solution, we gained valuable time working on problems more central to our research. Additionally, this lesson can be seen in our choice to assemble the fluidic drive cylinders from numerous commercially available components, with only a custom-design frame and coupler to integrate these parts. There were 16 drive cylinders used in the spatial manipulation system. At this scale, had we not assembled these from standard components, we would have spent considerable time maintaining, debugging, and otherwise tending to these hardware systems.

6.4 Looking to the Future

At a high level, we have spent considerable time pondering the future directions of soft robotics research. The following discussion is meant to inform future research directions and summarizes what we feel are important open topics in the field.

6.4.1 How Soft is Too Soft?

This thesis has explored extremely soft robots made almost entirely from low durometer rubber, and these represent one extreme on the softness spectrum (see Fig. 1-2). It is not to say that factories of the future will consist of soft rubber tentacles assembling automobiles. Rather, it is likely that robots of the future will integrate softness when and where necessary. For example, perhaps rigid-body robots will have soft elastomer-based skins to assist in sensing and increase resiliency. Perhaps humanoid robots may integrate softer limbs or joints to combat hard, unexpected falls. Or perhaps fixed-wing drones can utilize soft legs to store enough potential energy for executing whole body take-offs without runways. We believe further research is needed to investigate the rich possibilities of hybrid soft-hard systems.

6.4.2 3D Printing Soft Materials

As mentioned in the related work, Umedachi et al. [2013] 3D printed the first soft robot. However, at the moment soft 3D printing technology is still unable to deliver elastomer materials with properties comparable to casted rubber. This limitation prevents such printing technology from being used to fabricate fluidic elastomer robots. The ability to 3D print low durometer rubber with high percent elongation at break would enable more complex fluidic elastomer robot morphologies to be realized as well as increase the consistency with which we produce the robots within this thesis. Accordingly, further development of 3D printing technology offers the potential to revolutionize the field of soft robotics.

6.4.3 Proprioceptive Sensing

The high compliances, larger deformations, and continuum motions of soft robots prevent the use of many traditional proprioceptive sensors including encoders, strain gauges, and IMUs. In this thesis, closed-loop processing and control systems used exterioceptive sensing. This provided a reliable means of localizing our robots in order to achieve the presented tasks. However, as soft robots transition out of laboratory

environments and become a pervasive part of our every day life, they will likely need to incorporate proprioceptive sensing. The ability for a soft robot to internally measure bending deformation will be immensely valuable for future soft robotic systems. Recently, soft elastomer-based sensors with embedded micro-fluidic channels filled with liquid conductors have been developed by Park et al. [2012]. This sensor shows immense promise for soft robots in that it can reliably measure multi-axis strain and applied normal pressures, while being functional up to 250% strain. Future work could integrate such sensors in combination with the algorithms presented in this thesis to provide closed-loop configuration control via proprioceptive sensing.

6.4.4 Contact Modeling

Soft robots are well-suited for environmental contact as seen when executing tasks like whole arm grasping and bracing. For a discussion and preliminary results on bracing without contact modeling see Appendix A. The modeling assumptions within this thesis may not suffice under such contact conditions. An important area of future work is in developing more accurate models of soft actuated materials in contact with objects. We envision these models will need to be accurate enough to capture the complex physical deformation of soft actuated material conforming to both rigid and compliant surfaces, as well as be computationally efficient enough to fit within the time constraints of real-time planning algorithms. Such developments would greatly extend the capabilities of current soft robotic systems.

Appendix A

Bracing

Static bracing is a motion primitive enabled by the development of an identified dynamic model for the soft manipulation system in Section 5.2.4. By understanding the system's dynamics we can devise a planning algorithm that searches for and executes an environmental brace during a manipulation task. This is similar to the way humans rest their wrists against a table while writing. By statically bracing against nearby objects, we are able to ground the manipulator at a point between its base and end-effector, effectively reducing the contribution of dynamic forces and uncertainty from some number of manipulator segments on the primary manipulation task, e.g. end effector movement.

The concept of bracing for manipulation was first introduced in the 1980s [Book et al., 1985]. Bracing strategies with rigid body manipulators can involve physically fixing a distal point on the manipulator to a bracing surface (e.g. using suction, mechanical clamps, or magnets), but these approaches require additional hardware limiting the surfaces against which the manipulator can brace. Alternatively, normal force, or the component of contact force normal to the bracing surface, can be used to form braces as in Lew and Book [1994]. Here, a hybrid force-position controller is developed to mitigate the control complexity arising from normal force bracing. Despite the complexity, normal force bracing is a more universal strategy in that it only requires a suitable bracing surface to lie within the null space of the manipulator. Additionally, bracing strategies with probabilistic contact estimation [Petrovskaya et al.,

2007] and multiple contact controllers [Park and Khatib, 2008] have been developed for rigid body manipulators. It is evident that for such manipulators contact force must be controlled to prevent damage to both the robot and the environment.

Here, we show that bracing is also a feasible and effective strategy for soft fluidic elastomer manipulators. With such a soft manipulator the required bracing force is generally small relative to that of a traditional rigid body manipulator for a given task. For tasks requiring high bracing forces, a soft fluidic elastomer manipulator will safely undergo elastic deformation before the bracing surface for a wider range of surfaces than a more traditional manipulator.

A.1 Limitations

We do not provide a dynamic model that considers contact, nor do we have knowledge of the contact forces through sensing. Rather, the intent of this section is to show that the presented dynamic model and planning infrastructure is sufficient to accomplish simple normal force bracing at an arm segment's endpoint. Specifically, we make the assumption that the piece-wise constant curvature modeling assumption remains valid despite the kinematic constraints imposed by the brace.

A.2 Bracing Conditions

We outline three criteria for the static bracing strategy and these conditions help to illustrate the differences in employing this manipulation primitive with a soft elastomer robot as opposed to with a more traditional rigid body manipulator.

A.2.1 Condition 1

The contact force between the robot and object must be of sufficient magnitude to form a static brace. Figure A-1 illustrates this concept. We assume the surface of the robot and the surface of the object come into contact and that they are non-moving. We can relate the normal force F_n, or the component of contact force normal to the

bracing surface at $\mathring{\mathbf{a}}$, the brace position and orientation, and the friction force F_f as

$$F_f \leq \mu_s F_n,$$

where μ_s is the static coefficient of friction between the two surfaces. Accordingly, $\mu_s F_n$ is the threshold below which the robot's tangential force F_t will not break the static brace. That is,

$$F_t < \mu_s F_n.$$

Components of force due to end-effector interactions as well as due to the robot's actuators compose F_n and F_t.

In general, the soft fluidic elastomer robot presented in this work can statically brace with less contact force than a hard robot. The soft robot is composed of low durometer silicone rubber and this material has a high coefficient of static friction when in contact with solids [ToolBox]. During normal force bracing, a rigid body robot is coated with a wear resistant surface [Book et al., 1985], which has a low coefficient of static friction when in contact with solids [ToolBox]. For example, teflon in contact with steel has a μ_s of between 0.05 and 0.20 [ToolBox] whereas soft silicone rubber in contact with steel has a μ_s of between 0.6 and 0.9 [Mesa Múnera et al., 2011]. It follows that for a given tangential force F_t, a hard robot will have to exert a force on the object of between 3 and 18 times greater than a soft robot to maintain a static brace.

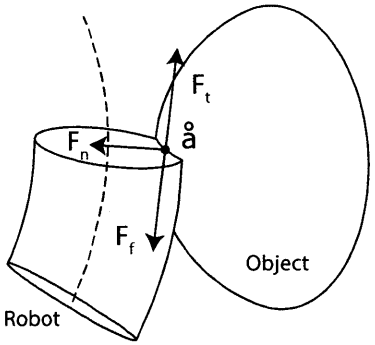
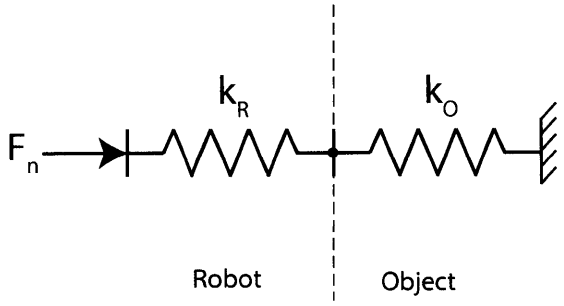


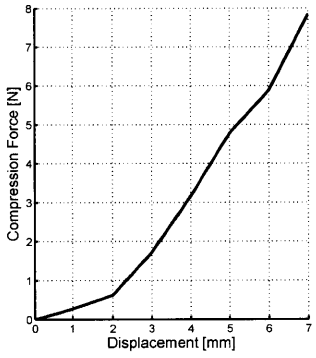
Figure A-1: Illustration of normal force bracing where the first condition is that contact force between the robot and object must be of sufficient magnitude to form a static brace.

A.2.2 Condition 2

The normal force at the static brace point should not deform the object. The motivation behind this condition is that the robot should not damage the environment by bracing. Figure A-2a schematically represents the local interaction between the robot and object. The normal force F_n radially compresses both the robot and the object, whose local stiffnesses are represented by k_R and k_O respectively. The condition can then be written as $k_O >> k_R$. This relationship implies that the robot will deform



(a)



(b)

Figure A-2: (a) A simplified model of the local interaction between the robot and object. (b) Radial stiffness, normal to the bending axis, of a robot made of silicone rubber.

well before the object deforms. For an entirely soft robot this condition is satisfied over a wider larger range of objects than for a rigid body manipulator with mechanical compliance. For example, Figure A-2b depicts the radial stiffness, normal to the bending axis, of a robot made entirely of silicone rubber. The robot's radial stiffness

is approximately 1 $\frac{N}{mm}$. Additionally, the torsional stiffness between the base and brace point is approximately 0.2 $\frac{N}{rad}$. To the best of our knowledge, the stiffness of a soft fluidic elastomer robot is lower than a rigid body manipulator with mechanical compliance.

A.2.3 Condition 3

There must exist a pose $\mathring{\mathbf{a}}$ on and tangential to an object's surface O and a set of joint space parameters $\boldsymbol{\theta}$ and $\boldsymbol{\gamma}$ such that the task space is contained within the workspace, or reachable envelope, of the manipulator and $\mathring{\mathbf{a}}$ is within the nullspace.

Figure A-3 illustrates the kinematic conditions for static bracing. Here the task space is shown as a square region, the bracing object O is shown as a sphere, and the soft robot is composed of multiple cylindrical bending segments.

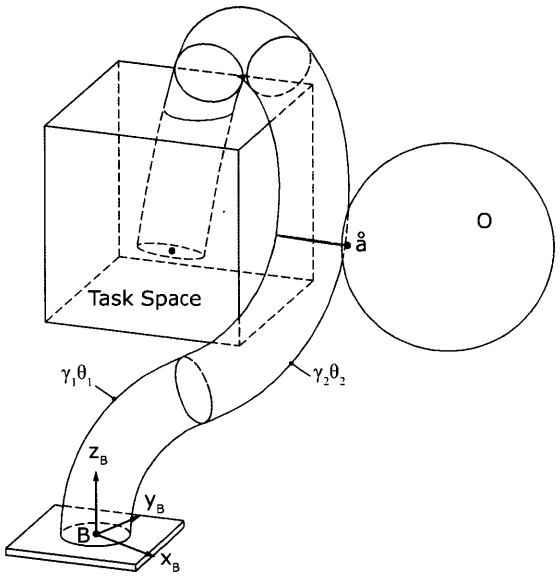


Figure A-3: A depiction of the third kinematic condition for static bracing

A.3 Bracing Algorithm

Having outlined the conditions required for bracing, we next devise a planning algorithm that satisfies these conditions and allows the soft elastomer manipulator to

execute an environmental brace. To begin, consider a soft manipulator whose dynamics can be represented in the manipulator form as outlined in Section 5.2.4

$$\begin{aligned} \mathbf{H}(\boldsymbol{\theta})\,\ddot{\boldsymbol{\theta}} + \mathbf{C}\left(\boldsymbol{\theta},\,\dot{\boldsymbol{\theta}}\right)\,\dot{\boldsymbol{\theta}} + \mathbf{G}(\boldsymbol{\theta}) &= \mathbf{B}\,\boldsymbol{\tau} + \frac{\partial \phi}{\partial \boldsymbol{\theta}}\,\boldsymbol{\lambda}, \\ \phi(\boldsymbol{\theta}) &= 0, \end{aligned}$$

where $\boldsymbol{\lambda}$ are external forces defined by static brace constraints. We propose finding a feasible static brace pose $\mathring{\mathbf{a}}$ by solving the following optimization,

$$\begin{aligned} \min_{\boldsymbol{\tau},\boldsymbol{\theta},\boldsymbol{\gamma},\mathring{\mathbf{a}}} \quad & \boldsymbol{\tau}^{\mathrm{T}}\,R\,\boldsymbol{\tau} \quad \Leftarrow \textbf{Minimal Effort} \\ \text{subject to} \quad & \mathbf{G} - \frac{\partial \phi}{\partial \boldsymbol{\theta}}\,\boldsymbol{\lambda} - \mathbf{B}\,\boldsymbol{\tau} = \mathbf{0}, \quad \Leftarrow \textbf{Gravity and Contact Force Comp} \\ & \|arm.\textsc{forKin}_N\,(\boldsymbol{\theta},\,\boldsymbol{\gamma}) - \mathrm{Goal}\| = \mathbf{0}, \quad \Leftarrow \textbf{End Effector Task} \\ & \|arm.\textsc{forKin}_{N-n}\,(\boldsymbol{\theta},\,\boldsymbol{\gamma}) - \mathring{\mathbf{a}}\| = \mathbf{0}, \quad \Leftarrow \textbf{Brace Pose Constraint} \\ & \tau_m^{\min} \le \tau_m \le \tau_m^{\max} \quad \forall m = 1\,..\,M, \\ & \theta_n^{\min} \le \theta_n \le \theta_n^{\max} \quad \forall n = 1\,..\,N, \\ & \gamma_n^{\min} \le \gamma_n \le \gamma_n^{\max}, \\ & \dot{\theta}_n = 0, \\ & \dot{\gamma}_n = 0. \end{aligned} \tag{A.1}$$

Algorithm 14 uses the optimization outlined in Equation A.1 as well as a classical controller for each generalized coordinate of the soft arm to perform the primary task of positioning the manipulator's end effector while accomplishing the secondary task of bracing an intermediate segment's endpoint against a nearby surface if possible. For simplicity, we again operate within a sagittal plane defined by γ_1 and fix $\gamma_2 \ldots \gamma_N = 0$. The task space is simply a *Goal* point in $\mathbb{R}^3$.

A.4 Bracing Simulations

To evaluate the strategy of static bracing for a soft elastomer manipulator, we simulated Algorithm 14 on an identified model of the manipulator but with increased

Algorithm 14: Static Brace Strategy

$n = 1$.
while *A feasible solution does not exist* **do**
  $\boldsymbol{\theta}^*$, $\mathring{\mathbf{a}}^*$ $\leftarrow$ Find an optimal static brace solution using (A.1) (refer to Section A.2.3).
  $n++$.
end
Move into contact with the object at $\mathring{\mathbf{a}}^*$ by servoing the proximal $N - n$ segments to $\theta_1^* \ldots \theta_n^*$.
Apply normal force to object (refer to Section A.2.1).
Replan optimal solution for reaching goal using (A.1) with the added constraints of $\theta_1^* \ldots \theta_n^*$ equaling the measured arc space configuration.
Move to goal by servoing the distal n segments to $\theta_{n+1}^* \ldots \theta_N^*$.

generalized input limits. The objective of the simulations is to demonstrate the formation of a simple environmental brace. The arm was servoed during steps 2 and 5 of the simulation using a PD controller for each arm segment. During step 3, we assume the arm is capable of satisfying Condition 1 without simulating the contact force, and in order to simulate the effect of contact we increase the friction, or damping coefficient, acting on the braced segments once the Brace Pose Constraint is satisfied. The results of the simulation are shown in Figures A-4. The top 3 panels of Figure A-4 show an example of a static brace formed at the endpoint of the second link of a four link manipulator. The left panel illustrate steps 1 through 3 of Algorithm 14 occurring over 5 seconds. Here, the black circle represents the object, the blue curves represent the neutral axis of the 4-link soft manipulator overlaid at one second intervals, and the small red circle represents the goal location of the end effecter. The center panel illustrates steps 4 and 5 of Algorithm 14 where the manipulator executes its primary task of moving the end effector to the goal. The neutral axes are overlaid at 0.5 second intervals. The right panel depicts the arm moving to the goal location in the absence of a nearby object, where a brace strategy is not feasible.

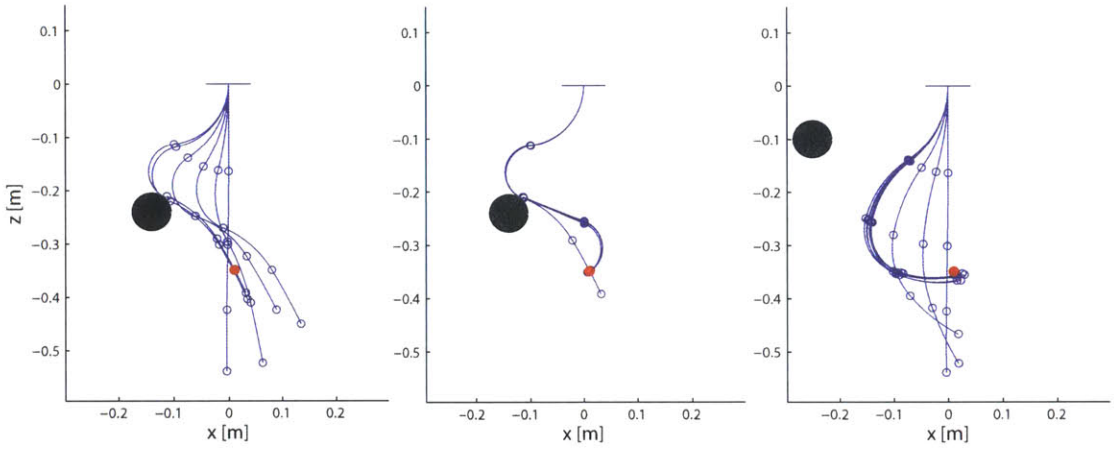


Figure A-4: Simulation of static bracing with a soft elastomer manipulator. Left panel: Steps 1 through 3 of Algorithm 14 are illustrated. Here, the black circle represents the object, the blue curves represent the neutral axis of the 4-link soft manipulator overlaid at one second intervals, and the small red circle represents the goal location of the end effecter. Center Panel: Steps 4 and 5 of Algorithm 14 are illustrated. Here, the manipulator executes its primary task of moving its end effector to the red goal location. The neutral axes are overlaid at 0.5 second intervals. Right Panel: A depiction of the arm moving to the goal location in the absence of a nearby object, where a brace strategy is not feasible.

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